

MatrixMultiplicationBias

January 23, 2026

1 Triton Kernel Profiling on AMD GPUs

1.1 Get the System Hardware Info

```
[1]: !amd-smi
```

```
+-----+
| AMD-SMI 26.0.2+39589fda      amdgpu version: 6.16.6   ROCm version: 7.0.3   |
| Platform: Linux Guest                                             |
+-----+
| BDF                        GPU-Name | Mem-Uti   Temp   UEC       Power-Usage |
| GPU  HIP-ID  OAM-ID  Partition-Mode | GFX-Uti   Fan    |           Mem-Usage |
+=====+=====+
| 0000:ff:00.0 AMD Instinct MI300X VF | 0 %       36 °C   0         150/750 W |
| 0      0      7      SPX/NPS1      | 0 %       N/A    |         285/196288 MB |
+-----+
+-----+
| Processes:                                                         |
| GPU      PID  Process Name          GTT_MEM  VRAM_MEM  MEM_USAGE  CU % |
+=====+=====+=====+=====+
| No running processes found                                         |
+-----+
```

1.2 Check profiling environment

```
[2]: !rocprof-compute --version
```

```
-----
rocprofiler-compute version: 3.2.3 (release)
Git revision:      1d722c14
-----
```

1.3 Triton MatMulBias Kernel Program

```
[3]: %%writefile triton/matmulbias.py
import argparse
import torch

import triton
```

```

import triton.language as tl
import triton.profiler as proton

def _matmul_launch_metadata(grid, kernel, args):
    ret = {}
    M, N, K, WS = args["M"], args["N"], args["K"], args.get("WARP_SPECIALIZE",
↪False)
    BM, BN, BK = args["BLOCK_SIZE_M"], args["BLOCK_SIZE_N"],
↪args["BLOCK_SIZE_K"]
    ws_str = "_ws" if WS else ""
    ret["name"] = f"{kernel.name}{ws_str} [M={M}, N={N}, K={K}] [BM={BM},
↪BN={BN} BK={BK}]"
    if "output_ptr" in args:
        bytes_per_elem = args["output_ptr"].element_size()
    else:
        bytes_per_elem = 2
    ret[f"flops{bytes_per_elem * 8}"] = 2. * M * N * K
    ret["bytes"] = bytes_per_elem * (M * K + N * K + M * N)
    return ret

def _bias_launch_metadata(grid, kernel, args):
    ret = {}
    M, N, WS = args["M"], args["N"], args.get("WARP_SPECIALIZE", False)
    BM, BN = args["BLOCK_SIZE_M"], args["BLOCK_SIZE_N"]
    ws_str = "_ws" if WS else ""
    ret["name"] = f"{kernel.name}{ws_str} [M={M}, N={N}] [BM={BM}, BN={BN}]"
    if "output_ptr" in args:
        bytes_per_elem = args["output_ptr"].element_size()
    else:
        bytes_per_elem = 2
    ret[f"flops{bytes_per_elem * 8}"] = 2. * M * N
    ret["bytes"] = bytes_per_elem * (M + N + M * N)
    return ret

def matmul_autotune_config(pre_hook=None):
    return [
        triton.Config({'BLOCK_SIZE_M': BM, 'BLOCK_SIZE_N': BN, "BLOCK_SIZE_K":
↪BK, "GROUP_SIZE_M": 8}, num_stages=s,
            num_warps=w, pre_hook=pre_hook)
        for BM in [128]
        for BN in [128, 256]
        for BK in [64, 128]
        for s in ([3, 4, 5])
        for w in [4, 8]
    ]

```

```

def bias_autotune_config(pre_hook=None):
    return [
        triton.Config({'BLOCK_SIZE_M': BM, 'BLOCK_SIZE_N': BN, "GROUP_SIZE_M": 8}, num_stages=s,
        ↪}, num_warps=w, pre_hook=pre_hook)
        for BM in [128]
        for BN in [128, 256]
        for s in ([3, 4, 5])
        for w in [4, 8]
    ]

# MatMul kernel
@triton.autotune(
    configs=matmul_autotune_config(),
    key=['M', 'N', 'K'],
)
@triton.jit(launch_metadata=_matmul_launch_metadata)
def matmul_kernel(
    a_ptr, b_ptr, c_ptr,
    M, N, K,
    stride_am, stride_ak,
    stride_bk, stride_bn,
    stride_cm, stride_cn,
    BLOCK_SIZE_M: tl.constexpr, BLOCK_SIZE_N: tl.constexpr, BLOCK_SIZE_K: 8,
    ↪tl.constexpr,
    GROUP_SIZE_M: tl.constexpr
):
    pid = tl.program_id(axis=0)
    num_pid_m = tl.cdiv(M, BLOCK_SIZE_M)
    num_pid_n = tl.cdiv(N, BLOCK_SIZE_N)
    num_pid_in_group = GROUP_SIZE_M * num_pid_n
    group_id = pid // num_pid_in_group
    first_pid_m = group_id * GROUP_SIZE_M
    group_size_m = min(num_pid_m - first_pid_m, GROUP_SIZE_M)
    pid_m = first_pid_m + ((pid % num_pid_in_group) % group_size_m)
    pid_n = (pid % num_pid_in_group) // group_size_m

    tl.assume(pid_m >= 0)
    tl.assume(pid_n >= 0)
    tl.assume(stride_am > 0)
    tl.assume(stride_ak > 0)
    tl.assume(stride_bn > 0)
    tl.assume(stride_bk > 0)
    tl.assume(stride_cm > 0)
    tl.assume(stride_cn > 0)

    offs_am = (pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)) % M

```

```

    offs_bn = (pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)) % N
    offs_k = tl.arange(0, BLOCK_SIZE_K)
    a_ptrs = a_ptr + (offs_am[:, None] * stride_am + offs_k[None, :] *
↪stride_ak)
    b_ptrs = b_ptr + (offs_k[:, None] * stride_bk + offs_bn[None, :] *
↪stride_bn)

    accumulator = tl.zeros((BLOCK_SIZE_M, BLOCK_SIZE_N), dtype=tl.float32)
    for k in range(0, tl.cdiv(K, BLOCK_SIZE_K)):
        a = tl.load(a_ptrs, mask=offs_k[None, :] < K - k * BLOCK_SIZE_K,
↪other=0.0)
        b = tl.load(b_ptrs, mask=offs_k[:, None] < K - k * BLOCK_SIZE_K,
↪other=0.0)
        accumulator = tl.dot(a, b, accumulator)
        a_ptrs += BLOCK_SIZE_K * stride_ak
        b_ptrs += BLOCK_SIZE_K * stride_bk
    c = accumulator.to(tl.float16)

    offs_cm = pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)
    offs_cn = pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)
    c_ptrs = c_ptr + stride_cm * offs_cm[:, None] + stride_cn * offs_cn[None, :]
    c_mask = (offs_cm[:, None] < M) & (offs_cn[None, :] < N)
    tl.store(c_ptrs, c, mask=c_mask)

# Bias kernel
@triton.autotune(
    configs=bias_autotune_config(),
    key=['M', 'K'],
)
@triton.jit(launch_metadata=_bias_launch_metadata)
def bias_kernel(
    a_ptr,
    bias_ptr,
    output_ptr,
    M, N,
    stride_m, stride_n,
    BLOCK_SIZE_M: tl.constexpr, BLOCK_SIZE_N: tl.constexpr, GROUP_SIZE_M:
↪tl.constexpr,
):
    pid = tl.program_id(axis=0)
    num_pid_m = tl.cdiv(M, BLOCK_SIZE_M)
    num_pid_n = tl.cdiv(N, BLOCK_SIZE_N)
    num_pid_in_group = GROUP_SIZE_M * num_pid_n
    group_id = pid // num_pid_in_group
    first_pid_m = group_id * GROUP_SIZE_M
    group_size_m = min(num_pid_m - first_pid_m, GROUP_SIZE_M)
    pid_m = first_pid_m + ((pid % num_pid_in_group) % group_size_m)

```

```

pid_n = (pid % num_pid_in_group) // group_size_m

offs_m = (pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)) % M
offs_n = (pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)) % N
a_ptrs = a_ptr + (offs_m[:, None] * stride_m + offs_n[None, :] * stride_n)

a = tl.load(a_ptrs)

bias = tl.load(bias_ptr + offs_n, mask=offs_n < N, other=0.0)

output = a + bias[None, :]

offs_m = pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)
offs_n = pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)
output_ptrs = output_ptr + stride_m * offs_m[:, None] + stride_n *
↪offs_n[None, :]
output_mask = (offs_m[:, None] < M) & (offs_n[None, :] < N)
tl.store(output_ptrs, output, mask=output_mask)

# MatMulBias Fusion kernel
@triton.autotune(
    configs=matmul_autotune_config(),
    key=['M', 'N', 'K'],
)
@triton.jit(launch_metadata=_matmul_launch_metadata)
def matmulbias_kernel(
    a_ptr, b_ptr, c_ptr,
    bias_ptr,
    M, N, K,
    stride_am, stride_ak,
    stride_bk, stride_bn,
    stride_cm, stride_cn,
    BLOCK_SIZE_M: tl.constexpr, BLOCK_SIZE_N: tl.constexpr, BLOCK_SIZE_K:
↪tl.constexpr,
    GROUP_SIZE_M: tl.constexpr
):
    pid = tl.program_id(axis=0)
    num_pid_m = tl.cdiv(M, BLOCK_SIZE_M)
    num_pid_n = tl.cdiv(N, BLOCK_SIZE_N)
    num_pid_in_group = GROUP_SIZE_M * num_pid_n
    group_id = pid // num_pid_in_group
    first_pid_m = group_id * GROUP_SIZE_M
    group_size_m = min(num_pid_m - first_pid_m, GROUP_SIZE_M)
    pid_m = first_pid_m + ((pid % num_pid_in_group) % group_size_m)
    pid_n = (pid % num_pid_in_group) // group_size_m

    tl.assume(pid_m >= 0)

```

```

tl.assume(pid_n >= 0)
tl.assume(stride_am > 0)
tl.assume(stride_ak > 0)
tl.assume(stride_bn > 0)
tl.assume(stride_bk > 0)
tl.assume(stride_cm > 0)
tl.assume(stride_cn > 0)

offs_am = (pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)) % M
offs_bn = (pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)) % N
offs_k = tl.arange(0, BLOCK_SIZE_K)
a_ptrs = a_ptr + (offs_am[:, None] * stride_am + offs_k[None, :] *
↪stride_ak)
b_ptrs = b_ptr + (offs_k[:, None] * stride_bk + offs_bn[None, :] *
↪stride_bn)

accumulator = tl.zeros((BLOCK_SIZE_M, BLOCK_SIZE_N), dtype=tl.float32)
for k in range(0, tl.cdiv(K, BLOCK_SIZE_K)):
    a = tl.load(a_ptrs, mask=offs_k[None, :] < K - k * BLOCK_SIZE_K,
↪other=0.0)
    b = tl.load(b_ptrs, mask=offs_k[:, None] < K - k * BLOCK_SIZE_K,
↪other=0.0)
    accumulator = tl.dot(a, b, accumulator)
    a_ptrs += BLOCK_SIZE_K * stride_ak
    b_ptrs += BLOCK_SIZE_K * stride_bk
c = accumulator.to(tl.float16)

bias = tl.load(bias_ptr + offs_bn, mask=offs_bn < N, other=0.0)
output = c + bias[None, :]

offs_cm = pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)
offs_cn = pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)
c_ptrs = c_ptr + stride_cm * offs_cm[:, None] + stride_cn * offs_cn[None, :]
c_mask = (offs_cm[:, None] < M) & (offs_cn[None, :] < N)
tl.store(c_ptrs, output, mask=c_mask)

# MatMulBias kernel wrapper function
def nonfusion_matmulbias(a: torch.Tensor, b: torch.Tensor, bias: torch.Tensor):
    assert a.shape[1] == b.shape[0], "Incompatible dimensions"
    assert a.is_contiguous(), "Matrix A must be contiguous"
    M, K = a.shape
    K, N = b.shape
    assert bias.shape[0] == N, "BIAS has incompatible dimensions"
    c = torch.empty((M, N), device=a.device, dtype=torch.float16)
    matmul_grid = lambda META: (triton.cdiv(M, META['BLOCK_SIZE_M']) * triton.
↪cdiv(N, META['BLOCK_SIZE_N']), )

```

```

matmul_kernel[matmul_grid](
    a, b, c,
    M, N, K,
    a.stride(0), a.stride(1),
    b.stride(0), b.stride(1),
    c.stride(0), c.stride(1),
)

o = torch.empty((M, N), device=c.device, dtype=torch.float16)
bias_grid = lambda META: (triton.cdiv(M, META['BLOCK_SIZE_M']) * triton.
↪cdiv(N, META['BLOCK_SIZE_N']), )
bias_kernel[bias_grid](
    c, bias, o,
    M, N,
    c.stride(0), c.stride(1),
)

return o

# MatMulBias kernel wrapper function
def fusion_matmulbias(a: torch.Tensor, b: torch.Tensor, bias: torch.Tensor):
    assert a.shape[1] == b.shape[0], "Incompatible dimensions"
    assert a.is_contiguous(), "Matrix A must be contiguous"
    M, K = a.shape
    K, N = b.shape
    assert bias.shape[0] == N, "BIAS has incompatible dimensions"
    o = torch.empty((M, N), device=a.device, dtype=torch.float16)
    grid = lambda META: (triton.cdiv(M, META['BLOCK_SIZE_M']) * triton.cdiv(N, ↪
↪META['BLOCK_SIZE_N']), )
    matmulbias_kernel[grid](
        a, b, o,
        bias,
        M, N, K,
        a.stride(0), a.stride(1),
        b.stride(0), b.stride(1),
        o.stride(0), o.stride(1),
    )

    return o

def get_rtol():
    target = triton.runtime.driver.active.get_current_target()
    if target.backend == "hip":
        if target.arch == "gfx90a":
            return 1e-2
        elif target.arch == "gfx942":

```

```

        return 1e-2
    else:
        return 0

def show_profile(profile_name):
    import triton.profiler.viewer as proton_viewer
    metric_names = ["time/ms", "tflop16/s"]
    file_name = f"profiles/{profile_name}.hatchet"
    tree, metrics = proton_viewer.parse(metric_names, file_name)

    print(f"Proton profile results for {profile_name}")
    proton_viewer.print_tree(tree, metrics)

if __name__ == "__main__":
    parser = argparse.ArgumentParser()
    parser.add_argument("--profile", action="store_true")
    parser.add_argument("--verify", action="store_true")
    parser.add_argument("--M", default=1024, type=int, help="M size of the
↪input matrices. Default: %(default)d")
    parser.add_argument("--N", default=1024, type=int, help="N size of the
↪input matrices. Default: %(default)d")
    parser.add_argument("kernel", choices=["fusion", "nonfusion"])
    args = parser.parse_args()

    # Test matrices
    torch.manual_seed(0)
    M = args.M
    N = args.N
    a = torch.randn((M, N), device='cuda', dtype=torch.float16)
    b = torch.randn((N, M), device='cuda', dtype=torch.float16)
    bias = torch.randn(N, device='cuda', dtype=torch.float16)

    if args.profile:
        print(f"Profiling the {args.kernel} matmulbias kernel (input size M:
↪{args.M} N:{args.N}) ...")
        proton.start(f"profiles/{args.kernel}_matmulbias_{args.M}x{args.N}",
↪hook="triton")
    else:
        print(f"Running the {args.kernel} matmulbias kernel (input size M:{args.
↪M} N:{args.N}) ...")

    if args.kernel == "fusion":
        triton_output = fusion_matmulbias(a, b, bias)
    elif args.kernel == "nonfusion":
        triton_output = nonfusion_matmulbias(a, b, bias)

```



```

if args.profile:
    proton.finalize()
print(f"triton_output: {triton_output}")

if args.profile:
    show_profile(f"{args.kernel}_matmulbias_{args.M}x{args.N}")

if args.verify:
    print(f"Running the torch matmulbias kernel ...")
    torch_output = torch.addmm(bias, a, b)
    print(f"torch_output: {torch_output}")
    print("Verifying triton results with torch ...")
    if torch.allclose(triton_output, torch_output, atol=1e-2,
↪rtol=get_rtol()):
        print("Verified")
    else:
        print("Differ")

```

Writing triton/matmulbias.py

1.4 Non-Fusion MatMulBias Kernel

1.4.1 Verify the Non-Fusion MatMulBias Kernel

This also ensures the kernel has been built and cached. The results from the triton kernel are compared against torch.

Small Input Default size is M: 1024 and N: 1024

```
[4]: !python triton/matmulbias.py --verify nonfusion
```

```

Running the nonfusion matmulbias kernel (input size M:1024 N:1024) ...
triton_output: tensor([[ 2.7480,  43.6250,  30.9531, ...,  26.2500,   8.8203,
  6.4609],
        [ 7.1562, -21.3906,  17.7969, ..., -16.8125,  29.6562, -48.3125],
        [ 13.8125, -6.4609,   0.5146, ...,  48.0000,  25.1562, -27.5156],
        ...,
        [-54.4375, -1.4375, -0.4214, ...,   5.4297,  17.2812, -39.0625],
        [ 9.6094, -52.2500,  43.0000, ..., -45.8438, -49.3125,  32.6875],
        [-12.8516,  9.5391,  53.3750, ..., -9.3438,   7.6797,  54.1250]],
        device='cuda:0', dtype=torch.float16)
Running the torch matmulbias kernel ...
torch_output: tensor([[ 2.7480,  43.6562,  30.9531, ...,  26.2500,   8.8203,
  6.4609],
        [ 7.1602, -21.3906,  17.7969, ..., -16.8125,  29.6406, -48.3125],
        [ 13.8125, -6.4609,   0.5142, ...,  48.0312,  25.1406, -27.5156],
        ...,
        [-54.4375, -1.4375, -0.4214, ...,   5.4297,  17.2812, -39.0625],
        [ 9.6094, -52.2500,  43.0000, ..., -45.8125, -49.3125,  32.7188],

```

```

        [-12.8516,  9.5391, 53.3750, ..., -9.3438,  7.6797, 54.1562]],
        device='cuda:0', dtype=torch.float16)
Verifying triton results with torch ...
Verified

```

Large Input

```
[5]: !python triton/matmulbias.py --M 32768 --N 32768 --verify nonfusion
```

```

Running the nonfusion matmulbias kernel (input size M:32768 N:32768) ...
triton_output: tensor([[ -60.8750,  56.6562,  57.0938, ..., 274.0000,
295.2500,
        14.8828],
[ 205.3750, -353.7500, -102.1875, ..., -24.4219,  57.0625,
-30.7969],
[  56.2188, 258.7500,  46.6562, ..., -366.0000, 395.2500,
285.2500],
...,
[  -5.0977,  99.9375, -73.5625, ..., -70.5000, 204.2500,
-156.8750],
[  86.8125, -26.3594, -128.8750, ..., -60.7500, -36.3750,
100.8750],
[ -27.6406,  93.5625,  77.9375, ..., -213.1250, -323.2500,
-16.8438]], device='cuda:0', dtype=torch.float16)

```

```

Running the torch matmulbias kernel ...
torch_output: tensor([[ -60.8750,  56.6562,  57.1250, ..., 274.0000,
295.2500,
        14.8828],
[ 205.3750, -353.7500, -102.1875, ..., -24.4219,  57.0312,
-30.7969],
[  56.2188, 258.7500,  46.6562, ..., -366.0000, 395.2500,
285.2500],
...,
[  -5.0977,  99.9375, -73.5625, ..., -70.5000, 204.2500,
-156.8750],
[  86.8125, -26.3594, -128.8750, ..., -60.7500, -36.3750,
100.8750],
[ -27.6406,  93.5625,  77.9375, ..., -213.0000, -323.2500,
-16.8438]], device='cuda:0', dtype=torch.float16)

```

```

Verifying triton results with torch ...
Verified

```

1.4.2 Profile the Non-Fusion MatMulBias Kernel

Using the Triton Proton CLI Generates a hatchet file, `nonfusion_matmulbias_x.hatchet`, under the profiles directory.

Small Input

```
[6]: !proton -n profiles/nonfusion_matmulbias_1024x1024 -k triton triton/matmulbias.
      ↪py nonfusion
```

Running the nonfusion matmulbias kernel (input size M:1024 N:1024) ...

```
triton_output: tensor([[ 2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203,
6.4609],
[ 7.1562, -21.3906, 17.7969, ..., -16.8125, 29.6562, -48.3125],
[ 13.8125, -6.4609, 0.5146, ..., 48.0000, 25.1562, -27.5156],
...,
[-54.4375, -1.4375, -0.4214, ..., 5.4297, 17.2812, -39.0625],
[ 9.6094, -52.2500, 43.0000, ..., -45.8438, -49.3125, 32.6875],
[-12.8516, 9.5391, 53.3750, ..., -9.3438, 7.6797, 54.1250]],
device='cuda:0', dtype=torch.float16)
```

Large Input

```
[7]: !proton -n profiles/nonfusion_matmulbias_32768x32768 -k triton triton/
      ↪matmulbias.py --M 32768 --N 32768 nonfusion
```

Running the nonfusion matmulbias kernel (input size M:32768 N:32768) ...

```
triton_output: tensor([[ -60.8750, 56.6562, 57.0938, ..., 274.0000,
295.2500,
14.8828],
[ 205.3750, -353.7500, -102.1875, ..., -24.4219, 57.0625,
-30.7969],
[ 56.2188, 258.7500, 46.6562, ..., -366.0000, 395.2500,
285.2500],
...,
[ -5.0977, 99.9375, -73.5625, ..., -70.5000, 204.2500,
-156.8750],
[ 86.8125, -26.3594, -128.8750, ..., -60.7500, -36.3750,
100.8750],
[ -27.6406, 93.5625, 77.9375, ..., -213.1250, -323.2500,
-16.8438]], device='cuda:0', dtype=torch.float16)
```

Using the Triton Proton API Generates a hatchet file, `nonfusion_matmulbias_x.hatchet`, under the profiles directory and outputs the profile.

Small Input

```
[8]: !python triton/matmulbias.py --profile nonfusion
```

Profiling the nonfusion matmulbias kernel (input size M:1024 N:1024) ...

```
triton_output: tensor([[ 2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203,
6.4609],
[ 7.1562, -21.3906, 17.7969, ..., -16.8125, 29.6562, -48.3125],
[ 13.8125, -6.4609, 0.5146, ..., 48.0000, 25.1562, -27.5156],
...,
[-54.4375, -1.4375, -0.4214, ..., 5.4297, 17.2812, -39.0625],
[ 9.6094, -52.2500, 43.0000, ..., -45.8438, -49.3125, 32.6875],
```

```

        [-12.8516,  9.5391, 53.3750, ..., -9.3438,  7.6797, 54.1250]],
        device='cuda:0', dtype=torch.float16)
Proton profile results for nonfusion_matmulbias_1024x1024
1020.102 3.647 ROOT
    889.389 nan _ZN2at6native29vectorized_elementwise_kern
elILi4ENS0_11FillFunctorIiEESt5arrayIPcLm1EEEEviT0_T1_
    44.652 0.584 bias_kernel [M=1024, N=1024] [BM=128,
BN=128]
    47.389 0.550 bias_kernel [M=1024, N=1024] [BM=128,
BN=256]
    38.672 94.846 matmul_kernel [M=1024, N=1024, K=1024]
[BM=128, BN=128 BK=64]

```

Legend (Metric: time/ms (inc) Min: 38.67 Max: 1020.10)

```

921.96 - 1020.10
725.67 - 921.96
529.39 - 725.67
333.10 - 529.39
136.82 - 333.10
38.67 - 136.82

```

```

name User code      Only in left graph
Only in right graph

```

Large Input

```
[9]: !python triton/matmulbias.py --M 32768 --N 32768 --profile nonfusion
```

```

Profiling the nonfusion matmulbias kernel (input size M:32768 N:32768) ...
triton_output: tensor([[ -60.8750,  56.6562,  57.0938, ..., 274.0000,
295.2500,

```

```

    14.8828],
    [ 205.3750, -353.7500, -102.1875, ..., -24.4219,  57.0625,
    -30.7969],
    [  56.2188, 258.7500,  46.6562, ..., -366.0000, 395.2500,
    285.2500],
    ...,
    [ -5.0977,  99.9375, -73.5625, ..., -70.5000, 204.2500,
    -156.8750],
    [  86.8125, -26.3594, -128.8750, ..., -60.7500, -36.3750,
    100.8750],
    [ -27.6406,  93.5625,  77.9375, ..., -213.1250, -323.2500,
    -16.8438]], device='cuda:0', dtype=torch.float16)

```

```

Proton profile results for nonfusion_matmulbias_32768x32768

```

```

5894.205 203.423 ROOT
    51.261 nan _ZN2at6native29vectorized_elementwise_kernel
ILi4ENS0_11FillFunctorIiEESt5arrayIPcLm1EEEEviT0_T1_
    806.546 1.725 bias_kernel [M=32768, N=32768] [BM=128,

```

```
BN=128]
  791.781 1.717 bias_kernel [M=32768, N=32768] [BM=128,
BN=256]
  4244.617 281.832 matmul_kernel [M=32768, N=32768,
K=32768] [BM=128, BN=128 BK=64]
```

Legend (Metric: time/ms (inc) Min: 51.26 Max: 5894.20)

5309.91 - 5894.20
4141.32 - 5309.91
2972.73 - 4141.32
1804.14 - 2972.73
635.56 - 1804.14
51.26 - 635.56

```
name User code      Only in left graph
Only in right graph
```

Using the ROCm Systems Compute profiler Generates files found under the workloads/nonfusion_matmulbias_x/ directory.

Small Input

```
[10]: !ROCPROF=rocprofiler-sdk rocprof-compute profile -n
      ↪ nonfusion_matmulbias_1024x1024 -- python triton/matmulbias.py nonfusion
```

[illegible]

WARNING DEPRECATION WARNING: rocm-smi is deprecated in ROCm 7.0 and will be removed from rocprof-compute in ROCm 7.1. Please migrate to amd-smi for compute partition parsing. For migration help, see <https://github.com/ROCm/amdsmi>

INFO Rocprofiler-Compute version: 3.2.3

INFO Profiler choice: rocprofiler-sdk

INFO Path:

```
/workspace/user/workloads/nonfusion_matmulbias_1024x1024/MI300X_A1
```

INFO Target: MI300X_A1

INFO Command: python triton/matmulbias.py nonfusion

INFO Kernel Selection: None

```

INFO Dispatch Selection: None
INFO Hardware Blocks: []
INFO Report Sections: []
INFO
INFO ~~~~~
INFO Collecting Performance Counters
INFO ~~~~~
INFO
INFO [Run 1/12][Approximate profiling time left: pending first
measurement...]
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/SQ_IFETCH_LEVEL.txt
INFO      |-> [rocprofiler-sdk] W20260123 18:22:30.998008
139907002773056 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.153518 sec
INFO      |-> [rocprofiler-sdk] W20260123 18:22:30.998111
139907002773056 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec
INFO      |-> [rocprofiler-sdk] W20260123 18:22:32.267460
139907002773056 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)
INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:22:35.866946
139907002773056 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.868836 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:22:35.881135
139907002773056 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/270_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:22:35.903108
139907002773056 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/270_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:22:37.213659
139907002773056 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/270_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:22:37.216754
139907002773056 simple_timer.cpp:55] [rocprofv3] output generation ::
1.343329 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:22:37.217819
139907002773056 simple_timer.cpp:55] [rocprofv3] tool finalization ::
1.344465 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 2/12] [Approximate profiling time left: 1 minute and 53
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/SQ_INST_LEVEL_LDS.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:22:42.535849
140211338866240 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.178143 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:22:42.535951
140211338866240 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:22:43.795095
140211338866240 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```



```

INFO      |-> [rocprofiler-sdk] W20260123 18:22:46.751547
140211338866240 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.215596 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:22:46.765200
140211338866240 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/291_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:22:46.786133
140211338866240 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/291_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:22:49.086826
140211338866240 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/291_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:22:49.089912
140211338866240 simple_timer.cpp:55] [rocprofv3] output generation ::
2.332705 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:22:49.091005
140211338866240 simple_timer.cpp:55] [rocprofv3] tool finalization ::
2.333867 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 3/12] [Approximate profiling time left: 1 minute and 57
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/SQ_INST_LEVEL_SMEM.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:22:57.282461
140254303350336 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.179786 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:22:57.282566
140254303350336 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:22:58.544083
140254303350336 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:23:01.505285
140254303350336 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.222719 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:23:01.518752
140254303350336 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/312_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:23:01.539752
140254303350336 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/312_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:23:03.836540
140254303350336 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/312_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:23:03.839473
140254303350336 simple_timer.cpp:55] [rocprofv3] output generation ::
2.328481 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:23:03.840548
140254303350336 simple_timer.cpp:55] [rocprofv3] tool finalization ::
2.329626 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 4/12] [Approximate profiling time left: 1 minute and 49
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/SQ_INST_LEVEL_VMEM.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:23:12.111645
140244264345152 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.178181 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:23:12.111743
140244264345152 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:23:13.375844
140244264345152 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:23:16.345620
140244264345152 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.233878 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:23:16.358971
140244264345152 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/334_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:23:16.379710
140244264345152 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/334_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:23:18.742447
140244264345152 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/334_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:23:18.745433
140244264345152 simple_timer.cpp:55] [rocprofv3] output generation ::
2.394238 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:23:18.746524
140244264345152 simple_timer.cpp:55] [rocprofv3] tool finalization ::
2.395398 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 5/12] [Approximate profiling time left: 1 minute and 37
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/SQ_LEVEL_WAVES.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:23:26.766869
140513993444928 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128837 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:23:26.766973
140513993444928 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:23:28.045536
140513993444928 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:23:30.807470
140513993444928 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.040497 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:23:30.820791
140513993444928 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/355_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:23:30.843283
140513993444928 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/355_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:23:30.979505
140513993444928 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/355_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:23:30.982481
140513993444928 simple_timer.cpp:55] [rocprofv3] output generation ::
0.169432 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:23:30.983465
140513993444928 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.170487 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 6/12] [Approximate profiling time left: 1 minute and 14
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_0.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:23:32.695695
139863071362624 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128177 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:23:32.695794
139863071362624 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:23:33.971908
139863071362624 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```



```

INFO      |-> [rocprofiler-sdk] W20260123 18:23:36.731651
139863071362624 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.035857 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:23:36.744980
139863071362624 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/376_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:23:36.767429
139863071362624 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/376_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:23:36.889578
139863071362624 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/376_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:23:36.892487
139863071362624 simple_timer.cpp:55] [rocprofv3] output generation ::
0.155210 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:23:36.893441
139863071362624 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.156244 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 7/12] [Approximate profiling time left: 56 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_1.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:23:38.553828
139743375322688 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128551 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:23:38.553925
139743375322688 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:23:39.808344
139743375322688 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:23:42.572753
139743375322688 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.018828 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:23:42.586219
139743375322688 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/396_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:23:42.608668
139743375322688 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/396_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:23:42.729098
139743375322688 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/396_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:23:42.732078
139743375322688 simple_timer.cpp:55] [rocprofv3] output generation ::
0.153695 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:23:42.733043
139743375322688 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.154733 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 8/12] [Approximate profiling time left: 42 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_2.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:23:44.398130
140202949160512 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128301 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:23:44.398228
140202949160512 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:23:45.661315
140202949160512 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:23:48.422014
140202949160512 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.023786 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:23:48.435256
140202949160512 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/417_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:23:48.457608
140202949160512 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/417_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:23:48.579003
140202949160512 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/417_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:23:48.581912
140202949160512 simple_timer.cpp:55] [rocprofv3] output generation ::
0.154406 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:23:48.582885
140202949160512 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.155444 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 9/12] [Approximate profiling time left: 29 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_3.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:23:50.249005
140586846129728 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128394 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:23:50.249105
140586846129728 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:23:51.507731
140586846129728 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:23:54.293289
140586846129728 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.044184 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:23:54.306597
140586846129728 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/437_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:23:54.332877
140586846129728 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/437_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:23:54.476509
140586846129728 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/437_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:23:54.479576
140586846129728 simple_timer.cpp:55] [rocprofv3] output generation ::
0.180696 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:23:54.480548
140586846129728 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.181737 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 10/12][Approximate profiling time left: 18 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_4.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:23:56.257027
140332043390528 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128374 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:23:56.257128
140332043390528 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:23:57.513302
140332043390528 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```



```

INFO      |-> [rocprofiler-sdk] W20260123 18:24:00.377682
140332043390528 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.120555 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:24:00.391165
140332043390528 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/458_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:24:00.427075
140332043390528 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/458_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:24:00.627608
140332043390528 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/458_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:24:00.630539
140332043390528 simple_timer.cpp:55] [rocprofv3] output generation ::
0.247089 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:24:00.631525
140332043390528 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.248145 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 11/12][Approximate profiling time left: 9 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_5.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:24:02.625624
139892060274240 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128336 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:24:02.625736
139892060274240 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:24:03.884588
139892060274240 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:24:06.756218
139892060274240 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.130482 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:24:06.769899
139892060274240 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/479_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:24:06.805909
139892060274240 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/479_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:24:07.012908
139892060274240 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/479_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:24:07.015863
139892060274240 simple_timer.cpp:55] [rocprofv3] output generation ::
0.253682 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:24:07.016819
139892060274240 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.254713 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 12/12][Approximate profiling time left: 0 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_6.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:24:09.057855
140714435472960 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128267 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:24:09.057951
140714435472960 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:24:10.314867
140714435472960 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:24:12.892320
140714435472960 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      3.834369 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:24:12.905970
140714435472960 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/499_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:24:12.951895
140714435472960 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/499_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:24:13.090954
140714435472960 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/499_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:24:13.094046
140714435472960 simple_timer.cpp:55] [rocprofv3] output generation ::
0.195802 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:24:13.095074
140714435472960 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.196900 sec

INFO      |-> [rocprofiler-sdk]
WARNING Detected differing Workgroup_Size values while joining
pmc_perf.csv
INFO [roofline] Checking for roofline.csv in
/workspace/user/workloads/nonfusion_matmulbias_1024x1024/MI300X_A1
INFO [roofline] No roofline data found. Generating...
Empirical Roofline Calculation
Copyright © 2025 Advanced Micro Devices, Inc. All rights reserved.
Total detected GPU devices: 1
GPU Device 0 (gfx942) with 304 CUs: Profiling..
 99% [||||| ]
HBM BW, GPU ID: 0, workgroupSize:256, workgroups:6225920, experiments:100,
traffic:25501368320 bytes, duration:6.1 ms, mean:4199.9 GB/sec, stdev=10.7
GB/sec
 99% [||||| ]
MALL BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,

```

traffic:2611340115968 bytes, duration:417.1 ms, mean:6245.3 GB/sec, stdev=29.7 GB/sec
 99% [|||||]
 L2 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
 traffic:1632087572480 bytes, duration:63.7 ms, mean:25550.4 GB/sec, stdev=187.0 GB/sec
 99% [|||||]
 L1 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
 traffic:127506841600 bytes, duration:3.8 ms, mean:34024.0 GB/sec, stdev=169.6 GB/sec
 99% [|||||]
 LDS BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
 traffic:79691776000 bytes, duration:1.3 ms, mean:61744.4 GB/sec, stdev=411.7 GB/sec
 99% [|||||]
 Peak FLOPs (FP8), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, FLOP:1081258016768, duration:13.0 ms, mean:82813.1 GFLOPS,
 stdev=1450.0 GFLOPS
 99% [|||||]
 Peak FLOPs (FP16), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, FLOP:530428461056, duration:13.4 ms, mean:39444.1 GFLOPS,
 stdev=647.7 GFLOPS
 99% [|||||]
 Peak FLOPs (BF16), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, FLOP:530428461056, duration:91.957 ms, mean:5755.9 GFLOPS,
 stdev=34.4 GFLOPS
 99% [|||||]
 Peak FLOPs (FP32), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, FLOP:265214230528, duration:2.031 ms, mean:130771.0 GFLOPS,
 stdev=1595.8 GFLOPS
 99% [|||||]
 Peak FLOPs (FP64), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, FLOP:122406567936, duration:1.1 ms, mean:105948.1 GFLOPS,
 stdev=1650.0 GFLOPS
 99% [|||||]
 Peak IOPs (INT8), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, IOP:1081258016768, duration:13.1 ms, mean:82549.4 GOPS,
 stdev=1392.7 GOPS
 99% [|||||]
 Peak IOPs (INT32), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, IOP:265214230528, duration:3.7 ms, mean:71750.3 GOPS,
 stdev=701.5 GOPS
 99% [|||||]
 Peak IOPs (INT64), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, IOP:122406567936, duration:6.2 ms, mean:19710.8 GOPS,
 stdev=58.1 GOPS
 100% [|||||]
 Peak MFMA FLOPs (F4), GPU ID: 0, workgroupSize:256, workgroups:38912,

```
INFO [roofline] Empirical Roofline PDFs saved!
```

```
[11]: !ROCPROF=rocprofiler-sdk rocprof-compute profile -n
      ↪ nonfusion_matmulbias_32768x32768 -- python triton/matmulbias.py --M 32768
      ↪ --N 32768 nonfusion
```

[illegible]

```

WARNING DEPRECATION WARNING: rocm-smi is deprecated in ROCm 7.0 and
will be removed from rocprof-compute in ROCm 7.1. Please migrate to amd-smi for
compute partition parsing. For migration help, see
https://github.com/ROCm/amdsmi
INFO Rocprofiler-Compute version: 3.2.3
INFO Profiler choice: rocprofiler-sdk
INFO Path:
/workspace/user/workloads/nonfusion_matmulbias_32768x32768/MI300X_A1
INFO Target: MI300X_A1
INFO Command: python triton/matmulbias.py --M 32768 --N 32768
nonfusion
INFO Kernel Selection: None
INFO Dispatch Selection: None
INFO Hardware Blocks: []
INFO Report Sections: []
INFO
INFO ~~~~~
INFO Collecting Performance Counters
INFO ~~~~~
INFO
INFO [Run 1/12][Approximate profiling time left: pending first
measurement...]
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/SQ_IFETCH_LEVEL.txt
INFO |-> [rocprofiler-sdk] W20260123 18:26:19.364755
139806363520576 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.152250 sec
INFO |-> [rocprofiler-sdk] W20260123 18:26:19.364858
139806363520576 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' :: 0.000000 sec
INFO |-> [rocprofiler-sdk] W20260123 18:26:20.622328
139806363520576 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)
INFO |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562, 57.0938, ..., 274.0000, 295.2500,
INFO |-> [rocprofiler-sdk] 14.8828],
INFO |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219, 57.0625,
INFO |-> [rocprofiler-sdk] -30.7969],
INFO |-> [rocprofiler-sdk] [ 56.2188, 258.7500, 46.6562,
..., -366.0000, 395.2500,
INFO |-> [rocprofiler-sdk] 285.2500],

```



```

INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000, 204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [ 86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)
INFO      |-> [rocprofiler-sdk] W20260123 18:26:27.624822
139806363520576 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.259964 sec
INFO      |-> [rocprofiler-sdk] E20260123 18:26:27.638309
139806363520576 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/608_kerne
l_trace.csv
INFO      |-> [rocprofiler-sdk] E20260123 18:26:27.655565
139806363520576 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/608_count
er_collection.csv
INFO      |-> [rocprofiler-sdk] E20260123 18:26:28.233839
139806363520576 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/608_agent
_info.csv
INFO      |-> [rocprofiler-sdk] W20260123 18:26:28.236385
139806363520576 simple_timer.cpp:55] [rocprofv3] output generation ::
0.605912 sec
INFO      |-> [rocprofiler-sdk] W20260123 18:26:28.237423
139806363520576 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.607019 sec
INFO      |-> [rocprofiler-sdk]
INFO [Run 2/12] [Approximate profiling time left: 2 minutes]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/SQ_INST_LEVEL_LDS.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:26:31.572510
140402663272000 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.179011 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:26:31.572618
140402663272000 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:26:32.836791
140402663272000 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:26:39.826195
140402663272000 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.253577 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:26:39.839445
140402663272000 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/630_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:26:39.855766
140402663272000 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/630_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:26:40.914392
140402663272000 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/630_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:26:40.917306
140402663272000 simple_timer.cpp:55] [rocprofv3] output generation ::
1.085618 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:26:40.918378
140402663272000 simple_timer.cpp:55] [rocprofv3] tool finalization ::
1.086754 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 3/12] [Approximate profiling time left: 1 minute and 56
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/SQ_INST_LEVEL_SMEM.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:26:45.468322
140525970857536 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.178676 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:26:45.468424
140525970857536 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:26:46.727164
140525970857536 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,   57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,   57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,   46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,   99.9375,  -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125,  -26.3594, -128.8750,
..., -60.7500,  -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,   93.5625,   77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:26:53.702123
140525970857536 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.233699 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:26:53.715457
140525970857536 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/655_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:26:53.731762
140525970857536 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/655_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:26:54.831434
140525970857536 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/655_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:26:54.834319
140525970857536 simple_timer.cpp:55] [rocprofv3] output generation ::
1.126685 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:26:54.835417
140525970857536 simple_timer.cpp:55] [rocprofv3] tool finalization ::
1.127851 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 4/12] [Approximate profiling time left: 1 minute and 47
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/SQ_INST_LEVEL_VMEM.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:26:59.688415
139838922124864 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.177227 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:26:59.688521
139838922124864 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:27:00.955030
139838922124864 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:27:07.930623
139838922124864 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.242102 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:27:07.943951
139838922124864 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/676_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:27:07.960224
139838922124864 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/676_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:27:09.048999
139838922124864 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/676_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:27:09.051878
139838922124864 simple_timer.cpp:55] [rocprofv3] output generation ::
1.115674 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:27:09.052941
139838922124864 simple_timer.cpp:55] [rocprofv3] tool finalization ::
1.116808 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 5/12] [Approximate profiling time left: 1 minute and 35
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/SQ_LEVEL_WAVES.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:27:13.585710
139685596655168 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128818 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:27:13.585820
139685596655168 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:27:14.852437
139685596655168 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,   57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,   57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,   46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,   99.9375,  -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125,  -26.3594, -128.8750,
..., -60.7500,  -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,   93.5625,   77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```



```

INFO      |-> [rocprofiler-sdk] W20260123 18:27:21.851853
139685596655168 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.266033 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:27:21.865167
139685596655168 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/698_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:27:21.881786
139685596655168 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/698_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:27:21.939282
139685596655168 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/698_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:27:21.942241
139685596655168 simple_timer.cpp:55] [rocprofv3] output generation ::
0.084837 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:27:21.943242
139685596655168 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.085905 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 6/12] [Approximate profiling time left: 1 minute and 17
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_0.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:27:23.405871
140418456559168 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.129673 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:27:23.405974
140418456559168 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:27:24.666766
140418456559168 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:27:31.662477
140418456559168 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.256503 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:27:31.675729
140418456559168 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/719_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:27:31.692313
140418456559168 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/719_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:27:31.744284
140418456559168 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/719_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:27:31.747193
140418456559168 simple_timer.cpp:55] [rocprofv3] output generation ::
0.079191 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:27:31.748158
140418456559168 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.080225 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 7/12] [Approximate profiling time left: 1 minute and 1
second]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_1.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:27:33.184666
140370771172928 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128226 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:27:33.184762
140370771172928 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:27:34.447544
140370771172928 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:27:41.391526
140370771172928 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.206764 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:27:41.404822
140370771172928 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/740_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:27:41.421328
140370771172928 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/740_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:27:41.472645
140370771172928 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/740_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:27:41.475561
140370771172928 simple_timer.cpp:55] [rocprofv3] output generation ::
0.078483 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:27:41.476551
140370771172928 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.079540 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 8/12] [Approximate profiling time left: 47 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_2.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:27:42.907074
140360106475072 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128492 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:27:42.907175
140360106475072 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:27:44.167451
140360106475072 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:27:51.167134
140360106475072 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.259959 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:27:51.180477
140360106475072 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/760_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:27:51.196930
140360106475072 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/760_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:27:51.250093
140360106475072 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/760_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:27:51.253072
140360106475072 simple_timer.cpp:55] [rocprofv3] output generation ::
0.080389 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:27:51.254098
140360106475072 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.081483 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 9/12] [Approximate profiling time left: 35 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_3.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:27:52.699435
140627465371200 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128393 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:27:52.699537
140627465371200 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:27:53.967932
140627465371200 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```



```

INFO      |-> [rocprofiler-sdk] W20260123 18:28:00.911602
140627465371200 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.212065 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:28:00.924908
140627465371200 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/781_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:28:00.941885
140627465371200 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/781_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:28:00.992279
140627465371200 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/781_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:28:00.995251
140627465371200 simple_timer.cpp:55] [rocprofv3] output generation ::
0.078250 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:28:00.996224
140627465371200 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.079289 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 10/12][Approximate profiling time left: 22 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_4.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:28:02.431063
140068441521728 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128361 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:28:02.431164
140068441521728 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:28:03.688830
140068441521728 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:28:10.660760
140068441521728 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.229596 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:28:10.674431
140068441521728 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/802_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:28:10.692303
140068441521728 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/802_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:28:10.740268
140068441521728 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/802_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:28:10.743323
140068441521728 simple_timer.cpp:55] [rocprofv3] output generation ::
0.077018 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:28:10.744298
140068441521728 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.078066 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 11/12][Approximate profiling time left: 11 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_5.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:28:12.176238
140423696068160 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128225 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:28:12.176339
140423696068160 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:28:13.435909
140423696068160 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:28:20.418894
140423696068160 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.242555 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:28:20.432166
140423696068160 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/823_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:28:20.449663
140423696068160 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/823_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:28:20.497603
140423696068160 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/823_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:28:20.500535
140423696068160 simple_timer.cpp:55] [rocprofv3] output generation ::
0.076145 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:28:20.501516
140423696068160 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.077194 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 12/12][Approximate profiling time left: 0 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_6.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:28:21.927991
140212855520832 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128256 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:28:21.928092
140212855520832 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:28:23.186071
140212855520832 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:28:30.165112
140212855520832 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 nonfusion' ::      8.237020 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:28:30.178510
140212855520832 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/844_kerne
l_trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:28:30.196669
140212855520832 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/844_count
er_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:28:30.223955
140212855520832 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/844_agent
_info.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:28:30.226997
140212855520832 simple_timer.cpp:55] [rocprofv3] output generation ::
0.056604 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:28:30.227955
140212855520832 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.057625 sec

INFO      |-> [rocprofiler-sdk]
WARNING Detected differing Workgroup_Size values while joining
pmc_perf.csv
INFO [roofline] Checking for roofline.csv in
/workspace/user/workloads/nonfusion_matmulbias_32768x32768/MI300X_A1
INFO [roofline] No roofline data found. Generating...
Empirical Roofline Calculation
Copyright © 2025 Advanced Micro Devices, Inc. All rights reserved.
Total detected GPU devices: 1
GPU Device 0 (gfx942) with 304 CUs: Profiling..
 99% [||||| ]
HBM BW, GPU ID: 0, workgroupSize:256, workgroups:6225920, experiments:100,
traffic:25501368320 bytes, duration:6.1 ms, mean:4164.4 GB/sec, stdev=16.0
GB/sec
 99% [||||| ]
MALL BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,

```

traffic:2611340115968 bytes, duration:423.8 ms, mean:6119.2 GB/sec, stdev=31.4 GB/sec
 99% [|||||]
 L2 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
 traffic:1632087572480 bytes, duration:64.1 ms, mean:25452.3 GB/sec, stdev=190.7 GB/sec
 99% [|||||]
 L1 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
 traffic:127506841600 bytes, duration:3.7 ms, mean:33652.4 GB/sec, stdev=361.1 GB/sec
 99% [|||||]
 LDS BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
 traffic:79691776000 bytes, duration:1.3 ms, mean:59641.7 GB/sec, stdev=186.6 GB/sec
 99% [|||||]
 Peak FLOPs (FP8), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, FLOP:1081258016768, duration:13.0 ms, mean:82750.0 GFLOPS,
 stdev=1443.1 GFLOPS
 99% [|||||]
 Peak FLOPs (FP16), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, FLOP:530428461056, duration:13.4 ms, mean:39448.9 GFLOPS,
 stdev=680.8 GFLOPS
 99% [|||||]
 Peak FLOPs (BF16), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, FLOP:530428461056, duration:92.048 ms, mean:5747.0 GFLOPS,
 stdev=34.9 GFLOPS
 99% [|||||]
 Peak FLOPs (FP32), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, FLOP:265214230528, duration:2.020 ms, mean:130554.4 GFLOPS,
 stdev=1366.4 GFLOPS
 99% [|||||]
 Peak FLOPs (FP64), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, FLOP:122406567936, duration:1.1 ms, mean:105985.9 GFLOPS,
 stdev=1564.5 GFLOPS
 99% [|||||]
 Peak IOPs (INT8), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, IOP:1081258016768, duration:13.1 ms, mean:82449.3 GOPS,
 stdev=1401.8 GOPS
 99% [|||||]
 Peak IOPs (INT32), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, IOP:265214230528, duration:3.7 ms, mean:71754.8 GOPS,
 stdev=561.5 GOPS
 99% [|||||]
 Peak IOPs (INT64), GPU ID: 0, workgroupSize:256, workgroups:38912,
 experiments:100, IOP:122406567936, duration:6.2 ms, mean:19692.9 GOPS,
 stdev=59.9 GOPS
 100% [|||||]
 Peak MFMA FLOPs (F4), GPU ID: 0, workgroupSize:256, workgroups:38912,


```

experiments:1, FLOP:0, duration:0.0 ms, mean:0.0 GFLOPS, stdev=0.0 GFLOPS
100% [|||||]
Peak MFMA FLOPs (F6), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:1, FLOP:0, duration:0.0 ms, mean:0.0 GFLOPS, stdev=0.0 GFLOPS
99% [|||||]
Peak MFMA FLOPs (F8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:10200547328000, duration:5.3 ms, mean:1916290.8 GFLOPS,
stdev=14275.8 GFLOPS
99% [|||||]
Peak MFMA FLOPs (F16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:5100273664000, duration:5.4 ms, mean:935847.2 GFLOPS,
stdev=34727.1 GFLOPS
99% [|||||]
Peak MFMA FLOPs (BF16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:2550136832000, duration:4.9 ms, mean:522179.2 GFLOPS,
stdev=2828.4 GFLOPS
99% [|||||]
Peak MFMA FLOPs (F32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:1275068416000, duration:9.1 ms, mean:139936.2 GFLOPS,
stdev=378.7 GFLOPS
99% [|||||]
Peak MFMA FLOPs (F64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:637534208000, duration:4.4 ms, mean:144787.5 GFLOPS,
stdev=5424.0 GFLOPS
99% [|||||]
Peak MFMA IOPs (I8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:5100273664000, duration:5.1 ms, mean:1000043.1 GOPS,
stdev=5525.1 GOPS

```

INFO [roofline] Empirical Roofline PDFs saved!

1.4.3 Analyze the Non-Fusion MatMulBias Kernel Profiles

Using the Triton Proton CLI

Small Input

```
[12]: !proton-viewer -m time/ms,flops16 -f full profiles/
      ↪nonfusion_matmulbias_1024x1024.hatchet
```

```

1020.102 3720049852416.000 ROOT
889.389 nan _ZN2at6native29vectorized_elementwise_kern
elILi4ENS0_11FillFunctorIiEESt5arrayIPCm1EEEEviT0_T1_
44.652 26073890816.000 bias_kernel [M=1024, N=1024]
[BM=128, BN=128]
47.389 26073890816.000 bias_kernel [M=1024, N=1024]
[BM=128, BN=256]
38.672 3667902070784.000 matmul_kernel [M=1024, N=1024,
K=1024] [BM=128, BN=128 BK=64]

```

Legend (Metric: time/ms (inc) Min: 38.67 Max: 1020.10)

921.96 - 1020.10
725.67 - 921.96
529.39 - 725.67
333.10 - 529.39
136.82 - 333.10
38.67 - 136.82

name User code Only in left graph
Only in right graph

Large Input

```
[13]: !proton-viewer -m time/ms,flops16 -f full profiles/  
      ↪nonfusion_matmulbias_32768x32768.hatchet
```

```
5894.205 1199019577573376.000 ROOT  
51.261 nan _ZN2at6native29vectorized_elementwise_kernel  
ILi4ENS0_11FillFunctorIiEESt5arrayIPcLm1EEEEviT0_T1_  
806.546 1391569403904.000 bias_kernel [M=32768,  
N=32768] [BM=128, BN=128]  
791.781 1359357149184.000 bias_kernel [M=32768,  
N=32768] [BM=128, BN=256]  
4244.617 1196268651020288.000 matmul_kernel [M=32768,  
N=32768, K=32768] [BM=128, BN=128 BK=64]
```

Legend (Metric: time/ms (inc) Min: 51.26 Max: 5894.20)

5309.91 - 5894.20
4141.32 - 5309.91
2972.73 - 4141.32
1804.14 - 2972.73
635.56 - 1804.14
51.26 - 635.56

name User code Only in left graph
Only in right graph

Using the ROCm Systems Compute analyzer

Small Input

```
[14]: !rocprof-compute analyze -p workloads/nonfusion_matmulbias_1024x1024/MI300X_A1_␣  
      ↪-k 0 1
```

----- / _ | ----- _ | _

```

---
|'___/ _ \ / __| ' \ | '___/ _ \ | | _____ / __/ _ \ | ' ^ _ \ | ' \ | | | | ___/ _
\
| | | ( ) | ( _ | | ) | | | ( ) | | _____ | ( | ( ) | | | | | | | | ) | | | | | |
___/
| | \___/ \___| .___/ | | \___/ | | \___\___/ | | | | | | .___/
\___, \___\___|
|_|
|_|

```

INFO Analysis mode = cli

INFO [analysis] deriving rocprofiler-compute metrics...

WARNING PC sampling: can not detect pc sampling method without /workspa
ce/user/workloads/nonfusion_matmulbias_1024x1024/MI300X_A1/ps_file_pc_sampling_h
ost_trap.csv

0. Top Stats

0.1 Top Kernels

Kernel_Name				Count	Sum(ns)
Mean(ns)	Median(ns)	Pct	S		
0	void at::native::vectorized_elementwise_	1418.00	60835235.38		
42902.14	42683.58	86.16	*		
kernel<4, at::native::FillFunctor<int>,					
std::array<char*, 1ul> >(int, at::nat...					
1	bias_kernel	1584.00	5633274.82		
3556.36	3374.42	7.98	*		
2	matmul_kernel	152.00	3772422.35		
24818.57	23769.00	5.34			
3	__amd_rocclr_copyBuffer	89.00	263248.50		
2957.85	2946.75	0.37			
4	void at::native::(anonymous namespace)::	3.00	19073.58		
6357.86	7360.08	0.03			
distribution_elementwise_grid_stride_ker					

```

    nel<float, 4, at::native::templates::...

5   void at::native::(anonymous namespace)::      7.00      16267.42
2323.92      2215.17    0.02
    CatArrayBatchedCopy_contig<at::native::(<br>
    anonymous namespace)::OpaqueType<2u>,...

6   void at::native::vectorized_elementwise_      3.00      10350.17
3450.06      3471.25    0.01
    kernel<4, at::native::compare_scalar_ker<br>
    nel<double>(at::TensorIteratorBase&, ...

7   void at::native::vectorized_elementwise_      2.00      6270.75
3135.38      3135.38    0.01
    kernel<8, at::native::AUnaryFunctor<c10:<br>
    :Half, c10::Half, bool, at::native::(...

8   void at::native::vectorized_elementwise_      2.00      5078.42
2539.21      2539.21    0.01
    kernel<8, at::native::AbsFunctor<c10::Ha<br>
    lf>, std::array<char*, 2ul> >(int, at...

9   void at::native::vectorized_elementwise_      2.00      5021.42
2510.71      2510.71    0.01
    kernel<8, at::native::BinaryFunctor<c10:<br>
    :Half, c10::Half, bool, at::native::(...

```

0.2 Dispatch List

Dispatch_ID Kernel_Name

GPU_ID

```
0          0 void at::native::(anonymous
namespace)::distribution_elementwise_grid_stride_ker      0
                        nel<float, 4,
at::native::templates::cuda::normal_and_transform<c10::Half, float
                        , at::CUDAGeneratorImpl*,
at::native::templates::cuda::normal_kernel<at::CUDAGen
                        eratorImpl*>(at::TensorBase const&, double, double,
at::CUDAGeneratorImpl*):...
```

```
1          1 void at::native::(anonymous
namespace)::distribution_elementwise_grid_stride_ker      0
                        nel<float, 4,
at::native::templates::cuda::normal_and_transform<c10::Half, float
                        , at::CUDAGeneratorImpl*,
at::native::templates::cuda::normal_kernel<at::CUDAGen
                        eratorImpl*>(at::TensorBase const&, double, double,
at::CUDAGeneratorImpl*):...
```

```
2          2 void at::native::(anonymous
namespace)::distribution_elementwise_grid_stride_ker      0
                        nel<float, 4,
at::native::templates::cuda::normal_and_transform<c10::Half, float
                        , at::CUDAGeneratorImpl*,
at::native::templates::cuda::normal_kernel<at::CUDAGen
                        eratorImpl*>(at::TensorBase const&, double, double,
at::CUDAGeneratorImpl*):...
```

```
3          3 matmul_kernel
0
```

```
4          4 void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,      0
                        std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
                        1>)
```

```
5          5 matmul_kernel
0
```

```

6          6 void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
          std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
          1>)

```

```

7          7 matmul_kernel
0

```

```

8          8 void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
          std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
          1>)

```

```

9          9 matmul_kernel
0

```

1. System Info

	Info
workload_name	nonfusion_matmulbias_1024x1024
command	python triton/matmulbias.py nonfusion
ip_blocks	SQ LDS SQC TA TD TCP TCC SPI CPC CPF roofline
timestamp	Fri Jan 23 18:22:27 2026 (UTC)
version	3
hostname	6b82f9811751
cpu_model	Intel(R) Xeon(R) Platinum 8470
sbios	SeaBIOS1.15.0-1

linux_distro	CentOS Stream 10 (Coughlan)
linux_kernel_version	5.15.0-164-generic
amd_gpu_kernel_version	nan
cpu_memory	230893524
gpu_memory	nan
rocm_version	7.0.3
vbios	113-M3000108-103
compute_partition	SPX
memory_partition	nan
gpu_series	MI300
gpu_model	MI300X_A1
gpu_arch	gfx942
gpu_chip_id	29877
gpu_l1	32
gpu_l2	4096
cu_per_gpu	304
simd_per_cu	4
se_per_gpu	32
wave_size	64
workgroup_max_size	1024
max_waves_per_cu	32
max_sclk	2100
max_mclk	1300
cur_sclk	2100

cur_mclk	1300
total_l2_chan	128
lds_banks_per_cu	32
sqc_per_gpu	160
pipes_per_gpu	4
num_xcd	8
num_hbm_channels	128

2. System Speed-of-Light

2.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit	Peak
Pct of Peak				
2.1.0	VALU FLOPs	102.01	Gflop/s	81715.2
0.12				
2.1.1	VALU IOPs	648.57	Giop/s	81715.2
0.79				
2.1.2	MFMA FLOPs (F8)	0.0	Gflop/s	
2614886.4	0.0			
2.1.3	MFMA FLOPs (BF16)	0.0	Gflop/s	
1307443.2	0.0			
2.1.4	MFMA FLOPs (F16)	0.0	Gflop/s	
1307443.2	0.0			
2.1.5	MFMA FLOPs (F32)	0.0	Gflop/s	
163430.4	0.0			

2.1.6 163430.4	MFMA FLOPs (F64) 0.0	0.0	Gflop/s	
2.1.7	MFMA FLOPs (F6F4)		Gflop/s	
2.1.8 2614886.4	MFMA IOPs (Int8) 0.0	0.0	Giop/s	
2.1.9 40.13	Active CUs	122.0	Cus	304.0
2.1.10 1.67	SALU Utilization	1.67	Pct	100.0
2.1.11 1.58	VALU Utilization	1.58	Pct	100.0
2.1.12 0.0	MFMA Utilization	0.0	Pct	100.0
2.1.13 0.42	VMEM Utilization	0.42	Pct	100.0
2.1.14 0.3	Branch Utilization	0.3	Pct	100.0
2.1.15 99.5	VALU Active Threads	63.68	Threads	64.0
2.1.16 5.5	IPC	0.28	Instr/cycle	5.0
2.1.17 75.33	Wavefront Occupancy	7328.32	Wavefronts	9728.0

2.1.18 0.01	Theoretical LDS Bandwidth	9.16	Gb/s	81715.2
2.1.19 3.12	LDS Bank Conflicts/Access	1.0	Conflicts/access	32.0
2.1.20 15.43	vL1D Cache Hit Rate	15.43	Pct	100.0
2.1.21 8.91	vL1D Cache BW	7278.12	Gb/s	81715.2
2.1.22 42.15	L2 Cache Hit Rate	42.15	Pct	100.0
2.1.23 20.1	L2 Cache BW	6915.71	Gb/s	34406.4
2.1.24 6.03	L2-Fabric Read BW	321.16	Gb/s	5324.8
2.1.25 61.61	L2-Fabric Write BW	3280.39	Gb/s	5324.8
2.1.26	L2-Fabric Read Latency	409.86	Cycles	
2.1.27	L2-Fabric Write Latency	380.04	Cycles	
2.1.28 99.74	sL1D Cache Hit Rate	99.74	Pct	100.0
2.1.29 0.22	sL1D Cache BW	46.33	Gb/s	21504.0

2.1.30	L1I Hit Rate	99.82	Pct	100.0
99.82				
2.1.31	L1I BW	487.22	Gb/s	21504.0
2.27				
2.1.32	L1I Fetch Latency		Cycles	

3. Memory Chart

3.1 Memory Chart

(Normalization: per_kernel)

Instr Buff	Instr Dispatch	Exec
LDS	L2 Cache	

SALU: 447223

			-----> Active
CUs	Req : 153	Util : 0 %	
		xGMI/PCIE	
Wave	0 Instr Buf		12
2: 304	<-----		

SMEM:

```

30956          Lat  :
N/A          Hit   :    42
%
-----
>
^
|
RVGPRseq:  36.0

      |  |
Wave N-1 Instr Buf      VALU:
374143      Vector L1 Cache
      Request          |
|
-----
>
|
v
SGPRs :    36

MFMA: 0
      Rd    : 78077 Hit   :    15 %      Rd    :
8717      Rd    : 11478

-----
>      <-----      <-----
-
Fabric

Wt    : 7993908 Lat   :    N/A      Wr    : 1998477 Wr    :
1998477      Rd    :
8991

VMEM: 126138      LDS Alloc:  -
----->      -----
>      <----- Latency (cycles)      Rd:
8986

----->
0      Atomic:    0 Coales:    25 %      Atomic:    0      Atomic:
0      Wr    : 1998477      <-----      HBM
      <
----->      <-----
>      ----->      Rd    :    410      Wr:
1998477

LDS: 153
Stall :    36
cycles      Atomic:    0      Wr    :
380      ----->

```

```

-----> Scratch Alloc:

-----> Atomic:      0
                                0

Wave Occupancy      GWS: 0
Scalar L1D
Cache      Latency (cycles)

60      -----
--->      Rd      :
76

Wave Life
      <-----
      ^

|

7316      BRANCH:
61912      Wavefronts:      Rd      : 30956 Hit      : 93.0 %      Wr
:      0      Rd      :      N/A      |
|
      ----->      0 <
-----
>      |
|

Lat      :      N/A      Atomic:      0      Wr      :
N/A      |
v
      ^
      <-----

>

|      Workgroups:

|
0      Instr L1 Cache
      GMI
|

|

```

| Fetch :
225509 Hit : 99 % Req
: 2679

----- <-----

Lat : 1048.0 cycles

4. Roofline

4.1 Roofline

139997.1
139997GFLOP/s

36222.0

9371.9

2424.8

@ @

627.4 61744 GB/s

34024 GB/s

162.3

42.0 4200 GB/s

0.01

0.18

3.16

56.23

1000.00

Performance (GFLOP/sec) Arithmetic Intensity (FLOP)s/Byte)

5. Command Processor (CPC/CPF)

5.1 Command Processor Fetcher

Metric_ID	Metric	Avg	Min	Max	Unit
5.1.0	CPF Utilization	100.00	100.00	100.00	Pct
5.1.1	CPF Stall	2.39	0.06	7.25	Pct
5.1.2	CPF-L2 Utilization	23.89	1.05	48.39	Pct
5.1.3	CPF-L2 Stall	0.00	0.00	0.00	Pct
5.1.4	CPF-UTCL1 Stall	2.39	0.06	7.25	Pct

5.2 Packet Processor

Metric_ID Unit	Metric	Avg	Min	Max	
5.2.0	CPC SYNC FIFO Full Rate				Pct
5.2.1	CPC CANE Stall Rate				Pct
5.2.2	CPC ADC Utilization				Pct
5.2.3	CPC Utilization	100.0	100.0	100.0	Pct
5.2.4	CPC Stall Rate	23.09	7.18	43.79	Pct
5.2.5	CPC Packet Decoding Utilization	53.2	23.81	104.99	Pct
5.2.6	CPC-Workgroup Manager Utilization	33.17	1.43	225.94	Pct
5.2.7	CPC-L2 Utilization	0.68	0.16	1.31	Pct
5.2.8	CPC-UTCL1 Stall	0.42	0.11	6.96	Pct
5.2.9	CPC-UTCL2 Utilization	0.4	0.1	4.86	Pct

6. Workgroup Manager (SPI)

6.1 Workgroup Manager Utilizations

Metric_ID	Metric	Avg	Min	Max	Unit
6.1.0	Schedule-Pipe Wave Occupancy				Wave
6.1.1	Accelerator Utilization	100.0	100.0	100.0	Pct
6.1.2	Scheduler-Pipe Utilization	0.0	0.0	0.0	Pct
6.1.3	Scheduler-Pipe Wave Utilization				Pct
6.1.4	Workgroup Manager Utilization	52.63	19.12	87.07	Pct
6.1.5	Shader Engine Utilization	47.38	12.69	83.36	Pct
6.1.6	SIMD Utilization	37.66	1.58	81.2	Pct
6.1.7 Workgroups	Dispatched Workgroups	0.0	0.0	0.0	
6.1.8 Wavefronts	Dispatched Wavefronts	0.0	0.0	0.0	
6.1.9	VGPR Writes				

Cycles/wave

6.1.10 SGPR Writes
Cycles/wave

6.2 Workgroup Manager - Resource Allocation

Metric_ID Unit	Metric	Avg	Min	Max
6.2.0 Pct	Not-scheduled Rate (Workgroup Manager)	0.0	0.0	0.0
6.2.1 Pct	Not-scheduled Rate (Scheduler-Pipe)	0.0	0.0	0.0
6.2.2 Pct	Scheduler-Pipe FIFO Full Rate			
6.2.3 Pct	Scheduler-Pipe Stall Rate	0.0	0.0	0.0
6.2.4 Pct	Scratch Stall Rate	0.0	0.0	0.0
6.2.5 Pct	Insufficient SIMD Waveslots	0.0	0.0	0.0
6.2.6 Pct	Insufficient SIMD VGPRs	0.0	0.0	0.0
6.2.7 Pct	Insufficient SIMD SGPRs	0.0	0.0	0.0
6.2.8 Pct	Insufficient CU LDS	0.0	0.0	0.0

6.2.9 Pct	Insufficient CU Barriers	0.0	0.0	0.0
6.2.10 Pct	Reached CU Workgroup Limit	0.0	0.0	0.0
6.2.11 Pct	Reached CU Wavefront Limit	0.0	0.0	0.0

7. Wavefront

7.1 Wavefront Launch Stats

Metric_ID	Metric	Avg	Min	Max	Unit
7.1.0 items	Grid Size	1990920.70	8192.00	4194304.00	Work
7.1.1 items	Workgroup Size	296.25	256.00	512.00	Work
7.1.2 Wavefronts	Total Wavefronts	0.00	0.00	0.00	
7.1.3 Wavefronts	Saved Wavefronts	0.00	0.00	0.00	
7.1.4 Wavefronts	Restored Wavefronts	0.00	0.00	0.00	
7.1.5 Registers	VGPRs	36.28	12.00	88.00	

7.1.6 Registers	AGPRs	1.89	0.00	4.00	
7.1.7 Registers	SGPRs	36.19	32.00	48.00	
7.1.8	LDS Allocation	0.00	0.00	0.00	Bytes
7.1.9 Bytes/workitem	Scratch Allocation	0.00	0.00	0.00	

7.2 Wavefront Runtime Stats

Metric_ID Max	Metric Unit		Avg	Min	
7.2.0 248644.92	Kernel Time (Nanosec) Ns		22141.41	2986.75	
7.2.1 135843.25	Kernel Time (Cycles) Cycle		61526.01	19622.38	
7.2.2 620.00	Instructions per wavefront Instr/wavefront		227.50	37.00	
7.2.3 921173328.00	Wave Cycles Cycles per kernel		316055607.05	709024.00	
7.2.4 202973988.00	Dependency Wait Cycles Cycles per kernel		71711023.04	291304.00	
7.2.5 711768872.00	Issue Wait Cycles Cycles per kernel		240405985.54	80192.00	

7.2.6	Active Cycles	4294225.01	310784.00
8650752.00	Cycles per kernel		

7.2.7	Wavefront Occupancy	7328.32	7.11
33580.72	Wavefronts		

10. Compute Units - Instruction Mix

10.1 Overall Instruction Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.1.0	VALU	374143.10	58240.00	720896.00	Instr per kernel
10.1.1	VMEM	126137.52	4224.00	262144.00	Instr per kernel
10.1.2	LDS	153.11	0.00	1152.00	Instr per kernel
10.1.3	MFMA	0.00	0.00	0.00	Instr per kernel
10.1.4	SALU	447222.83	12160.00	917504.00	Instr per kernel
10.1.5	SMEM	30956.05	0.00	65536.00	Instr per kernel
10.1.6	Branch	61912.09	0.00	131072.00	Instr per kernel

10.2 VALU Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.2.0	INT32	203424.83	31488.00	393216.00	Instr per kernel
10.2.1	INT64	30956.05	0.00	65536.00	Instr per kernel
10.2.2	F16-ADD	4322.49	0.00	8192.00	Instr per kernel

10.2.3	F16-MUL	0.00	0.00	0.00	Instr per kernel
10.2.4	F16-FMA	0.00	0.00	0.00	Instr per kernel
10.2.5	F16-Trans	0.00	0.00	0.00	Instr per kernel
10.2.6	F32-ADD	0.00	0.00	0.00	Instr per kernel
10.2.7	F32-MUL	608.36	0.00	2048.00	Instr per kernel
10.2.8	F32-FMA	0.00	0.00	0.00	Instr per kernel
10.2.9	F32-Trans	608.36	0.00	2048.00	Instr per kernel
10.2.10	F64-ADD	0.00	0.00	0.00	Instr per kernel
10.2.11	F64-MUL	0.00	0.00	0.00	Instr per kernel
10.2.12	F64-FMA	0.00	0.00	0.00	Instr per kernel
10.2.13	F64-Trans	0.00	0.00	0.00	Instr per kernel
10.2.14	Conversion	1216.72	0.00	4096.00	Instr per kernel

10.3 VMEM Instr Mix

Metric_ID Unit	Metric	Avg	Min	Max
10.3.0 Instr per kernel	Global/Generic Instr	123824.18	0.0	262144.0
10.3.1 Instr per kernel	Global/Generic Read	0.0	0.0	0.0
10.3.2 Instr per kernel	Global/Generic Write	123824.18	0.0	262144.0
10.3.3 Instr per kernel	Global/Generic Atomic	0.0	0.0	0.0
10.3.4 Instr per kernel	Spill/Stack Instr	2313.34	0.0	4608.0
10.3.5 Instr per kernel	Spill/Stack Coalesceable Instr			
10.3.6 Instr per kernel	Spill/Stack Read	1232.71	0.0	2560.0
10.3.7 Instr per kernel	Spill/Stack Write	1080.62	0.0	2048.0
10.3.8 Instr per kernel	Spill/Stack Atomic	0.0	0.0	0.0

10.4 MFMA Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
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10.4.0	MFMA-I8	0.0	0.0	0.0	Instr per kernel
10.4.1	MFMA-F8	0.0	0.0	0.0	Instr per kernel
10.4.2	MFMA-F16	0.0	0.0	0.0	Instr per kernel
10.4.3	MFMA-BF16	0.0	0.0	0.0	Instr per kernel
10.4.4	MFMA-F32	0.0	0.0	0.0	Instr per kernel
10.4.5	MFMA-F64	0.0	0.0	0.0	Instr per kernel
10.4.6	MFMA-F6F4				Instr per kernel

11. Compute Units - Compute Pipeline

11.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit	Peak	Pct of Peak
11.1.0	VALU FLOPs	102.01	Gflop	81715.2	0.12
11.1.1	VALU IOPs	648.57	Giop	81715.2	0.79
11.1.2	MFMA FLOPs (F8)	0.0	Gflop	2614886.4	0.0
11.1.3	MFMA FLOPs (BF16)	0.0	Gflop	1307443.2	0.0
11.1.4	MFMA FLOPs (F16)	0.0	Gflop	1307443.2	0.0
11.1.5	MFMA FLOPs (F32)	0.0	Gflop	163430.4	0.0

11.1.6	MFMA FLOPs (F64)	0.0	Gflop	163430.4	0.0
11.1.7	MFMA FLOPs (F6F4)		Gflop		
11.1.8	MFMA IOPs (INT8)	0.0	Giop	2614886.4	0.0

11.2 Pipeline Stats

Metric_ID	Metric	Avg	Min	Max	Unit
11.2.0	IPC	0.28	0.07	0.61	Instr/cycle
11.2.1	IPC (Issued)	0.98	0.95	1.0	Instr/cycle
11.2.2	SALU Utilization	1.67	0.14	3.2	Pct
11.2.3	VALU Utilization	1.58	0.63	2.35	Pct
11.2.4	VALU Co-Issue Efficiency				Pct
11.2.5	VMEM Utilization	0.42	0.04	0.85	Pct
11.2.6	Branch Utilization	0.3	0.0	0.64	Pct

11.2.7	VALU Active Threads	63.68	63.0	64.0	Threads
11.2.8	MFMA Utilization	0.0	0.0	0.0	Pct
11.2.9	MFMA Instr Cycles				Cycles/instr
11.2.10	VMEM Latency				Cycles
11.2.11	SMEM Latency				Cycles
11.3 Arithmetic Operations					
Metric_ID	Metric	Avg	Min	Max	Unit
11.3.0 kernel	FLOPs (Total)	354510.03	0.0	786432.0	Ops per
11.3.1 kernel	IOPs (Total)	15000376.11	2015232.0	29360128.0	Ops per
11.3.2 kernel	F8 OPs	0.0	0.0	0.0	Ops per
11.3.3 kernel	F16 OPs	276639.64	0.0	524288.0	Ops per
11.3.4 kernel	BF16 OPs	0.0	0.0	0.0	Ops per
11.3.5	F32 OPs	77870.39	0.0	262144.0	Ops per

kernel

11.3.6	F64 OPs	0.0	0.0	0.0	Ops per
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kernel

11.3.7	F6F4 OPs				Ops per
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kernel

11.3.8	INT8 OPs	0.0	0.0	0.0	Ops per
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kernel

12. Local Data Share (LDS)

12.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
12.1.0	Utilization	0.01	Pct of peak
12.1.1	Access Rate	0.00	Pct of peak
12.1.2	Theoretical Bandwidth (% of Peak)	0.01	Pct of peak
12.1.3	Bank Conflict Rate	3.12	Pct of peak

12.2 LDS Stats

Metric_ID	Metric	Avg	Min	Max	Unit
12.2.0 per kernel	LDS Instrs	153.11	0.0	1152.0	Instr
12.2.1 per kernel	LDS LOAD				Instr
12.2.2 per kernel	LDS STORE				Instr

12.2.3	LDS ATOMIC				Instr
per kernel					
12.2.4	LDS LOAD Bandwidth				Gbps
12.2.5	LDS STORE Bandwidth				Gbps
12.2.6	LDS ATOMIC Bandwidth				Gbps
12.2.7	Theoretical Bandwidth	39197.16	0.0	294912.0	Bytes
per kernel					
12.2.8	LDS Latency				
Cycles					
12.2.9	Bank Conflicts/Access	1.0	1.0	1.0	
Conflicts/access					
12.2.10	Index Accesses	612.46	0.0	4608.0	
Cycles per kernel					
12.2.11	Atomic Return Cycles	0.0	0.0	0.0	
Cycles per kernel					
12.2.12	Bank Conflict	306.23	0.0	2304.0	
Cycles per kernel					
12.2.13	Addr Conflict	0.0	0.0	0.0	
Cycles per kernel					
12.2.14	Unaligned Stall	0.0	0.0	0.0	
Cycles per kernel					

12.2.15	Mem Violations	0.0	0.0	0.0
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Accesses per kernel

12.2.16	LDS Command FIFO Full Rate
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Cycles per kernel

12.2.17	LDS Data FIFO Full Rate
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Cycles per kernel

13. Instruction Cache

13.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
13.1.0	Bandwidth	2.27	Pct of peak
13.1.1	Cache Hit Rate	98.68	Pct of peak
13.1.2	L1I-L2 Bandwidth	0.04	Pct of peak

13.2 Instruction Cache Accesses

Metric_ID	Metric	Avg	Min	Max
13.2.0	Req	225509.48	14080.0	458752.0
	Unit			
13.2.0	Req	225509.48	14080.0	458752.0
	per kernel			
13.2.1	Hits	223871.08	13766.0	455764.0
	Hits per kernel			
13.2.2	Misses - Non Duplicated	101.38	32.0	160.0
	Misses per kernel			
13.2.3	Misses - Duplicated	1556.78	192.0	6950.0

Misses per kernel

13.2.4	Cache Hit Rate	98.68	96.98	99.35	Pct
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13.2.5 Instruction Fetch Latency
Cycles

13.3 Instruction Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
13.3.0	L1I-L2 Bandwidth	171464.40	65536.00	245760.00	Bytes per kernel

14. Scalar L1 Data Cache

14.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
14.1.0	Bandwidth	0.22	Pct of peak
14.1.1	Cache Hit Rate	92.74	Pct of peak
14.1.2	sL1D-L2 BW	0.00	Pct of peak

14.2 Scalar L1D Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
14.2.0	Req	30956.05	0.00	65536.00	Req per kernel
14.2.1	Hits	28712.96	0.00	61117.00	Req per

kernel

14.2.2	Misses - Non Duplicated	75.58	0.00	160.00	Req per
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kernel

14.2.3	Misses- Duplicated	2172.96	0.00	4738.00	Req per
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kernel

14.2.4	Cache Hit Rate	92.74	92.55	93.68	Pct
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14.2.5	Read Req (Total)	30956.05	0.00	65536.00	Req per
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kernel

14.2.6	Atomic Req	0.00	0.00	0.00	Req per
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kernel

14.2.7	Read Req (1 DWord)	0.00	0.00	0.00	Req per
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kernel

14.2.8	Read Req (2 DWord)	0.00	0.00	0.00	Req per
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kernel

14.2.9	Read Req (4 DWord)	30956.05	0.00	65536.00	Req per
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kernel

14.2.10	Read Req (8 DWord)	0.00	0.00	0.00	Req per
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kernel

14.2.11	Read Req (16 DWord)	0.00	0.00	0.00	Req per
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kernel

14.3 Scalar L1D Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
14.3.0	sL1D-L2 BW	4836.88	0.00	10240.00	Bytes per kernel

14.3.1	Read Req	75.58	0.00	160.00	Req per kernel
14.3.2	Write Req	0.00	0.00	0.00	Req per kernel
14.3.3	Atomic Req	0.00	0.00	0.00	Req per kernel
14.3.4	Stall Cycles	0.00	0.00	0.00	Cycles per kernel

15. Address Processing Unit and Data Return Path (TA/TD)

15.1 Address Processing Unit

Metric_ID Max	Metric Unit		Avg	Min
15.1.0 81.42	Address Processing Unit Busy Pct		35.68	0.86
15.1.1 1.27	Address Stall Pct		0.12	0.0
15.1.2 68.58	Data Stall Pct		28.78	0.01
15.1.3 0.0	Data-Processor → Address Stall Pct		0.0	0.0
15.1.4	Sequencer → TA Address Stall Cycles per kernel			
15.1.5	Sequencer → TA Command Stall Cycles per kernel			
15.1.6	Sequencer → TA Data Stall Cycles per kernel			

15.1.7	Total Instructions	126137.52	4224.0
262144.0	Instructions per kernel		
15.1.8	Global/Generic Instructions	123824.18	0.0
262144.0	Instructions per kernel		
15.1.9	Global/Generic Read Instructions	0.0	0.0
0.0	Instructions per kernel		
15.1.10	Global/Generic Read Instructions for LDS		
	Instructions per kernel		
15.1.11	Global/Generic Write Instructions	123824.18	0.0
262144.0	Instructions per kernel		
15.1.12	Global/Generic Atomic Instructions	0.0	0.0
0.0	Instructions per kernel		
15.1.13	Spill/Stack Instructions	2313.34	0.0
4608.0	Instructions per kernel		
15.1.14	Spill/Stack Read Instructions	1232.71	0.0
2560.0	Instructions per kernel		
15.1.15	Spill/Stack Read Instructions for LDS		
	Instructions per kernel		
15.1.16	Spill/Stack Write Instructions	1080.62	0.0
2048.0	Instructions per kernel		
15.1.17	Spill/Stack Atomic Instructions	0.0	0.0
0.0	Instructions per kernel		
15.1.18	Spill/Stack Total Cycles	36809.25	0.0
73728.0	Cycles per kernel		

15.1.19	Spill/Stack Coalesced Read	68.05	0.0
512.0	Cycles per kernel		

15.1.20	Spill/Stack Coalesced Write	0.0	0.0
0.0	Cycles per kernel		

15.2 Data-Return Path

Metric_ID Unit	Metric	Avg	Min	Max
15.2.0 79.92	Data-Return Busy Pct	36.38	1.06	
15.2.1 79.25	Cache RAM → Data-Return Stall Pct	35.67	0.64	
15.2.2	Workgroup manager → Data-Return Stall Pct	0.0	0.0	0.0
15.2.3 128.0	Coalescable Instructions Instructions per kernel	17.01	0.0	
15.2.4 2560.0	Read Instructions Instructions per kernel	1232.71	0.0	
15.2.5 262144.0	Write Instructions Instructions per kernel	124904.8	2048.0	
15.2.6	Atomic Instructions Instructions per kernel	0.0	0.0	0.0
15.2.7	Write Ack Instructions Instructions per kernel			

INFO Not showing table with empty column(s): 16.6 L1D Addr
Translation Stalls

16. Vector L1 Data Cache

16.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
16.1.0	Hit rate	15.43	Pct of peak
16.1.1	Bandwidth	8.91	Pct of peak
16.1.2	Utilization	47.71	Pct of peak
16.1.3	Coalescing	25.08	Pct of peak

16.2 L1D Cache Stalls (%)

Metric_ID	Metric	Min	Q1	Median	Q3	Max
16.2.0 108.52	Stalled on L2 Data	1.49	3.97	5.58	93.22	
16.2.1 84.04	Stalled on L2 Req	0.0	0.0	0.0	76.81	
16.2.2	Stalled on Address					
16.2.3	Stalled on Data					
16.2.4	Stalled on Latency FIFO					
16.2.5	Stalled on Request FIFO					

16.2.6 Stalled on Read Return

16.2.7 Tag RAM Stall (Read) 0.0 0.0 0.0 0.0 0.0

16.2.8 Tag RAM Stall (Write) 0.0 0.0 0.0 0.0 0.0

16.2.9 Tag RAM Stall (Atomic) 0.0 0.0 0.0 0.0 0.0

16.3 L1D Cache Accesses

Metric_ID Unit	Metric	Avg	Min	Max
16.3.0 Req per kernel	Total Req	8071984.61	264192.0	16777216.0
16.3.1 Req per kernel	Read Req	78077.1	0.0	163840.0
16.3.2 Req per kernel	Write Req	7993907.51	131072.0	16777216.0
16.3.3 Req per kernel	Atomic Req	0.0	0.0	0.0
16.3.4 Bytes per kernel	Cache BW	258303507.44	8454144.0	536870912.0
16.3.5 Pct	Cache Hit Rate	15.43	0.0	33.16
16.3.6	Cache Accesses	2017996.15	66048.0	4194304.0

Req per kernel

16.3.7	Cache Hits	10802.69	0.0	24448.0
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Req per kernel

16.3.8	Invalidations	170.09	32.0	304.0
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Req per kernel

16.3.9	L1-L2 BW	129018242.98	4210688.0	268435456.0
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Bytes per kernel

16.3.10	Tag RAM 0 Req			
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Req per kernel

16.3.11	Tag RAM 1 Req			
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Req per kernel

16.3.12	Tag RAM 2 Req			
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Req per kernel

16.3.13	Tag RAM 3 Req			
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Req per kernel

16.3.14	L1-L2 Read	8716.58	0.0	16640.0
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Req per kernel

16.3.15	L1-L2 Write	1998476.88	32768.0	4194304.0
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Req per kernel

16.3.16	L1-L2 Atomic	0.0	0.0	0.0
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Req per kernel

16.3.17	L1 Access Latency			
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Cycles

16.3.18	L1-L2 Read Latency			
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Cycles

16.3.19 L1-L2 Write Latency
Cycles

16.4 L1D - L2 Transactions

Metric_ID	Metric	Xfer	Coherency	Avg	Min
Max Unit					
16.4.0	NC - Read	Read	NC	0.00	0.00
0.00	Req per kernel				
16.4.1	UC - Read	Read	UC	0.00	0.00
0.00	Req per kernel				
16.4.2	CC - Read	Read	CC	0.00	0.00
0.00	Req per kernel				
16.4.3	RW - Read	Read	RW	8717.43	0.00
16636.00	Req per kernel				
16.4.4	RW - Write	Write	RW	1998476.88	32768.00
4194304.00	Req per kernel				
16.4.5	NC - Write	Write	NC	0.00	0.00
0.00	Req per kernel				
16.4.6	UC - Write	Write	UC	0.00	0.00
0.00	Req per kernel				
16.4.7	CC - Write	Write	CC	0.00	0.00
0.00	Req per kernel				
16.4.8	NC - Atomic	Atomic	NC	0.00	0.00
0.00	Req per kernel				

16.4.9	UC - Atomic	Atomic	UC	0.00	0.00
0.00	Req per kernel				

16.4.10	CC - Atomic	Atomic	CC	0.00	0.00
0.00	Req per kernel				

16.4.11	RW - Atomic	Atomic	RW	0.00	0.00
0.00	Req per kernel				

16.5 L1D Addr Translation

Metric_ID	Metric	Avg	Min	Max
Units				
16.5.0	Req	2017996.15	66048.0	4194304.0
	Req per kernel			
16.5.1	Inflight Req			
	Req per kernel			
16.5.2	Hit Ratio	94.27	91.13	100.0
	Pct			
16.5.3	Hits	1853615.93	64512.0	3862979.0
	Req per kernel			
16.5.4	Translation Misses	7800.7	0.0	16384.0
	Req per kernel			
16.5.5	Misses under Translation Miss			
	Req per kernel			
16.5.6	Permission Misses	0.0	0.0	0.0
	Req per kernel			

INFO Not showing table with empty column(s): 17.4 L2 Cache Stalls

17. L2 Cache

17.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
17.1.0	Utilization	51.99	Pct
17.1.1	Bandwidth	20.10	Pct
17.1.2	Hit Rate	42.15	Pct
17.1.3	L2-Fabric Read BW	321.16	Gb/s
17.1.4	L2-Fabric Write and Atomic BW	3280.39	Gb/s
17.1.5	HBM Bandwidth	5324.80	Gb/s

17.2 L2 - Fabric Transactions

Metric_ID Max	Metric Unit	Avg	Min
17.2.0 2151424.0	Read BW Bytes per kernel	1133025.81	12800.0
17.2.1 108.0	HBM Read Traffic Pct	98.85	88.25
17.2.2 11.75	Remote Read Traffic Pct	1.37	0.0
17.2.3 117.96	Uncached Read Traffic Pct	4.79	0.0
17.2.4 268435456.0	Write and Atomic BW Bytes per kernel	127902520.11	2097152.0

17.2.5 100.0	HBM Write and Atomic Traffic Pct	100.0	100.0
17.2.6 0.0	Remote Write and Atomic Traffic Pct	0.0	0.0
17.2.7 0.0	Atomic Traffic Pct	0.0	0.0
17.2.8 0.0	Uncached Write and Atomic Traffic Pct	0.0	0.0
17.2.9 1594.12	Read Latency Cycles	409.86	242.22
17.2.10 561.55	Write and Atomic Latency Cycles	380.04	316.07
17.2.11 Cycles	Atomic Latency		
17.2.12 Pct	Read Stall		
17.2.13 Pct	Write Stall		

17.3 L2 Cache Accesses

Metric_ID Unit	Metric	Avg	Min	Max
17.3.0 Bytes per kernel	Bandwidth	257274897.06	6438912.0	537410048.0

17.3.1 Read Bandwidth
Bytes per kernel

17.3.2 Write Bandwidth
Bytes per kernel

17.3.3 Atomic Bandwidth
Bytes per kernel

17.3.4 Req	2009960.13	50304.0	4198516.0
Req per kernel			

17.3.5 Read Req	11478.39	4000.0	18800.0
Req per kernel			

17.3.6 Write Req	1998476.88	32768.0	4194304.0
Req per kernel			

17.3.7 Atomic Req	0.0	0.0	0.0
Req per kernel			

17.3.8 Streaming Req	0.0	0.0	0.0
Req per kernel			

17.3.9 Bypassss Req			
Req per kernel			

17.3.10 Probe Req	0.0	0.0	0.0
Req per kernel			

17.3.11 Input Buffer Req			
Req per kernel			

17.3.12 Cache Hit	42.15	34.35	50.05
Pct			

17.3.13	Hits	1001864.32	17280.0	2101048.0
Hits per kernel				
17.3.14	Misses	1008089.14	33024.0	2097694.0
Misses per kernel				
17.3.15	Writeback	999322.54	16496.0	2097208.0
Cachelines per kernel				
17.3.16	Writeback (Internal)	928681.37	0.0	1966091.0
Cachelines per kernel				
17.3.17	Writeback (vL1D Req)	70557.07	16384.0	131072.0
Cachelines per kernel				
17.3.18	Evict (Internal)	866822.03	0.0	1835120.0
Cachelines per kernel				
17.3.19	Evict (vL1D Req)	0.0	0.0	0.0
Cachelines per kernel				
17.3.20	NC Req	0.0	0.0	0.0
Req per kernel				
17.3.21	UC Req	2.68	0.0	198.0
Req per kernel				
17.3.22	CC Req	0.0	0.0	0.0
Req per kernel				
17.3.23	RW Req	2009948.17	50304.0	4198304.0
Req per kernel				

17.5 L2 - Fabric Interface Stalls

Metric_ID	Metric	Type
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Transaction	Avg	Min	Max	Unit		
17.5.0	Read - PCIe Stall Pct				PCIe Stall	Read
17.5.1	Read - Infinity Fabric Stall Pct				Infinity Fabric Stall	Read
17.5.2	Read - HBM Stall Pct				HBM Stall	Read
17.5.3	Write - PCIe Stall Pct				PCIe Stall	Write
17.5.4	Write - Infinity Fabric Stall Pct				Infinity Fabric Stall	Write
17.5.5	Write - HBM Stall Pct				HBM Stall	Write
17.5.6	Write - Credit Starvation 0.0 0.0 0.0 Pct				Credit Starvation	Write

17.6 L2 - Fabric Detailed Transaction Breakdown

Metric_ID	Metric	Avg	Min	Max
Unit				
17.6.0	Read (32B)	0.0	0.0	0.0
Req per kernel				
17.6.1	Read (64B)	278.47	200.0	
592.0	Req per kernel			
17.6.2	Read (128B)	8712.53	0.0	
16512.0	Req per kernel			

17.6.3	Read (Uncached)	17.79	0.0	
532.0	Req per kernel			
17.6.4	HBM Read	8986.22	200.0	
17096.0	Req per kernel			
17.6.5	Remote Read	5.61	0.0	
53.0	Req per kernel			
17.6.6	Read Bandwidth - PCIe			
	Bytes per kernel			
17.6.7	Read Bandwidth - Infinity Fabric			
	Bytes per kernel			
17.6.8	Read Bandwidth - HBM			
	Bytes per kernel			
17.6.9	Write and Atomic (32B)	0.0	0.0	0.0
	Req per kernel			
17.6.10	Write and Atomic (Uncached)	0.0	0.0	0.0
	Req per kernel			
17.6.11	Write and Atomic (64B)	1998476.88	32768.0	
4194304.0	Req per kernel			
17.6.12	HBM Write and Atomic	1998476.88	32768.0	
4194304.0	Req per kernel			
17.6.13	Remote Write and Atomic	0.0	0.0	0.0
	Req per kernel			
17.6.14	Write Bandwidth - PCIe			
	Bytes per kernel			

17.6.15 Write Bandwidth - Infinity Fabric
Bytes per kernel

17.6.16 Write Bandwidth - HBM
Bytes per kernel

17.6.17 Atomic 0.0 0.0 0.0
Req per kernel

17.6.18 Atomic - HBM
Req per kernel

17.6.19 Atomic Bandwidth - PCIe
Bytes per kernel

17.6.20 Atomic Bandwidth - Infinity Fabric
Bytes per kernel

17.6.21 Atomic Bandwidth - HBM
Bytes per kernel

INFO Not showing table with empty column(s): 18.9 L2-Fabric Read
Stall (Cycles per normUnit)

INFO Not showing table with empty column(s): 18.10 L2-Fabric Write
and Atomic Stall (Cycles per normUnit)

18. L2 Cache (per Channel)

18.1 Aggregate Stats (All channels)

Metric_ID	Metric	Avg	Std Dev	Min	Max	Unit
18.1.0	L2 Cache Hit Rate	42.11	7.51	34.33	50.05	Pct

18.2 L2 Cache Hit Rate (pct)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
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18.2.0	0.00	32.99	33.33	37.25	50.12	50.14
18.2.1	1.00	33.07	33.33	37.23	50.12	50.14
18.2.2	2.00	33.07	33.33	37.27	50.12	50.14
18.2.3	3.00	33.07	33.33	37.27	50.12	50.14
18.2.4	4.00	33.07	33.33	37.27	50.12	50.15
18.2.5	5.00	32.99	33.33	37.14	50.00	50.03
18.2.6	6.00	33.07	33.33	37.27	50.00	50.03
18.2.7	7.00	33.07	33.33	37.32	50.00	50.03
18.2.8	8.00	32.82	35.32	37.38	50.00	50.03
18.2.9	9.00	32.90	35.32	37.38	50.00	50.03
18.2.10	10.00	32.90	35.32	37.38	50.00	50.03
18.2.11	11.00	32.90	35.32	37.38	50.12	50.15
18.2.12	12.00	32.90	35.32	37.38	50.00	50.03
18.2.13	13.00	32.82	35.32	37.50	50.00	50.03
18.2.14	14.00	32.90	35.50	37.50	50.00	50.03
18.2.15	15.00	32.90	35.50	37.50	50.00	50.03
18.2.16	16.00	32.82	35.32	37.38	50.00	50.03
18.2.17	17.00	32.90	35.32	37.38	50.00	50.03
18.2.18	18.00	32.90	35.32	37.38	50.00	50.03
18.2.19	19.00	32.90	35.32	37.38	50.12	50.15
18.2.20	20.00	32.90	35.32	37.38	50.00	50.03
18.2.21	21.00	32.82	35.32	37.50	50.00	50.03
18.2.22	22.00	32.90	35.50	37.50	50.00	50.03
18.2.23	23.00	32.90	35.50	37.50	50.00	50.03

18.2.24	24.00	32.99	33.33	37.12	50.12	50.14
18.2.25	25.00	33.07	33.33	37.25	50.12	50.14
18.2.26	26.00	33.07	33.33	37.25	50.12	50.14
18.2.27	27.00	33.07	33.33	37.25	50.12	50.14
18.2.28	28.00	33.07	33.33	37.07	50.12	50.15
18.2.29	29.00	32.99	33.33	37.03	50.00	50.03
18.2.30	30.00	33.07	33.33	37.25	50.00	50.03
18.2.31	31.00	33.07	33.33	36.96	50.00	50.03
18.2.32	32.00	32.99	33.33	37.38	50.12	50.14
18.2.33	33.00	33.07	33.33	37.38	50.12	50.14
18.2.34	34.00	33.07	33.33	37.38	50.12	50.14
18.2.35	35.00	33.07	33.33	37.38	50.12	50.14
18.2.36	36.00	33.07	33.33	37.38	50.12	50.15
18.2.37	37.00	32.99	33.33	37.38	50.00	50.03
18.2.38	38.00	33.07	33.33	37.38	50.00	50.03
18.2.39	39.00	33.07	33.33	37.38	50.00	50.03
18.2.40	40.00	32.82	35.32	37.38	50.00	50.03
18.2.41	41.00	32.90	35.32	37.38	50.00	50.03
18.2.42	42.00	32.90	35.32	37.38	50.00	50.03
18.2.43	43.00	32.90	35.32	37.38	50.12	50.15
18.2.44	44.00	32.90	35.32	37.38	50.00	50.03
18.2.45	45.00	32.82	35.32	37.50	50.00	50.03
18.2.46	46.00	32.90	35.50	37.50	50.00	50.03
18.2.47	47.00	32.90	35.50	37.50	50.00	50.03

18.2.48	48.00	32.82	35.32	37.38	50.00	50.03
18.2.49	49.00	32.90	35.32	37.38	50.00	50.03
18.2.50	50.00	32.90	35.32	37.38	50.00	50.03
18.2.51	51.00	32.90	35.32	37.38	50.12	50.15
18.2.52	52.00	32.90	35.32	37.38	50.00	50.03
18.2.53	53.00	32.82	35.32	37.50	50.00	50.03
18.2.54	54.00	32.90	35.50	37.50	50.00	50.03
18.2.55	55.00	32.90	35.50	37.50	50.00	50.03
18.2.56	56.00	32.99	33.33	37.29	50.12	50.14
18.2.57	57.00	33.07	33.33	37.38	50.12	50.14
18.2.58	58.00	33.07	33.33	37.33	50.12	50.14
18.2.59	59.00	33.07	33.33	37.38	50.12	50.14
18.2.60	60.00	33.07	33.33	37.38	50.12	50.15
18.2.61	61.00	32.99	33.33	37.29	50.00	50.03
18.2.62	62.00	33.07	33.33	37.38	50.00	50.03
18.2.63	63.00	33.07	33.33	37.38	50.00	50.03
18.2.64	64.00	32.82	35.32	37.38	50.00	50.03
18.2.65	65.00	32.90	35.32	37.38	50.00	50.03
18.2.66	66.00	32.90	35.32	37.38	50.00	50.03
18.2.67	67.00	32.90	35.32	37.38	50.12	50.15
18.2.68	68.00	32.90	35.32	37.38	50.00	50.03
18.2.69	69.00	32.82	35.32	37.50	50.00	50.03
18.2.70	70.00	32.90	35.50	37.50	50.00	50.03
18.2.71	71.00	32.90	35.50	37.50	50.00	50.03

18.2.72	72.00	32.99	33.33	37.38	50.12	50.14
18.2.73	73.00	33.07	33.33	37.38	50.12	50.14
18.2.74	74.00	33.07	33.33	37.38	50.12	50.14
18.2.75	75.00	33.07	33.33	37.38	50.12	50.14
18.2.76	76.00	33.07	33.33	37.38	50.12	50.15
18.2.77	77.00	32.99	33.33	37.38	50.00	50.03
18.2.78	78.00	33.07	33.33	37.38	50.00	50.03
18.2.79	79.00	33.07	33.33	37.38	50.00	50.03
18.2.80	80.00	32.99	33.33	37.29	50.12	50.14
18.2.81	81.00	33.07	33.33	37.38	50.12	50.14
18.2.82	82.00	33.07	33.33	37.38	50.12	50.14
18.2.83	83.00	33.07	33.33	37.33	50.12	50.14
18.2.84	84.00	33.07	33.33	37.38	50.12	50.15
18.2.85	85.00	32.99	33.33	37.33	50.00	50.03
18.2.86	86.00	33.07	33.51	37.38	50.00	50.03
18.2.87	87.00	33.07	33.42	37.38	50.00	50.03
18.2.88	88.00	32.82	35.32	37.38	50.00	50.03
18.2.89	89.00	32.90	35.32	37.38	50.00	50.03
18.2.90	90.00	32.90	35.32	37.38	50.00	50.03
18.2.91	91.00	32.90	35.32	37.38	50.12	50.15
18.2.92	92.00	32.90	35.32	37.38	50.00	50.03
18.2.93	93.00	32.82	35.32	37.50	50.00	50.03
18.2.94	94.00	32.90	35.50	37.50	50.00	50.03
18.2.95	95.00	32.90	35.50	37.50	50.00	50.03

18.2.96	96.00	32.82	35.32	37.38	50.00	50.03
18.2.97	97.00	32.90	35.32	37.38	50.00	50.03
18.2.98	98.00	32.90	35.32	37.38	50.00	50.03
18.2.99	99.00	32.90	35.32	37.38	50.12	50.15
18.2.100	100.00	32.90	35.32	37.38	50.00	50.03
18.2.101	101.00	32.82	35.32	37.50	50.00	50.03
18.2.102	102.00	32.90	35.50	37.50	50.00	50.03
18.2.103	103.00	32.90	35.50	37.50	50.00	50.03
18.2.104	104.00	32.99	33.33	37.38	50.12	50.14
18.2.105	105.00	33.07	33.33	37.38	50.12	50.14
18.2.106	106.00	33.07	33.33	37.38	50.12	50.14
18.2.107	107.00	33.07	33.33	37.38	50.12	50.14
18.2.108	108.00	33.07	33.33	37.38	50.12	50.15
18.2.109	109.00	32.99	33.33	37.38	50.00	50.03
18.2.110	110.00	33.07	33.33	37.38	50.00	50.03
18.2.111	111.00	33.07	33.33	37.38	50.00	50.03
18.2.112	112.00	32.99	33.33	37.38	50.12	50.14
18.2.113	113.00	33.07	33.33	37.38	50.12	50.14
18.2.114	114.00	33.07	33.33	37.38	50.12	50.14
18.2.115	115.00	33.07	33.33	37.38	50.12	50.14
18.2.116	116.00	33.07	33.33	37.38	50.12	50.15
18.2.117	117.00	32.99	33.33	37.38	50.00	50.03
18.2.118	118.00	33.07	33.33	37.38	50.00	50.03
18.2.119	119.00	33.07	33.33	37.38	50.00	50.03

18.2.120	120.00	32.82	35.32	37.29	50.00	50.03
18.2.121	121.00	32.90	35.32	37.38	50.00	50.03
18.2.122	122.00	32.90	35.32	37.38	50.00	50.03
18.2.123	123.00	32.90	35.32	37.38	50.12	50.15
18.2.124	124.00	32.90	35.32	37.38	50.00	50.03
18.2.125	125.00	32.82	35.32	37.50	50.00	50.03
18.2.126	126.00	32.90	35.50	37.50	50.00	50.03
18.2.127	127.00	32.90	35.50	37.50	50.00	50.03

18.3 L2 Requests (per normUnit)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.3.0	0.00	384.00	400.00	412.00	32848.00	32870.00
18.3.1	1.00	384.00	400.00	412.00	32848.00	32869.00
18.3.2	2.00	384.00	400.00	412.00	32848.00	32869.00
18.3.3	3.00	384.00	400.00	412.00	32848.00	32869.00
18.3.4	4.00	384.00	400.00	412.00	32768.00	32789.00
18.3.5	5.00	384.00	400.00	412.00	32768.00	32790.00
18.3.6	6.00	384.00	400.00	412.00	32768.00	32789.00
18.3.7	7.00	384.00	400.00	416.00	32848.00	32869.00
18.3.8	8.00	386.00	402.00	418.00	32848.00	32870.00
18.3.9	9.00	386.00	402.00	418.00	32768.00	32789.00
18.3.10	10.00	386.00	402.00	414.00	32768.00	32789.00
18.3.11	11.00	386.00	402.00	414.00	32768.00	32790.00
18.3.12	12.00	386.00	402.00	414.00	32768.00	32789.00
18.3.13	13.00	386.00	402.00	414.00	32768.00	32789.00

18.3.14	14.00	386.00	402.00	414.00	32768.00	32789.00
18.3.15	15.00	386.00	402.00	414.00	32768.00	32789.00
18.3.16	16.00	386.00	402.00	418.00	32848.00	32870.00
18.3.17	17.00	386.00	402.00	418.00	32768.00	32789.00
18.3.18	18.00	386.00	402.00	410.00	32768.00	32789.00
18.3.19	19.00	386.00	402.00	410.50	32768.00	32790.00
18.3.20	20.00	386.00	402.00	411.00	32768.00	32789.00
18.3.21	21.00	386.00	402.00	411.00	32768.00	32789.00
18.3.22	22.00	386.00	402.00	411.00	32768.00	32789.00
18.3.23	23.00	386.00	402.00	411.00	32768.00	32789.00
18.3.24	24.00	384.00	400.00	412.00	32848.00	32870.00
18.3.25	25.00	384.00	400.00	412.00	32848.00	32869.00
18.3.26	26.00	384.00	400.00	412.00	32848.00	32869.00
18.3.27	27.00	384.00	400.00	412.00	32848.00	32869.00
18.3.28	28.00	384.00	400.00	412.00	32768.00	32789.00
18.3.29	29.00	384.00	400.00	412.00	32768.00	32790.00
18.3.30	30.00	384.00	400.00	412.00	32768.00	32789.00
18.3.31	31.00	384.00	400.00	416.00	32848.00	32869.00
18.3.32	32.00	384.00	400.00	412.00	32848.00	32870.00
18.3.33	33.00	384.00	400.00	412.00	32848.00	32869.00
18.3.34	34.00	384.00	400.00	412.00	32848.00	32869.00
18.3.35	35.00	384.00	400.00	412.00	32848.00	32869.00
18.3.36	36.00	384.00	400.00	412.00	32768.00	32789.00
18.3.37	37.00	384.00	400.00	412.00	32768.00	32790.00

18.3.38	38.00	384.00	400.00	412.00	32768.00	32789.00
18.3.39	39.00	384.00	400.00	416.00	32848.00	32869.00
18.3.40	40.00	386.00	402.00	418.00	32848.00	32870.00
18.3.41	41.00	386.00	402.00	418.00	32768.00	32789.00
18.3.42	42.00	386.00	402.00	410.00	32768.00	32789.00
18.3.43	43.00	386.00	402.00	411.00	32768.00	32790.00
18.3.44	44.00	386.00	402.00	411.00	32768.00	32789.00
18.3.45	45.00	386.00	402.00	411.00	32768.00	32789.00
18.3.46	46.00	386.00	402.00	411.00	32768.00	32789.00
18.3.47	47.00	386.00	402.00	410.50	32768.00	32789.00
18.3.48	48.00	386.00	402.00	418.00	32848.00	32870.00
18.3.49	49.00	386.00	402.00	418.00	32768.00	32789.00
18.3.50	50.00	386.00	402.00	410.00	32768.00	32789.00
18.3.51	51.00	386.00	402.00	411.00	32768.00	32790.00
18.3.52	52.00	386.00	402.00	412.00	32768.00	32789.00
18.3.53	53.00	386.00	402.00	412.00	32768.00	32789.00
18.3.54	54.00	386.00	402.00	411.00	32768.00	32789.00
18.3.55	55.00	386.00	402.00	412.00	32768.00	32789.00
18.3.56	56.00	384.00	400.00	413.00	32848.00	32870.00
18.3.57	57.00	384.00	400.00	412.00	32848.00	32869.00
18.3.58	58.00	384.00	400.00	412.00	32848.00	32869.00
18.3.59	59.00	384.00	400.00	412.00	32848.00	32869.00
18.3.60	60.00	384.00	400.00	412.00	32768.00	32789.00
18.3.61	61.00	384.00	400.00	412.50	32768.00	32790.00

18.3.62	62.00	384.00	400.00	412.00	32768.00	32789.00
18.3.63	63.00	384.00	400.00	416.00	32848.00	32869.00
18.3.64	64.00	386.00	402.00	418.00	32848.00	32870.00
18.3.65	65.00	386.00	402.00	418.00	32768.00	32789.00
18.3.66	66.00	386.00	402.00	411.00	32768.00	32789.00
18.3.67	67.00	386.00	402.00	411.00	32768.00	32790.00
18.3.68	68.00	386.00	402.00	412.00	32768.00	32789.00
18.3.69	69.00	386.00	402.00	412.00	32768.00	32789.00
18.3.70	70.00	386.00	402.00	412.00	32768.00	32789.00
18.3.71	71.00	386.00	402.00	412.00	32768.00	32789.00
18.3.72	72.00	384.00	400.00	412.00	32848.00	32870.00
18.3.73	73.00	384.00	400.00	412.00	32848.00	32869.00
18.3.74	74.00	384.00	400.00	412.00	32848.00	32869.00
18.3.75	75.00	384.00	400.00	412.00	32848.00	32869.00
18.3.76	76.00	384.00	400.00	412.00	32768.00	32789.00
18.3.77	77.00	384.00	400.00	412.00	32768.00	32790.00
18.3.78	78.00	384.00	400.00	412.00	32768.00	32789.00
18.3.79	79.00	384.00	400.00	416.00	32848.00	32869.00
18.3.80	80.00	384.00	400.00	412.00	32848.00	32870.00
18.3.81	81.00	384.00	400.00	412.00	32848.00	32869.00
18.3.82	82.00	384.00	400.00	412.00	32848.00	32869.00
18.3.83	83.00	384.00	400.00	412.00	32848.00	32869.00
18.3.84	84.00	384.00	400.00	412.00	32768.00	32789.00
18.3.85	85.00	384.00	400.00	412.00	32768.00	32790.00

18.3.86	86.00	384.00	400.00	412.00	32768.00	32789.00
18.3.87	87.00	384.00	400.00	416.00	32848.00	32869.00
18.3.88	88.00	386.00	402.00	418.00	32848.00	32870.00
18.3.89	89.00	386.00	402.00	418.00	32768.00	32789.00
18.3.90	90.00	386.00	402.00	414.00	32768.00	32789.00
18.3.91	91.00	386.00	402.00	414.00	32768.00	32790.00
18.3.92	92.00	386.00	402.00	414.00	32768.00	32789.00
18.3.93	93.00	386.00	402.00	414.00	32768.00	32789.00
18.3.94	94.00	386.00	402.00	414.00	32768.00	32789.00
18.3.95	95.00	386.00	402.00	414.00	32768.00	32789.00
18.3.96	96.00	386.00	402.00	418.00	32848.00	32870.00
18.3.97	97.00	386.00	402.00	418.00	32768.00	32789.00
18.3.98	98.00	386.00	402.00	414.00	32768.00	32789.00
18.3.99	99.00	386.00	402.00	414.00	32768.00	32790.00
18.3.100	100.00	386.00	402.00	414.00	32768.00	32789.00
18.3.101	101.00	386.00	402.00	414.00	32768.00	32789.00
18.3.102	102.00	386.00	402.00	414.00	32768.00	32789.00
18.3.103	103.00	386.00	402.00	414.00	32768.00	32789.00
18.3.104	104.00	384.00	400.00	412.00	32848.00	32870.00
18.3.105	105.00	384.00	400.00	412.00	32848.00	32869.00
18.3.106	106.00	384.00	400.00	412.00	32848.00	32869.00
18.3.107	107.00	384.00	400.00	412.00	32848.00	32869.00
18.3.108	108.00	384.00	400.00	412.00	32768.00	32789.00
18.3.109	109.00	384.00	400.00	412.00	32768.00	32790.00

18.3.110	110.00	384.00	400.00	412.00	32768.00	32789.00
18.3.111	111.00	384.00	400.00	416.00	32848.00	32869.00
18.3.112	112.00	384.00	400.00	412.00	32848.00	32870.00
18.3.113	113.00	384.00	400.00	412.00	32848.00	32869.00
18.3.114	114.00	384.00	400.00	412.00	32848.00	32869.00
18.3.115	115.00	384.00	400.00	412.00	32848.00	32869.00
18.3.116	116.00	384.00	400.00	412.00	32768.00	32789.00
18.3.117	117.00	384.00	400.00	412.00	32768.00	32790.00
18.3.118	118.00	384.00	400.00	412.00	32768.00	32789.00
18.3.119	119.00	384.00	400.00	416.00	32848.00	32869.00
18.3.120	120.00	386.00	402.00	418.00	32848.00	32870.00
18.3.121	121.00	386.00	402.00	418.00	32768.00	32789.00
18.3.122	122.00	386.00	402.00	414.00	32768.00	32789.00
18.3.123	123.00	386.00	402.00	414.00	32768.00	32790.00
18.3.124	124.00	386.00	402.00	414.00	32768.00	32789.00
18.3.125	125.00	386.00	402.00	414.00	32768.00	32789.00
18.3.126	126.00	386.00	402.00	414.00	32768.00	32789.00
18.3.127	127.00	386.00	402.00	414.00	32768.00	32789.00

18.4 L2 Requests (per normUnit)

Metric_ID	Channel	L2 Read	L2 Write	L2 Atomic
18.4.0	0.00	111.45	15613.10	0.00
18.4.1	1.00	111.44	15613.10	0.00
18.4.2	2.00	111.59	15613.10	0.00
18.4.3	3.00	111.44	15613.10	0.00

18.4.4	4.00	111.38	15613.10	0.00
18.4.5	5.00	73.79	15613.10	0.00
18.4.6	6.00	73.69	15613.10	0.00
18.4.7	7.00	73.59	15613.10	0.00
18.4.8	8.00	76.61	15613.10	0.00
18.4.9	9.00	76.60	15613.10	0.00
18.4.10	10.00	76.69	15613.10	0.00
18.4.11	11.00	114.48	15613.10	0.00
18.4.12	12.00	77.14	15613.10	0.00
18.4.13	13.00	77.32	15613.10	0.00
18.4.14	14.00	77.85	15613.10	0.00
18.4.15	15.00	77.96	15613.10	0.00
18.4.16	16.00	76.70	15613.10	0.00
18.4.17	17.00	76.69	15613.10	0.00
18.4.18	18.00	76.77	15613.10	0.00
18.4.19	19.00	114.56	15613.10	0.00
18.4.20	20.00	77.22	15613.10	0.00
18.4.21	21.00	77.40	15613.10	0.00
18.4.22	22.00	77.93	15613.10	0.00
18.4.23	23.00	78.05	15613.10	0.00
18.4.24	24.00	111.47	15613.10	0.00
18.4.25	25.00	111.46	15613.10	0.00
18.4.26	26.00	111.59	15613.10	0.00
18.4.27	27.00	111.44	15613.10	0.00

18.4.28	28.00	111.35	15613.10	0.00
18.4.29	29.00	73.78	15613.10	0.00
18.4.30	30.00	73.67	15613.10	0.00
18.4.31	31.00	73.60	15613.10	0.00
18.4.32	32.00	111.52	15613.10	0.00
18.4.33	33.00	111.52	15613.10	0.00
18.4.34	34.00	111.66	15613.10	0.00
18.4.35	35.00	111.51	15613.10	0.00
18.4.36	36.00	111.43	15613.10	0.00
18.4.37	37.00	73.83	15613.10	0.00
18.4.38	38.00	73.75	15613.10	0.00
18.4.39	39.00	73.64	15613.10	0.00
18.4.40	40.00	76.69	15613.10	0.00
18.4.41	41.00	76.68	15613.10	0.00
18.4.42	42.00	76.77	15613.10	0.00
18.4.43	43.00	114.55	15613.10	0.00
18.4.44	44.00	77.22	15613.10	0.00
18.4.45	45.00	77.39	15613.10	0.00
18.4.46	46.00	77.93	15613.10	0.00
18.4.47	47.00	78.04	15613.10	0.00
18.4.48	48.00	76.65	15613.10	0.00
18.4.49	49.00	76.64	15613.10	0.00
18.4.50	50.00	76.72	15613.10	0.00
18.4.51	51.00	114.51	15613.10	0.00

18.4.52	52.00	77.17	15613.10	0.00
18.4.53	53.00	77.35	15613.10	0.00
18.4.54	54.00	77.88	15613.10	0.00
18.4.55	55.00	78.00	15613.10	0.00
18.4.56	56.00	111.45	15613.10	0.00
18.4.57	57.00	111.45	15613.10	0.00
18.4.58	58.00	111.59	15613.10	0.00
18.4.59	59.00	111.45	15613.10	0.00
18.4.60	60.00	111.39	15613.10	0.00
18.4.61	61.00	73.82	15613.10	0.00
18.4.62	62.00	73.67	15613.10	0.00
18.4.63	63.00	73.60	15613.10	0.00
18.4.64	64.00	76.62	15613.10	0.00
18.4.65	65.00	76.61	15613.10	0.00
18.4.66	66.00	76.70	15613.10	0.00
18.4.67	67.00	114.49	15613.10	0.00
18.4.68	68.00	77.15	15613.10	0.00
18.4.69	69.00	77.32	15613.10	0.00
18.4.70	70.00	77.86	15613.10	0.00
18.4.71	71.00	77.97	15613.10	0.00
18.4.72	72.00	111.52	15613.10	0.00
18.4.73	73.00	111.53	15613.10	0.00
18.4.74	74.00	111.66	15613.10	0.00
18.4.75	75.00	111.51	15613.10	0.00

18.4.76	76.00	111.43	15613.10	0.00
18.4.77	77.00	73.87	15613.10	0.00
18.4.78	78.00	73.75	15613.10	0.00
18.4.79	79.00	73.67	15613.10	0.00
18.4.80	80.00	111.48	15613.10	0.00
18.4.81	81.00	111.48	15613.10	0.00
18.4.82	82.00	111.62	15613.10	0.00
18.4.83	83.00	111.47	15613.10	0.00
18.4.84	84.00	111.41	15613.10	0.00
18.4.85	85.00	73.83	15613.10	0.00
18.4.86	86.00	73.74	15613.10	0.00
18.4.87	87.00	73.65	15613.10	0.00
18.4.88	88.00	76.54	15613.10	0.00
18.4.89	89.00	76.54	15613.10	0.00
18.4.90	90.00	76.62	15613.10	0.00
18.4.91	91.00	114.41	15613.10	0.00
18.4.92	92.00	77.07	15613.10	0.00
18.4.93	93.00	77.25	15613.10	0.00
18.4.94	94.00	77.78	15613.10	0.00
18.4.95	95.00	77.89	15613.10	0.00
18.4.96	96.00	76.74	15613.10	0.00
18.4.97	97.00	76.74	15613.10	0.00
18.4.98	98.00	76.82	15613.10	0.00
18.4.99	99.00	114.61	15613.10	0.00

18.4.100	100.00	77.27	15613.10	0.00
18.4.101	101.00	77.45	15613.10	0.00
18.4.102	102.00	77.98	15613.10	0.00
18.4.103	103.00	78.09	15613.10	0.00
18.4.104	104.00	111.52	15613.10	0.00
18.4.105	105.00	111.51	15613.10	0.00
18.4.106	106.00	111.67	15613.10	0.00
18.4.107	107.00	111.52	15613.10	0.00
18.4.108	108.00	111.45	15613.10	0.00
18.4.109	109.00	73.86	15613.10	0.00
18.4.110	110.00	73.75	15613.10	0.00
18.4.111	111.00	73.66	15613.10	0.00
18.4.112	112.00	111.57	15613.10	0.00
18.4.113	113.00	111.57	15613.10	0.00
18.4.114	114.00	111.70	15613.10	0.00
18.4.115	115.00	111.56	15613.10	0.00
18.4.116	116.00	111.48	15613.10	0.00
18.4.117	117.00	73.89	15613.10	0.00
18.4.118	118.00	73.80	15613.10	0.00
18.4.119	119.00	73.71	15613.10	0.00
18.4.120	120.00	76.60	15613.10	0.00
18.4.121	121.00	76.59	15613.10	0.00
18.4.122	122.00	76.68	15613.10	0.00
18.4.123	123.00	114.47	15613.10	0.00

18.4.124	124.00	77.13	15613.10	0.00
18.4.125	125.00	77.31	15613.10	0.00
18.4.126	126.00	77.84	15613.10	0.00
18.4.127	127.00	77.95	15613.10	0.00

18.5 L2-Fabric Requests (per normUnit)

Metric_ID L2-Fabric Atomic	Channel	L2-Fabric Read	L2-Fabric Write and Atomic
18.5.0 0.00	0.00	71.77	15613.10
18.5.1 0.00	1.00	71.43	15613.10
18.5.2 0.00	2.00	71.26	15613.10
18.5.3 0.00	3.00	71.27	15613.10
18.5.4 0.00	4.00	69.39	15613.10
18.5.5 0.00	5.00	69.38	15613.10
18.5.6 0.00	6.00	69.69	15613.10
18.5.7 0.00	7.00	71.57	15613.10
18.5.8	8.00	71.46	15613.10

0.00

18.5.9	9.00	69.56	15613.10
0.00			

18.5.10	10.00	69.91	15613.10
0.00			

18.5.11	11.00	69.57	15613.10
0.00			

18.5.12	12.00	69.58	15613.10
0.00			

18.5.13	13.00	69.56	15613.10
0.00			

18.5.14	14.00	69.68	15613.10
0.00			

18.5.15	15.00	69.56	15613.10
0.00			

18.5.16	16.00	71.46	15613.10
0.00			

18.5.17	17.00	69.56	15613.10
0.00			

18.5.18	18.00	69.91	15613.10
0.00			

18.5.19	19.00	69.57	15613.10
0.00			

18.5.20	20.00	69.58	15613.10
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0.00

18.5.21	21.00	69.56	15613.10
0.00			

18.5.22	22.00	69.68	15613.10
0.00			

18.5.23	23.00	69.56	15613.10
0.00			

18.5.24	24.00	71.77	15613.10
0.00			

18.5.25	25.00	71.43	15613.10
0.00			

18.5.26	26.00	71.26	15613.10
0.00			

18.5.27	27.00	71.27	15613.10
0.00			

18.5.28	28.00	69.39	15613.10
0.00			

18.5.29	29.00	69.38	15613.10
0.00			

18.5.30	30.00	69.69	15613.10
0.00			

18.5.31	31.00	71.57	15613.10
0.00			

18.5.32	32.00	71.77	15613.10
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0.00

18.5.33	33.00	71.43	15613.10
0.00			

18.5.34	34.00	71.26	15613.10
0.00			

18.5.35	35.00	71.27	15613.10
0.00			

18.5.36	36.00	69.39	15613.10
0.00			

18.5.37	37.00	69.38	15613.10
0.00			

18.5.38	38.00	69.69	15613.10
0.00			

18.5.39	39.00	71.57	15613.10
0.00			

18.5.40	40.00	71.46	15613.10
0.00			

18.5.41	41.00	69.56	15613.10
0.00			

18.5.42	42.00	69.91	15613.10
0.00			

18.5.43	43.00	69.57	15613.10
0.00			

18.5.44	44.00	69.58	15613.10
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0.00

18.5.45	45.00	69.56	15613.10
0.00			

18.5.46	46.00	69.68	15613.10
0.00			

18.5.47	47.00	69.56	15613.10
0.00			

18.5.48	48.00	71.46	15613.10
0.00			

18.5.49	49.00	69.56	15613.10
0.00			

18.5.50	50.00	69.91	15613.10
0.00			

18.5.51	51.00	69.57	15613.10
0.00			

18.5.52	52.00	69.58	15613.10
0.00			

18.5.53	53.00	69.56	15613.10
0.00			

18.5.54	54.00	69.68	15613.10
0.00			

18.5.55	55.00	69.56	15613.10
0.00			

18.5.56	56.00	71.77	15613.10
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0.00

18.5.57	57.00	71.43	15613.10
0.00			

18.5.58	58.00	71.26	15613.10
0.00			

18.5.59	59.00	71.27	15613.10
0.00			

18.5.60	60.00	69.39	15613.10
0.00			

18.5.61	61.00	69.38	15613.10
0.00			

18.5.62	62.00	69.69	15613.10
0.00			

18.5.63	63.00	71.57	15613.10
0.00			

18.5.64	64.00	71.46	15613.10
0.00			

18.5.65	65.00	69.56	15613.10
0.00			

18.5.66	66.00	69.91	15613.10
0.00			

18.5.67	67.00	69.57	15613.10
0.00			

18.5.68	68.00	69.58	15613.10
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0.00

18.5.69	69.00	69.56	15613.10
0.00			

18.5.70	70.00	69.68	15613.10
0.00			

18.5.71	71.00	69.56	15613.10
0.00			

18.5.72	72.00	71.77	15613.10
0.00			

18.5.73	73.00	71.43	15613.10
0.00			

18.5.74	74.00	71.26	15613.10
0.00			

18.5.75	75.00	71.27	15613.10
0.00			

18.5.76	76.00	69.39	15613.10
0.00			

18.5.77	77.00	69.38	15613.10
0.00			

18.5.78	78.00	69.69	15613.10
0.00			

18.5.79	79.00	71.57	15613.10
0.00			

18.5.80	80.00	71.77	15613.10
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0.00

18.5.81	81.00	71.43	15613.10
0.00			

18.5.82	82.00	71.26	15613.10
0.00			

18.5.83	83.00	71.27	15613.10
0.00			

18.5.84	84.00	69.39	15613.10
0.00			

18.5.85	85.00	69.38	15613.10
0.00			

18.5.86	86.00	69.69	15613.10
0.00			

18.5.87	87.00	71.57	15613.10
0.00			

18.5.88	88.00	71.46	15613.10
0.00			

18.5.89	89.00	69.56	15613.10
0.00			

18.5.90	90.00	69.91	15613.10
0.00			

18.5.91	91.00	69.57	15613.10
0.00			

18.5.92	92.00	69.58	15613.10
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0.00

18.5.93	93.00	69.56	15613.10
0.00			

18.5.94	94.00	69.68	15613.10
0.00			

18.5.95	95.00	69.56	15613.10
0.00			

18.5.96	96.00	71.46	15613.10
0.00			

18.5.97	97.00	69.56	15613.10
0.00			

18.5.98	98.00	69.91	15613.10
0.00			

18.5.99	99.00	69.57	15613.10
0.00			

18.5.100	100.00	69.58	15613.10
0.00			

18.5.101	101.00	69.56	15613.10
0.00			

18.5.102	102.00	69.68	15613.10
0.00			

18.5.103	103.00	69.56	15613.10
0.00			

18.5.104	104.00	71.77	15613.10
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0.00

18.5.105	105.00	71.43	15613.10
0.00			

18.5.106	106.00	71.26	15613.10
0.00			

18.5.107	107.00	71.27	15613.10
0.00			

18.5.108	108.00	69.39	15613.10
0.00			

18.5.109	109.00	69.38	15613.10
0.00			

18.5.110	110.00	69.69	15613.10
0.00			

18.5.111	111.00	71.57	15613.10
0.00			

18.5.112	112.00	71.77	15613.10
0.00			

18.5.113	113.00	71.43	15613.10
0.00			

18.5.114	114.00	71.26	15613.10
0.00			

18.5.115	115.00	71.27	15613.10
0.00			

18.5.116	116.00	69.39	15613.10
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0.00

18.5.117	117.00	69.38	15613.10
0.00			

18.5.118	118.00	69.69	15613.10
0.00			

18.5.119	119.00	71.57	15613.10
0.00			

18.5.120	120.00	71.46	15613.10
0.00			

18.5.121	121.00	69.56	15613.10
0.00			

18.5.122	122.00	69.91	15613.10
0.00			

18.5.123	123.00	69.57	15613.10
0.00			

18.5.124	124.00	69.58	15613.10
0.00			

18.5.125	125.00	69.56	15613.10
0.00			

18.5.126	126.00	69.68	15613.10
0.00			

18.5.127	127.00	69.56	15613.10
0.00			

18.6 L2-Fabric Read Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.6.0	0.00	200.43	346.50	466.50	500.24	761.72
18.6.1	1.00	296.20	374.75	479.83	509.24	665.00
18.6.2	2.00	254.00	380.50	496.56	527.05	729.00
18.6.3	3.00	252.00	380.00	497.03	517.23	710.69
18.6.4	4.00	484.20	499.67	509.82	551.58	2549.00
18.6.5	5.00	0.00	495.09	505.09	523.72	827.00
18.6.6	6.00	0.00	525.26	538.31	565.66	719.76
18.6.7	7.00	0.00	0.00	501.49	526.99	676.22
18.6.8	8.00	0.00	0.00	465.47	481.96	5399.00
18.6.9	9.00	0.00	489.92	498.51	509.32	658.00
18.6.10	10.00	0.00	229.50	490.39	501.12	10565.00
18.6.11	11.00	481.90	497.13	504.23	520.12	2835.00
18.6.12	12.00	0.00	491.22	502.11	517.60	805.00
18.6.13	13.00	0.00	488.58	498.22	511.30	676.06
18.6.14	14.00	0.00	510.58	520.89	536.04	702.46
18.6.15	15.00	0.00	495.66	506.72	522.67	672.82
18.6.16	16.00	0.00	0.00	464.60	485.75	5556.25
18.6.17	17.00	0.00	488.96	496.67	514.07	719.02
18.6.18	18.00	0.00	477.87	489.33	506.56	12087.00
18.6.19	19.00	481.99	495.30	505.02	520.44	2919.00
18.6.20	20.00	0.00	491.01	500.44	516.97	726.00
18.6.21	21.00	0.00	487.54	497.32	515.15	653.27
18.6.22	22.00	0.00	508.93	519.65	542.55	697.00

18.6.23	23.00	0.00	493.98	506.70	533.71	676.25
18.6.24	24.00	196.86	344.00	464.12	502.70	694.91
18.6.25	25.00	292.40	376.50	478.52	517.89	668.73
18.6.26	26.00	253.17	398.25	494.24	531.74	713.90
18.6.27	27.00	250.50	398.00	493.43	517.18	698.27
18.6.28	28.00	483.11	499.24	507.37	553.94	2386.00
18.6.29	29.00	0.00	493.65	504.31	528.56	1019.00
18.6.30	30.00	0.00	523.41	539.13	578.62	772.00
18.6.31	31.00	0.00	0.00	497.90	529.97	718.45
18.6.32	32.00	146.29	260.00	472.09	508.52	716.86
18.6.33	33.00	215.00	348.75	491.70	528.51	686.93
18.6.34	34.00	181.83	359.25	490.12	525.26	758.95
18.6.35	35.00	183.83	364.25	500.97	525.29	754.25
18.6.36	36.00	488.12	506.54	516.04	560.66	2332.00
18.6.37	37.00	0.00	500.86	512.12	536.51	783.00
18.6.38	38.00	0.00	519.87	537.52	567.69	770.68
18.6.39	39.00	0.00	0.00	497.59	524.19	729.91
18.6.40	40.00	0.00	0.00	464.34	483.68	5676.75
18.6.41	41.00	0.00	487.07	494.81	510.98	718.00
18.6.42	42.00	0.00	440.00	501.66	514.82	9764.50
18.6.43	43.00	483.44	499.65	508.81	524.68	2472.00
18.6.44	44.00	0.00	496.24	507.52	524.80	743.42
18.6.45	45.00	0.00	493.71	504.57	525.19	720.22
18.6.46	46.00	0.00	526.60	535.56	557.07	691.45

18.6.47	47.00	0.00	511.59	522.85	547.91	697.72
18.6.48	48.00	0.00	0.00	464.22	481.90	5118.75
18.6.49	49.00	0.00	484.57	492.11	504.71	710.00
18.6.50	50.00	0.00	469.50	497.67	508.70	9717.00
18.6.51	51.00	482.51	496.32	506.07	521.32	2390.00
18.6.52	52.00	0.00	493.66	504.28	530.62	701.00
18.6.53	53.00	0.00	492.21	504.27	522.88	667.00
18.6.54	54.00	0.00	522.57	532.15	554.54	691.85
18.6.55	55.00	0.00	507.92	520.76	547.08	700.94
18.6.56	56.00	138.57	253.00	467.96	508.02	711.27
18.6.57	57.00	213.80	340.25	488.16	523.13	651.86
18.6.58	58.00	207.50	355.56	490.30	522.99	726.68
18.6.59	59.00	183.00	355.76	496.05	519.07	684.23
18.6.60	60.00	484.72	502.64	510.53	543.69	2254.00
18.6.61	61.00	0.00	498.09	507.21	526.35	778.00
18.6.62	62.00	0.00	516.76	530.56	556.21	728.00
18.6.63	63.00	0.00	0.00	494.00	517.00	620.40
18.6.64	64.00	0.00	0.00	465.92	485.08	5393.50
18.6.65	65.00	0.00	486.55	493.51	506.25	664.12
18.6.66	66.00	0.00	487.11	500.48	515.29	5876.50
18.6.67	67.00	252.50	500.80	509.34	521.85	1797.00
18.6.68	68.00	0.00	490.49	503.66	517.17	628.92
18.6.69	69.00	0.00	488.60	501.39	513.56	604.53
18.6.70	70.00	0.00	518.99	532.25	547.88	667.00

18.6.71	71.00	0.00	504.51	517.79	535.21	678.56
18.6.72	72.00	76.33	131.00	466.55	495.51	646.76
18.6.73	73.00	102.60	210.00	488.32	519.11	677.69
18.6.74	74.00	104.60	219.00	484.53	514.61	734.06
18.6.75	75.00	107.20	222.75	493.80	514.02	676.51
18.6.76	76.00	291.00	502.88	511.27	555.48	1764.00
18.6.77	77.00	0.00	498.54	507.66	534.14	758.00
18.6.78	78.00	0.00	518.32	532.43	556.96	865.00
18.6.79	79.00	0.00	0.00	493.47	515.81	682.67
18.6.80	80.00	76.43	139.25	469.17	501.45	661.80
18.6.81	81.00	103.60	240.31	487.03	519.21	668.55
18.6.82	82.00	105.20	247.56	483.44	512.87	685.64
18.6.83	83.00	108.40	254.50	493.54	513.98	683.79
18.6.84	84.00	345.50	501.58	510.23	544.28	1775.00
18.6.85	85.00	0.00	498.20	506.41	523.25	801.00
18.6.86	86.00	0.00	516.94	530.67	556.09	864.00
18.6.87	87.00	0.00	0.00	494.01	517.28	636.28
18.6.88	88.00	0.00	0.00	463.92	483.12	5321.75
18.6.89	89.00	0.00	487.01	494.12	506.72	618.28
18.6.90	90.00	0.00	485.76	500.62	513.56	5523.50
18.6.91	91.00	253.50	500.52	509.14	524.12	1798.00
18.6.92	92.00	0.00	490.96	504.67	521.24	623.75
18.6.93	93.00	0.00	487.92	499.80	517.55	658.73
18.6.94	94.00	0.00	519.31	530.28	552.11	647.00

18.6.95	95.00	0.00	504.66	516.10	536.78	640.10
18.6.96	96.00	0.00	0.00	461.47	481.52	5642.25
18.6.97	97.00	0.00	487.30	494.87	509.39	589.07
18.6.98	98.00	0.00	477.80	487.53	499.54	5289.00
18.6.99	99.00	466.50	494.43	503.47	517.69	2227.00
18.6.100	100.00	0.00	486.73	499.25	520.09	653.58
18.6.101	101.00	0.00	484.13	494.03	512.02	619.96
18.6.102	102.00	0.00	504.94	516.77	537.21	677.91
18.6.103	103.00	0.00	491.12	503.72	523.53	700.97
18.6.104	104.00	135.57	240.25	461.55	498.13	657.61
18.6.105	105.00	191.40	341.44	476.85	511.51	660.66
18.6.106	106.00	177.17	351.31	488.73	526.50	696.38
18.6.107	107.00	191.50	361.25	490.54	514.90	706.59
18.6.108	108.00	478.39	497.96	505.96	546.49	2118.00
18.6.109	109.00	0.00	492.43	501.98	523.37	922.00
18.6.110	110.00	0.00	522.91	536.32	564.74	902.00
18.6.111	111.00	0.00	0.00	498.15	519.84	671.72
18.6.112	112.00	136.57	235.81	463.27	495.83	664.77
18.6.113	113.00	190.80	314.50	476.88	510.11	664.59
18.6.114	114.00	189.40	330.38	488.16	521.51	700.93
18.6.115	115.00	162.50	339.00	490.41	513.40	654.62
18.6.116	116.00	481.41	497.44	505.33	543.54	2369.00
18.6.117	117.00	0.00	493.74	503.39	527.50	934.00
18.6.118	118.00	0.00	521.83	534.82	565.36	997.00

18.6.119	119.00	0.00	0.00	497.76	521.43	690.05
18.6.120	120.00	0.00	0.00	458.51	476.19	5341.75
18.6.121	121.00	0.00	487.10	494.62	507.21	700.11
18.6.122	122.00	0.00	475.52	487.01	502.11	6310.00
18.6.123	123.00	466.50	493.24	501.04	519.43	2090.00
18.6.124	124.00	0.00	487.84	497.89	517.32	694.68
18.6.125	125.00	0.00	484.79	494.86	509.29	625.83
18.6.126	126.00	0.00	505.50	515.87	538.55	736.30
18.6.127	127.00	0.00	490.35	502.99	523.79	670.04

18.7 L2-Fabric Write and Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.7.0	0.00	302.14	332.81	345.00	355.06	491.53
18.7.1	1.00	308.04	329.37	337.63	354.71	470.48
18.7.2	2.00	313.73	345.69	355.00	366.59	498.74
18.7.3	3.00	314.68	344.32	352.68	364.52	497.11
18.7.4	4.00	309.94	350.80	373.88	383.91	546.13
18.7.5	5.00	311.05	350.18	365.79	374.77	526.49
18.7.6	6.00	323.87	370.71	388.00	397.31	552.61
18.7.7	7.00	319.98	362.55	374.64	386.50	515.13
18.7.8	8.00	303.18	345.54	381.92	470.60	667.33
18.7.9	9.00	311.15	362.53	403.90	470.54	664.14
18.7.10	10.00	308.26	352.02	386.66	460.85	662.77
18.7.11	11.00	315.51	365.14	401.84	453.77	665.97
18.7.12	12.00	308.90	354.76	392.99	419.46	589.40

18.7.13	13.00	311.85	356.40	393.20	413.50	579.46
18.7.14	14.00	319.75	359.86	394.60	436.99	606.57
18.7.15	15.00	312.89	361.47	400.64	459.63	663.53
18.7.16	16.00	301.69	348.10	377.94	465.69	664.00
18.7.17	17.00	310.51	368.45	401.09	466.20	661.58
18.7.18	18.00	306.70	355.31	382.84	456.36	651.75
18.7.19	19.00	313.94	369.44	398.35	448.23	654.02
18.7.20	20.00	309.00	359.32	389.43	415.05	581.40
18.7.21	21.00	310.62	361.82	389.63	409.90	573.64
18.7.22	22.00	317.63	363.07	391.03	432.71	606.69
18.7.23	23.00	311.18	366.33	396.79	454.54	654.31
18.7.24	24.00	301.98	333.73	343.93	354.15	491.43
18.7.25	25.00	308.53	327.83	336.31	365.85	468.97
18.7.26	26.00	315.11	345.17	353.61	372.10	497.05
18.7.27	27.00	314.11	342.56	350.65	370.12	495.38
18.7.28	28.00	311.30	357.45	371.58	380.87	540.50
18.7.29	29.00	312.53	352.48	363.82	372.87	509.94
18.7.30	30.00	324.22	374.06	386.17	394.91	547.14
18.7.31	31.00	320.46	363.35	372.99	388.00	512.00
18.7.32	32.00	303.82	348.65	381.47	464.40	664.10
18.7.33	33.00	314.07	366.71	404.56	470.41	660.58
18.7.34	34.00	308.93	353.64	385.34	456.88	643.81
18.7.35	35.00	317.20	368.78	404.08	452.69	652.94
18.7.36	36.00	310.40	359.22	393.36	418.82	603.02

18.7.37	37.00	312.86	361.01	392.74	412.58	594.83
18.7.38	38.00	319.55	372.10	405.33	426.47	604.44
18.7.39	39.00	313.78	358.01	389.86	445.16	656.83
18.7.40	40.00	304.51	336.91	347.80	357.16	482.55
18.7.41	41.00	309.65	331.53	339.80	366.59	464.57
18.7.42	42.00	317.57	346.33	355.85	370.25	489.89
18.7.43	43.00	317.27	344.59	353.44	369.41	486.90
18.7.44	44.00	312.13	360.25	375.14	384.48	523.45
18.7.45	45.00	314.42	355.85	367.39	376.76	516.62
18.7.46	46.00	326.11	377.53	391.13	399.91	544.88
18.7.47	47.00	321.38	361.81	371.56	386.18	509.85
18.7.48	48.00	302.08	332.72	343.67	353.15	479.85
18.7.49	49.00	308.13	328.04	336.46	360.92	459.06
18.7.50	50.00	312.68	342.60	351.71	365.22	485.03
18.7.51	51.00	314.36	341.59	349.96	364.26	480.53
18.7.52	52.00	308.98	354.55	371.33	380.55	509.41
18.7.53	53.00	310.36	351.86	363.80	372.96	498.72
18.7.54	54.00	323.46	371.69	387.48	395.68	531.89
18.7.55	55.00	318.09	358.27	368.12	380.93	501.93
18.7.56	56.00	302.37	347.78	379.49	457.66	656.65
18.7.57	57.00	311.40	365.76	401.23	465.60	653.51
18.7.58	58.00	308.20	353.05	383.16	450.34	633.87
18.7.59	59.00	315.62	367.44	401.42	445.86	655.08
18.7.60	60.00	309.97	358.00	390.37	414.12	596.36

18.7.61	61.00	312.18	360.58	389.65	408.00	589.42
18.7.62	62.00	318.89	371.09	401.40	421.69	593.10
18.7.63	63.00	313.07	357.41	387.96	440.35	654.91
18.7.64	64.00	302.20	334.54	343.93	356.90	492.86
18.7.65	65.00	308.71	327.88	335.35	369.95	475.57
18.7.66	66.00	313.46	342.74	350.95	373.57	498.30
18.7.67	67.00	313.89	341.46	349.44	371.60	493.58
18.7.68	68.00	308.71	358.85	373.04	384.35	555.10
18.7.69	69.00	310.91	354.25	364.08	375.81	527.16
18.7.70	70.00	323.83	375.76	388.40	398.66	558.76
18.7.71	71.00	319.28	359.16	367.62	387.48	519.54
18.7.72	72.00	301.77	352.34	384.87	458.66	664.27
18.7.73	73.00	311.17	370.38	408.19	464.42	659.16
18.7.74	74.00	307.93	356.73	388.13	455.85	661.78
18.7.75	75.00	314.95	371.80	406.15	445.54	655.65
18.7.76	76.00	309.03	361.96	393.68	416.75	635.72
18.7.77	77.00	311.18	364.67	393.04	408.42	615.75
18.7.78	78.00	319.62	375.56	405.52	422.14	621.21
18.7.79	79.00	313.23	361.04	392.64	448.00	651.05
18.7.80	80.00	302.14	349.01	383.65	460.46	665.65
18.7.81	81.00	312.16	366.97	406.96	464.08	659.13
18.7.82	82.00	307.63	353.52	386.64	458.40	658.73
18.7.83	83.00	316.95	369.29	405.18	449.32	658.94
18.7.84	84.00	308.98	359.33	393.20	419.21	629.47

18.7.85	85.00	312.15	361.30	392.89	411.18	618.40
18.7.86	86.00	319.52	371.81	405.07	425.01	624.74
18.7.87	87.00	313.53	357.94	390.57	449.45	651.42
18.7.88	88.00	304.12	336.50	346.35	358.77	499.94
18.7.89	89.00	310.07	330.21	338.52	370.71	477.86
18.7.90	90.00	316.77	345.69	353.98	374.15	505.89
18.7.91	91.00	316.21	343.61	351.99	372.53	502.33
18.7.92	92.00	312.08	359.30	374.80	385.14	545.51
18.7.93	93.00	314.02	355.03	366.26	376.71	526.04
18.7.94	94.00	326.50	376.84	390.23	399.73	555.26
18.7.95	95.00	322.50	360.29	369.33	388.71	526.84
18.7.96	96.00	302.98	353.05	382.72	449.35	654.22
18.7.97	97.00	312.68	374.27	405.94	463.49	654.01
18.7.98	98.00	307.68	359.98	387.59	444.03	657.07
18.7.99	99.00	317.02	374.58	400.74	434.39	659.63
18.7.100	100.00	309.24	363.92	390.17	406.89	587.71
18.7.101	101.00	311.98	366.82	389.00	400.29	572.19
18.7.102	102.00	317.74	368.19	394.64	426.91	627.38
18.7.103	103.00	313.14	373.09	402.40	447.69	651.11
18.7.104	104.00	304.61	332.80	341.32	357.57	475.12
18.7.105	105.00	309.98	326.05	333.59	371.67	457.09
18.7.106	106.00	317.57	342.53	350.78	376.71	482.78
18.7.107	107.00	316.50	339.88	348.17	374.64	477.33
18.7.108	108.00	312.24	358.02	368.59	378.67	508.81

18.7.109	109.00	314.28	351.28	359.75	373.60	496.67
18.7.110	110.00	325.52	375.28	384.50	395.18	528.63
18.7.111	111.00	321.95	358.33	365.88	393.40	500.34
18.7.112	112.00	302.83	332.09	341.71	354.56	482.39
18.7.113	113.00	308.95	326.81	334.88	366.37	458.56
18.7.114	114.00	314.95	342.79	351.35	372.42	485.02
18.7.115	115.00	315.13	340.76	349.53	370.80	478.85
18.7.116	116.00	309.84	356.25	369.08	378.69	515.93
18.7.117	117.00	310.77	350.87	360.72	372.02	505.92
18.7.118	118.00	324.34	374.41	385.34	394.67	535.11
18.7.119	119.00	319.69	359.07	367.18	388.36	504.02
18.7.120	120.00	303.12	351.96	382.49	450.31	653.34
18.7.121	121.00	313.25	371.52	405.38	463.29	653.14
18.7.122	122.00	309.59	359.11	387.46	447.20	656.74
18.7.123	123.00	316.97	372.68	401.52	438.72	662.76
18.7.124	124.00	309.88	361.75	390.51	409.41	580.78
18.7.125	125.00	312.28	365.44	389.24	402.58	571.08
18.7.126	126.00	320.06	366.87	395.02	429.09	628.26
18.7.127	127.00	314.12	370.65	402.58	449.61	648.62

18.8 L2-Fabric Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.8.0	0.00	0.00	0.00	0.00	0.00	0.00
18.8.1	1.00	0.00	0.00	0.00	0.00	0.00
18.8.2	2.00	0.00	0.00	0.00	0.00	0.00

18.8.3	3.00	0.00	0.00	0.00	0.00	0.00
18.8.4	4.00	0.00	0.00	0.00	0.00	0.00
18.8.5	5.00	0.00	0.00	0.00	0.00	0.00
18.8.6	6.00	0.00	0.00	0.00	0.00	0.00
18.8.7	7.00	0.00	0.00	0.00	0.00	0.00
18.8.8	8.00	0.00	0.00	0.00	0.00	0.00
18.8.9	9.00	0.00	0.00	0.00	0.00	0.00
18.8.10	10.00	0.00	0.00	0.00	0.00	0.00
18.8.11	11.00	0.00	0.00	0.00	0.00	0.00
18.8.12	12.00	0.00	0.00	0.00	0.00	0.00
18.8.13	13.00	0.00	0.00	0.00	0.00	0.00
18.8.14	14.00	0.00	0.00	0.00	0.00	0.00
18.8.15	15.00	0.00	0.00	0.00	0.00	0.00
18.8.16	16.00	0.00	0.00	0.00	0.00	0.00
18.8.17	17.00	0.00	0.00	0.00	0.00	0.00
18.8.18	18.00	0.00	0.00	0.00	0.00	0.00
18.8.19	19.00	0.00	0.00	0.00	0.00	0.00
18.8.20	20.00	0.00	0.00	0.00	0.00	0.00
18.8.21	21.00	0.00	0.00	0.00	0.00	0.00
18.8.22	22.00	0.00	0.00	0.00	0.00	0.00
18.8.23	23.00	0.00	0.00	0.00	0.00	0.00
18.8.24	24.00	0.00	0.00	0.00	0.00	0.00
18.8.25	25.00	0.00	0.00	0.00	0.00	0.00
18.8.26	26.00	0.00	0.00	0.00	0.00	0.00

18.8.27	27.00	0.00	0.00	0.00	0.00	0.00
18.8.28	28.00	0.00	0.00	0.00	0.00	0.00
18.8.29	29.00	0.00	0.00	0.00	0.00	0.00
18.8.30	30.00	0.00	0.00	0.00	0.00	0.00
18.8.31	31.00	0.00	0.00	0.00	0.00	0.00
18.8.32	32.00	0.00	0.00	0.00	0.00	0.00
18.8.33	33.00	0.00	0.00	0.00	0.00	0.00
18.8.34	34.00	0.00	0.00	0.00	0.00	0.00
18.8.35	35.00	0.00	0.00	0.00	0.00	0.00
18.8.36	36.00	0.00	0.00	0.00	0.00	0.00
18.8.37	37.00	0.00	0.00	0.00	0.00	0.00
18.8.38	38.00	0.00	0.00	0.00	0.00	0.00
18.8.39	39.00	0.00	0.00	0.00	0.00	0.00
18.8.40	40.00	0.00	0.00	0.00	0.00	0.00
18.8.41	41.00	0.00	0.00	0.00	0.00	0.00
18.8.42	42.00	0.00	0.00	0.00	0.00	0.00
18.8.43	43.00	0.00	0.00	0.00	0.00	0.00
18.8.44	44.00	0.00	0.00	0.00	0.00	0.00
18.8.45	45.00	0.00	0.00	0.00	0.00	0.00
18.8.46	46.00	0.00	0.00	0.00	0.00	0.00
18.8.47	47.00	0.00	0.00	0.00	0.00	0.00
18.8.48	48.00	0.00	0.00	0.00	0.00	0.00
18.8.49	49.00	0.00	0.00	0.00	0.00	0.00
18.8.50	50.00	0.00	0.00	0.00	0.00	0.00

18.8.51	51.00	0.00	0.00	0.00	0.00	0.00
18.8.52	52.00	0.00	0.00	0.00	0.00	0.00
18.8.53	53.00	0.00	0.00	0.00	0.00	0.00
18.8.54	54.00	0.00	0.00	0.00	0.00	0.00
18.8.55	55.00	0.00	0.00	0.00	0.00	0.00
18.8.56	56.00	0.00	0.00	0.00	0.00	0.00
18.8.57	57.00	0.00	0.00	0.00	0.00	0.00
18.8.58	58.00	0.00	0.00	0.00	0.00	0.00
18.8.59	59.00	0.00	0.00	0.00	0.00	0.00
18.8.60	60.00	0.00	0.00	0.00	0.00	0.00
18.8.61	61.00	0.00	0.00	0.00	0.00	0.00
18.8.62	62.00	0.00	0.00	0.00	0.00	0.00
18.8.63	63.00	0.00	0.00	0.00	0.00	0.00
18.8.64	64.00	0.00	0.00	0.00	0.00	0.00
18.8.65	65.00	0.00	0.00	0.00	0.00	0.00
18.8.66	66.00	0.00	0.00	0.00	0.00	0.00
18.8.67	67.00	0.00	0.00	0.00	0.00	0.00
18.8.68	68.00	0.00	0.00	0.00	0.00	0.00
18.8.69	69.00	0.00	0.00	0.00	0.00	0.00
18.8.70	70.00	0.00	0.00	0.00	0.00	0.00
18.8.71	71.00	0.00	0.00	0.00	0.00	0.00
18.8.72	72.00	0.00	0.00	0.00	0.00	0.00
18.8.73	73.00	0.00	0.00	0.00	0.00	0.00
18.8.74	74.00	0.00	0.00	0.00	0.00	0.00

18.8.75	75.00	0.00	0.00	0.00	0.00	0.00
18.8.76	76.00	0.00	0.00	0.00	0.00	0.00
18.8.77	77.00	0.00	0.00	0.00	0.00	0.00
18.8.78	78.00	0.00	0.00	0.00	0.00	0.00
18.8.79	79.00	0.00	0.00	0.00	0.00	0.00
18.8.80	80.00	0.00	0.00	0.00	0.00	0.00
18.8.81	81.00	0.00	0.00	0.00	0.00	0.00
18.8.82	82.00	0.00	0.00	0.00	0.00	0.00
18.8.83	83.00	0.00	0.00	0.00	0.00	0.00
18.8.84	84.00	0.00	0.00	0.00	0.00	0.00
18.8.85	85.00	0.00	0.00	0.00	0.00	0.00
18.8.86	86.00	0.00	0.00	0.00	0.00	0.00
18.8.87	87.00	0.00	0.00	0.00	0.00	0.00
18.8.88	88.00	0.00	0.00	0.00	0.00	0.00
18.8.89	89.00	0.00	0.00	0.00	0.00	0.00
18.8.90	90.00	0.00	0.00	0.00	0.00	0.00
18.8.91	91.00	0.00	0.00	0.00	0.00	0.00
18.8.92	92.00	0.00	0.00	0.00	0.00	0.00
18.8.93	93.00	0.00	0.00	0.00	0.00	0.00
18.8.94	94.00	0.00	0.00	0.00	0.00	0.00
18.8.95	95.00	0.00	0.00	0.00	0.00	0.00
18.8.96	96.00	0.00	0.00	0.00	0.00	0.00
18.8.97	97.00	0.00	0.00	0.00	0.00	0.00
18.8.98	98.00	0.00	0.00	0.00	0.00	0.00

18.8.99	99.00	0.00	0.00	0.00	0.00	0.00
18.8.100	100.00	0.00	0.00	0.00	0.00	0.00
18.8.101	101.00	0.00	0.00	0.00	0.00	0.00
18.8.102	102.00	0.00	0.00	0.00	0.00	0.00
18.8.103	103.00	0.00	0.00	0.00	0.00	0.00
18.8.104	104.00	0.00	0.00	0.00	0.00	0.00
18.8.105	105.00	0.00	0.00	0.00	0.00	0.00
18.8.106	106.00	0.00	0.00	0.00	0.00	0.00
18.8.107	107.00	0.00	0.00	0.00	0.00	0.00
18.8.108	108.00	0.00	0.00	0.00	0.00	0.00
18.8.109	109.00	0.00	0.00	0.00	0.00	0.00
18.8.110	110.00	0.00	0.00	0.00	0.00	0.00
18.8.111	111.00	0.00	0.00	0.00	0.00	0.00
18.8.112	112.00	0.00	0.00	0.00	0.00	0.00
18.8.113	113.00	0.00	0.00	0.00	0.00	0.00
18.8.114	114.00	0.00	0.00	0.00	0.00	0.00
18.8.115	115.00	0.00	0.00	0.00	0.00	0.00
18.8.116	116.00	0.00	0.00	0.00	0.00	0.00
18.8.117	117.00	0.00	0.00	0.00	0.00	0.00
18.8.118	118.00	0.00	0.00	0.00	0.00	0.00
18.8.119	119.00	0.00	0.00	0.00	0.00	0.00
18.8.120	120.00	0.00	0.00	0.00	0.00	0.00
18.8.121	121.00	0.00	0.00	0.00	0.00	0.00
18.8.122	122.00	0.00	0.00	0.00	0.00	0.00

18.8.123	123.00	0.00	0.00	0.00	0.00	0.00
18.8.124	124.00	0.00	0.00	0.00	0.00	0.00
18.8.125	125.00	0.00	0.00	0.00	0.00	0.00
18.8.126	126.00	0.00	0.00	0.00	0.00	0.00
18.8.127	127.00	0.00	0.00	0.00	0.00	0.00

18.12 L2-Fabric (128B read requests per normUnit)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.12.0	0.00	0.00	0.00	128.00	128.00	128.00
18.12.1	1.00	0.00	0.00	128.00	128.00	128.00
18.12.2	2.00	0.00	0.00	128.00	128.00	128.00
18.12.3	3.00	0.00	0.00	128.00	128.00	128.00
18.12.4	4.00	0.00	0.00	128.00	128.00	128.00
18.12.5	5.00	0.00	0.00	128.00	128.00	128.00
18.12.6	6.00	0.00	0.00	128.00	128.00	128.00
18.12.7	7.00	0.00	0.00	128.00	128.00	128.00
18.12.8	8.00	0.00	0.00	130.00	130.00	130.00
18.12.9	9.00	0.00	0.00	130.00	130.00	130.00
18.12.10	10.00	0.00	0.00	130.00	130.00	130.00
18.12.11	11.00	0.00	0.00	130.00	130.00	130.00
18.12.12	12.00	0.00	0.00	130.00	130.00	130.00
18.12.13	13.00	0.00	0.00	130.00	130.00	130.00
18.12.14	14.00	0.00	0.00	130.00	130.00	130.00
18.12.15	15.00	0.00	0.00	130.00	130.00	130.00
18.12.16	16.00	0.00	0.00	130.00	130.00	130.00

18.12.17	17.00	0.00	0.00	130.00	130.00	130.00
18.12.18	18.00	0.00	0.00	130.00	130.00	130.00
18.12.19	19.00	0.00	0.00	130.00	130.00	130.00
18.12.20	20.00	0.00	0.00	130.00	130.00	130.00
18.12.21	21.00	0.00	0.00	130.00	130.00	130.00
18.12.22	22.00	0.00	0.00	130.00	130.00	130.00
18.12.23	23.00	0.00	0.00	130.00	130.00	130.00
18.12.24	24.00	0.00	0.00	128.00	128.00	128.00
18.12.25	25.00	0.00	0.00	128.00	128.00	128.00
18.12.26	26.00	0.00	0.00	128.00	128.00	128.00
18.12.27	27.00	0.00	0.00	128.00	128.00	128.00
18.12.28	28.00	0.00	0.00	128.00	128.00	128.00
18.12.29	29.00	0.00	0.00	128.00	128.00	128.00
18.12.30	30.00	0.00	0.00	128.00	128.00	128.00
18.12.31	31.00	0.00	0.00	128.00	128.00	128.00
18.12.32	32.00	0.00	0.00	128.00	128.00	128.00
18.12.33	33.00	0.00	0.00	128.00	128.00	128.00
18.12.34	34.00	0.00	0.00	128.00	128.00	128.00
18.12.35	35.00	0.00	0.00	128.00	128.00	128.00
18.12.36	36.00	0.00	0.00	128.00	128.00	128.00
18.12.37	37.00	0.00	0.00	128.00	128.00	128.00
18.12.38	38.00	0.00	0.00	128.00	128.00	128.00
18.12.39	39.00	0.00	0.00	128.00	128.00	128.00
18.12.40	40.00	0.00	0.00	130.00	130.00	130.00

18.12.41	41.00	0.00	0.00	130.00	130.00	130.00
18.12.42	42.00	0.00	0.00	130.00	130.00	130.00
18.12.43	43.00	0.00	0.00	130.00	130.00	130.00
18.12.44	44.00	0.00	0.00	130.00	130.00	130.00
18.12.45	45.00	0.00	0.00	130.00	130.00	130.00
18.12.46	46.00	0.00	0.00	130.00	130.00	130.00
18.12.47	47.00	0.00	0.00	130.00	130.00	130.00
18.12.48	48.00	0.00	0.00	130.00	130.00	130.00
18.12.49	49.00	0.00	0.00	130.00	130.00	130.00
18.12.50	50.00	0.00	0.00	130.00	130.00	130.00
18.12.51	51.00	0.00	0.00	130.00	130.00	130.00
18.12.52	52.00	0.00	0.00	130.00	130.00	130.00
18.12.53	53.00	0.00	0.00	130.00	130.00	130.00
18.12.54	54.00	0.00	0.00	130.00	130.00	130.00
18.12.55	55.00	0.00	0.00	130.00	130.00	130.00
18.12.56	56.00	0.00	0.00	128.00	128.00	128.00
18.12.57	57.00	0.00	0.00	128.00	128.00	128.00
18.12.58	58.00	0.00	0.00	128.00	128.00	128.00
18.12.59	59.00	0.00	0.00	128.00	128.00	128.00
18.12.60	60.00	0.00	0.00	128.00	128.00	128.00
18.12.61	61.00	0.00	0.00	128.00	128.00	128.00
18.12.62	62.00	0.00	0.00	128.00	128.00	128.00
18.12.63	63.00	0.00	0.00	128.00	128.00	128.00
18.12.64	64.00	0.00	0.00	130.00	130.00	130.00

18.12.65	65.00	0.00	0.00	130.00	130.00	130.00
18.12.66	66.00	0.00	0.00	130.00	130.00	130.00
18.12.67	67.00	0.00	0.00	130.00	130.00	130.00
18.12.68	68.00	0.00	0.00	130.00	130.00	130.00
18.12.69	69.00	0.00	0.00	130.00	130.00	130.00
18.12.70	70.00	0.00	0.00	130.00	130.00	130.00
18.12.71	71.00	0.00	0.00	130.00	130.00	130.00
18.12.72	72.00	0.00	0.00	128.00	128.00	128.00
18.12.73	73.00	0.00	0.00	128.00	128.00	128.00
18.12.74	74.00	0.00	0.00	128.00	128.00	128.00
18.12.75	75.00	0.00	0.00	128.00	128.00	128.00
18.12.76	76.00	0.00	0.00	128.00	128.00	128.00
18.12.77	77.00	0.00	0.00	128.00	128.00	128.00
18.12.78	78.00	0.00	0.00	128.00	128.00	128.00
18.12.79	79.00	0.00	0.00	128.00	128.00	128.00
18.12.80	80.00	0.00	0.00	128.00	128.00	128.00
18.12.81	81.00	0.00	0.00	128.00	128.00	128.00
18.12.82	82.00	0.00	0.00	128.00	128.00	128.00
18.12.83	83.00	0.00	0.00	128.00	128.00	128.00
18.12.84	84.00	0.00	0.00	128.00	128.00	128.00
18.12.85	85.00	0.00	0.00	128.00	128.00	128.00
18.12.86	86.00	0.00	0.00	128.00	128.00	128.00
18.12.87	87.00	0.00	0.00	128.00	128.00	128.00
18.12.88	88.00	0.00	0.00	130.00	130.00	130.00

18.12.89	89.00	0.00	0.00	130.00	130.00	130.00
18.12.90	90.00	0.00	0.00	130.00	130.00	130.00
18.12.91	91.00	0.00	0.00	130.00	130.00	130.00
18.12.92	92.00	0.00	0.00	130.00	130.00	130.00
18.12.93	93.00	0.00	0.00	130.00	130.00	130.00
18.12.94	94.00	0.00	0.00	130.00	130.00	130.00
18.12.95	95.00	0.00	0.00	130.00	130.00	130.00
18.12.96	96.00	0.00	0.00	130.00	130.00	130.00
18.12.97	97.00	0.00	0.00	130.00	130.00	130.00
18.12.98	98.00	0.00	0.00	130.00	130.00	130.00
18.12.99	99.00	0.00	0.00	130.00	130.00	130.00
18.12.100	100.00	0.00	0.00	130.00	130.00	130.00
18.12.101	101.00	0.00	0.00	130.00	130.00	130.00
18.12.102	102.00	0.00	0.00	130.00	130.00	130.00
18.12.103	103.00	0.00	0.00	130.00	130.00	130.00
18.12.104	104.00	0.00	0.00	128.00	128.00	128.00
18.12.105	105.00	0.00	0.00	128.00	128.00	128.00
18.12.106	106.00	0.00	0.00	128.00	128.00	128.00
18.12.107	107.00	0.00	0.00	128.00	128.00	128.00
18.12.108	108.00	0.00	0.00	128.00	128.00	128.00
18.12.109	109.00	0.00	0.00	128.00	128.00	128.00
18.12.110	110.00	0.00	0.00	128.00	128.00	128.00
18.12.111	111.00	0.00	0.00	128.00	128.00	128.00
18.12.112	112.00	0.00	0.00	128.00	128.00	128.00

Large Input

[illegible]

```

INFO Analysis mode = cli
INFO [analysis] deriving rocprofiler-compute metrics...
WARNING PC sampling: can not detect pc sampling method without /workspa
ce/user/workloads/nonfusion_matmulbias_32768x32768/MI300X_A1/ps_file_pc_sampling
_host_trap.csv

```

0. Top Stats

0.1 Top Kernels

Kernel_Name	Count	Sum(ns)
Mean(ns) Median(ns) Pct S		
0 matmul_kernel 251829353.73 248242519.75 82.68 *	17.00	4281099013.34
1 bias_kernel 1167862.41 1160099.46 16.73 *	742.00	866553904.99
2 void at::native::vectorized_elementwise_ 43530.49 43358.58 0.52 kernel<4, at::native::FillFunctor<int>, std::array<char*, 1ul> >(int, at::nat...	617.00	26858311.75
3 void at::native::(anonymous namespace):: 1114831.64 1582553.25 0.06 distribution_elementwise_grid_stride_ker nel<float, 4, at::native::templates::...	3.00	3344494.92
4 __amd_rocclr_copyBuffer 3003.98 2956.58 0.01	89.00	267354.00
5 void at::native::(anonymous namespace):: 2395.85 2311.75 0.00 CatArrayBatchedCopy_contig<at::native::(	7.00	16770.92


```

anonymous namespace)::OpaqueType<2u>,...

6 void at::native::vectorized_elementwise_      3.00      10527.17
3509.06      3524.67      0.00
kernel<4, at::native::compare_scalar_ker
nel<double>(at::TensorIteratorBase&, ...

7 void at::native::vectorized_elementwise_      2.00      6250.92
3125.46      3125.46      0.00
kernel<8, at::native::UnaryFunctor<c10:
:Half, c10::Half, bool, at::native::(...

8 void at::native::vectorized_elementwise_      2.00      5044.67
2522.33      2522.33      0.00
kernel<8, at::native::AbsFunctor<c10::Ha
lf>, std::array<char*, 2ul> >(int, at...

9 void at::native::vectorized_elementwise_      2.00      5027.92
2513.96      2513.96      0.00
kernel<8, at::native::BinaryFunctor<c10:
:Half, c10::Half, bool, at::native::(...

```

0.2 Dispatch List

```

Dispatch_ID  Kernel_Name
GPU_ID

0            0 void at::native::(anonymous
namespace)::distribution_elementwise_grid_stride_ker
nel<float, 4,
at::native::templates::cuda::normal_and_transform<c10::Half, float
, at::CUDAGeneratorImpl*,

```

```

at::native::templates::cuda::normal_kernel<at::CUDAGen
        eratorImpl*>(at::TensorBase const&, double, double,
at::CUDAGeneratorImpl*)::...

```

```

1          1 void at::native::(anonymous
namespace)::distribution_elementwise_grid_stride_ker          0
        nel<float, 4,
at::native::templates::cuda::normal_and_transform<c10::Half, float
        , at::CUDAGeneratorImpl*,
at::native::templates::cuda::normal_kernel<at::CUDAGen
        eratorImpl*>(at::TensorBase const&, double, double,
at::CUDAGeneratorImpl*)::...

```

```

2          2 void at::native::(anonymous
namespace)::distribution_elementwise_grid_stride_ker          0
        nel<float, 4,
at::native::templates::cuda::normal_and_transform<c10::Half, float
        , at::CUDAGeneratorImpl*,
at::native::templates::cuda::normal_kernel<at::CUDAGen
        eratorImpl*>(at::TensorBase const&, double, double,
at::CUDAGeneratorImpl*)::...

```

```

3          3 matmul_kernel
0

```

```

4          4 void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
        std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
        1>)

```

```

5          5 matmul_kernel
0

```

```

6          6 void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
        std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
        1>)

```

```

7          7  matmul_kernel
0

8          8  void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
      std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
      1>)

```

```

9          9  matmul_kernel
0

```

1. System Info

	Info
workload_name	nonfusion_matmulbias_32768x32768
command nonfusion	python triton/matmulbias.py --M 32768 --N 32768
ip_blocks	SQ LDS SQC TA TD TCP TCC SPI CPC CPF roofline
timestamp	Fri Jan 23 18:26:16 2026 (UTC)
version	3
hostname	6b82f9811751

cpu_model	Intel(R) Xeon(R) Platinum 8470
sbios	SeaBIOS1.15.0-1
linux_distro	CentOS Stream 10 (Coughlan)
linux_kernel_version	5.15.0-164-generic
amd_gpu_kernel_version	nan
cpu_memory	230893524
gpu_memory	nan
rocm_version	7.0.3
vbios	113-M3000108-103
compute_partition	SPX
memory_partition	nan
gpu_series	MI300

gpu_model	MI300X_A1
-----------	-----------

gpu_arch	gfx942
----------	--------

gpu_chip_id	29877
-------------	-------

gpu_l1	32
--------	----

gpu_l2	4096
--------	------

cu_per_gpu	304
------------	-----

simd_per_cu	4
-------------	---

se_per_gpu	32
------------	----

wave_size	64
-----------	----

workgroup_max_size	1024
--------------------	------

max_waves_per_cu	32
------------------	----

max_sclk	2100
----------	------

max_mclk	1300
cur_sclk	2100
cur_mclk	1300
total_l2_chan	128
lds_banks_per_cu	32
sqc_per_gpu	160
pipes_per_gpu	4
num_xcd	8
num_hbm_channels	128

2. System Speed-of-Light

2.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit	Peak
Pct of Peak				

2.1.0 0.7	VALU FLOPs	574.57	Gflop/s	81715.2
2.1.1 2.75	VALU IOPs	2251.08	Giop/s	81715.2
2.1.2 2614886.4	MFMA FLOPs (F8) 0.0	0.0	Gflop/s	
2.1.3 1307443.2	MFMA FLOPs (BF16) 0.0	0.0	Gflop/s	
2.1.4 1307443.2	MFMA FLOPs (F16) 0.48	6263.64	Gflop/s	
2.1.5 163430.4	MFMA FLOPs (F32) 0.0	0.0	Gflop/s	
2.1.6 163430.4	MFMA FLOPs (F64) 0.0	0.0	Gflop/s	
2.1.7	MFMA FLOPs (F6F4)		Gflop/s	
2.1.8 2614886.4	MFMA IOPs (Int8) 0.0	0.0	Giop/s	
2.1.9 98.03	Active CUs	298.0	Cus	304.0
2.1.10 4.57	SALU Utilization	4.57	Pct	100.0
2.1.11 12.22	VALU Utilization	12.22	Pct	100.0

2.1.12 0.88	MFMA Utilization	0.88	Pct	100.0
2.1.13 0.75	VMEM Utilization	0.75	Pct	100.0
2.1.14 0.4	Branch Utilization	0.4	Pct	100.0
2.1.15 99.25	VALU Active Threads	63.52	Threads	64.0
2.1.16 3.92	IPC	0.2	Instr/cycle	5.0
2.1.17 45.9	Wavefront Occupancy	4465.52	Wavefronts	9728.0
2.1.18 0.5	Theoretical LDS Bandwidth	405.32	Gb/s	81715.2
2.1.19 2.87	LDS Bank Conflicts/Access	0.92	Conflicts/access	32.0
2.1.20 29.56	vL1D Cache Hit Rate	29.56	Pct	100.0
2.1.21 9.61	vL1D Cache BW	7850.68	Gb/s	81715.2
2.1.22 33.64	L2 Cache Hit Rate	33.64	Pct	100.0
2.1.23 16.04	L2 Cache BW	5519.3	Gb/s	34406.4

2.1.24 35.04	L2-Fabric Read BW	1865.79	Gb/s	5324.8
2.1.25 33.8	L2-Fabric Write BW	1799.64	Gb/s	5324.8
2.1.26	L2-Fabric Read Latency	2415.7	Cycles	
2.1.27	L2-Fabric Write Latency	478.61	Cycles	
2.1.28 0.0	sL1D Cache Hit Rate	0.0	Pct	100.0
2.1.29 0.0	sL1D Cache BW	0.0	Gb/s	21504.0
2.1.30 100.0	L1I Hit Rate	100.0	Pct	100.0
2.1.31 4.74	L1I BW	1019.89	Gb/s	21504.0
2.1.32	L1I Fetch Latency		Cycles	

3. Memory Chart

3.1 Memory Chart

(Normalization: per_kernel)

Instr Buff	Instr Dispatch	Exec
LDS	L2 Cache	

SALU: 141855160

CUs	Req	: 11846627	Util	:	1 %	Active
			xGMI/PCIE			

Wave	0	Instr Buf	29
8: 304		<-----	

		SMEM: 188
	Lat	:
N/A	Hit	: 34
%		

>		
		^
		RVGPRseq: 59.0

Wave N-1	Instr Buf	VALU:	
465545723		Vector L1 Cache	
Request			

>			
v			
		SGPRs :	32

		MFMA:	
96198213	Rd	: 16870407	Hit : 30 % Rd

: 208990313 Rd : 208997335

> <-----<-----
-

Fabric

Wt : 13724191Lat : N/A Wr : 35812956 Wr :
35813042 Rd :
119490544

VMEM: 28530390 LDS Alloc: -
----->-----

> <----- Latency (cycles) Rd:
118770225

----->
0 Atomic: 0 Coales: 25 % Atomic: 0 Atomic:
0 Wr : 33558538 <----- HBM
<

-----> <-----
> -----> Rd : 2416 Wr:
33558379

LDS:
118466272 Stall : 41
cycles Atomic: 0 Wr :
479 ----->
-----> Scratch Alloc:

-----> Atomic: 0
0

Wave Occupancy GWS: 0
Scalar L1D
Cache Latency (cycles)

15
---> Rd :
76

Wave Life
<-----
^

|
82473
BRANCH: 34708896 Wavefronts: Rd : 194 Hit : 0.0
% Wr : 157 Rd : N/A |
|
-----> 0 <

```

-----
> |
|
Lat : N/A Atomic: 0 Wr :
N/A |
v
^
<-----
>
| Workgroups:
|
| Instr L1 Cache
0 | GMI
|
|
| Fetch :
154424758 Hit : 100 % Req
: 8758
-----
-----
<-----

Lat : 0.0 cycles

```

4. Roofline

4.1 Roofline

279431.1

139936GFLOP/s

64341.0

14815.0

3411.2

785.5

59642 GB/s

33652 GB/s

180.9 25452 GB/s

41.6 4164 GB/s

0.01	0.18	3.16	56.23	1000.00
Performance (GFLOP/sec)		Arithmetic Intensity (FLOP)s/Byte)		

5. Command Processor (CPC/CPF)

5.1 Command Processor Fetcher

Metric_ID	Metric	Avg	Min	Max	Unit
5.1.0	CPF Utilization	100.00	100.00	100.00	Pct
5.1.1	CPF Stall	0.03	0.00	0.20	Pct
5.1.2	CPF-L2 Utilization	0.44	0.00	0.96	Pct
5.1.3	CPF-L2 Stall	0.00	0.00	0.00	Pct
5.1.4	CPF-UTCL1 Stall	0.03	0.00	0.20	Pct

5.2 Packet Processor

Metric_ID	Metric	Avg	Min	Max	Unit
5.2.0	CPC SYNC FIFO Full Rate				Pct
5.2.1	CPC CANE Stall Rate				Pct
5.2.2	CPC ADC Utilization				Pct
5.2.3	CPC Utilization	100.0	98.37	100.0	Pct
5.2.4	CPC Stall Rate	0.85	0.1	10.12	Pct

5.2.5	CPC Packet Decoding Utilization	3.19	0.38	5.34	Pct
5.2.6	CPC-Workgroup Manager Utilization	96.97	70.83	113.13	Pct
5.2.7	CPC-L2 Utilization	0.01	0.0	0.03	Pct
5.2.8	CPC-UTCL1 Stall	0.01	0.0	0.36	Pct
5.2.9	CPC-UTCL2 Utilization	0.01	0.0	0.1	Pct

6. Workgroup Manager (SPI)

6.1 Workgroup Manager Utilizations

Metric_ID	Metric	Avg	Min	Max	Unit
6.1.0	Schedule-Pipe Wave Occupancy				Wave
6.1.1	Accelerator Utilization	100.27	28.07	117.49	Pct
6.1.2	Scheduler-Pipe Utilization	0.0	0.0	0.0	Pct
6.1.3	Scheduler-Pipe Wave Utilization				Pct

6.1.4	Workgroup Manager Utilization	98.16	95.94	99.99	Pct
6.1.5	Shader Engine Utilization	97.58	81.93	113.66	Pct
6.1.6	SIMD Utilization	95.54	75.01	130.56	Pct
6.1.7 Workgroups	Dispatched Workgroups	0.0	0.0	0.0	
6.1.8 Wavefronts	Dispatched Wavefronts	0.0	0.0	0.0	
6.1.9 Cycles/wave	VGPR Writes				
6.1.10 Cycles/wave	SGPR Writes				

6.2 Workgroup Manager - Resource Allocation

Metric_ID Unit	Metric	Avg	Min	Max
6.2.0 Pct	Not-scheduled Rate (Workgroup Manager)	0.0	0.0	0.0
6.2.1 Pct	Not-scheduled Rate (Scheduler-Pipe)	0.0	0.0	0.0
6.2.2 Pct	Scheduler-Pipe FIFO Full Rate			
6.2.3	Scheduler-Pipe Stall Rate	0.0	0.0	0.0

Pct

6.2.4	Scratch Stall Rate	0.0	0.0	0.0
-------	--------------------	-----	-----	-----

Pct

6.2.5	Insufficient SIMD Waveslots	0.0	0.0	0.0
-------	-----------------------------	-----	-----	-----

Pct

6.2.6	Insufficient SIMD VGPRs	0.0	0.0	0.0
-------	-------------------------	-----	-----	-----

Pct

6.2.7	Insufficient SIMD SGPRs	0.0	0.0	0.0
-------	-------------------------	-----	-----	-----

Pct

6.2.8	Insufficient CU LDS	0.0	0.0	0.0
-------	---------------------	-----	-----	-----

Pct

6.2.9	Insufficient CU Barriers	0.0	0.0	0.0
-------	--------------------------	-----	-----	-----

Pct

6.2.10	Reached CU Workgroup Limit	0.0	0.0	0.0
--------	----------------------------	-----	-----	-----

Pct

6.2.11	Reached CU Wavefront Limit	0.0	0.0	0.0
--------	----------------------------	-----	-----	-----

Pct

7. Wavefront

7.1 Wavefront Launch Stats

Metric_ID	Metric	Avg	Min	Max
Unit				
7.1.0	Grid Size	18777661.39	8388608.00	33554432.00
Work items				

7.1.1 Work items	Workgroup Size	365.96	256.00	512.00
7.1.2 Wavefronts	Total Wavefronts	0.00	0.00	0.00
7.1.3 Wavefronts	Saved Wavefronts	4.63	0.00	2298.00
7.1.4 Wavefronts	Restored Wavefronts	3.25	0.00	1250.00
7.1.5 Registers	VGPRs	58.56	12.00	96.00
7.1.6 Registers	AGPRs	4.61	0.00	192.00
7.1.7 Registers	SGPRs	32.36	32.00	48.00
7.1.8 Bytes	LDS Allocation	0.00	0.00	0.00
7.1.9 Bytes/workitem	Scratch Allocation	0.00	0.00	0.00

7.2 Wavefront Runtime Stats

Metric_ID Max Unit	Metric	Avg	Min
7.2.0 260441671.17	Kernel Time (Nanosec) Ns	6782151.41	1112981.00

7.2.1	Kernel Time (Cycles)	8463174.86	1674118.50
393267573.12	Cycle		
7.2.2	Instructions per wavefront	2474.83	262.00
109377.89	Instr/wavefront		
7.2.3	Wave Cycles	23878574490.76	8938035940.00
623941939328.00	Cycles per kernel		
7.2.4	Dependency Wait Cycles	9473213138.26	2008166088.00
187038516520.00	Cycles per kernel		
7.2.5	Issue Wait Cycles	10299955323.46	3121581688.00
283325154024.00	Cycles per kernel		
7.2.6	Active Cycles	3606377497.97	341311968.00
161075697392.00	Cycles per kernel		
7.2.7	Wavefront Occupancy	4465.52	256.31
9491.53	Wavefronts		

10. Compute Units - Instruction Mix

10.1 Overall Instruction Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.1.0	VALU	369347509.52	63569920.00	15156260976.00	Instr
	per kernel				
10.1.1	VMEM	28530390.18	4325376.00	1078654208.00	Instr
	per kernel				

10.1.2 per kernel	LDS	118466271.96	0.00	6979359232.00	Instr
10.1.3 per kernel	MFMA	96198213.48	0.00	4294967296.00	Instr
10.1.4 per kernel	SALU	141855159.66	13238272.00	6219932358.00	Instr
10.1.5 per kernel	SMEM	188.20	0.00	92988.00	Instr
10.1.6 per kernel	Branch	34708896.45	1048576.00	1612374198.00	Instr

10.2 VALU Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.2.0 per kernel	INT32	156398960.48	30146560.00	7536180462.00	Instr
10.2.1 per kernel	INT64	32340418.61	8388608.00	1078460416.00	Instr
10.2.2 per kernel	F16-ADD	8200720.86	0.00	8388608.00	Instr
10.2.3 per kernel	F16-MUL	0.00	0.00	0.00	Instr
10.2.4 per kernel	F16-FMA	0.00	0.00	0.00	Instr
10.2.5	F16-Trans	0.00	0.00	0.00	Instr

per kernel

10.2.6	F32-ADD	0.00	0.00	0.00	Instr
--------	---------	------	------	------	-------

per kernel

10.2.7	F32-MUL	1173603.84	524288.00	2097152.00	Instr
--------	---------	------------	-----------	------------	-------

per kernel

10.2.8	F32-FMA	0.00	0.00	0.00	Instr
--------	---------	------	------	------	-------

per kernel

10.2.9	F32-Trans	1173603.84	524288.00	2097152.00	Instr
--------	-----------	------------	-----------	------------	-------

per kernel

10.2.10	F64-ADD	0.00	0.00	0.00	Instr
---------	---------	------	------	------	-------

per kernel

10.2.11	F64-MUL	0.00	0.00	0.00	Instr
---------	---------	------	------	------	-------

per kernel

10.2.12	F64-FMA	0.00	0.00	0.00	Instr
---------	---------	------	------	------	-------

per kernel

10.2.13	F64-Trans	0.00	0.00	0.00	Instr
---------	-----------	------	------	------	-------

per kernel

10.2.14	Conversion	2722981.94	1048576.00	20971520.00	Instr
---------	------------	------------	------------	-------------	-------

per kernel

10.3 VMEM Instr Mix

Metric_ID	Metric	Avg	Min	Max
Unit				
10.3.0	Global/Generic Instr	28243857.37	4194304.0	
1077936128.0	Instr per kernel			

10.3.1	Global/Generic Read	26099733.59	2097152.0	
1073741824.0	Instr per kernel			
10.3.2	Global/Generic Write	2144123.78	2097152.0	
4194304.0	Instr per kernel			
10.3.3	Global/Generic Atomic	0.0	0.0	0.0
	Instr per kernel			
10.3.4	Spill/Stack Instr	286617.3	0.0	
782208.0	Instr per kernel			
10.3.5	Spill/Stack Coalesceable Instr			
	Instr per kernel			
10.3.6	Spill/Stack Read	285623.84	0.0	
524288.0	Instr per kernel			
10.3.7	Spill/Stack Write	897.18	0.0	
389120.0	Instr per kernel			
10.3.8	Spill/Stack Atomic	0.0	0.0	0.0
	Instr per kernel			

10.4 MFMA Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.4.0	MFMA-I8	0.0	0.0	0.0	Instr per
kernel					
10.4.1	MFMA-F8	0.0	0.0	0.0	Instr per
kernel					

10.4.2 kernel	MFMA-F16	96198213.48	0.0	4294967296.0	Instr per
10.4.3 kernel	MFMA-BF16	0.0	0.0	0.0	Instr per
10.4.4 kernel	MFMA-F32	0.0	0.0	0.0	Instr per
10.4.5 kernel	MFMA-F64	0.0	0.0	0.0	Instr per
10.4.6 kernel	MFMA-F6F4				Instr per

11. Compute Units - Compute Pipeline

11.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit	Peak	Pct of Peak
11.1.0	VALU FLOPs	574.57	Gflop	81715.2	0.7
11.1.1	VALU IOPs	2251.08	Giop	81715.2	2.75
11.1.2	MFMA FLOPs (F8)	0.0	Gflop	2614886.4	0.0
11.1.3	MFMA FLOPs (BF16)	0.0	Gflop	1307443.2	0.0

11.1.4	MFMA FLOPs (F16)	6263.64	Gflop	1307443.2	0.48
11.1.5	MFMA FLOPs (F32)	0.0	Gflop	163430.4	0.0
11.1.6	MFMA FLOPs (F64)	0.0	Gflop	163430.4	0.0
11.1.7	MFMA FLOPs (F6F4)		Gflop		
11.1.8	MFMA IOPs (INT8)	0.0	Giop	2614886.4	0.0

11.2 Pipeline Stats

Metric_ID	Metric	Avg	Min	Max	Unit
11.2.0	IPC	0.2	0.15	0.48	Instr/cycle
11.2.1	IPC (Issued)	0.97	0.85	0.98	Instr/cycle
11.2.2	SALU Utilization	4.57	2.02	8.52	Pct
11.2.3	VALU Utilization	12.22	9.51	24.47	Pct
11.2.4	VALU Co-Issue Efficiency				Pct
11.2.5	VMEM Utilization	0.75	0.61	1.36	Pct

11.2.6	Branch Utilization	0.4	0.2	2.7	Pct
11.2.7	VALU Active Threads	63.52	63.23	64.0	Threads
11.2.8	MFMA Utilization	0.88	0.0	43.21	Pct
11.2.9	MFMA Instr Cycles	32.0	32.0	32.0	Cycles/instr
11.2.10	VMEM Latency				Cycles
11.2.11	SMEM Latency				Cycles
11.3 Arithmetic Operations					
Metric_ID	Metric	Avg	Min	Max	Unit
11.3.0	FLOPs (Total)	1576786597097.4	603979776.0		
70369012613120.0	Ops per kernel				
11.3.1	IOPs (Total)	12079320262.16	2466250752.0	551337016192.0	
	Ops per kernel				
11.3.2	F8 OPs	0.0	0.0	0.0	
	Ops per kernel				
11.3.3	F16 OPs	1576636375806.31	536870912.0		
70368744177664.0	Ops per kernel				

11.3.4	BF16 OPs	0.0	0.0	0.0
Ops per kernel				
11.3.5	F32 OPs	150221291.09	67108864.0	268435456.0
Ops per kernel				
11.3.6	F64 OPs	0.0	0.0	0.0
Ops per kernel				
11.3.7	F6F4 OPs			
Ops per kernel				
11.3.8	INT8 OPs	0.0	0.0	0.0
Ops per kernel				

12. Local Data Share (LDS)

12.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
12.1.0	Utilization	0.98	Pct of peak
12.1.1	Access Rate	0.62	Pct of peak
12.1.2	Theoretical Bandwidth (% of Peak)	0.50	Pct of peak
12.1.3	Bank Conflict Rate	2.87	Pct of peak

12.2 LDS Stats

Metric_ID	Metric	Avg	Min	Max
Unit				
12.2.0	LDS Instrs	118466271.96	0.0	
6979359232.0	Instr per kernel			

12.2.1	LDS LOAD			
	Instr per kernel			
12.2.2	LDS STORE			
	Instr per kernel			
12.2.3	LDS ATOMIC			
	Instr per kernel			
12.2.4	LDS LOAD Bandwidth			
	Gbps			
12.2.5	LDS STORE Bandwidth			
	Gbps			
12.2.6	LDS ATOMIC Bandwidth			
	Gbps			
12.2.7	Theoretical Bandwidth	85545328046.38	0.0	
4398064467968.0	Bytes per kernel			
12.2.8	LDS Latency			
	Cycles			
12.2.9	Bank Conflicts/Access	0.92	0.0	1.0
	Conflicts/access			
12.2.10	Index Accesses	668916584.9	0.0	
34359878656.0	Cycles per kernel			
12.2.11	Atomic Return Cycles	0.0	0.0	0.0
	Cycles per kernel			
12.2.12	Bank Conflict	593709.53	0.0	2359296.0
	Cycles per kernel			

12.2.13	Addr Conflict	0.0	0.0	0.0
Cycles per kernel				

12.2.14	Unaligned Stall	0.0	0.0	0.0
Cycles per kernel				

12.2.15	Mem Violations	0.0	0.0	0.0
Accesses per kernel				

12.2.16	LDS Command FIFO Full Rate			
Cycles per kernel				

12.2.17	LDS Data FIFO Full Rate			
Cycles per kernel				

13. Instruction Cache

13.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
13.1.0	Bandwidth	4.74	Pct of peak
13.1.1	Cache Hit Rate	99.97	Pct of peak
13.1.2	L1I-L2 Bandwidth	0.00	Pct of peak

13.2 Instruction Cache Accesses

Metric_ID	Metric	Avg	Min	Max
Unit				
13.2.0	Req	154424757.92	15597568.0	
7007826290.0	Req per kernel			
13.2.1	Hits	154418798.06	15591153.0	
7007675006.0	Hits per kernel			

13.2.2	Misses - Non Duplicated Misses per kernel	290.93	160.0	2212.0	
13.2.3	Misses - Duplicated Misses per kernel	4331.37	2028.0	16593.0	
13.2.4	Cache Hit Rate Pct	99.97	99.93	100.0	
13.2.5	Instruction Fetch Latency Cycles				

13.3 Instruction Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
13.3.0	L1I-L2 Bandwidth per kernel	560524.90	327680.00	4638720.00	Bytes

14. Scalar L1 Data Cache

14.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
14.1.0	Bandwidth	0.00	Pct of peak
14.1.1	Cache Hit Rate	0.00	Pct of peak
14.1.2	sL1D-L2 BW	0.00	Pct of peak

14.2 Scalar L1D Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
-----------	--------	-----	-----	-----	------

14.2.0 kernel	Req	194.03	0.00	97416.00	Req per
14.2.1 kernel	Hits	0.00	0.00	0.00	Req per
14.2.2 kernel	Misses - Non Duplicated	194.03	0.00	97416.00	Req per
14.2.3 kernel	Misses- Duplicated	0.00	0.00	0.00	Req per
14.2.4	Cache Hit Rate	0.00	0.00	0.00	Pct
14.2.5 kernel	Read Req (Total)	59.55	0.00	13750.00	Req per
14.2.6 kernel	Atomic Req	0.00	0.00	0.00	Req per
14.2.7 kernel	Read Req (1 DWord)	35.74	0.00	13750.00	Req per
14.2.8 kernel	Read Req (2 DWord)	0.00	0.00	0.00	Req per
14.2.9 kernel	Read Req (4 DWord)	9.61	0.00	4864.00	Req per
14.2.10 kernel	Read Req (8 DWord)	0.00	0.00	0.00	Req per
14.2.11 kernel	Read Req (16 DWord)	14.20	0.00	7128.00	Req per

14.3 Scalar L1D Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
14.3.0	sL1D-L2 BW	14897.24	0.00	3737600.00	Bytes per kernel
14.3.1	Read Req	75.72	0.00	38016.00	Req per kernel
14.3.2	Write Req	157.05	0.00	58400.00	Req per kernel
14.3.3	Atomic Req	0.00	0.00	0.00	Req per kernel
14.3.4	Stall Cycles	1.20	0.00	626.00	Cycles per kernel

15. Address Processing Unit and Data Return Path (TA/TD)

15.1 Address Processing Unit

Metric_ID	Metric	Avg	Min
Max	Unit		
15.1.0	Address Processing Unit Busy	87.8	34.86
124.86	Pct		
15.1.1	Address Stall	2.46	1.44
14.94	Pct		

15.1.2	Data Stall	56.0	0.65
65.92	Pct		
15.1.3	Data-Processor → Address Stall	0.0	0.0
0.0	Pct		
15.1.4	Sequencer → TA Address Stall		
	Cycles per kernel		
15.1.5	Sequencer → TA Command Stall		
	Cycles per kernel		
15.1.6	Sequencer → TA Data Stall		
	Cycles per kernel		
15.1.7	Total Instructions	28530319.98	
4325376.0	1078600928.0 Instructions per kernel		
15.1.8	Global/Generic Instructions	28243857.37	
4194304.0	1077936128.0 Instructions per kernel		
15.1.9	Global/Generic Read Instructions	26099733.59	
2097152.0	1073741824.0 Instructions per kernel		
15.1.10	Global/Generic Read Instructions for LDS		
	Instructions per kernel		
15.1.11	Global/Generic Write Instructions	2144123.78	
2097152.0	4194304.0 Instructions per kernel		
15.1.12	Global/Generic Atomic Instructions	0.0	0.0
0.0	Instructions per kernel		
15.1.13	Spill/Stack Instructions	286617.3	0.0
782208.0	Instructions per kernel		

15.1.14 524288.0	Spill/Stack Read Instructions Instructions per kernel	285623.84	0.0
15.1.15	Spill/Stack Read Instructions for LDS Instructions per kernel		
15.1.16 389120.0	Spill/Stack Write Instructions Instructions per kernel	897.18	0.0
15.1.17 0.0	Spill/Stack Atomic Instructions Instructions per kernel	0.0	0.0
15.1.18 8388608.0	Spill/Stack Total Cycles Cycles per kernel	4166901.88	0.0
15.1.19 1638912.0	Spill/Stack Coalesced Read Cycles per kernel	135837.85	0.0
15.1.20 1368576.0	Spill/Stack Coalesced Write Cycles per kernel	3136.08	0.0

15.2 Data-Return Path

Metric_ID Max	Metric Unit	Avg	Min
15.2.0 97.79	Data-Return Busy Pct	95.83	68.22
15.2.1 103.14	Cache RAM → Data-Return Stall Pct	88.13	45.54
15.2.2 0.0	Workgroup manager → Data-Return Stall Pct	0.0	0.0
15.2.3	Coalescable Instructions	34722.66	0.0

736064.0	Instructions per kernel		
15.2.4	Read Instructions	26385465.51	2228224.0
1074524032.0	Instructions per kernel		
15.2.5	Write Instructions	2145009.16	2097152.0
4574464.0	Instructions per kernel		
15.2.6	Atomic Instructions	0.0	0.0
0.0	Instructions per kernel		
15.2.7	Write Ack Instructions		
	Instructions per kernel		

INFO Not showing table with empty column(s): 16.6 L1D Addr Translation Stalls

16. Vector L1 Data Cache

16.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
16.1.0	Hit rate	29.56	Pct of peak
16.1.1	Bandwidth	9.61	Pct of peak
16.1.2	Utilization	98.03	Pct of peak
16.1.3	Coalescing	25.15	Pct of peak

16.2 L1D Cache Stalls (%)

Metric_ID	Metric	Min	Q1	Median	Q3	Max
16.2.0	Stalled on L2 Data	38.98	87.78	91.34	95.1	
114.84						

16.2.1 96.59	Stalled on L2 Req	0.0	7.78	9.46	74.49	
16.2.2	Stalled on Address					
16.2.3	Stalled on Data					
16.2.4	Stalled on Latency FIFO					
16.2.5	Stalled on Request FIFO					
16.2.6	Stalled on Read Return					
16.2.7	Tag RAM Stall (Read)	0.0	0.0	0.0	0.0	0.0
16.2.8 0.26	Tag RAM Stall (Write)	0.0	0.0	0.0	0.0	
16.2.9	Tag RAM Stall (Atomic)	0.0	0.0	0.0	0.0	0.0

16.3 L1D Cache Accesses

Metric_ID Unit	Metric	Avg	Min	Max
16.3.0 69000536576.0	Total Req Req per kernel	1824278968.5	270532608.0	
16.3.1	Read Req	1687040771.71	136314880.0	

68726325248.0	Req per kernel			
16.3.2	Write Req	137241911.65	134217728.0	
275552256.0	Req per kernel			
16.3.3	Atomic Req	0.0	0.0	0.0
Req per kernel				
16.3.4	Cache BW	58954276335.14	8657043456.0	
2233861144576.0	Bytes per kernel			
16.3.5	Cache Hit Rate	29.56	25.39	50.01
Pct				
16.3.6	Cache Accesses	460580283.87	67633152.0	
17452040192.0	Req per kernel			
16.3.7	Cache Hits	215777014.69	17170432.0	
8727887872.0	Req per kernel			
16.3.8	Invalidations	310.85	304.0	2736.0
Req per kernel				
16.3.9	L1-L2 BW	29042789273.97	4311744128.0	
1108208695936.0	Bytes per kernel			
16.3.10	Tag RAM 0 Req			
Req per kernel				
16.3.11	Tag RAM 1 Req			
Req per kernel				
16.3.12	Tag RAM 2 Req			
Req per kernel				
16.3.13	Tag RAM 3 Req			

Req per kernel

16.3.14	L1-L2 Read	208990313.23	16908285.0
8590771573.0	Req per kernel		

16.3.15	L1-L2 Write	35812955.94	33554432.0
135838360.0	Req per kernel		

16.3.16	L1-L2 Atomic	0.0	0.0	0.0
Req per kernel				

16.3.17	L1 Access Latency
Cycles	

16.3.18	L1-L2 Read Latency
Cycles	

16.3.19	L1-L2 Write Latency
Cycles	

16.4 L1D - L2 Transactions

Metric_ID	Metric	Xfer	Coherency	Avg	Min
Max	Unit				
16.4.0	NC - Read	Read	NC	0.00	0.00
0.00	Req per kernel				
16.4.1	UC - Read	Read	UC	0.00	0.00
0.00	Req per kernel				
16.4.2	CC - Read	Read	CC	0.00	0.00
0.00	Req per kernel				
16.4.3	RW - Read	Read	RW	208986119.98	16908286.00
8590687421.00	Req per kernel				

16.4.4	RW - Write	Write	RW	35812656.87	33554432.00
135611368.00	Req per kernel				
16.4.5	NC - Write	Write	NC	0.00	0.00
0.00	Req per kernel				
16.4.6	UC - Write	Write	UC	0.00	0.00
0.00	Req per kernel				
16.4.7	CC - Write	Write	CC	0.00	0.00
0.00	Req per kernel				
16.4.8	NC - Atomic	Atomic	NC	0.00	0.00
0.00	Req per kernel				
16.4.9	UC - Atomic	Atomic	UC	0.00	0.00
0.00	Req per kernel				
16.4.10	CC - Atomic	Atomic	CC	0.00	0.00
0.00	Req per kernel				
16.4.11	RW - Atomic	Atomic	RW	0.00	0.00
0.00	Req per kernel				

16.5 L1D Addr Translation

Metric_ID	Metric	Avg	Min	Max
Units				
16.5.0	Req	460580170.54	67633152.0	
17451954176.0	Req per kernel			
16.5.1	Inflight Req			
	Req per kernel			

16.5.2 Pct	Hit Ratio	99.82	99.59	99.9
16.5.3 17383841053.0	Hits Req per kernel	458946902.24	67474174.0	
16.5.4 67411891.0	Translation Misses Req per kernel	1586172.24	77824.0	
16.5.5	Misses under Translation Miss Req per kernel			
16.5.6	Permission Misses Req per kernel	0.0	0.0	0.0

INFO Not showing table with empty column(s): 17.4 L2 Cache Stalls

17. L2 Cache

17.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
17.1.0	Utilization	90.64	Pct
17.1.1	Bandwidth	16.04	Pct
17.1.2	Hit Rate	33.64	Pct
17.1.3	L2-Fabric Read BW	1865.79	Gb/s
17.1.4	L2-Fabric Write and Atomic BW	1799.64	Gb/s
17.1.5	HBM Bandwidth	5324.80	Gb/s

17.2 L2 - Fabric Transactions

Metric_ID	Metric	Avg	Min
Max	Unit		

17.2.0	Read BW	15268917439.16	
2163812480.0	625316607616.0 Bytes per kernel		
17.2.1	HBM Read Traffic	99.99	93.35
105.22	Pct		
17.2.2	Remote Read Traffic	0.03	0.0
6.65	Pct		
17.2.3	Uncached Read Traffic	0.0	0.0
0.01	Pct		
17.2.4	Write and Atomic BW	2147799474.72	
2147483648.0	2201149696.0 Bytes per kernel		
17.2.5	HBM Write and Atomic Traffic	100.0	95.25
104.7	Pct		
17.2.6	Remote Write and Atomic Traffic	0.01	0.0
4.75	Pct		
17.2.7	Atomic Traffic	0.0	0.0
0.0	Pct		
17.2.8	Uncached Write and Atomic Traffic	0.0	0.0
0.15	Pct		
17.2.9	Read Latency	2415.7	457.31
3008.21	Cycles		
17.2.10	Write and Atomic Latency	478.61	155.02
588.43	Cycles		
17.2.11	Atomic Latency		
	Cycles		

17.2.12 Read Stall
 Pct

17.2.13 Write Stall
 Pct

17.3 L2 Cache Accesses

Metric_ID Unit	Metric	Avg	Min	Max
17.3.0 1117039304832.0	Bandwidth Bytes per kernel	31335810325.75	6460784384.0	
17.3.1 Bytes per kernel	Read Bandwidth			
17.3.2 Bytes per kernel	Write Bandwidth			
17.3.3 Bytes per kernel	Atomic Bandwidth			
17.3.4 8726869569.0	Req Req per kernel	244811018.17	50474878.0	
17.3.5 8590738680.0	Read Req Req per kernel	208997334.82	16920447.0	
17.3.6 135871612.0	Write Req Req per kernel	35813042.11	33554432.0	
17.3.7 Req per kernel	Atomic Req	0.0	0.0	0.0

17.3.8	Streaming Req	5514.96	0.0	2356992.0
Req per kernel				
17.3.9	Bypasss Req			
Req per kernel				
17.3.10	Probe Req	111.56	0.0	54275.0
Req per kernel				
17.3.11	Input Buffer Req			
Req per kernel				
17.3.12	Cache Hit	33.64	33.27	49.1
Pct				
17.3.13	Hits	108741100.67	16791510.0	
4283935032.0	Hits per kernel			
17.3.14	Misses	135865953.37	33681477.0	
4877132896.0	Misses per kernel			
17.3.15	Writeback	16780467.08	16777328.0	
17679526.0	Cachelines per kernel			
17.3.16	Writeback (Internal)	16663652.53	16474772.0	
17486728.0	Cachelines per kernel			
17.3.17	Writeback (vL1D Req)	115487.97	26807.0	129321.0
Cachelines per kernel				
17.3.18	Evict (Internal)	135483535.79	33419754.0	
4893631391.0	Cachelines per kernel			
17.3.19	Evict (vL1D Req)	0.0	0.0	0.0
Cachelines per kernel				

17.3.20	NC Req	0.0	0.0	0.0
Req per kernel				
17.3.21	UC Req	54.97	0.0	17424.0
Req per kernel				
17.3.22	CC Req	0.0	0.0	0.0
Req per kernel				
17.3.23	RW Req	244807867.03	50474878.0	
8726529573.0	Req per kernel			

17.5 L2 - Fabric Interface Stalls

Metric_ID	Metric	Min	Max	Unit	Type
Transaction	Avg				
17.5.0	Read - PCIe Stall Pct				PCIe Stall Read
17.5.1	Read - Infinity Fabric Stall Pct				Infinity Fabric Stall Read
17.5.2	Read - HBM Stall Pct				HBM Stall Read
17.5.3	Write - PCIe Stall Pct				PCIe Stall Write
17.5.4	Write - Infinity Fabric Stall Pct				Infinity Fabric Stall Write
17.5.5	Write - HBM Stall Pct				HBM Stall Write
17.5.6	Write - Credit Starvation				Credit Starvation Write

0.0 0.0 0.0 Pct

17.6 L2 - Fabric Detailed Transaction Breakdown

Metric_ID Max	Metric Unit	Avg	Min
17.6.0 0.0	Read (32B) Req per kernel	0.0	0.0
17.6.1 -244195552.0	Read (64B) 261294630.0 Req per kernel	404252.48	
17.6.2 4860355328.0	Read (128B) Req per kernel	119086291.25	16903941.0
17.6.3 39018.0	Read (Uncached) Req per kernel	150.3	0.0
17.6.4 4947291346.0	HBM Read Req per kernel	118770225.17	16905006.0
17.6.5 327742705.0	Remote Read Req per kernel	1279351.95	0.0
17.6.6	Read Bandwidth - PCIe Bytes per kernel		
17.6.7	Read Bandwidth - Infinity Fabric Bytes per kernel		
17.6.8	Read Bandwidth - HBM Bytes per kernel		
17.6.9 1677064.0	Write and Atomic (32B) Req per kernel	-1657.84	-1659420.0

17.6.10	Write and Atomic (Uncached)	102.54	0.0
51225.0	Req per kernel		
17.6.11	Write and Atomic (64B)	33560195.71	33554432.0
35218319.0	Req per kernel		
17.6.12	HBM Write and Atomic	33558379.34	33554432.0
35134235.0	Req per kernel		
17.6.13	Remote Write and Atomic	2245.35	0.0
1673026.0	Req per kernel		
17.6.14	Write Bandwidth - PCIe		
	Bytes per kernel		
17.6.15	Write Bandwidth - Infinity Fabric		
	Bytes per kernel		
17.6.16	Write Bandwidth - HBM		
	Bytes per kernel		
17.6.17	Atomic	0.0	0.0
0.0	Req per kernel		
17.6.18	Atomic - HBM		
	Req per kernel		
17.6.19	Atomic Bandwidth - PCIe		
	Bytes per kernel		
17.6.20	Atomic Bandwidth - Infinity Fabric		
	Bytes per kernel		
17.6.21	Atomic Bandwidth - HBM		
	Bytes per kernel		

INFO Not showing table with empty column(s): 18.9 L2-Fabric Read Stall (Cycles per normUnit)

INFO Not showing table with empty column(s): 18.10 L2-Fabric Write and Atomic Stall (Cycles per normUnit)

18. L2 Cache (per Channel)

18.1 Aggregate Stats (All channels)

Metric_ID	Metric	Avg	Std Dev	Min	Max	Unit
18.1.0	L2 Cache Hit Rate	33.65	2.09	33.26	49.10	Pct

18.2 L2 Cache Hit Rate (pct)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.2.0	0.00	33.26	33.28	33.31	33.40	49.10
18.2.1	1.00	33.27	33.28	33.31	33.40	49.10
18.2.2	2.00	33.26	33.28	33.31	33.40	49.10
18.2.3	3.00	33.26	33.28	33.31	33.40	49.10
18.2.4	4.00	33.26	33.27	33.31	33.40	49.10
18.2.5	5.00	33.26	33.27	33.32	33.40	49.10
18.2.6	6.00	33.26	33.27	33.31	33.40	49.10
18.2.7	7.00	33.26	33.27	33.31	33.40	49.10
18.2.8	8.00	33.26	33.27	33.31	33.41	49.10
18.2.9	9.00	33.26	33.27	33.32	33.41	49.10
18.2.10	10.00	33.26	33.27	33.32	33.41	49.10
18.2.11	11.00	33.26	33.27	33.31	33.41	49.10
18.2.12	12.00	33.26	33.27	33.31	33.40	49.10
18.2.13	13.00	33.26	33.27	33.31	33.41	49.10

18.2.14	14.00	33.26	33.27	33.32	33.41	49.10
18.2.15	15.00	33.26	33.27	33.32	33.41	49.10
18.2.16	16.00	33.26	33.27	33.31	33.41	49.10
18.2.17	17.00	33.26	33.27	33.31	33.41	49.10
18.2.18	18.00	33.26	33.27	33.31	33.40	49.10
18.2.19	19.00	33.26	33.27	33.31	33.41	49.10
18.2.20	20.00	33.26	33.27	33.30	33.41	49.10
18.2.21	21.00	33.26	33.27	33.31	33.41	49.10
18.2.22	22.00	33.26	33.27	33.31	33.41	49.10
18.2.23	23.00	33.26	33.27	33.31	33.41	49.10
18.2.24	24.00	33.26	33.28	33.30	33.40	49.10
18.2.25	25.00	33.27	33.28	33.31	33.40	49.10
18.2.26	26.00	33.26	33.28	33.30	33.40	49.10
18.2.27	27.00	33.26	33.28	33.31	33.40	49.10
18.2.28	28.00	33.26	33.27	33.30	33.40	49.10
18.2.29	29.00	33.26	33.27	33.31	33.40	49.10
18.2.30	30.00	33.26	33.27	33.30	33.40	49.10
18.2.31	31.00	33.26	33.27	33.31	33.40	49.10
18.2.32	32.00	33.26	33.28	33.31	33.40	49.10
18.2.33	33.00	33.27	33.28	33.31	33.40	49.10
18.2.34	34.00	33.26	33.28	33.31	33.40	49.10
18.2.35	35.00	33.26	33.28	33.30	33.40	49.10
18.2.36	36.00	33.26	33.27	33.31	33.40	49.10
18.2.37	37.00	33.26	33.27	33.31	33.40	49.10

18.2.38	38.00	33.26	33.27	33.31	33.40	49.10
18.2.39	39.00	33.26	33.27	33.31	33.41	49.10
18.2.40	40.00	33.26	33.27	33.31	33.41	49.10
18.2.41	41.00	33.26	33.27	33.31	33.40	49.10
18.2.42	42.00	33.26	33.27	33.31	33.40	49.10
18.2.43	43.00	33.26	33.27	33.31	33.41	49.10
18.2.44	44.00	33.26	33.27	33.31	33.41	49.10
18.2.45	45.00	33.26	33.27	33.31	33.41	49.10
18.2.46	46.00	33.26	33.27	33.31	33.41	49.10
18.2.47	47.00	33.26	33.27	33.31	33.41	49.10
18.2.48	48.00	33.26	33.27	33.32	33.41	49.10
18.2.49	49.00	33.26	33.27	33.32	33.41	49.10
18.2.50	50.00	33.26	33.27	33.32	33.41	49.10
18.2.51	51.00	33.26	33.27	33.32	33.41	49.10
18.2.52	52.00	33.26	33.27	33.32	33.41	49.10
18.2.53	53.00	33.26	33.27	33.32	33.41	49.10
18.2.54	54.00	33.26	33.27	33.32	33.41	49.10
18.2.55	55.00	33.26	33.27	33.32	33.41	49.10
18.2.56	56.00	33.26	33.28	33.32	33.41	49.10
18.2.57	57.00	33.27	33.28	33.32	33.41	49.10
18.2.58	58.00	33.26	33.28	33.31	33.41	49.10
18.2.59	59.00	33.26	33.28	33.32	33.40	49.10
18.2.60	60.00	33.26	33.27	33.32	33.40	49.10
18.2.61	61.00	33.26	33.27	33.32	33.41	49.10

18.2.62	62.00	33.26	33.27	33.32	33.40	49.10
18.2.63	63.00	33.26	33.27	33.32	33.41	49.10
18.2.64	64.00	33.27	33.29	33.31	33.41	49.10
18.2.65	65.00	33.27	33.29	33.31	33.41	49.10
18.2.66	66.00	33.27	33.29	33.31	33.41	49.10
18.2.67	67.00	33.27	33.29	33.31	33.41	49.10
18.2.68	68.00	33.27	33.29	33.31	33.41	49.10
18.2.69	69.00	33.27	33.29	33.30	33.41	49.10
18.2.70	70.00	33.27	33.29	33.31	33.41	49.10
18.2.71	71.00	33.27	33.29	33.31	33.41	49.10
18.2.72	72.00	33.27	33.29	33.30	33.41	49.10
18.2.73	73.00	33.27	33.29	33.31	33.41	49.10
18.2.74	74.00	33.27	33.29	33.31	33.41	49.10
18.2.75	75.00	33.27	33.29	33.31	33.41	49.10
18.2.76	76.00	33.27	33.28	33.31	33.41	49.10
18.2.77	77.00	33.27	33.28	33.31	33.41	49.10
18.2.78	78.00	33.27	33.28	33.31	33.41	49.10
18.2.79	79.00	33.27	33.28	33.30	33.41	49.10
18.2.80	80.00	33.26	33.28	33.32	33.41	49.10
18.2.81	81.00	33.27	33.28	33.32	33.41	49.10
18.2.82	82.00	33.26	33.28	33.32	33.41	49.10
18.2.83	83.00	33.26	33.28	33.32	33.41	49.10
18.2.84	84.00	33.26	33.27	33.32	33.40	49.10
18.2.85	85.00	33.26	33.27	33.31	33.41	49.10

18.2.86	86.00	33.26	33.27	33.31	33.41	49.10
18.2.87	87.00	33.26	33.27	33.31	33.41	49.10
18.2.88	88.00	33.26	33.27	33.32	33.41	49.10
18.2.89	89.00	33.26	33.27	33.32	33.41	49.10
18.2.90	90.00	33.26	33.27	33.32	33.41	49.10
18.2.91	91.00	33.26	33.27	33.32	33.41	49.10
18.2.92	92.00	33.26	33.27	33.32	33.41	49.10
18.2.93	93.00	33.26	33.27	33.32	33.41	49.10
18.2.94	94.00	33.26	33.27	33.32	33.41	49.10
18.2.95	95.00	33.26	33.27	33.32	33.41	49.10
18.2.96	96.00	33.27	33.29	33.31	33.41	49.10
18.2.97	97.00	33.27	33.29	33.31	33.41	49.10
18.2.98	98.00	33.27	33.29	33.30	33.41	49.10
18.2.99	99.00	33.27	33.29	33.31	33.41	49.10
18.2.100	100.00	33.27	33.29	33.31	33.41	49.10
18.2.101	101.00	33.27	33.29	33.31	33.41	49.10
18.2.102	102.00	33.27	33.29	33.30	33.41	49.10
18.2.103	103.00	33.27	33.29	33.30	33.41	49.10
18.2.104	104.00	33.27	33.29	33.30	33.41	49.10
18.2.105	105.00	33.27	33.29	33.31	33.41	49.10
18.2.106	106.00	33.27	33.29	33.31	33.41	49.10
18.2.107	107.00	33.27	33.29	33.30	33.41	49.10
18.2.108	108.00	33.27	33.28	33.31	33.41	49.10
18.2.109	109.00	33.27	33.28	33.31	33.41	49.10

18.2.110	110.00	33.26	33.28	33.31	33.41	49.10
18.2.111	111.00	33.27	33.28	33.31	33.41	49.10
18.2.112	112.00	33.26	33.28	33.32	33.41	49.10
18.2.113	113.00	33.27	33.28	33.33	33.40	49.10
18.2.114	114.00	33.26	33.28	33.32	33.41	49.10
18.2.115	115.00	33.26	33.28	33.32	33.40	49.10
18.2.116	116.00	33.26	33.27	33.32	33.41	49.10
18.2.117	117.00	33.26	33.27	33.32	33.41	49.10
18.2.118	118.00	33.26	33.27	33.31	33.41	49.10
18.2.119	119.00	33.26	33.27	33.31	33.41	49.10
18.2.120	120.00	33.26	33.27	33.32	33.41	49.10
18.2.121	121.00	33.26	33.27	33.33	33.41	49.10
18.2.122	122.00	33.26	33.27	33.32	33.41	49.10
18.2.123	123.00	33.26	33.27	33.32	33.41	49.10
18.2.124	124.00	33.26	33.27	33.33	33.41	49.10
18.2.125	125.00	33.26	33.27	33.32	33.41	49.10
18.2.126	126.00	33.26	33.27	33.32	33.41	49.10
18.2.127	127.00	33.26	33.27	33.32	33.41	49.10

18.3 L2 Requests (per normUnit)

Metric_ID Max	Channel	Min	Q1	Median	Q3
18.3.0 68178743.00	0.00	394320.00	394400.00	394660.00	395194.00
18.3.1	1.00	394400.00	394401.00	394668.00	395196.00

68178740.00

18.3.2	2.00	394320.00	394400.50	394670.00	395194.00
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68178738.00

18.3.3	3.00	394320.00	394400.50	394650.00	395196.00
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68178681.00

18.3.4	4.00	394319.00	394321.50	394660.00	395194.00
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68178434.00

18.3.5	5.00	394319.00	394321.00	394666.00	395196.00
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68178500.00

18.3.6	6.00	394320.00	394321.00	394652.00	395200.00
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68178707.00

18.3.7	7.00	394320.00	394321.00	394654.00	395200.00
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68178611.00

18.3.8	8.00	394320.00	394321.00	394690.00	395179.00
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68178666.00

18.3.9	9.00	394320.00	394321.00	394668.00	395179.00
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68178625.00

18.3.10	10.00	394320.00	394320.50	394666.00	395186.00
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68178752.00

18.3.11	11.00	394320.00	394321.00	394684.00	395179.00
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68178726.00

18.3.12	12.00	394320.00	394320.50	394622.00	395192.00
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68178644.00

18.3.13	13.00	394320.00	394321.00	394622.00	395192.00
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68178855.00

18.3.14	14.00	394320.00	394320.00	394640.00	395196.00
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68178686.00

18.3.15	15.00	394320.00	394321.00	394646.00	395195.00
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68178632.00

18.3.16	16.00	394320.00	394321.00	394588.00	395174.00
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68178479.00

18.3.17	17.00	394320.00	394321.00	394606.00	395172.00
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68178603.00

18.3.18	18.00	394320.00	394320.50	394606.00	395177.50
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68178517.00

18.3.19	19.00	394320.00	394321.00	394636.00	395167.00
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68178625.00

18.3.20	20.00	394320.00	394320.50	394572.00	395188.00
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68178818.00

18.3.21	21.00	394320.00	394321.00	394591.00	395188.00
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68178831.00

18.3.22	22.00	394320.00	394320.00	394592.00	395192.00
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68178751.00

18.3.23	23.00	394320.00	394321.00	394592.00	395190.00
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68178714.00

18.3.24	24.00	394320.00	394400.00	394600.00	395193.00
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68178729.00

18.3.25	25.00	394400.00	394401.00	394586.00	395193.00
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68178692.00

18.3.26	26.00	394320.00	394400.50	394595.00	395193.00
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68178610.00

18.3.27	27.00	394320.00	394400.50	394616.00	395193.50
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68178696.00

18.3.28	28.00	394320.00	394321.50	394590.00	395194.00
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68178614.00

18.3.29	29.00	394320.00	394321.00	394596.00	395195.00
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68178533.00

18.3.30	30.00	394320.00	394321.00	394592.00	395196.00
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68178911.00

18.3.31	31.00	394320.00	394321.00	394578.00	395194.00
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68178636.00

18.3.32	32.00	394320.00	394400.00	394616.00	395192.00
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68178899.00

18.3.33	33.00	394400.00	394401.00	394620.00	395192.00
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68178784.00

18.3.34	34.00	394320.00	394400.50	394587.00	395194.00
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68178782.00

18.3.35	35.00	394320.00	394400.50	394618.00	395196.00
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68178849.00

18.3.36	36.00	394320.00	394321.50	394604.00	395193.50
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68178750.00

18.3.37	37.00	394320.00	394321.00	394598.00	395192.50
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68178624.00

18.3.38	38.00	394320.00	394321.00	394624.00	395194.00
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68178833.00

18.3.39	39.00	394320.00	394321.00	394580.00	395192.00
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68178422.00

18.3.40	40.00	394320.00	394321.00	394604.00	395174.00
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68178579.00

18.3.41	41.00	394320.00	394321.00	394600.00	395179.00
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68178615.00

18.3.42	42.00	394320.00	394320.50	394600.00	395174.00
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68178447.00

18.3.43	43.00	394320.00	394321.00	394588.00	395176.00
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68178696.00

18.3.44	44.00	394320.00	394320.50	394606.00	395188.00
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68178435.00

18.3.45	45.00	394320.00	394321.00	394584.00	395188.00
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68178757.00

18.3.46	46.00	394320.00	394320.00	394582.00	395192.00
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68178625.00

18.3.47	47.00	394320.00	394321.00	394590.00	395192.00
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68178640.00

18.3.48	48.00	394320.00	394321.00	394677.00	395179.00
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68174972.00

18.3.49	49.00	394320.00	394321.00	394666.00	395174.00
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68175100.00

18.3.50	50.00	394320.00	394320.50	394686.00	395177.00
68175024.00					

18.3.51	51.00	394320.00	394321.00	394702.00	395176.00
68174957.00					

18.3.52	52.00	394320.00	394320.50	394670.00	395191.00
68174785.00					

18.3.53	53.00	394320.00	394321.00	394644.00	395194.00
68174820.00					

18.3.54	54.00	394320.00	394320.00	394644.00	395194.00
68174845.00					

18.3.55	55.00	394320.00	394321.00	394652.00	395196.00
68174943.00					

18.3.56	56.00	394320.00	394400.00	394678.00	395192.00
68175136.00					

18.3.57	57.00	394399.00	394401.00	394694.00	395194.00
68175208.00					

18.3.58	58.00	394320.00	394400.50	394674.00	395194.00
68175192.00					

18.3.59	59.00	394320.00	394400.50	394646.00	395196.00
68175136.00					

18.3.60	60.00	394320.00	394321.50	394668.00	395193.00
68174985.00					

18.3.61	61.00	394319.00	394321.00	394696.00	395194.00
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68174832.00

18.3.62	62.00	394319.00	394321.00	394664.00	395195.00
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68174931.00

18.3.63	63.00	394320.00	394321.00	394690.00	395196.00
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68174853.00

18.3.64	64.00	394320.00	394321.00	394628.00	395188.00
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68178575.00

18.3.65	65.00	394320.00	394321.00	394604.00	395188.00
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68178692.00

18.3.66	66.00	394320.00	394320.50	394610.00	395180.00
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68178509.00

18.3.67	67.00	394320.00	394321.00	394608.00	395186.00
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68178687.00

18.3.68	68.00	394320.00	394320.50	394612.00	395215.50
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68178704.00

18.3.69	69.00	394320.00	394321.00	394584.00	395216.00
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68178674.00

18.3.70	70.00	394320.00	394320.00	394626.00	395212.00
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68178679.00

18.3.71	71.00	394319.00	394321.00	394596.00	395214.00
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68178551.00

18.3.72	72.00	394320.00	394400.00	394624.00	395224.00
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68178745.00

18.3.73	73.00	394400.00	394401.00	394604.00	395224.00
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68178649.00

18.3.74	74.00	394320.00	394400.50	394610.00	395226.00
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68178669.00

18.3.75	75.00	394320.00	394400.50	394598.00	395226.00
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68178933.00

18.3.76	76.00	394320.00	394321.50	394610.00	395226.00
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68178627.00

18.3.77	77.00	394320.00	394321.00	394634.00	395226.00
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68178606.00

18.3.78	78.00	394320.00	394321.00	394582.00	395226.00
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68178950.00

18.3.79	79.00	394320.00	394321.00	394598.00	395227.00
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68178513.00

18.3.80	80.00	394320.00	394400.00	394676.00	395202.00
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68177275.00

18.3.81	81.00	394400.00	394401.00	394696.00	395202.00
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68177191.00

18.3.82	82.00	394320.00	394400.50	394656.00	395198.00
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68177328.00

18.3.83	83.00	394320.00	394400.50	394654.00	395200.00
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68177293.00

18.3.84	84.00	394320.00	394321.50	394676.00	395200.00
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68177181.00

18.3.85	85.00	394320.00	394321.00	394668.00	395198.00
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68177194.00

18.3.86	86.00	394320.00	394321.00	394692.00	395200.00
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68177369.00

18.3.87	87.00	394320.00	394321.00	394670.00	395203.00
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68177020.00

18.3.88	88.00	394320.00	394321.00	394674.00	395181.00
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68177169.00

18.3.89	89.00	394320.00	394321.00	394674.00	395179.50
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68177184.00

18.3.90	90.00	394320.00	394320.50	394658.00	395178.00
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68177171.00

18.3.91	91.00	394320.00	394321.00	394706.00	395179.00
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68177331.00

18.3.92	92.00	394320.00	394320.50	394652.00	395194.00
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68176953.00

18.3.93	93.00	394320.00	394321.00	394632.00	395191.00
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68177195.00

18.3.94	94.00	394319.00	394320.00	394632.00	395196.00
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68177011.00

18.3.95	95.00	394319.00	394321.00	394644.00	395196.00
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68177067.00

18.3.96	96.00	394320.00	394321.00	394660.00	395194.00
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68178541.00

18.3.97	97.00	394319.00	394321.00	394598.00	395193.00
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68178641.00

18.3.98	98.00	394320.00	394320.50	394614.00	395181.50
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68178720.00

18.3.99	99.00	394320.00	394321.00	394612.00	395194.00
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68178562.00

18.3.100	100.00	394320.00	394320.50	394600.00	395222.00
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68178550.00

18.3.101	101.00	394320.00	394321.00	394576.00	395222.00
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68178634.00

18.3.102	102.00	394320.00	394320.00	394592.00	395220.00
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68178590.00

18.3.103	103.00	394320.00	394321.00	394640.00	395220.00
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68178422.00

18.3.104	104.00	394320.00	394400.00	394606.00	395227.00
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68178726.00

18.3.105	105.00	394400.00	394401.00	394616.00	395226.00
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68178701.00

18.3.106	106.00	394320.00	394400.50	394626.00	395229.00
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68178798.00

18.3.107	107.00	394320.00	394400.50	394638.00	395229.50
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68178915.00

18.3.108	108.00	394320.00	394321.50	394632.00	395224.00
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68178711.00

18.3.109	109.00	394320.00	394321.00	394598.00	395224.00
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68178691.00

18.3.110	110.00	394320.00	394321.00	394628.00	395228.00
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68178819.00

18.3.111	111.00	394320.00	394321.00	394646.00	395228.00
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68178665.00

18.3.112	112.00	394320.00	394401.00	394674.00	395195.00
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68175172.00

18.3.113	113.00	394400.00	394401.00	394684.00	395196.00
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68175015.00

18.3.114	114.00	394320.00	394400.50	394674.00	395190.00
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68175063.00

18.3.115	115.00	394320.00	394400.50	394656.00	395196.00
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68175166.00

18.3.116	116.00	394319.00	394321.50	394672.00	395194.50
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68174792.00

18.3.117	117.00	394319.00	394321.00	394702.00	395192.00
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68174839.00

18.3.118	118.00	394320.00	394321.00	394644.00	395194.00
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68174890.00

18.3.119	119.00	394320.00	394321.00	394650.00	395196.00
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68174643.00

18.3.120	120.00	394320.00	394321.00	394684.00	395178.00
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68174932.00

18.3.121	121.00	394320.00	394321.00	394690.00	395183.00
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68174874.00

18.3.122	122.00	394320.00	394320.50	394646.00	395176.00
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68174945.00

18.3.123	123.00	394320.00	394321.00	394680.00	395178.00
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68174909.00

18.3.124	124.00	394320.00	394320.50	394644.00	395190.00
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68174860.00

18.3.125	125.00	394320.00	394321.00	394656.00	395194.50
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68174944.00

18.3.126	126.00	394320.00	394320.00	394632.00	395192.00
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68174886.00

18.3.127	127.00	394320.00	394321.00	394668.00	395194.00
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68174852.00

18.4 L2 Requests (per normUnit)

Metric_ID	Channel	L2 Read	L2 Write	L2 Atomic
18.4.0	0.00	1632834.24	279781.84	0.00
18.4.1	1.00	1632840.31	279781.68	0.00
18.4.2	2.00	1632834.91	279781.65	0.00
18.4.3	3.00	1632834.42	279781.57	0.00
18.4.4	4.00	1632812.19	279781.76	0.00
18.4.5	5.00	1632812.23	279781.68	0.00
18.4.6	6.00	1632818.06	279781.71	0.00
18.4.7	7.00	1632813.43	279781.85	0.00
18.4.8	8.00	1632796.75	279781.64	0.00

18.4.9	9.00	1632796.50	279781.75	0.00
18.4.10	10.00	1632794.82	279781.80	0.00
18.4.11	11.00	1632798.52	279781.82	0.00
18.4.12	12.00	1632800.18	279781.76	0.00
18.4.13	13.00	1632800.84	279781.70	0.00
18.4.14	14.00	1632804.41	279781.74	0.00
18.4.15	15.00	1632803.81	279781.70	0.00
18.4.16	16.00	1632783.17	279790.61	0.00
18.4.17	17.00	1632783.72	279790.66	0.00
18.4.18	18.00	1632780.44	279790.75	0.00
18.4.19	19.00	1632782.37	279790.68	0.00
18.4.20	20.00	1632787.01	279790.59	0.00
18.4.21	21.00	1632787.80	279790.50	0.00
18.4.22	22.00	1632791.76	279790.48	0.00
18.4.23	23.00	1632790.28	279790.55	0.00
18.4.24	24.00	1632819.60	279790.77	0.00
18.4.25	25.00	1632825.89	279790.64	0.00
18.4.26	26.00	1632820.23	279790.79	0.00
18.4.27	27.00	1632820.59	279790.65	0.00
18.4.28	28.00	1632799.50	279790.40	0.00
18.4.29	29.00	1632798.94	279790.55	0.00
18.4.30	30.00	1632804.04	279790.42	0.00
18.4.31	31.00	1632799.70	279790.56	0.00
18.4.32	32.00	1632819.62	279772.35	0.00

18.4.33	33.00	1632826.06	279772.32	0.00
18.4.34	34.00	1632822.09	279772.36	0.00
18.4.35	35.00	1632822.03	279772.42	0.00
18.4.36	36.00	1632798.62	279772.39	0.00
18.4.37	37.00	1632797.76	279772.40	0.00
18.4.38	38.00	1632803.87	279772.34	0.00
18.4.39	39.00	1632798.47	279772.37	0.00
18.4.40	40.00	1632786.36	279772.28	0.00
18.4.41	41.00	1632786.10	279772.36	0.00
18.4.42	42.00	1632783.92	279772.35	0.00
18.4.43	43.00	1632786.22	279772.36	0.00
18.4.44	44.00	1632787.36	279772.37	0.00
18.4.45	45.00	1632788.05	279772.29	0.00
18.4.46	46.00	1632793.67	279772.31	0.00
18.4.47	47.00	1632792.77	279772.34	0.00
18.4.48	48.00	1632797.27	279790.67	0.00
18.4.49	49.00	1632796.58	279790.68	0.00
18.4.50	50.00	1632794.77	279790.42	0.00
18.4.51	51.00	1632797.29	279790.74	0.00
18.4.52	52.00	1632799.10	279790.59	0.00
18.4.53	53.00	1632799.79	279790.50	0.00
18.4.54	54.00	1632803.30	279790.66	0.00
18.4.55	55.00	1632805.23	279790.60	0.00
18.4.56	56.00	1632829.98	279790.59	0.00

18.4.57	57.00	1632837.76	279790.52	0.00
18.4.58	58.00	1632832.18	279790.67	0.00
18.4.59	59.00	1632830.57	279790.58	0.00
18.4.60	60.00	1632810.70	279790.62	0.00
18.4.61	61.00	1632809.55	279790.66	0.00
18.4.62	62.00	1632814.03	279790.57	0.00
18.4.63	63.00	1632809.09	279790.52	0.00
18.4.64	64.00	1632794.28	279772.32	0.00
18.4.65	65.00	1632795.23	279772.39	0.00
18.4.66	66.00	1632792.46	279772.39	0.00
18.4.67	67.00	1632796.04	279772.30	0.00
18.4.68	68.00	1632799.21	279772.43	0.00
18.4.69	69.00	1632800.17	279772.26	0.00
18.4.70	70.00	1632802.81	279772.36	0.00
18.4.71	71.00	1632802.84	279772.37	0.00
18.4.72	72.00	1632832.30	279772.41	0.00
18.4.73	73.00	1632839.85	279772.30	0.00
18.4.74	74.00	1632832.97	279772.32	0.00
18.4.75	75.00	1632831.90	279772.40	0.00
18.4.76	76.00	1632811.93	279772.29	0.00
18.4.77	77.00	1632811.56	279772.43	0.00
18.4.78	78.00	1632815.86	279772.33	0.00
18.4.79	79.00	1632811.50	279772.30	0.00
18.4.80	80.00	1632834.27	279786.93	0.00

18.4.81	81.00	1632840.50	279786.99	0.00
18.4.82	82.00	1632835.28	279787.00	0.00
18.4.83	83.00	1632835.82	279787.01	0.00
18.4.84	84.00	1632813.37	279787.07	0.00
18.4.85	85.00	1632812.30	279787.04	0.00
18.4.86	86.00	1632818.50	279787.04	0.00
18.4.87	87.00	1632813.53	279787.20	0.00
18.4.88	88.00	1632796.97	279787.03	0.00
18.4.89	89.00	1632795.65	279786.99	0.00
18.4.90	90.00	1632794.95	279787.02	0.00
18.4.91	91.00	1632799.05	279786.99	0.00
18.4.92	92.00	1632800.25	279787.07	0.00
18.4.93	93.00	1632800.89	279786.99	0.00
18.4.94	94.00	1632805.34	279786.99	0.00
18.4.95	95.00	1632805.51	279787.02	0.00
18.4.96	96.00	1632798.94	279772.36	0.00
18.4.97	97.00	1632799.26	279772.40	0.00
18.4.98	98.00	1632798.01	279772.23	0.00
18.4.99	99.00	1632800.33	279772.43	0.00
18.4.100	100.00	1632801.85	279772.39	0.00
18.4.101	101.00	1632803.38	279772.31	0.00
18.4.102	102.00	1632805.05	279772.37	0.00
18.4.103	103.00	1632804.25	279772.26	0.00
18.4.104	104.00	1632833.10	279772.38	0.00

18.4.105	105.00	1632839.40	279772.29	0.00
18.4.106	106.00	1632834.39	279772.41	0.00
18.4.107	107.00	1632833.86	279772.29	0.00
18.4.108	108.00	1632812.47	279772.30	0.00
18.4.109	109.00	1632811.12	279772.37	0.00
18.4.110	110.00	1632816.07	279772.36	0.00
18.4.111	111.00	1632812.72	279772.45	0.00
18.4.112	112.00	1632830.49	279787.18	0.00
18.4.113	113.00	1632837.85	279787.20	0.00
18.4.114	114.00	1632830.46	279787.34	0.00
18.4.115	115.00	1632831.03	279787.10	0.00
18.4.116	116.00	1632809.17	279787.32	0.00
18.4.117	117.00	1632808.45	279787.15	0.00
18.4.118	118.00	1632812.84	279787.23	0.00
18.4.119	119.00	1632809.25	279787.29	0.00
18.4.120	120.00	1632796.51	279787.26	0.00
18.4.121	121.00	1632797.38	279787.28	0.00
18.4.122	122.00	1632793.04	279787.09	0.00
18.4.123	123.00	1632796.01	279787.31	0.00
18.4.124	124.00	1632797.64	279787.36	0.00
18.4.125	125.00	1632800.28	279787.17	0.00
18.4.126	126.00	1632802.84	279787.29	0.00
18.4.127	127.00	1632803.80	279787.25	0.00

18.5 L2-Fabric Requests (per normUnit)

Metric_ID L2-Fabric Atomic	Channel	L2-Fabric Read	L2-Fabric Write and Atomic
18.5.0 0.00	0.00	932168.59	262176.97
18.5.1 0.00	1.00	932169.28	262177.00
18.5.2 0.00	2.00	932167.87	262176.96
18.5.3 0.00	3.00	932169.02	262176.94
18.5.4 0.00	4.00	932167.27	262176.85
18.5.5 0.00	5.00	932167.35	262176.99
18.5.6 0.00	6.00	932167.18	262176.96
18.5.7 0.00	7.00	932168.13	262177.13
18.5.8 0.00	8.00	932169.71	262176.92
18.5.9 0.00	9.00	932170.19	262177.00
18.5.10 0.00	10.00	932170.22	262177.03

18.5.11 0.00	11.00	932170.82	262177.09
18.5.12 0.00	12.00	932169.91	262177.11
18.5.13 0.00	13.00	932171.17	262177.10
18.5.14 0.00	14.00	932170.52	262177.08
18.5.15 0.00	15.00	932170.86	262177.00
18.5.16 0.00	16.00	929156.84	262176.81
18.5.17 0.00	17.00	929159.57	262177.07
18.5.18 0.00	18.00	929156.93	262176.78
18.5.19 0.00	19.00	929157.67	262177.12
18.5.20 0.00	20.00	929157.34	262177.17
18.5.21 0.00	21.00	929160.68	262177.16
18.5.22 0.00	22.00	929158.00	262177.16

18.5.23 0.00	23.00	929158.08	262177.17
18.5.24 0.00	24.00	929151.87	262176.93
18.5.25 0.00	25.00	929154.84	262176.99
18.5.26 0.00	26.00	929152.99	262176.80
18.5.27 0.00	27.00	929153.12	262177.05
18.5.28 0.00	28.00	929150.28	262176.96
18.5.29 0.00	29.00	929152.62	262177.16
18.5.30 0.00	30.00	929151.40	262177.07
18.5.31 0.00	31.00	929151.12	262177.18
18.5.32 0.00	32.00	940563.14	262177.50
18.5.33 0.00	33.00	940565.09	262177.44
18.5.34 0.00	34.00	940565.80	262177.37

18.5.35 0.00	35.00	940567.87	262177.50
18.5.36 0.00	36.00	940561.86	262177.50
18.5.37 0.00	37.00	940563.40	262177.53
18.5.38 0.00	38.00	940563.79	262177.49
18.5.39 0.00	39.00	940565.67	262177.40
18.5.40 0.00	40.00	940549.60	262177.35
18.5.41 0.00	41.00	940551.51	262177.46
18.5.42 0.00	42.00	940551.81	262177.19
18.5.43 0.00	43.00	940554.08	262177.46
18.5.44 0.00	44.00	940549.47	262177.29
18.5.45 0.00	45.00	940552.45	262177.44
18.5.46 0.00	46.00	940551.81	262177.44

18.5.47 0.00	47.00	940553.80	262177.56
18.5.48 0.00	48.00	918291.82	262173.66
18.5.49 0.00	49.00	918288.93	262173.79
18.5.50 0.00	50.00	918290.72	262173.64
18.5.51 0.00	51.00	918289.53	262173.68
18.5.52 0.00	52.00	918291.05	262173.67
18.5.53 0.00	53.00	918289.07	262173.52
18.5.54 0.00	54.00	918291.49	262173.72
18.5.55 0.00	55.00	918290.14	262173.74
18.5.56 0.00	56.00	918292.10	262173.81
18.5.57 0.00	57.00	918290.09	262173.73
18.5.58 0.00	58.00	918292.34	262173.83

18.5.59 0.00	59.00	918291.49	262173.81
18.5.60 0.00	60.00	918290.25	262173.83
18.5.61 0.00	61.00	918287.66	262173.74
18.5.62 0.00	62.00	918289.82	262173.55
18.5.63 0.00	63.00	918288.94	262173.88
18.5.64 0.00	64.00	928810.73	262176.88
18.5.65 0.00	65.00	928810.39	262177.10
18.5.66 0.00	66.00	928811.18	262176.89
18.5.67 0.00	67.00	928812.97	262177.00
18.5.68 0.00	68.00	928811.83	262177.17
18.5.69 0.00	69.00	928812.21	262176.94
18.5.70 0.00	70.00	928811.09	262177.11

18.5.71 0.00	71.00	928812.40	262177.02
18.5.72 0.00	72.00	928812.15	262177.09
18.5.73 0.00	73.00	928813.78	262176.88
18.5.74 0.00	74.00	928812.83	262176.98
18.5.75 0.00	75.00	928813.81	262177.20
18.5.76 0.00	76.00	928809.72	262177.08
18.5.77 0.00	77.00	928810.58	262177.15
18.5.78 0.00	78.00	928811.91	262177.31
18.5.79 0.00	79.00	928812.70	262176.99
18.5.80 0.00	80.00	934801.22	262175.91
18.5.81 0.00	81.00	934800.97	262175.89
18.5.82 0.00	82.00	934802.76	262175.99

18.5.83 0.00	83.00	934800.49	262175.90
18.5.84 0.00	84.00	934800.08	262176.09
18.5.85 0.00	85.00	934799.43	262176.01
18.5.86 0.00	86.00	934800.25	262176.06
18.5.87 0.00	87.00	934797.69	262175.95
18.5.88 0.00	88.00	934790.48	262175.89
18.5.89 0.00	89.00	934790.30	262175.86
18.5.90 0.00	90.00	934791.24	262175.86
18.5.91 0.00	91.00	934789.11	262176.03
18.5.92 0.00	92.00	934789.47	262175.82
18.5.93 0.00	93.00	934790.12	262175.81
18.5.94 0.00	94.00	934791.86	262175.93

18.5.95 0.00	95.00	934789.29	262175.92
18.5.96 0.00	96.00	923909.71	262176.90
18.5.97 0.00	97.00	923909.39	262176.93
18.5.98 0.00	98.00	923911.71	262177.07
18.5.99 0.00	99.00	923912.42	262176.88
18.5.100 0.00	100.00	923910.01	262176.85
18.5.101 0.00	101.00	923909.78	262176.96
18.5.102 0.00	102.00	923910.87	262176.98
18.5.103 0.00	103.00	923911.41	262176.97
18.5.104 0.00	104.00	923908.94	262176.96
18.5.105 0.00	105.00	923908.66	262176.97
18.5.106 0.00	106.00	923909.59	262176.86

18.5.107 0.00	107.00	923910.29	262177.18
18.5.108 0.00	108.00	923908.03	262177.16
18.5.109 0.00	109.00	923906.96	262177.04
18.5.110 0.00	110.00	923909.08	262177.10
18.5.111 0.00	111.00	923909.42	262176.97
18.5.112 0.00	112.00	920813.65	262173.79
18.5.113 0.00	113.00	920815.75	262173.64
18.5.114 0.00	114.00	920815.16	262173.60
18.5.115 0.00	115.00	920815.47	262173.86
18.5.116 0.00	116.00	920812.88	262173.62
18.5.117 0.00	117.00	920814.72	262173.75
18.5.118 0.00	118.00	920813.28	262173.62

18.5.119 0.00	119.00	920812.97	262173.64
18.5.120 0.00	120.00	920811.47	262173.62
18.5.121 0.00	121.00	920813.11	262173.62
18.5.122 0.00	122.00	920811.90	262173.71
18.5.123 0.00	123.00	920812.17	262173.62
18.5.124 0.00	124.00	920811.51	262173.71
18.5.125 0.00	125.00	920813.90	262173.66
18.5.126 0.00	126.00	920812.68	262173.73
18.5.127 0.00	127.00	920813.16	262173.74

18.6 L2-Fabric Read Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.6.0	0.00	358.35	2035.03	2219.45	2462.48	3013.50
18.6.1	1.00	365.01	2068.65	2253.72	2495.94	3060.27
18.6.2	2.00	368.05	2077.01	2273.85	2540.47	3032.36

18.6.3	3.00	368.02	2070.52	2271.39	2511.92	3093.87
18.6.4	4.00	376.77	2063.10	2242.23	2447.62	3010.67
18.6.5	5.00	369.08	2064.05	2248.51	2448.61	2991.68
18.6.6	6.00	393.80	2134.04	2329.62	2571.85	3041.44
18.6.7	7.00	379.44	2117.83	2331.04	2558.61	3079.04
18.6.8	8.00	359.68	2516.59	2640.88	2773.38	3204.90
18.6.9	9.00	368.72	2620.79	2755.62	2894.89	3365.19
18.6.10	10.00	362.97	2549.96	2734.47	2926.38	3406.22
18.6.11	11.00	367.74	2602.75	2781.75	2961.88	3409.84
18.6.12	12.00	377.57	2216.33	2427.71	2629.87	3169.20
18.6.13	13.00	367.92	2216.13	2432.45	2647.60	3188.64
18.6.14	14.00	388.01	2312.15	2569.16	2844.25	3164.08
18.6.15	15.00	373.34	2321.11	2568.50	2830.72	3267.24
18.6.16	16.00	469.16	2508.94	2624.30	2749.96	3183.66
18.6.17	17.00	479.39	2593.44	2730.47	2874.61	3369.66
18.6.18	18.00	474.72	2542.93	2708.38	2905.05	3316.34
18.6.19	19.00	479.39	2582.80	2753.87	2934.20	3357.54
18.6.20	20.00	485.94	2212.24	2416.23	2608.43	3113.64
18.6.21	21.00	478.89	2206.09	2423.07	2626.87	3104.57
18.6.22	22.00	498.65	2303.42	2551.11	2810.81	3166.28
18.6.23	23.00	484.07	2315.15	2557.46	2810.84	3237.02
18.6.24	24.00	465.69	2031.69	2195.81	2437.93	2984.46
18.6.25	25.00	475.53	2057.06	2232.20	2467.88	3037.08
18.6.26	26.00	479.23	2070.44	2261.34	2520.79	2996.37

18.6.27	27.00	479.28	2062.08	2252.47	2489.84	3031.44
18.6.28	28.00	485.49	2056.69	2226.02	2422.31	2996.39
18.6.29	29.00	479.18	2062.18	2231.11	2430.76	2993.78
18.6.30	30.00	503.82	2125.99	2322.35	2548.92	3045.93
18.6.31	31.00	489.70	2110.15	2316.52	2534.21	3122.08
18.6.32	32.00	498.69	2520.59	2641.40	2754.07	3231.45
18.6.33	33.00	510.62	2610.67	2746.19	2886.76	3390.81
18.6.34	34.00	505.63	2580.72	2749.10	2906.12	3448.48
18.6.35	35.00	510.48	2628.47	2790.22	2959.52	3437.65
18.6.36	36.00	516.15	2229.98	2449.37	2628.04	3216.71
18.6.37	37.00	510.07	2222.90	2437.67	2653.48	3131.02
18.6.38	38.00	530.33	2318.82	2570.04	2841.83	3228.46
18.6.39	39.00	515.25	2250.72	2488.79	2711.27	3209.39
18.6.40	40.00	496.14	2040.83	2217.43	2460.75	2980.32
18.6.41	41.00	505.30	2080.66	2258.09	2511.40	3033.04
18.6.42	42.00	507.91	2076.83	2273.54	2531.18	3011.16
18.6.43	43.00	509.11	2067.55	2257.43	2510.25	3017.63
18.6.44	44.00	515.83	2076.16	2246.63	2460.45	2956.04
18.6.45	45.00	510.55	2080.66	2252.77	2480.97	2958.22
18.6.46	46.00	535.21	2139.79	2337.71	2565.29	3045.52
18.6.47	47.00	521.25	2125.67	2323.47	2548.49	3060.68
18.6.48	48.00	347.76	1996.78	2181.47	2434.26	2958.62
18.6.49	49.00	354.87	2031.16	2222.90	2475.13	3028.56
18.6.50	50.00	357.28	2024.31	2232.74	2494.85	2944.56

18.6.51	51.00	358.55	2022.34	2226.48	2483.76	2920.98
18.6.52	52.00	366.08	2027.57	2217.06	2412.90	2908.68
18.6.53	53.00	360.57	2031.88	2232.19	2427.02	2946.98
18.6.54	54.00	386.32	2092.68	2300.49	2539.27	3030.36
18.6.55	55.00	371.48	2077.82	2291.89	2518.14	3030.76
18.6.56	56.00	345.35	2472.54	2596.61	2718.82	3218.71
18.6.57	57.00	360.18	2571.32	2710.36	2851.94	3382.62
18.6.58	58.00	354.56	2535.94	2704.26	2876.34	3358.25
18.6.59	59.00	359.69	2582.39	2746.38	2926.72	3382.58
18.6.60	60.00	366.27	2184.96	2414.66	2599.91	3092.65
18.6.61	61.00	360.12	2176.03	2405.51	2622.07	3073.96
18.6.62	62.00	380.99	2269.30	2524.34	2818.76	3223.48
18.6.63	63.00	365.12	2205.27	2445.14	2684.26	3168.35
18.6.64	64.00	370.21	2038.32	2202.33	2426.55	2926.01
18.6.65	65.00	378.83	2069.87	2244.50	2473.61	3007.40
18.6.66	66.00	380.04	2071.41	2260.57	2464.83	2896.77
18.6.67	67.00	381.51	2064.58	2252.34	2453.52	2948.43
18.6.68	68.00	385.37	2061.36	2224.56	2417.71	2862.57
18.6.69	69.00	379.99	2067.99	2230.05	2421.85	2920.95
18.6.70	70.00	404.73	2125.98	2311.94	2509.65	2936.95
18.6.71	71.00	390.58	2109.49	2308.23	2496.79	3003.46
18.6.72	72.00	365.85	2510.72	2629.44	2737.77	3238.82
18.6.73	73.00	379.80	2603.85	2729.54	2869.66	3310.45
18.6.74	74.00	374.80	2562.90	2708.98	2881.91	3301.54

18.6.75	75.00	379.67	2613.28	2764.37	2934.84	3363.90
18.6.76	76.00	385.58	2225.48	2421.92	2599.09	3038.85
18.6.77	77.00	379.64	2222.32	2422.25	2610.92	3043.20
18.6.78	78.00	399.68	2311.02	2557.14	2807.14	3157.89
18.6.79	79.00	384.74	2252.16	2460.68	2695.52	3060.69
18.6.80	80.00	480.77	2474.40	2606.22	2728.63	3267.15
18.6.81	81.00	492.57	2577.86	2708.81	2853.12	3392.54
18.6.82	82.00	487.63	2531.41	2685.24	2875.56	3282.89
18.6.83	83.00	492.43	2584.55	2747.87	2933.25	3330.04
18.6.84	84.00	498.00	2182.60	2417.98	2611.63	3105.92
18.6.85	85.00	492.04	2180.72	2418.82	2624.04	3099.88
18.6.86	86.00	512.00	2266.70	2522.40	2814.60	3199.82
18.6.87	87.00	497.15	2208.38	2449.40	2697.22	3079.34
18.6.88	88.00	482.23	2002.14	2185.99	2437.03	2938.10
18.6.89	89.00	491.46	2043.75	2237.62	2490.06	3077.74
18.6.90	90.00	492.88	2039.22	2244.25	2493.52	2943.28
18.6.91	91.00	494.03	2030.94	2239.96	2474.70	3023.29
18.6.92	92.00	497.78	2023.99	2212.38	2419.79	2938.62
18.6.93	93.00	492.48	2032.77	2219.80	2437.80	2986.99
18.6.94	94.00	516.84	2091.21	2297.41	2528.81	3011.62
18.6.95	95.00	502.95	2077.67	2301.12	2518.10	3048.13
18.6.96	96.00	350.85	2509.42	2635.64	2765.43	3160.93
18.6.97	97.00	359.82	2616.50	2745.47	2891.03	3353.19
18.6.98	98.00	351.68	2558.30	2711.96	2905.95	3415.04

18.6.99	99.00	357.30	2594.76	2757.92	2947.42	3403.14
18.6.100	100.00	368.68	2212.07	2425.05	2601.01	3104.98
18.6.101	101.00	358.19	2215.67	2435.45	2611.85	3064.09
18.6.102	102.00	377.45	2310.29	2540.50	2812.45	3166.95
18.6.103	103.00	363.03	2320.97	2550.17	2810.44	3236.85
18.6.104	104.00	349.06	2033.77	2208.26	2413.29	2935.10
18.6.105	105.00	357.28	2071.82	2245.79	2471.89	2974.25
18.6.106	106.00	358.34	2085.52	2276.64	2493.40	2958.00
18.6.107	107.00	359.47	2081.92	2272.22	2477.78	2929.77
18.6.108	108.00	368.65	2058.26	2232.66	2422.15	2930.61
18.6.109	109.00	358.58	2057.77	2245.62	2426.53	2898.66
18.6.110	110.00	383.64	2133.16	2343.20	2531.79	2978.68
18.6.111	111.00	369.36	2117.92	2332.09	2521.26	2957.24
18.6.112	112.00	426.66	2007.40	2197.85	2442.00	2935.75
18.6.113	113.00	433.51	2042.37	2224.09	2485.29	3016.53
18.6.114	114.00	436.12	2054.56	2266.31	2508.56	3012.58
18.6.115	115.00	437.11	2050.85	2259.24	2490.02	3013.43
18.6.116	116.00	444.59	2025.86	2227.31	2440.20	2990.17
18.6.117	117.00	435.69	2026.80	2220.13	2452.36	2947.13
18.6.118	118.00	460.77	2102.40	2329.40	2555.48	3072.84
18.6.119	119.00	446.18	2084.98	2304.37	2538.79	3062.53
18.6.120	120.00	423.89	2473.73	2617.13	2756.59	3218.63
18.6.121	121.00	436.61	2585.25	2731.87	2886.25	3383.39
18.6.122	122.00	431.05	2527.66	2701.88	2897.32	3348.98

18.6.123	123.00	435.66	2569.82	2753.74	2940.77	3397.96
18.6.124	124.00	444.77	2176.75	2404.51	2610.89	3093.19
18.6.125	125.00	435.83	2179.47	2405.37	2622.34	3065.92
18.6.126	126.00	455.37	2274.85	2522.94	2810.76	3133.92
18.6.127	127.00	440.56	2279.12	2543.08	2818.00	3198.88

18.7 L2-Fabric Write and Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.7.0	0.00	141.31	372.86	407.26	453.62	574.77
18.7.1	1.00	145.72	383.83	418.97	469.74	603.94
18.7.2	2.00	146.71	382.60	432.06	481.19	597.14
18.7.3	3.00	149.05	383.39	428.38	478.27	594.85
18.7.4	4.00	150.56	390.58	426.62	467.53	579.53
18.7.5	5.00	148.38	391.38	427.93	470.98	585.45
18.7.6	6.00	159.40	417.86	468.91	504.67	626.55
18.7.7	7.00	153.35	412.16	463.20	501.70	619.99
18.7.8	8.00	141.97	512.71	541.94	572.38	680.88
18.7.9	9.00	147.88	542.81	573.21	605.97	744.00
18.7.10	10.00	143.77	527.00	562.45	609.54	768.30
18.7.11	11.00	147.74	543.31	577.72	624.24	779.14
18.7.12	12.00	149.17	445.22	481.88	523.18	630.71
18.7.13	13.00	146.68	445.50	480.74	529.27	651.31
18.7.14	14.00	155.74	475.09	516.82	584.62	715.67
18.7.15	15.00	148.83	473.62	524.31	586.93	682.50
18.7.16	16.00	139.63	508.68	536.15	566.12	684.89

18.7.17	17.00	145.79	536.73	567.95	598.73	724.38
18.7.18	18.00	141.47	520.98	555.91	603.99	786.01
18.7.19	19.00	146.56	538.60	571.79	619.55	756.45
18.7.20	20.00	147.83	443.21	477.17	519.21	630.10
18.7.21	21.00	145.66	440.94	475.61	522.00	640.35
18.7.22	22.00	154.18	471.01	514.90	581.13	710.38
18.7.23	23.00	147.59	468.68	516.33	579.34	680.67
18.7.24	24.00	139.05	366.19	402.67	450.52	586.32
18.7.25	25.00	143.66	379.75	413.55	465.78	601.64
18.7.26	26.00	144.41	377.34	428.82	474.73	606.22
18.7.27	27.00	145.76	378.50	425.27	470.72	598.16
18.7.28	28.00	147.34	383.29	422.18	459.79	583.01
18.7.29	29.00	145.29	384.98	422.29	464.20	579.89
18.7.30	30.00	156.05	410.04	462.37	499.47	595.61
18.7.31	31.00	150.00	406.92	457.35	496.50	602.30
18.7.32	32.00	141.09	514.12	541.56	570.31	696.46
18.7.33	33.00	147.27	544.09	575.40	606.26	734.00
18.7.34	34.00	143.35	530.96	563.69	613.48	769.60
18.7.35	35.00	148.23	548.69	583.53	630.16	780.46
18.7.36	36.00	149.69	447.12	484.91	526.63	615.47
18.7.37	37.00	147.54	447.71	487.85	533.55	628.39
18.7.38	38.00	156.14	473.64	520.74	586.82	691.98
18.7.39	39.00	148.91	451.48	492.52	557.00	674.78
18.7.40	40.00	139.67	368.39	406.42	455.46	570.44

18.7.41	41.00	144.67	382.11	420.82	471.01	596.84
18.7.42	42.00	145.19	378.28	427.38	475.37	592.20
18.7.43	43.00	146.53	378.59	422.94	472.33	582.61
18.7.44	44.00	149.00	390.30	426.34	466.06	590.08
18.7.45	45.00	147.33	392.97	432.65	473.24	596.54
18.7.46	46.00	158.11	411.75	462.67	502.23	627.23
18.7.47	47.00	151.89	406.47	456.07	498.77	606.11
18.7.48	48.00	139.69	362.44	400.86	453.03	568.48
18.7.49	49.00	144.68	375.92	416.11	465.85	604.99
18.7.50	50.00	144.92	372.36	421.34	470.13	574.73
18.7.51	51.00	147.68	373.67	419.99	468.46	576.15
18.7.52	52.00	150.01	384.24	424.48	460.67	590.26
18.7.53	53.00	148.41	387.42	428.21	468.26	588.32
18.7.54	54.00	158.95	405.38	457.97	498.10	609.21
18.7.55	55.00	152.95	401.02	450.64	494.89	601.88
18.7.56	56.00	141.33	507.32	534.78	565.15	674.53
18.7.57	57.00	147.39	537.92	567.98	600.31	721.25
18.7.58	58.00	143.48	522.76	557.14	605.20	752.29
18.7.59	59.00	147.33	540.69	577.35	622.16	772.10
18.7.60	60.00	148.76	438.76	479.56	520.75	601.28
18.7.61	61.00	146.52	439.13	483.38	525.70	625.16
18.7.62	62.00	155.41	465.76	512.94	582.65	682.57
18.7.63	63.00	147.85	444.04	486.89	548.78	658.86
18.7.64	64.00	141.42	372.09	407.07	454.83	559.90

18.7.65	65.00	146.49	384.81	423.07	470.80	582.74
18.7.66	66.00	146.93	381.54	430.09	472.61	572.01
18.7.67	67.00	149.44	382.55	425.54	471.84	569.77
18.7.68	68.00	150.85	390.73	426.20	466.59	570.71
18.7.69	69.00	149.15	392.63	431.15	472.46	566.90
18.7.70	70.00	160.16	411.59	460.63	499.72	593.02
18.7.71	71.00	153.68	407.66	455.81	496.78	590.87
18.7.72	72.00	142.01	513.15	541.79	570.81	715.88
18.7.73	73.00	148.36	542.17	572.48	606.84	768.45
18.7.74	74.00	144.44	528.39	560.03	612.81	783.98
18.7.75	75.00	148.03	545.21	580.97	627.26	785.28
18.7.76	76.00	149.50	445.86	484.83	526.29	617.96
18.7.77	77.00	147.44	447.30	488.65	530.76	626.31
18.7.78	78.00	156.09	472.87	522.64	585.76	703.34
18.7.79	79.00	148.85	453.37	492.31	552.88	661.55
18.7.80	80.00	139.96	508.89	536.31	564.18	709.45
18.7.81	81.00	146.13	538.14	568.88	601.04	756.18
18.7.82	82.00	142.13	522.97	554.92	607.88	756.29
18.7.83	83.00	146.96	540.23	576.90	623.55	777.75
18.7.84	84.00	148.38	440.48	479.85	521.35	609.51
18.7.85	85.00	146.39	440.47	485.05	527.99	627.55
18.7.86	86.00	154.79	467.47	516.19	583.11	704.15
18.7.87	87.00	147.57	448.67	488.58	550.82	651.17
18.7.88	88.00	139.31	364.89	405.10	453.68	577.04

18.7.89	89.00	144.46	379.65	421.04	469.49	600.13
18.7.90	90.00	144.45	376.61	427.05	473.38	588.36
18.7.91	91.00	146.28	377.40	423.28	470.60	585.09
18.7.92	92.00	147.75	382.81	422.24	463.68	582.87
18.7.93	93.00	146.13	386.03	428.07	469.79	578.84
18.7.94	94.00	156.62	405.23	456.11	497.23	610.52
18.7.95	95.00	150.52	401.23	451.79	494.43	602.23
18.7.96	96.00	143.04	510.80	537.73	568.50	699.21
18.7.97	97.00	148.99	539.09	569.19	603.84	725.96
18.7.98	98.00	144.70	524.19	555.40	608.71	794.88
18.7.99	99.00	149.68	542.47	574.06	623.48	768.66
18.7.100	100.00	151.22	442.54	479.85	520.53	622.21
18.7.101	101.00	149.15	443.45	479.62	526.60	637.28
18.7.102	102.00	157.60	473.41	513.85	581.42	707.11
18.7.103	103.00	150.81	472.03	519.71	582.09	716.88
18.7.104	104.00	142.08	369.54	409.33	453.86	574.74
18.7.105	105.00	147.01	382.30	423.37	469.26	603.16
18.7.106	106.00	147.50	381.40	432.41	475.81	591.74
18.7.107	107.00	148.91	382.66	429.50	474.59	577.54
18.7.108	108.00	150.70	387.43	425.15	462.89	556.36
18.7.109	109.00	148.54	387.77	426.23	467.60	565.27
18.7.110	110.00	159.58	414.41	464.01	500.12	608.00
18.7.111	111.00	153.13	406.89	457.44	496.47	610.30
18.7.112	112.00	140.59	364.31	406.30	453.68	566.06

18.7.113	113.00	145.45	377.74	419.23	467.66	598.90
18.7.114	114.00	145.98	376.40	427.75	477.22	605.84
18.7.115	115.00	148.40	378.38	424.98	474.87	586.06
18.7.116	116.00	150.24	381.59	423.13	461.15	564.35
18.7.117	117.00	147.90	382.98	424.73	465.73	570.06
18.7.118	118.00	159.20	408.66	463.90	501.32	602.51
18.7.119	119.00	152.54	402.26	456.55	496.36	605.32
18.7.120	120.00	141.44	506.01	534.04	563.49	683.85
18.7.121	121.00	147.72	536.11	565.01	597.17	724.53
18.7.122	122.00	143.41	519.46	551.16	602.83	773.78
18.7.123	123.00	147.21	537.43	570.66	617.89	755.95
18.7.124	124.00	148.72	434.86	474.23	515.88	621.20
18.7.125	125.00	146.44	436.02	476.24	522.17	623.91
18.7.126	126.00	155.43	464.64	510.57	577.71	683.73
18.7.127	127.00	148.34	463.86	513.77	575.99	702.91

18.8 L2-Fabric Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.8.0	0.00	0.00	0.00	0.00	0.00	0.00
18.8.1	1.00	0.00	0.00	0.00	0.00	0.00
18.8.2	2.00	0.00	0.00	0.00	0.00	0.00
18.8.3	3.00	0.00	0.00	0.00	0.00	0.00
18.8.4	4.00	0.00	0.00	0.00	0.00	0.00
18.8.5	5.00	0.00	0.00	0.00	0.00	0.00
18.8.6	6.00	0.00	0.00	0.00	0.00	0.00

18.8.7	7.00	0.00	0.00	0.00	0.00	0.00
18.8.8	8.00	0.00	0.00	0.00	0.00	0.00
18.8.9	9.00	0.00	0.00	0.00	0.00	0.00
18.8.10	10.00	0.00	0.00	0.00	0.00	0.00
18.8.11	11.00	0.00	0.00	0.00	0.00	0.00
18.8.12	12.00	0.00	0.00	0.00	0.00	0.00
18.8.13	13.00	0.00	0.00	0.00	0.00	0.00
18.8.14	14.00	0.00	0.00	0.00	0.00	0.00
18.8.15	15.00	0.00	0.00	0.00	0.00	0.00
18.8.16	16.00	0.00	0.00	0.00	0.00	0.00
18.8.17	17.00	0.00	0.00	0.00	0.00	0.00
18.8.18	18.00	0.00	0.00	0.00	0.00	0.00
18.8.19	19.00	0.00	0.00	0.00	0.00	0.00
18.8.20	20.00	0.00	0.00	0.00	0.00	0.00
18.8.21	21.00	0.00	0.00	0.00	0.00	0.00
18.8.22	22.00	0.00	0.00	0.00	0.00	0.00
18.8.23	23.00	0.00	0.00	0.00	0.00	0.00
18.8.24	24.00	0.00	0.00	0.00	0.00	0.00
18.8.25	25.00	0.00	0.00	0.00	0.00	0.00
18.8.26	26.00	0.00	0.00	0.00	0.00	0.00
18.8.27	27.00	0.00	0.00	0.00	0.00	0.00
18.8.28	28.00	0.00	0.00	0.00	0.00	0.00
18.8.29	29.00	0.00	0.00	0.00	0.00	0.00
18.8.30	30.00	0.00	0.00	0.00	0.00	0.00

18.8.31	31.00	0.00	0.00	0.00	0.00	0.00
18.8.32	32.00	0.00	0.00	0.00	0.00	0.00
18.8.33	33.00	0.00	0.00	0.00	0.00	0.00
18.8.34	34.00	0.00	0.00	0.00	0.00	0.00
18.8.35	35.00	0.00	0.00	0.00	0.00	0.00
18.8.36	36.00	0.00	0.00	0.00	0.00	0.00
18.8.37	37.00	0.00	0.00	0.00	0.00	0.00
18.8.38	38.00	0.00	0.00	0.00	0.00	0.00
18.8.39	39.00	0.00	0.00	0.00	0.00	0.00
18.8.40	40.00	0.00	0.00	0.00	0.00	0.00
18.8.41	41.00	0.00	0.00	0.00	0.00	0.00
18.8.42	42.00	0.00	0.00	0.00	0.00	0.00
18.8.43	43.00	0.00	0.00	0.00	0.00	0.00
18.8.44	44.00	0.00	0.00	0.00	0.00	0.00
18.8.45	45.00	0.00	0.00	0.00	0.00	0.00
18.8.46	46.00	0.00	0.00	0.00	0.00	0.00
18.8.47	47.00	0.00	0.00	0.00	0.00	0.00
18.8.48	48.00	0.00	0.00	0.00	0.00	0.00
18.8.49	49.00	0.00	0.00	0.00	0.00	0.00
18.8.50	50.00	0.00	0.00	0.00	0.00	0.00
18.8.51	51.00	0.00	0.00	0.00	0.00	0.00
18.8.52	52.00	0.00	0.00	0.00	0.00	0.00
18.8.53	53.00	0.00	0.00	0.00	0.00	0.00
18.8.54	54.00	0.00	0.00	0.00	0.00	0.00

18.8.55	55.00	0.00	0.00	0.00	0.00	0.00
18.8.56	56.00	0.00	0.00	0.00	0.00	0.00
18.8.57	57.00	0.00	0.00	0.00	0.00	0.00
18.8.58	58.00	0.00	0.00	0.00	0.00	0.00
18.8.59	59.00	0.00	0.00	0.00	0.00	0.00
18.8.60	60.00	0.00	0.00	0.00	0.00	0.00
18.8.61	61.00	0.00	0.00	0.00	0.00	0.00
18.8.62	62.00	0.00	0.00	0.00	0.00	0.00
18.8.63	63.00	0.00	0.00	0.00	0.00	0.00
18.8.64	64.00	0.00	0.00	0.00	0.00	0.00
18.8.65	65.00	0.00	0.00	0.00	0.00	0.00
18.8.66	66.00	0.00	0.00	0.00	0.00	0.00
18.8.67	67.00	0.00	0.00	0.00	0.00	0.00
18.8.68	68.00	0.00	0.00	0.00	0.00	0.00
18.8.69	69.00	0.00	0.00	0.00	0.00	0.00
18.8.70	70.00	0.00	0.00	0.00	0.00	0.00
18.8.71	71.00	0.00	0.00	0.00	0.00	0.00
18.8.72	72.00	0.00	0.00	0.00	0.00	0.00
18.8.73	73.00	0.00	0.00	0.00	0.00	0.00
18.8.74	74.00	0.00	0.00	0.00	0.00	0.00
18.8.75	75.00	0.00	0.00	0.00	0.00	0.00
18.8.76	76.00	0.00	0.00	0.00	0.00	0.00
18.8.77	77.00	0.00	0.00	0.00	0.00	0.00
18.8.78	78.00	0.00	0.00	0.00	0.00	0.00

18.8.79	79.00	0.00	0.00	0.00	0.00	0.00
18.8.80	80.00	0.00	0.00	0.00	0.00	0.00
18.8.81	81.00	0.00	0.00	0.00	0.00	0.00
18.8.82	82.00	0.00	0.00	0.00	0.00	0.00
18.8.83	83.00	0.00	0.00	0.00	0.00	0.00
18.8.84	84.00	0.00	0.00	0.00	0.00	0.00
18.8.85	85.00	0.00	0.00	0.00	0.00	0.00
18.8.86	86.00	0.00	0.00	0.00	0.00	0.00
18.8.87	87.00	0.00	0.00	0.00	0.00	0.00
18.8.88	88.00	0.00	0.00	0.00	0.00	0.00
18.8.89	89.00	0.00	0.00	0.00	0.00	0.00
18.8.90	90.00	0.00	0.00	0.00	0.00	0.00
18.8.91	91.00	0.00	0.00	0.00	0.00	0.00
18.8.92	92.00	0.00	0.00	0.00	0.00	0.00
18.8.93	93.00	0.00	0.00	0.00	0.00	0.00
18.8.94	94.00	0.00	0.00	0.00	0.00	0.00
18.8.95	95.00	0.00	0.00	0.00	0.00	0.00
18.8.96	96.00	0.00	0.00	0.00	0.00	0.00
18.8.97	97.00	0.00	0.00	0.00	0.00	0.00
18.8.98	98.00	0.00	0.00	0.00	0.00	0.00
18.8.99	99.00	0.00	0.00	0.00	0.00	0.00
18.8.100	100.00	0.00	0.00	0.00	0.00	0.00
18.8.101	101.00	0.00	0.00	0.00	0.00	0.00
18.8.102	102.00	0.00	0.00	0.00	0.00	0.00

18.8.103	103.00	0.00	0.00	0.00	0.00	0.00
18.8.104	104.00	0.00	0.00	0.00	0.00	0.00
18.8.105	105.00	0.00	0.00	0.00	0.00	0.00
18.8.106	106.00	0.00	0.00	0.00	0.00	0.00
18.8.107	107.00	0.00	0.00	0.00	0.00	0.00
18.8.108	108.00	0.00	0.00	0.00	0.00	0.00
18.8.109	109.00	0.00	0.00	0.00	0.00	0.00
18.8.110	110.00	0.00	0.00	0.00	0.00	0.00
18.8.111	111.00	0.00	0.00	0.00	0.00	0.00
18.8.112	112.00	0.00	0.00	0.00	0.00	0.00
18.8.113	113.00	0.00	0.00	0.00	0.00	0.00
18.8.114	114.00	0.00	0.00	0.00	0.00	0.00
18.8.115	115.00	0.00	0.00	0.00	0.00	0.00
18.8.116	116.00	0.00	0.00	0.00	0.00	0.00
18.8.117	117.00	0.00	0.00	0.00	0.00	0.00
18.8.118	118.00	0.00	0.00	0.00	0.00	0.00
18.8.119	119.00	0.00	0.00	0.00	0.00	0.00
18.8.120	120.00	0.00	0.00	0.00	0.00	0.00
18.8.121	121.00	0.00	0.00	0.00	0.00	0.00
18.8.122	122.00	0.00	0.00	0.00	0.00	0.00
18.8.123	123.00	0.00	0.00	0.00	0.00	0.00
18.8.124	124.00	0.00	0.00	0.00	0.00	0.00
18.8.125	125.00	0.00	0.00	0.00	0.00	0.00
18.8.126	126.00	0.00	0.00	0.00	0.00	0.00

18.8.127	127.00	0.00	0.00	0.00	0.00	0.00
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18.12 L2-Fabric (128B read requests per normUnit)

Metric_ID Max	Channel	Min	Q1	Median	Q3
18.12.0 40285119.00	0.00	132047.00	132096.00	132096.00	132122.00
18.12.1 40283856.00	1.00	132045.00	132096.00	132096.00	132122.00
18.12.2 40285351.00	2.00	132051.00	132096.00	132096.00	132130.00
18.12.3 40283357.00	3.00	132049.00	132096.00	132096.00	132130.00
18.12.4 40285143.00	4.00	132043.00	132096.00	132096.00	132122.00
18.12.5 40283828.00	5.00	132050.00	132096.00	132096.00	132122.00
18.12.6 40285307.00	6.00	132051.00	132096.00	132096.00	132128.00
18.12.7 40283316.00	7.00	132051.00	132096.00	132096.00	132128.00
18.12.8 40286565.00	8.00	132046.00	132096.00	132096.00	132127.50
18.12.9 40284919.00	9.00	132048.00	132096.00	132096.00	132128.00

18.12.10 40286341.00	10.00	132049.00	132096.00	132096.00	132130.50
18.12.11 40284528.00	11.00	132052.00	132096.00	132096.00	132130.50
18.12.12 40286578.00	12.00	132043.00	132096.00	132096.00	132125.00
18.12.13 40284917.00	13.00	132053.00	132096.00	132096.00	132125.00
18.12.14 40286362.00	14.00	132050.00	132096.00	132096.00	132128.00
18.12.15 40284501.00	15.00	132052.00	132096.00	132096.00	132128.00
18.12.16 38810019.00	16.00	132064.00	132096.00	132096.00	132129.00
18.12.17 38809448.00	17.00	132064.00	132096.00	132096.00	132129.00
18.12.18 38808938.00	18.00	132067.00	132096.00	132096.00	132131.00
18.12.19 38809293.00	19.00	132068.00	132096.00	132096.00	132130.50
18.12.20 38809980.00	20.00	132064.00	132096.00	132096.00	132129.00
18.12.21 38809458.00	21.00	132061.00	132096.00	132096.00	132129.50

18.12.22 38808935.00	22.00	132066.00	132096.00	132096.00	132132.50
18.12.23 38809312.00	23.00	132062.00	132096.00	132096.00	132133.00
18.12.24 38811173.00	24.00	132065.00	132096.00	132096.00	132128.50
18.12.25 38810860.00	25.00	132068.00	132096.00	132096.00	132128.50
18.12.26 38810224.00	26.00	132070.00	132096.00	132096.00	132132.00
18.12.27 38810637.00	27.00	132065.00	132096.00	132096.00	132132.00
18.12.28 38811178.00	28.00	132063.00	132096.00	132096.00	132124.00
18.12.29 38810819.00	29.00	132065.00	132096.00	132096.00	132124.00
18.12.30 38810231.00	30.00	132069.00	132096.00	132096.00	132126.50
18.12.31 38810624.00	31.00	132066.00	132095.50	132096.00	132127.00
18.12.32 39453996.00	32.00	132044.00	132096.00	132096.00	132125.50
18.12.33 39455287.00	33.00	132057.00	132096.00	132096.00	132125.00

18.12.34 39453840.00	34.00	132039.00	132096.00	132096.00	132129.00
18.12.35 39455368.00	35.00	132031.00	132096.00	132096.00	132129.00
18.12.36 39453991.00	36.00	132049.00	132096.00	132096.00	132124.00
18.12.37 39455316.00	37.00	132054.00	132096.00	132096.00	132124.00
18.12.38 39453862.00	38.00	132051.00	132096.00	132096.00	132131.00
18.12.39 39455343.00	39.00	132038.00	132096.00	132096.00	132130.50
18.12.40 39453959.00	40.00	132047.00	132096.00	132096.00	132122.50
18.12.41 39455503.00	41.00	132050.00	132096.00	132096.00	132122.50
18.12.42 39453793.00	42.00	132043.00	132096.00	132096.00	132128.00
18.12.43 39455416.00	43.00	132045.00	132096.00	132096.00	132128.00
18.12.44 39453954.00	44.00	132044.00	132096.00	132096.00	132123.50
18.12.45 39455491.00	45.00	132039.00	132096.00	132096.00	132123.50

18.12.46 39453783.00	46.00	132039.00	132096.00	132096.00	132130.00
18.12.47 39455388.00	47.00	132049.00	132096.00	132096.00	132129.50
18.12.48 37592791.00	48.00	132031.00	132096.00	132096.00	132122.00
18.12.49 37593290.00	49.00	132037.00	132096.00	132096.00	132122.00
18.12.50 37592825.00	50.00	132032.00	132096.00	132096.00	132131.00
18.12.51 37593169.00	51.00	132033.00	132096.00	132096.00	132131.50
18.12.52 37592802.00	52.00	132050.00	132096.00	132096.00	132120.00
18.12.53 37593297.00	53.00	132041.00	132096.00	132096.00	132120.00
18.12.54 37592847.00	54.00	132032.00	132096.00	132096.00	132128.50
18.12.55 37593208.00	55.00	132030.00	132096.00	132096.00	132128.50
18.12.56 37592829.00	56.00	132038.00	132096.00	132096.00	132120.50
18.12.57 37593119.00	57.00	132043.00	132096.00	132096.00	132120.50

18.12.58 37592741.00	58.00	132029.00	132096.00	132096.00	132133.00
18.12.59 37592888.00	59.00	132036.00	132096.00	132096.00	132133.00
18.12.60 37592832.00	60.00	132049.00	132096.00	132096.00	132121.50
18.12.61 37593122.00	61.00	132048.00	132096.00	132096.00	132121.00
18.12.62 37592694.00	62.00	132037.00	132096.00	132096.00	132133.00
18.12.63 37592857.00	63.00	132036.00	132096.00	132096.00	132133.00
18.12.64 38131512.00	64.00	131962.00	132083.50	132096.00	132116.00
18.12.65 38130836.00	65.00	131964.00	132084.50	132096.00	132116.00
18.12.66 38131305.00	66.00	131966.00	132083.50	132096.00	132117.00
18.12.67 38130604.00	67.00	131960.00	132080.00	132096.00	132117.00
18.12.68 38131476.00	68.00	131966.00	132083.00	132096.00	132118.00
18.12.69 38130810.00	69.00	131962.00	132086.50	132096.00	132118.00

18.12.70 38131292.00	70.00	131968.00	132077.00	132096.00	132118.00
18.12.71 38130607.00	71.00	131965.00	132079.00	132096.00	132118.00
18.12.72 38131590.00	72.00	131973.00	132083.50	132096.00	132118.00
18.12.73 38130895.00	73.00	131964.00	132087.00	132096.00	132118.00
18.12.74 38131330.00	74.00	131971.00	132078.50	132096.00	132118.50
18.12.75 38130571.00	75.00	131966.00	132081.50	132096.00	132119.00
18.12.76 38131583.00	76.00	131969.00	132087.50	132096.00	132116.50
18.12.77 38130893.00	77.00	131968.00	132086.50	132096.00	132116.50
18.12.78 38131296.00	78.00	131968.00	132084.00	132096.00	132117.50
18.12.79 38130572.00	79.00	131970.00	132080.00	132096.00	132118.00
18.12.80 39424050.00	80.00	132062.00	132096.00	132096.00	132124.50
18.12.81 39423755.00	81.00	132056.00	132096.00	132096.00	132125.00

18.12.82 39423587.00	82.00	132054.00	132096.00	132096.00	132130.50
18.12.83 39423719.00	83.00	132049.00	132096.00	132096.00	132130.50
18.12.84 39424048.00	84.00	132056.00	132096.00	132096.00	132123.50
18.12.85 39423795.00	85.00	132059.00	132096.00	132096.00	132123.50
18.12.86 39423637.00	86.00	132052.00	132096.00	132096.00	132126.50
18.12.87 39423741.00	87.00	132050.00	132096.00	132096.00	132126.00
18.12.88 39424393.00	88.00	132058.00	132096.00	132096.00	132124.00
18.12.89 39423952.00	89.00	132061.00	132096.00	132096.00	132124.00
18.12.90 39423928.00	90.00	132055.00	132095.50	132096.00	132126.00
18.12.91 39424078.00	91.00	132043.00	132096.00	132096.00	132126.00
18.12.92 39424416.00	92.00	132062.00	132096.00	132096.00	132124.50
18.12.93 39423942.00	93.00	132057.00	132096.00	132096.00	132124.50

18.12.94 39423855.00	94.00	132049.00	132096.00	132096.00	132131.00
18.12.95 39424119.00	95.00	132049.00	132096.00	132096.00	132130.50
18.12.96 38214653.00	96.00	131969.00	132073.00	132096.00	132121.00
18.12.97 38213361.00	97.00	131960.00	132075.00	132096.00	132121.00
18.12.98 38214810.00	98.00	131961.00	132075.50	132096.00	132124.00
18.12.99 38213100.00	99.00	131965.00	132071.00	132096.00	132125.00
18.12.100 38214634.00	100.00	131966.00	132072.00	132096.00	132117.00
18.12.101 38213348.00	101.00	131959.00	132076.50	132096.00	132117.00
18.12.102 38214783.00	102.00	131967.00	132075.00	132096.00	132119.00
18.12.103 38213085.00	103.00	131961.00	132076.00	132096.00	132119.00
18.12.104 38214126.00	104.00	131966.00	132075.50	132096.00	132120.00
18.12.105 38212517.00	105.00	131974.00	132071.00	132096.00	132120.00

18.12.106 38214189.00	106.00	131954.00	132069.00	132096.00	132117.50
18.12.107 38212361.00	107.00	131963.00	132072.50	132096.00	132117.50
18.12.108 38214096.00	108.00	131969.00	132073.50	132096.00	132120.00
18.12.109 38212546.00	109.00	131970.00	132074.00	132096.00	132120.00
18.12.110 38214215.00	110.00	131969.00	132079.50	132096.00	132120.00
18.12.111 38212383.00	111.00	131967.00	132075.50	132096.00	132120.00
18.12.112 38300703.00	112.00	132058.00	132096.00	132096.00	132121.00
18.12.113 38299628.00	113.00	132057.00	132096.00	132096.00	132121.00
18.12.114 38300427.00	114.00	132062.00	132096.00	132096.00	132128.00
18.12.115 38299535.00	115.00	132051.00	132096.00	132096.00	132128.00
18.12.116 38300728.00	116.00	132065.00	132096.00	132096.00	132124.00
18.12.117 38299621.00	117.00	132064.00	132096.00	132096.00	132124.00

18.12.118 38300456.00	118.00	132058.00	132096.00	132096.00	132132.00
18.12.119 38299518.00	119.00	132061.00	132096.00	132096.00	132132.00
18.12.120 38302097.00	120.00	132065.00	132096.00	132096.00	132120.00
18.12.121 38301017.00	121.00	132065.00	132096.00	132096.00	132120.00
18.12.122 38301917.00	122.00	132061.00	132096.00	132096.00	132134.00
18.12.123 38300940.00	123.00	132063.00	132096.00	132096.00	132134.00
18.12.124 38302120.00	124.00	132058.00	132096.00	132096.00	132120.00
18.12.125 38301019.00	125.00	132058.00	132096.00	132096.00	132120.00
18.12.126 38301818.00	126.00	132063.00	132096.00	132096.00	132130.50
18.12.127 38300895.00	127.00	132057.00	132096.00	132096.00	132130.50

1.5 Fusion MatMulBias Kernel

1.5.1 Verify the Fusion MatMulBias Kernel

This also ensures the kernel has been built and cached. The results from the triton kernel are compared against torch.

Small Input

```
[16]: !python triton/matmulbias.py --verify fusion
```

```
Running the fusion matmulbias kernel (input size M:1024 N:1024) ...
triton_output: tensor([[ 2.7480,  43.6250,  30.9531, ...,  26.2500,   8.8203,
  6.4609],
                        [ 7.1562, -21.3906,  17.7969, ..., -16.8125,  29.6562, -48.3125],
                        [ 13.8125, -6.4609,   0.5146, ...,  48.0000,  25.1562, -27.5156],
                        ...,
                        [-54.4375, -1.4375, -0.4214, ...,   5.4297,  17.2812, -39.0625],
                        [ 9.6094, -52.2500,  43.0000, ..., -45.8438, -49.3125,  32.6875],
                        [-12.8516,  9.5391,  53.3750, ..., -9.3438,   7.6797,  54.1250]],
                        device='cuda:0', dtype=torch.float16)
Running the torch matmulbias kernel ...
torch_output: tensor([[ 2.7480,  43.6562,  30.9531, ...,  26.2500,   8.8203,
  6.4609],
                      [ 7.1602, -21.3906,  17.7969, ..., -16.8125,  29.6406, -48.3125],
                      [ 13.8125, -6.4609,   0.5142, ...,  48.0312,  25.1406, -27.5156],
                      ...,
                      [-54.4375, -1.4375, -0.4214, ...,   5.4297,  17.2812, -39.0625],
                      [ 9.6094, -52.2500,  43.0000, ..., -45.8125, -49.3125,  32.7188],
                      [-12.8516,  9.5391,  53.3750, ..., -9.3438,   7.6797,  54.1562]],
                      device='cuda:0', dtype=torch.float16)
Verifying triton results with torch ...
Verified
```

Large Input

```
[17]: !python triton/matmulbias.py --M 32768 --N 32768 --verify fusion
```

```
Running the fusion matmulbias kernel (input size M:32768 N:32768) ...
triton_output: tensor([[ -60.8750,  56.6562,  57.0938, ...,  274.0000,
 295.2500,
 14.8828],
                      [ 205.3750, -353.7500, -102.1875, ..., -24.4219,   57.0625,
 -30.7969],
                      [ 56.2188,  258.7500,  46.6562, ..., -366.0000,  395.2500,
 285.2500],
                      ...,
                      [ -5.0977,   99.9375, -73.5625, ..., -70.5000,  204.2500,
 -156.8750],
                      [ 86.8125, -26.3594, -128.8750, ..., -60.7500, -36.3750,
 100.8750],
```

```

    [-27.6406,  93.5625,  77.9375, ..., -213.1250, -323.2500,
    -16.8438]], device='cuda:0', dtype=torch.float16)
Running the torch matmulbias kernel ...
torch_output: tensor([[ -60.8750,  56.6562,  57.1250, ..., 274.0000,
295.2500,
    14.8828],
    [ 205.3750, -353.7500, -102.1875, ..., -24.4219,  57.0312,
    -30.7969],
    [  56.2188, 258.7500,  46.6562, ..., -366.0000, 395.2500,
    285.2500],
    ...,
    [ -5.0977,  99.9375, -73.5625, ..., -70.5000, 204.2500,
    -156.8750],
    [  86.8125, -26.3594, -128.8750, ..., -60.7500, -36.3750,
    100.8750],
    [-27.6406,  93.5625,  77.9375, ..., -213.0000, -323.2500,
    -16.8438]], device='cuda:0', dtype=torch.float16)
Verifying triton results with torch ...
Verified

```

1.5.2 Profile the Fusion MatMulBias Kernel

Using the Triton Proton CLI Generates a hatchet file, `fusion_matmulbias_x.hatchet`, under the profiles directory.

Small Input

```
[18]: !proton -n profiles/fusion_matmulbias_1024x1024 -k triton triton/matmulbias.py
      ↪fusion
```

```

Running the fusion matmulbias kernel (input size M:1024 N:1024) ...
triton_output: tensor([[ 2.7480, 43.6250, 30.9531, ..., 26.2500,  8.8203,
6.4609],
    [ 7.1562, -21.3906, 17.7969, ..., -16.8125, 29.6562, -48.3125],
    [ 13.8125, -6.4609,  0.5146, ..., 48.0000, 25.1562, -27.5156],
    ...,
    [-54.4375, -1.4375, -0.4214, ...,  5.4297, 17.2812, -39.0625],
    [  9.6094, -52.2500, 43.0000, ..., -45.8438, -49.3125, 32.6875],
    [-12.8516,  9.5391, 53.3750, ..., -9.3438,  7.6797, 54.1250]],
device='cuda:0', dtype=torch.float16)

```

Large Input

```
[19]: !proton -n profiles/fusion_matmulbias_32768x32768 -k triton triton/matmulbias.
      ↪py --M 32768 --N 32768 fusion
```

```

Running the fusion matmulbias kernel (input size M:32768 N:32768) ...
triton_output: tensor([[ -60.8750,  56.6562,  57.0938, ..., 274.0000,
295.2500,
    14.8828],

```

```
[ 205.3750, -353.7500, -102.1875, ..., -24.4219, 57.0625,
  -30.7969],
[ 56.2188, 258.7500, 46.6562, ..., -366.0000, 395.2500,
  285.2500],
...,
[ -5.0977, 99.9375, -73.5625, ..., -70.5000, 204.2500,
  -156.8750],
[ 86.8125, -26.3594, -128.8750, ..., -60.7500, -36.3750,
  100.8750],
[ -27.6406, 93.5625, 77.9375, ..., -213.1250, -323.2500,
  -16.8438]], device='cuda:0', dtype=torch.float16)
```

Using the Triton Proton API Generates a hatchet file, fusion_matmulbias_x.hatchet, under the profiles directory and outputs the profile.

Small Input

```
[20]: !python triton/matmulbias.py --profile fusion
```

```
Profiling the fusion matmulbias kernel (input size M:1024 N:1024) ...
triton_output: tensor([[ 2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203,
  6.4609],
  [ 7.1562, -21.3906, 17.7969, ..., -16.8125, 29.6562, -48.3125],
  [ 13.8125, -6.4609, 0.5146, ..., 48.0000, 25.1562, -27.5156],
  ...,
  [-54.4375, -1.4375, -0.4214, ..., 5.4297, 17.2812, -39.0625],
  [ 9.6094, -52.2500, 43.0000, ..., -45.8438, -49.3125, 32.6875],
  [-12.8516, 9.5391, 53.3750, ..., -9.3438, 7.6797, 54.1250]]],
  device='cuda:0', dtype=torch.float16)
Proton profile results for fusion_matmulbias_1024x1024
95.959 37.687 ROOT
56.559 nan_ZN2at6native29vectorized_elementwise_kernel
ILi4ENS0_11FillFunctorIiEEST5arrayIPcLm1EEEEviT0_T1_
39.399 91.787 matmulbias_kernel [M=1024, N=1024,
K=1024] [BM=128, BN=128 BK=64]
```

Legend (Metric: time/ms (inc) Min: 39.40 Max: 95.96)

```
90.30 - 95.96
78.99 - 90.30
67.68 - 78.99
56.37 - 67.68
45.06 - 56.37
39.40 - 45.06
```

```
name User code      Only in left graph
Only in right graph
```

Large Input

```
[21]: !python triton/matmulbias.py --M 32768 --N 32768 --profile fusion
```

```

Profiling the fusion matmulbias kernel (input size M:32768 N:32768) ...
triton_output: tensor([[ -60.8750,   56.6562,   57.0938, ..., 274.0000,
295.2500,
      14.8828],
 [ 205.3750, -353.7500, -102.1875, ..., -24.4219,   57.0625,
 -30.7969],
 [  56.2188,  258.7500,   46.6562, ..., -366.0000,  395.2500,
 285.2500],
 ...,
 [  -5.0977,   99.9375,  -73.5625, ..., -70.5000,  204.2500,
 -156.8750],
 [  86.8125, -26.3594, -128.8750, ..., -60.7500,  -36.3750,
 100.8750],
 [ -27.6406,   93.5625,   77.9375, ..., -213.1250, -323.2500,
 -16.8438]], device='cuda:0', dtype=torch.float16)

```

Proton profile results for fusion_matmulbias_32768x32768

4251.103 281.402 ROOT

```
0.557 nan _ZN2at6native29vectorized_elementwise_kernelI
Li4ENS0_11FillFunctorIiEESt5arrayIPcLm1EEEEviT0_T1_
```

```
4250.545 281.439 matmulbias_kernel [M=32768, N=32768,
K=32768] [BM=128, BN=128 BK=64]
```

Legend (Metric: time/ms (inc) Min: 0.56 Max: 4251.10)

3826.05 - 4251.10
2975.94 - 3826.05
2125.83 - 2975.94
1275.72 - 2125.83
425.61 - 1275.72
0.56 - 425.61

```
name User code      Only in left graph
Only in right graph
```

Using the ROCm Systems Compute profiler Generates files found under the workloads/fusion_matmulbias_x/ directory.

Small Input

```
[22]: !ROCPROF=rocprofiler-sdk rocprof-compute profile -n fusion_matmulbias_1024x1024
      ↪ -- python triton/matmulbias.py fusion
```

----- /-| ----- -| |
 |' / - \ / -| ' \ | ' / - \ | | ----- / - / - \ | ' - \ | ' \ | | | | - /

```

\
| | | ( ) | ( _ | | ) | | | ( ) | _ | _ _ _ | ( _ | ( ) | | | | | | | ) | | _ | | | |
_ _ /
| _ | \ _ _ / \ _ _ | . _ / | _ | \ _ _ / | _ | \ _ _ \ _ _ / | _ | | _ | | _ | . _ _ /
\ _ _ , _ | \ _ _ \ _ _ |
| _ |
| _ |

```

WARNING DEPRECATION WARNING: rocm-smi is deprecated in ROCm 7.0 and will be removed from rocprof-compute in ROCm 7.1. Please migrate to amd-smi for compute partition parsing. For migration help, see

<https://github.com/ROCm/amdsmi>

INFO Rocprofiler-Compute version: 3.2.3

INFO Profiler choice: rocprofiler-sdk

INFO Path:

/workspace/user/workloads/fusion_matmulbias_1024x1024/MI300X_A1

INFO Target: MI300X_A1

INFO Command: python triton/matmulbias.py fusion

INFO Kernel Selection: None

INFO Dispatch Selection: None

INFO Hardware Blocks: []

INFO Report Sections: []

INFO

INFO ~~~~~

INFO Collecting Performance Counters

INFO ~~~~~

INFO

INFO [Run 1/12] [Approximate profiling time left: pending first measurement...]

INFO [profiling] Current input file: /workspace/user/workloads/fusion_matmulbias_1024x1024/MI300X_A1/perfmon/SQ_IFETCH_LEVEL.txt

INFO |-> [rocprofiler-sdk] W20260123 18:46:00.435227

139849949089344 simple_timer.cpp:55] [rocprofv3] tool initialization ::

0.177975 sec

INFO |-> [rocprofiler-sdk] W20260123 18:46:00.435332

139849949089344 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py

fusion' :: 0.000000 sec

INFO |-> [rocprofiler-sdk] W20260123 18:46:01.699561

139849949089344 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO |-> [rocprofiler-sdk] Running the fusion matmulbias kernel (input size M:1024 N:1024) ...

INFO |-> [rocprofiler-sdk] triton_output: tensor([[2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],

INFO |-> [rocprofiler-sdk] [7.1562, -21.3906, 17.7969, ...,

```

-16.8125, 29.6562, -48.3125],
INFO    |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO    |-> [rocprofiler-sdk] ...,
INFO    |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO    |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO    |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO    |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO    |-> [rocprofiler-sdk] W20260123 18:46:02.968010

139849949089344 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.532678 sec

INFO    |-> [rocprofiler-sdk] E20260123 18:46:02.980908

139849949089344 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1239_kernel_tr
ace.csv

INFO    |-> [rocprofiler-sdk] E20260123 18:46:02.994566

139849949089344 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1239_counter_c
ollection.csv

INFO    |-> [rocprofiler-sdk] E20260123 18:46:03.307165

139849949089344 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1239_agent_inf
o.csv

INFO    |-> [rocprofiler-sdk] W20260123 18:46:03.310001

139849949089344 simple_timer.cpp:55] [rocprofv3] output generation ::
0.336830 sec

INFO    |-> [rocprofiler-sdk] W20260123 18:46:03.311150

139849949089344 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.338049 sec

INFO    |-> [rocprofiler-sdk]
INFO [Run 2/12][Approximate profiling time left: 53 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/SQ_INST_LEVEL_LDS.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:05.817482
139797254225472 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.177965 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:05.817585
139797254225472 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:07.083043
139797254225472 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```



```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:08.355564
139797254225472 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.537979 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:46:08.368840
139797254225472 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1259_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:08.382963
139797254225472 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1259_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:08.694308
139797254225472 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1259_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:46:08.697372
139797254225472 simple_timer.cpp:55] [rocprofv3] output generation ::
0.336616 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:08.698441
139797254225472 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.337750 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 3/12][Approximate profiling time left: 48 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/SQ_INST_LEVEL_SMEM.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:10.810943
140352685297216 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128123 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:10.811041
140352685297216 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:12.067476
140352685297216 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:13.314776
140352685297216 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.503735 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:46:13.327792
140352685297216 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1280_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:13.341607
140352685297216 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1280_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:13.363121
140352685297216 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1280_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:46:13.366130
140352685297216 simple_timer.cpp:55] [rocprofv3] output generation ::
0.046104 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:13.367092
140352685297216 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.047137 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 4/12][Approximate profiling time left: 38 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/SQ_INST_LEVEL_VMEM.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:14.846835
140576695504448 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.153485 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:14.846938
140576695504448 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:16.105303
140576695504448 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:17.362240
140576695504448 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.515303 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:46:17.375205
140576695504448 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1300_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:17.388921
140576695504448 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1300_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:17.566316
140576695504448 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1300_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:46:17.569185
140576695504448 simple_timer.cpp:55] [rocprofv3] output generation ::
0.201743 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:17.570222
140576695504448 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.202849 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 5/12] [Approximate profiling time left: 33 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/SQ_LEVEL_WAVES.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:19.668218
140701040270912 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.177602 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:19.668319
140701040270912 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:20.927103
140701040270912 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:22.193994
140701040270912 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.525675 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:46:22.206925
140701040270912 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1321_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:22.220648
140701040270912 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1321_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:22.540826
140701040270912 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1321_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:46:22.543712
140701040270912 simple_timer.cpp:55] [rocprofv3] output generation ::
0.344578 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:22.544829
140701040270912 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.345763 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 6/12][Approximate profiling time left: 29 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_0.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:24.706268
140132924616256 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128412 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:24.706370
140132924616256 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:25.967431
140132924616256 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```



```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:27.216186
140132924616256 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.509816 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:46:27.229219
140132924616256 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1341_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:27.243130
140132924616256 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1341_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:27.262881
140132924616256 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1341_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:46:27.265864
140132924616256 simple_timer.cpp:55] [rocprofv3] output generation ::
0.044414 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:27.266853
140132924616256 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.045472 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 7/12] [Approximate profiling time left: 23 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_1.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:28.573265
140345171930688 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128325 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:28.573367
140345171930688 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:29.836916
140345171930688 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:31.083075
140345171930688 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.509708 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:46:31.096051
140345171930688 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1361_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:31.109904
140345171930688 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1361_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:31.129953
140345171930688 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1361_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:46:31.132928
140345171930688 simple_timer.cpp:55] [rocprofv3] output generation ::
0.044644 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:31.133896
140345171930688 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.045682 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 8/12][Approximate profiling time left: 18 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_2.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:32.459309
140060370652736 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.129729 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:32.459411
140060370652736 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:33.717807
140060370652736 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:34.970545
140060370652736 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.511134 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:46:34.983465
140060370652736 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1382_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:34.997332
140060370652736 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1382_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:35.017160
140060370652736 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1382_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:46:35.020134
140060370652736 simple_timer.cpp:55] [rocprofv3] output generation ::
0.044413 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:35.021087
140060370652736 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.045436 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 9/12][Approximate profiling time left: 13 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_3.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:36.341788
139740556783168 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128566 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:36.341887
139740556783168 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:37.600713
139740556783168 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:38.836230
139740556783168 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.494344 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:46:38.849276
139740556783168 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1402_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:38.863518
139740556783168 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1402_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:38.884283
139740556783168 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1402_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:46:38.887298
139740556783168 simple_timer.cpp:55] [rocprofv3] output generation ::
0.045801 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:38.888283
139740556783168 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.046856 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 10/12][Approximate profiling time left: 8 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_4.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:40.205550
139951723380288 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128317 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:40.205696
139951723380288 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000048 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:41.469893
139951723380288 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```



```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:42.695614
139951723380288 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.489967 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:46:42.708730
139951723380288 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1427_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:42.723846
139951723380288 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1427_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:42.748635
139951723380288 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1427_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:46:42.751744
139951723380288 simple_timer.cpp:55] [rocprofv3] output generation ::
0.050968 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:42.752689
139951723380288 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.051981 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 11/12][Approximate profiling time left: 4 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_5.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:44.089462
139966960180800 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128096 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:44.089556
139966960180800 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:45.349683
139966960180800 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:46.571237
139966960180800 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.481681 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:46:46.584257
139966960180800 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1447_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:46.599062
139966960180800 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1447_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:46.623840
139966960180800 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1447_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:46:46.626769
139966960180800 simple_timer.cpp:55] [rocprofv3] output generation ::
0.050248 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:46.627711
139966960180800 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.051259 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 12/12][Approximate profiling time left: 0 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_1024x1024/MI300X_A1/perfmon/pmc_perf_6.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:47.960651
140560116121152 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128541 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:47.960751
140560116121152 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:49.225495
140560116121152 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:1024 N:1024) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:46:50.416043
140560116121152 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.455292 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:46:50.429076
140560116121152 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1467_kernel_tr
ace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:50.444603
140560116121152 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1467_counter_c
ollection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:46:50.462308
140560116121152 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_1024x1024/MI300X_A1/out/pmc_1/6b82f9811751/1467_agent_inf
o.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:46:50.465248
140560116121152 simple_timer.cpp:55] [rocprofv3] output generation ::
0.044212 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:46:50.466194
140560116121152 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.045222 sec

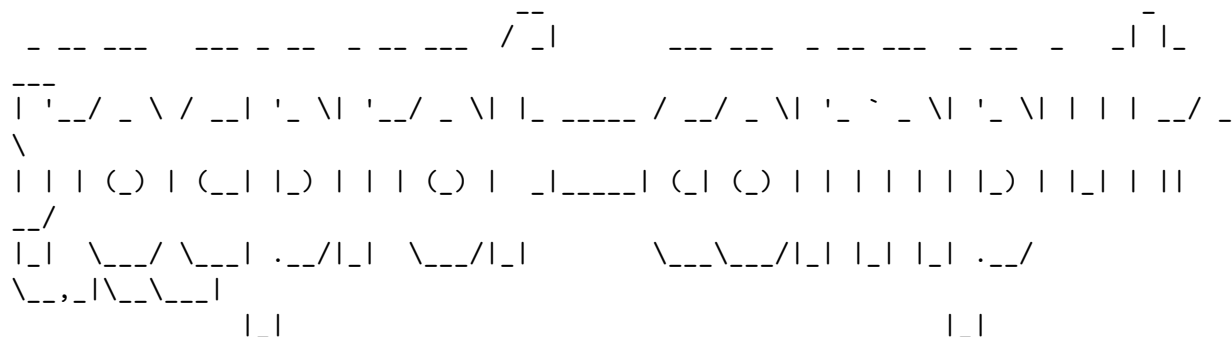
INFO      |-> [rocprofiler-sdk]
INFO [roofline] Checking for roofline.csv in
/workspace/user/workloads/fusion_matmulbias_1024x1024/MI300X_A1
INFO [roofline] No roofline data found. Generating...
Empirical Roofline Calculation
Copyright © 2025 Advanced Micro Devices, Inc. All rights reserved.
Total detected GPU devices: 1
GPU Device 0 (gfx942) with 304 CUs: Profiling...
99% [|||||||||||||||||||||||||||||||||||||||||||||||||||||||||||| ]
HBM BW, GPU ID: 0, workgroupSize:256, workgroups:6225920, experiments:100,
traffic:25501368320 bytes, duration:6.1 ms, mean:4211.2 GB/sec, stdev=8.7 GB/sec
99% [|||||||||||||||||||||||||||||||||||||||||||||||||||||||||||| ]
MALL BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:2611340115968 bytes, duration:411.2 ms, mean:6316.7 GB/sec, stdev=32.3
GB/sec
99% [|||||||||||||||||||||||||||||||||||||||||||||||||||||||||||| ]

```

L2 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:1632087572480 bytes, duration:64.0 ms, mean:25491.6 GB/sec, stdev=195.4
GB/sec
99% [|||||]
L1 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:127506841600 bytes, duration:3.8 ms, mean:33778.6 GB/sec, stdev=1447.4
GB/sec
99% [|||||]
LDS BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:79691776000 bytes, duration:1.3 ms, mean:63365.7 GB/sec, stdev=79.7
GB/sec
99% [|||||]
Peak FLOPs (FP8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:1081258016768, duration:13.0 ms, mean:82984.8 GFLOPS,
stdev=172.6 GFLOPS
99% [|||||]
Peak FLOPs (FP16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:530428461056, duration:13.4 ms, mean:39471.6 GFLOPS,
stdev=647.2 GFLOPS
99% [|||||]
Peak FLOPs (BF16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:530428461056, duration:91.862 ms, mean:5759.1 GFLOPS,
stdev=35.2 GFLOPS
99% [|||||]
Peak FLOPs (FP32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:265214230528, duration:2.024 ms, mean:130855.0 GFLOPS,
stdev=1460.2 GFLOPS
99% [|||||]
Peak FLOPs (FP64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:122406567936, duration:1.1 ms, mean:106041.7 GFLOPS,
stdev=1699.7 GFLOPS
99% [|||||]
Peak IOPs (INT8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:1081258016768, duration:13.1 ms, mean:82569.4 GOPS,
stdev=1454.7 GOPS
99% [|||||]
Peak IOPs (INT32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:265214230528, duration:3.8 ms, mean:70638.9 GOPS,
stdev=288.3 GOPS
99% [|||||]
Peak IOPs (INT64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:122406567936, duration:6.2 ms, mean:19715.7 GOPS,
stdev=66.3 GOPS
100% [|||||]
Peak MFMA FLOPs (F4), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:1, FLOP:0, duration:0.0 ms, mean:0.0 GFLOPS, stdev=0.0 GFLOPS
100% [|||||]
Peak MFMA FLOPs (F6), GPU ID: 0, workgroupSize:256, workgroups:38912,

```
INFO [roofline] Empirical Roofline PDFs saved!
```

```
[23]: !ROCPROF=rocprofiler-sdk rocprof-compute profile -n
      ↪ fusion_matmulbias_32768x32768 -- python triton/matmulbias.py --M 32768 --N
      ↪ 32768 fusion
```



```

https://github.com/ROCm/amdsmi
INFO Rocprofiler-Compute version: 3.2.3
INFO Profiler choice: rocprofiler-sdk
INFO Path:
/workspace/user/workloads/fusion_matmulbias_32768x32768/MI300X_A1
INFO Target: MI300X_A1
INFO Command: python triton/matmulbias.py --M 32768 --N 32768 fusion
INFO Kernel Selection: None
INFO Dispatch Selection: None
INFO Hardware Blocks: []
INFO Report Sections: []
INFO
INFO ~~~~~
INFO Collecting Performance Counters
INFO ~~~~~
INFO
INFO [Run 1/12][Approximate profiling time left: pending first
measurement...]
INFO [profiling] Current input file: /workspace/user/workloads/fusion
n_matmulbias_32768x32768/MI300X_A1/perfmon/SQ_IFETCH_LEVEL.txt
INFO |-> [rocprofiler-sdk] W20260123 18:48:21.942145
140208311238208 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.176095 sec
INFO |-> [rocprofiler-sdk] W20260123 18:48:21.942245
140208311238208 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' :: 0.000000 sec
INFO |-> [rocprofiler-sdk] W20260123 18:48:23.204599
140208311238208 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)
INFO |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562, 57.0938, ..., 274.0000, 295.2500,
INFO |-> [rocprofiler-sdk] 14.8828],
INFO |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219, 57.0625,
INFO |-> [rocprofiler-sdk] -30.7969],
INFO |-> [rocprofiler-sdk] [ 56.2188, 258.7500, 46.6562,
..., -366.0000, 395.2500,
INFO |-> [rocprofiler-sdk] 285.2500],
INFO |-> [rocprofiler-sdk] ...,
INFO |-> [rocprofiler-sdk] [ -5.0977, 99.9375, -73.5625,
..., -70.5000, 204.2500,
INFO |-> [rocprofiler-sdk] -156.8750],

```



```

INFO      |-> [rocprofiler-sdk] [ 86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406, 93.5625, 77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)
INFO      |-> [rocprofiler-sdk] W20260123 18:48:28.424021
140208311238208 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' :: 6.481777 sec
INFO      |-> [rocprofiler-sdk] E20260123 18:48:28.437480
140208311238208 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1572_kernel_
trace.csv
INFO      |-> [rocprofiler-sdk] E20260123 18:48:28.450488
140208311238208 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1572_counter
_collection.csv
INFO      |-> [rocprofiler-sdk] E20260123 18:48:28.565928
140208311238208 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1572_agent_i
nfo.csv
INFO      |-> [rocprofiler-sdk] W20260123 18:48:28.568753
140208311238208 simple_timer.cpp:55] [rocprofv3] output generation ::
0.138981 sec
INFO      |-> [rocprofiler-sdk] W20260123 18:48:28.569833
140208311238208 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.140121 sec
INFO      |-> [rocprofiler-sdk]
INFO [Run 2/12] [Approximate profiling time left: 1 minute and 25
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/SQ_INST_LEVEL_LDS.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:48:30.463336
140008418577984 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.167286 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:48:30.463435
140008418577984 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:48:31.723276
140008418577984 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:48:36.976493
140008418577984 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.513059 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:48:36.989857
140008418577984 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1593_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:48:37.002898
140008418577984 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1593_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:48:37.089542
140008418577984 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1593_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:48:37.092349
140008418577984 simple_timer.cpp:55] [rocprofv3] output generation ::
0.110230 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:48:37.093383
140008418577984 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.111329 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 3/12] [Approximate profiling time left: 1 minute and 16
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/SQ_INST_LEVEL_SMEM.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:48:38.853738
140055698910784 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.163813 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:48:38.853844
140055698910784 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:48:40.114598
140055698910784 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:48:45.333505
140055698910784 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.479662 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:48:45.346942
140055698910784 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1613_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:48:45.359990
140055698910784 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1613_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:48:45.447861
140055698910784 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1613_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:48:45.450721
140055698910784 simple_timer.cpp:55] [rocprofv3] output generation ::
0.111537 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:48:45.451752
140055698910784 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.112633 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 4/12] [Approximate profiling time left: 1 minute and 7
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/SQ_INST_LEVEL_VMEM.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:48:46.986283
140132216942144 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.129159 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:48:46.986387
140132216942144 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:48:48.246800
140132216942144 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:48:53.453572
140132216942144 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.467185 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:48:53.466995
140132216942144 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1638_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:48:53.480463
140132216942144 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1638_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:48:53.491620
140132216942144 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1638_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:48:53.494609
140132216942144 simple_timer.cpp:55] [rocprofv3] output generation ::
0.035390 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:48:53.495590
140132216942144 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.036435 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 5/12][Approximate profiling time left: 58 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/SQ_LEVEL_WAVES.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:48:55.140879
139744736374336 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.179319 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:48:55.140980
139744736374336 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:48:56.409011
139744736374336 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,   57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,   57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,   46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [  -5.0977,   99.9375,  -73.5625,
...,  -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125,  -26.3594, -128.8750,
...,  -60.7500,  -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,   93.5625,   77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```



```

INFO      |-> [rocprofiler-sdk] W20260123 18:49:01.658934
139744736374336 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.517954 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:49:01.672247
139744736374336 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1659_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:49:01.685208
139744736374336 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1659_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:49:01.800633
139744736374336 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1659_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:49:01.803522
139744736374336 simple_timer.cpp:55] [rocprofv3] output generation ::
0.138996 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:01.804575
139744736374336 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.140113 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 6/12][Approximate profiling time left: 50 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_0.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:49:03.410426
140371077946944 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128196 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:03.410524
140371077946944 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:04.672260
140371077946944 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:49:09.909308
140371077946944 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.498784 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:49:09.922644
140371077946944 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1680_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:49:09.935706
140371077946944 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1680_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:49:09.945832
140371077946944 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1680_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:49:09.948773
140371077946944 simple_timer.cpp:55] [rocprofv3] output generation ::
0.033840 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:09.949772
140371077946944 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.034894 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 7/12][Approximate profiling time left: 41 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_1.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:49:11.244631
139621899546176 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128305 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:11.244733
139621899546176 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:12.501902
139621899546176 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,   57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,   57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,   46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [  -5.0977,   99.9375,  -73.5625,
...,  -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125,  -26.3594, -128.8750,
...,  -60.7500,  -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,   93.5625,   77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:49:17.749874
139621899546176 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.505142 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:49:17.763274
139621899546176 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1701_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:49:17.776257
139621899546176 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1701_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:49:17.786391
139621899546176 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1701_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:49:17.789286
139621899546176 simple_timer.cpp:55] [rocprofv3] output generation ::
0.033747 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:17.790257
139621899546176 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.034783 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 8/12] [Approximate profiling time left: 32 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_2.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:49:19.087247
140221194174016 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128140 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:19.087345
140221194174016 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:20.350476
140221194174016 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:49:25.563888
140221194174016 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.476544 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:49:25.577193
140221194174016 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1721_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:49:25.590534
140221194174016 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1721_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:49:25.600792
140221194174016 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1721_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:49:25.603784
140221194174016 simple_timer.cpp:55] [rocprofv3] output generation ::
0.034319 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:25.604746
140221194174016 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.035346 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 9/12][Approximate profiling time left: 24 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_3.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:49:26.904699
140646562700864 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128165 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:26.904802
140646562700864 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:28.168443
140646562700864 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```



```

INFO      |-> [rocprofiler-sdk] W20260123 18:49:33.354550
140646562700864 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.449749 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:49:33.367843
140646562700864 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1742_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:49:33.380909
140646562700864 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1742_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:49:33.390393
140646562700864 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1742_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:49:33.393365
140646562700864 simple_timer.cpp:55] [rocprofv3] output generation ::
0.033246 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:33.394336
140646562700864 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.034281 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 10/12][Approximate profiling time left: 16 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_4.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:49:34.684989
140309280321088 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128106 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:34.685087
140309280321088 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:35.943493
140309280321088 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:49:41.073777
140309280321088 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.388690 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:49:41.087151
140309280321088 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1763_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:49:41.100157
140309280321088 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1763_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:49:41.109068
140309280321088 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1763_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:49:41.112063
140309280321088 simple_timer.cpp:55] [rocprofv3] output generation ::
0.032612 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:41.113023
140309280321088 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.033636 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 11/12][Approximate profiling time left: 8 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_5.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:49:42.399957
140594439523904 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128007 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:42.400054
140594439523904 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:43.663134
140594439523904 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,   57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,   57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,   46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [  -5.0977,   99.9375,  -73.5625,
...,  -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125,  -26.3594, -128.8750,
...,  -60.7500,  -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,   93.5625,   77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:49:48.813205
140594439523904 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.413151 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:49:48.826530
140594439523904 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1784_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:49:48.839561
140594439523904 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1784_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:49:48.848375
140594439523904 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1784_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:49:48.851372
140594439523904 simple_timer.cpp:55] [rocprofv3] output generation ::
0.032546 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:48.852334
140594439523904 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.033560 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 12/12][Approximate profiling time left: 0 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias_32768x32768/MI300X_A1/perfmon/pmc_perf_6.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:49:50.140261
140084846517824 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128126 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:50.140358
140084846517824 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:51.398788
140084846517824 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel (input size M:32768 N:32768) ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ -60.8750,
56.6562,  57.0938, ..., 274.0000, 295.2500,
INFO      |-> [rocprofiler-sdk] 14.8828],
INFO      |-> [rocprofiler-sdk] [ 205.3750, -353.7500, -102.1875,
..., -24.4219,  57.0625,
INFO      |-> [rocprofiler-sdk] -30.7969],
INFO      |-> [rocprofiler-sdk] [  56.2188,  258.7500,  46.6562,
..., -366.0000,  395.2500,
INFO      |-> [rocprofiler-sdk] 285.2500],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [ -5.0977,  99.9375, -73.5625,
..., -70.5000,  204.2500,
INFO      |-> [rocprofiler-sdk] -156.8750],
INFO      |-> [rocprofiler-sdk] [  86.8125, -26.3594, -128.8750,
..., -60.7500, -36.3750,
INFO      |-> [rocprofiler-sdk] 100.8750],
INFO      |-> [rocprofiler-sdk] [ -27.6406,  93.5625,  77.9375,
..., -213.1250, -323.2500,
INFO      |-> [rocprofiler-sdk] -16.8438]], device='cuda:0',
dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260123 18:49:56.559442
140084846517824 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
--M 32768 --N 32768 fusion' ::      6.419085 sec

INFO      |-> [rocprofiler-sdk] E20260123 18:49:56.572883
140084846517824 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1804_kernel_
trace.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:49:56.585880
140084846517824 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1804_counter
_collection.csv

INFO      |-> [rocprofiler-sdk] E20260123 18:49:56.592952
140084846517824 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias_32768x32768/MI300X_A1/out/pmc_1/6b82f9811751/1804_agent_i
nfo.csv

INFO      |-> [rocprofiler-sdk] W20260123 18:49:56.595918
140084846517824 simple_timer.cpp:55] [rocprofv3] output generation ::
0.030799 sec

INFO      |-> [rocprofiler-sdk] W20260123 18:49:56.596868
140084846517824 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.031815 sec

INFO      |-> [rocprofiler-sdk]
INFO [roofline] Checking for roofline.csv in
/workspace/user/workloads/fusion_matmulbias_32768x32768/MI300X_A1
INFO [roofline] No roofline data found. Generating...
Empirical Roofline Calculation
Copyright © 2025 Advanced Micro Devices, Inc. All rights reserved.
Total detected GPU devices: 1
GPU Device 0 (gfx942) with 304 CUs: Profiling...
99% [||||||||||||||||||||||||||||||||||||||||||||||||||||||||| ]
HBM BW, GPU ID: 0, workgroupSize:256, workgroups:6225920, experiments:100,
traffic:25501368320 bytes, duration:6.1 ms, mean:4198.3 GB/sec, stdev=8.9 GB/sec
99% [||||||||||||||||||||||||||||||||||||||||||||||||||||||||| ]
MALL BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:2611340115968 bytes, duration:420.3 ms, mean:6245.5 GB/sec, stdev=30.8
GB/sec
99% [||||||||||||||||||||||||||||||||||||||||||||||||||||||||| ]

```

L2 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:1632087572480 bytes, duration:63.9 ms, mean:25499.3 GB/sec, stdev=179.6
GB/sec
99% [|||||]
L1 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:127506841600 bytes, duration:3.8 ms, mean:33435.6 GB/sec, stdev=135.9
GB/sec
99% [|||||]
LDS BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:79691776000 bytes, duration:1.3 ms, mean:63612.5 GB/sec, stdev=302.6
GB/sec
99% [|||||]
Peak FLOPs (FP8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:1081258016768, duration:13.0 ms, mean:82769.6 GFLOPS,
stdev=1455.7 GFLOPS
99% [|||||]
Peak FLOPs (FP16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:530428461056, duration:13.4 ms, mean:39438.4 GFLOPS,
stdev=680.1 GFLOPS
99% [|||||]
Peak FLOPs (BF16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:530428461056, duration:92.021 ms, mean:5747.0 GFLOPS,
stdev=35.4 GFLOPS
99% [|||||]
Peak FLOPs (FP32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:265214230528, duration:2.023 ms, mean:130645.2 GFLOPS,
stdev=1437.8 GFLOPS
99% [|||||]
Peak FLOPs (FP64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:122406567936, duration:1.1 ms, mean:105962.6 GFLOPS,
stdev=1436.2 GFLOPS
99% [|||||]
Peak IOPs (INT8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:1081258016768, duration:13.1 ms, mean:82512.9 GOPS,
stdev=1397.0 GOPS
99% [|||||]
Peak IOPs (INT32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:265214230528, duration:3.7 ms, mean:71657.8 GOPS,
stdev=628.5 GOPS
99% [|||||]
Peak IOPs (INT64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:122406567936, duration:6.2 ms, mean:19698.0 GOPS,
stdev=57.5 GOPS
100% [|||||]
Peak MFMA FLOPs (F4), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:1, FLOP:0, duration:0.0 ms, mean:0.0 GFLOPS, stdev=0.0 GFLOPS
100% [|||||]
Peak MFMA FLOPs (F6), GPU ID: 0, workgroupSize:256, workgroups:38912,


```

experiments:1, FLOP:0, duration:0.0 ms, mean:0.0 GFLOPS, stdev=0.0 GFLOPS
 99% [||||| ]
Peak MFMA FLOPs (F8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:10200547328000, duration:8.7 ms, mean:1906647.5 GFLOPS,
stdev=75824.5 GFLOPS
 99% [||||| ]
Peak MFMA FLOPs (F16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:5100273664000, duration:5.4 ms, mean:939110.0 GFLOPS,
stdev=5345.6 GFLOPS
 99% [||||| ]
Peak MFMA FLOPs (BF16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:2550136832000, duration:4.8 ms, mean:528632.2 GFLOPS,
stdev=2206.3 GFLOPS
 99% [||||| ]
Peak MFMA FLOPs (F32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:1275068416000, duration:9.1 ms, mean:139812.4 GFLOPS,
stdev=487.5 GFLOPS
 99% [||||| ]
Peak MFMA FLOPs (F64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:637534208000, duration:4.4 ms, mean:144800.9 GFLOPS,
stdev=5372.5 GFLOPS
 99% [||||| ]
Peak MFMA IOPs (I8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:5100273664000, duration:5.1 ms, mean:1000116.6 GOPS,
stdev=6066.7 GOPS

```

INFO [roofline] Empirical Roofline PDFs saved!

1.5.3 Analyze the Fusion MatMulBias Kernel Profiles

Using the Triton Proton

Small Input

```
[24]: !proton-viewer -m time/ms,flops16 -f full profiles/fusion_matmulbias_1024x1024.
      ↪hatchet
```

```

95.959 3616362463232.000 ROOT
 56.559 nan _ZN2at6native29vectorized_elementwise_kernel
ILi4ENS0_11FillFunctorIiEEST5arrayIPcLm1EEEEviT0_T1_
 39.399 3616362463232.000 matmulbias_kernel [M=1024,
N=1024, K=1024] [BM=128, BN=128 BK=64]

```

Legend (Metric: time/ms (inc) Min: 39.40 Max: 95.96)

```

90.30 - 95.96
78.99 - 90.30
67.68 - 78.99
56.37 - 67.68
45.06 - 56.37

```

```
name User code      Only in left graph
Only in right graph
```

```
[25]: !proton-viewer -m time/ms,flops16 -f full profiles/  
      ↪ fusion_matmulbias_32768x32768.hatchet
```

Legend (Metric: time/ms (inc) Min: 0.56 Max: 4251.10)

```
name User code      Only in left graph
Only in right graph
```

```
[26]: !rocprow-compute analyze -p workloads/fusion_matmulbias_1024x1024/MI300X_A1 -k 0
```

```
INFO Analysis mode = cli
INFO [analysis] deriving rocprofiler-compute metrics...
WARNING PC sampling: can not detect pc sampling method without /workspa
```

ce/user/workloads/fusion_matmulbias_1024x1024/MI300X_A1/ps_file_pc_sampling_host_trap.csv

0. Top Stats

0.1 Top Kernels

Kernel_Name				Count	Sum(ns)
Mean(ns)	Median(ns)	Pct	S		
0	void at::native::vectorized_elementwise_43552.43	42733.83	55.20 *	139.00	6053788.25
	kernel<4, at::native::FillFunctor<int>, std::array<char*, 1ul> >(int, at::nat...				
1	matmulbias_kernel			172.00	4546417.75
26432.66	24669.33	41.46			
2	__amd_rocclr_copyBuffer			89.00	266058.58
2989.42	2956.83	2.43			
3	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4, at::native::templates::...			3.00	18869.83
6289.94	7373.42	0.17			
4	void at::native::(anonymous namespace)::CatArrayBatchedCopy_contig<at::native::(anonymous namespace)::OpaqueType<2u>,...			7.00	16367.58
2338.23	2221.67	0.15			
5	void at::native::vectorized_elementwise_3423.25	3421.08	0.09	3.00	10269.75
	kernel<4, at::native::compare_scalar_ker				

```

    nel<double>(at::TensorIteratorBase&, ...

6   void at::native::vectorized_elementwise_      2.00      6154.08
3077.04      3077.04      0.06
    kernel<8, at::native::UnaryFunctor<c10:
        :Half, c10::Half, bool, at::native::(...)

7   void at::native::vectorized_elementwise_      2.00      5111.75
2555.87      2555.87      0.05
    kernel<8, at::native::AbsFunctor<c10::Ha
        lf>, std::array<char*, 2ul> >(int, at...

8   void at::native::vectorized_elementwise_      2.00      5054.67
2527.33      2527.33      0.05
    kernel<8, at::native::BinaryFunctor<c10:
        :Half, c10::Half, bool, at::native::(...)

9   void at::native::reduce_kernel<512, 1, a      1.00      4286.33
4286.33      4286.33      0.04
    t::native::ReduceOp<double, at::native::
        func_wrapper_t<double, at::native::Ma...

```

0.2 Dispatch List

```

    Dispatch_ID  Kernel_Name
GPU_ID
0              0   void at::native::(anonymous
namespace)::distribution_elementwise_grid_stride_ker
    nel<float, 4,
at::native::templates::cuda::normal_and_transform<c10::Half, float
    , at::CUDAGeneratorImpl*,

```

```

at::native::templates::cuda::normal_kernel<at::CUDAGen
        eratorImpl*>(at::TensorBase const&, double, double,
at::CUDAGeneratorImpl*)::...

```

```

1          1 void at::native::(anonymous
namespace)::distribution_elementwise_grid_stride_ker      0
        nel<float, 4,
at::native::templates::cuda::normal_and_transform<c10::Half, float
        , at::CUDAGeneratorImpl*,
at::native::templates::cuda::normal_kernel<at::CUDAGen
        eratorImpl*>(at::TensorBase const&, double, double,
at::CUDAGeneratorImpl*)::...

```

```

2          2 void at::native::(anonymous
namespace)::distribution_elementwise_grid_stride_ker      0
        nel<float, 4,
at::native::templates::cuda::normal_and_transform<c10::Half, float
        , at::CUDAGeneratorImpl*,
at::native::templates::cuda::normal_kernel<at::CUDAGen
        eratorImpl*>(at::TensorBase const&, double, double,
at::CUDAGeneratorImpl*)::...

```

```

3          3 matmulbias_kernel
0

```

```

4          4 void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,      0
        std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
        1>)

```

```

5          5 matmulbias_kernel
0

```

```

6          6 void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,      0
        std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
        1>)

```

```

7          7  matmulbias_kernel
0

8          8  void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
          std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
          l>)

9          9  matmulbias_kernel
0

```

1. System Info

	Info
workload_name	fusion_matmulbias_1024x1024
command	python triton/matmulbias.py fusion
ip_blocks	SQ LDS SQC TA TD TCP TCC SPI CPC CPF roofline
timestamp	Fri Jan 23 18:45:57 2026 (UTC)
version	3
hostname	6b82f9811751
cpu_model	Intel(R) Xeon(R) Platinum 8470
sbios	SeaBIOS1.15.0-1
linux_distro	CentOS Stream 10 (Coughlan)
linux_kernel_version	5.15.0-164-generic
amd_gpu_kernel_version	nan
cpu_memory	230893524

gpu_memory	nan
rocm_version	7.0.3
vbios	113-M3000108-103
compute_partition	SPX
memory_partition	nan
gpu_series	MI300
gpu_model	MI300X_A1
gpu_arch	gfx942
gpu_chip_id	29877
gpu_l1	32
gpu_l2	4096
cu_per_gpu	304
simd_per_cu	4
se_per_gpu	32
wave_size	64
workgroup_max_size	1024
max_waves_per_cu	32
max_sclk	2100
max_mclk	1300
cur_sclk	2100
cur_mclk	1300
total_l2_chan	128
lds_banks_per_cu	32
sqc_per_gpu	160

pipes_per_gpu	4
num_xcd	8
num_hbm_channels	128

2. System Speed-of-Light

2.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit	Peak
Pct of Peak				
2.1.0	VALU FLOPs	0.0	Gflop/s	
81715.2	0.0			
2.1.1	VALU IOPs	677.2	Giop/s	
81715.2	0.83			
2.1.2	MFMA FLOPs (F8)	0.0	Gflop/s	
2614886.4	0.0			
2.1.3	MFMA FLOPs (BF16)	0.0	Gflop/s	
1307443.2	0.0			
2.1.4	MFMA FLOPs (F16)	0.0	Gflop/s	
1307443.2	0.0			
2.1.5	MFMA FLOPs (F32)	0.0	Gflop/s	
163430.4	0.0			
2.1.6	MFMA FLOPs (F64)	0.0	Gflop/s	
163430.4	0.0			
2.1.7	MFMA FLOPs (F6F4)		Gflop/s	

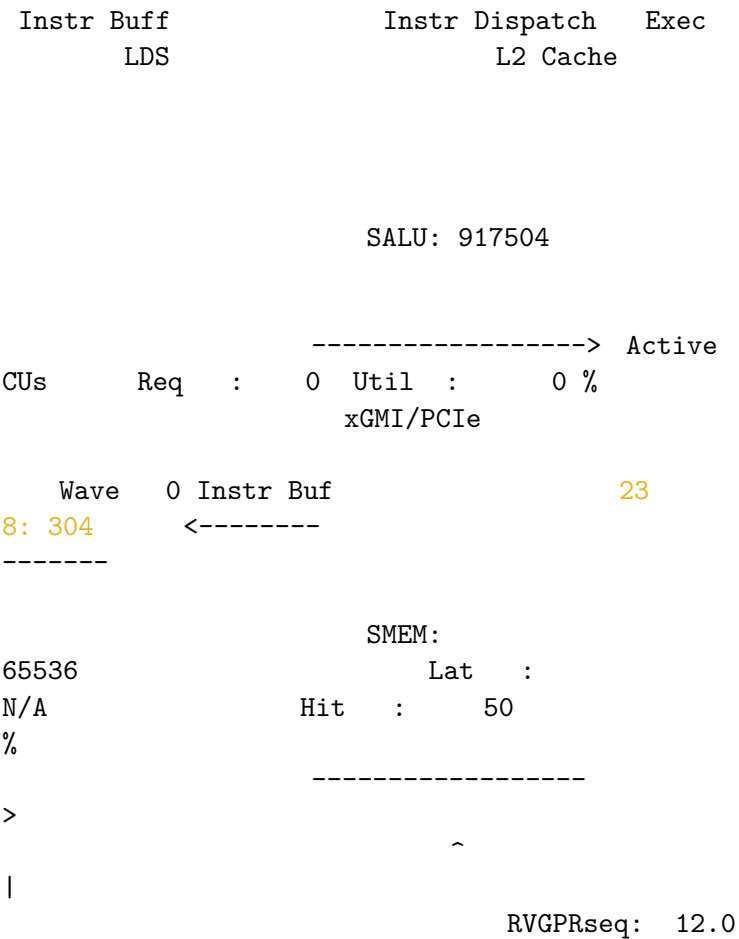
2.1.8 2614886.4	MFMA IOPs (Int8) 0.0	0.0	Giop/s	
2.1.9 78.29	Active CUs	238.0	Cus	304.0
2.1.10 3.09	SALU Utilization	3.09	Pct	100.0
2.1.11 2.27	VALU Utilization	2.27	Pct	100.0
2.1.12 0.0	MFMA Utilization	0.0	Pct	100.0
2.1.13 0.82	VMEM Utilization	0.82	Pct	100.0
2.1.14 0.62	Branch Utilization	0.62	Pct	100.0
2.1.15 100.0	VALU Active Threads	64.0	Threads	64.0
2.1.16 1.99	IPC	0.1	Instr/cycle	5.0
2.1.17 21.11	Wavefront Occupancy	2053.85	Wavefronts	9728.0
2.1.18 81715.2	Theoretical LDS Bandwidth 0.0	0.0	Gb/s	
2.1.19	LDS Bank Conflicts/Access		Conflicts/access	32.0

2.1.20 0.0	vL1D Cache Hit Rate	0.0	Pct	100.0
2.1.21 81715.2	vL1D Cache BW 15.15	12383.1	Gb/s	
2.1.22 50.05	L2 Cache Hit Rate	50.05	Pct	100.0
2.1.23 34406.4	L2 Cache BW 36.03	12394.96	Gb/s	
2.1.24 0.01	L2-Fabric Read BW	0.31	Gb/s	5324.8
2.1.25 116.28	L2-Fabric Write BW	6191.55	Gb/s	5324.8
2.1.26	L2-Fabric Read Latency	317.43	Cycles	
2.1.27	L2-Fabric Write Latency	404.7	Cycles	
2.1.28 99.74	sL1D Cache Hit Rate	99.74	Pct	100.0
2.1.29 21504.0	sL1D Cache BW 0.45	96.74	Gb/s	
2.1.30 99.96	L1I Hit Rate	99.96	Pct	100.0
2.1.31 21504.0	L1I BW 3.15	677.2	Gb/s	

3. Memory Chart

3.1 Memory Chart

(Normalization: per_kernel)



```

      | |
Wave N-1 Instr Buf      VALU:
720896      Vector L1 Cache
      Request      |
|
|-----|
>
|
v
      SGPRs :      32

      MFMA: 0
      Rd      :      0 Hit      :      0 %      Rd      :
0      Rd      :      4005

|-----|
>      <-----|      <-----|
-
Fabric

Wt      : 16777216Lat      :      N/A      Wr      : 4194304 Wr      :
4194304      Rd      :
214

      VMEM: 262144      LDS Alloc:      -
----->
>      <----- Latency (cycles)      Rd:
214

----->
0      Atomic:      0 Coales:      25 %      Atomic:      0      Atomic:
0      Wr      : 4194304      <----- HBM
<
----->      <-----
>      ----->      Rd      :      317      Wr:
4194304

      LDS: 0
      Stall :      77
cycles      Atomic:      0      Wr      :
405      ----->
----->      Scratch Alloc:

----->      Atomic:      0
0

Wave Occupancy      GWS: 0
Scalar L1D

```

Cache Latency (cycles)

9
---> Rd : 1
60

Wave Life

<-----
^
|

10283
BRANCH: 131072 Wavefronts: Rd : 65536 Hit : 93.0
% Wr : 0 Rd : N/A |
|
-----> 0 <

> |
|

Lat : N/A Atomic: 0 Wr :
N/A |
v

^
<-----
>

| Workgroups:

|
0 Instr L1 Cache
GMI
|

| Fetch :
458752 Hit : 99 % Req
: 3840

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Lat : 2261.0 cycles

4. Roofline

4.1 Roofline

139887.4

139887GFLOP/s

36214.6

9375.4

2427.1

628.3 63366 GB/s

33779 GB/s

162.7

42.1 4211 GB/s

0.01 0.18 3.16 56.23 1000.00

Performance (GFLOP/sec) Arithmetic Intensity (FLOP)s/Byte)

5. Command Processor (CPC/CPF)

5.1 Command Processor Fetcher

Metric_ID	Metric	Avg	Min	Max	Unit
5.1.0	CPF Utilization	100.00	100.00	100.00	Pct
5.1.1	CPF Stall	0.74	0.02	1.44	Pct
5.1.2	CPF-L2 Utilization	8.57	1.09	10.98	Pct
5.1.3	CPF-L2 Stall	0.00	0.00	0.00	Pct
5.1.4	CPF-UTCL1 Stall	0.87	0.02	1.45	Pct

5.2 Packet Processor

Metric_ID	Metric	Avg	Min	Max	Unit
5.2.0	CPC SYNC FIFO Full Rate				Pct

5.2.1	CPC CANE Stall Rate				Pct
5.2.2	CPC ADC Utilization				Pct
5.2.3	CPC Utilization	100.0	100.0	100.0	Pct
5.2.4	CPC Stall Rate	9.14	2.45	11.38	Pct
5.2.5	CPC Packet Decoding Utilization	27.57	8.57	38.54	Pct
5.2.6	CPC-Workgroup Manager Utilization	67.26	21.28	71.24	Pct
5.2.7	CPC-L2 Utilization	0.22	0.16	0.23	Pct
5.2.8	CPC-UTCL1 Stall	0.14	0.04	0.23	Pct
5.2.9	CPC-UTCL2 Utilization	0.14	0.04	0.22	Pct

6. Workgroup Manager (SPI)

6.1 Workgroup Manager Utilizations

Metric_ID	Metric	Avg	Min	Max	Unit
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6.1.0	Schedule-Pipe Wave Occupancy				Wave
6.1.1	Accelerator Utilization	101.04	93.52	291.95	Pct
6.1.2	Scheduler-Pipe Utilization	0.0	0.0	0.0	Pct
6.1.3	Scheduler-Pipe Wave Utilization				Pct
6.1.4	Workgroup Manager Utilization	78.81	29.49	82.79	Pct
6.1.5	Shader Engine Utilization	76.71	32.08	81.15	Pct
6.1.6	SIMD Utilization	76.72	73.48	104.56	Pct
6.1.7 Workgroups	Dispatched Workgroups	0.0	0.0	0.0	
6.1.8 Wavefronts	Dispatched Wavefronts	0.0	0.0	0.0	
6.1.9 Cycles/wave	VGPR Writes				
6.1.10 Cycles/wave	SGPR Writes				

6.2 Workgroup Manager - Resource Allocation

Metric_ID	Metric	Avg	Min	Max
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Unit

6.2.0 Pct	Not-scheduled Rate (Workgroup Manager)	0.0	0.0	0.0
6.2.1 Pct	Not-scheduled Rate (Scheduler-Pipe)	0.0	0.0	0.0
6.2.2 Pct	Scheduler-Pipe FIFO Full Rate			
6.2.3 Pct	Scheduler-Pipe Stall Rate	0.0	0.0	0.0
6.2.4 Pct	Scratch Stall Rate	0.0	0.0	0.0
6.2.5 Pct	Insufficient SIMD Waveslots	0.0	0.0	0.0
6.2.6 Pct	Insufficient SIMD VGPRs	0.0	0.0	0.0
6.2.7 Pct	Insufficient SIMD SGPRs	0.0	0.0	0.0
6.2.8 Pct	Insufficient CU LDS	0.0	0.0	0.0
6.2.9 Pct	Insufficient CU Barriers	0.0	0.0	0.0
6.2.10 Pct	Reached CU Workgroup Limit	0.0	0.0	0.0
6.2.11	Reached CU Wavefront Limit	0.0	0.0	0.0

Pct

7. Wavefront

7.1 Wavefront Launch Stats

Metric_ID Unit	Metric	Avg	Min	Max
7.1.0 Work items	Grid Size	4194304.00	4194304.00	4194304.00
7.1.1 Work items	Workgroup Size	256.00	256.00	256.00
7.1.2 Wavefronts	Total Wavefronts	0.00	0.00	0.00
7.1.3 Wavefronts	Saved Wavefronts	0.00	0.00	0.00
7.1.4 Wavefronts	Restored Wavefronts	0.00	0.00	0.00
7.1.5 Registers	VGPRs	12.00	12.00	12.00
7.1.6 Registers	AGPRs	4.00	4.00	4.00
7.1.7 Registers	SGPRs	32.00	32.00	32.00
7.1.8 Bytes	LDS Allocation	0.00	0.00	0.00

7.1.9	Scratch Allocation	0.00	0.00	0.00
Bytes/workitem				

7.2 Wavefront Runtime Stats

Metric_ID	Metric	Avg	Min
Max	Unit		
7.2.0	Kernel Time (Nanosec)	43552.43	42526.67
67576.83	Ns		
7.2.1	Kernel Time (Cycles)	106047.57	100857.25
340209.88	Cycle		
7.2.2	Instructions per wavefront	37.00	37.00
37.00	Instr/wavefront		
7.2.3	Wave Cycles	673889561.87	657418204.00
918528108.00	Cycles per kernel		
7.2.4	Dependency Wait Cycles	152946107.17	149329256.00
203439924.00	Cycles per kernel		
7.2.5	Issue Wait Cycles	514271429.21	500772036.00
720424476.00	Cycles per kernel		
7.2.6	Active Cycles	8650752.00	8650752.00
8650752.00	Cycles per kernel		
7.2.7	Wavefront Occupancy	2053.85	171.79
6869.48	Wavefronts		

10. Compute Units - Instruction Mix

10.1 Overall Instruction Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.1.0	VALU	720896.00	720896.00	720896.00	Instr per kernel
10.1.1	VMEM	262144.00	262144.00	262144.00	Instr per kernel
10.1.2	LDS	0.00	0.00	0.00	Instr per kernel
10.1.3	MFMA	0.00	0.00	0.00	Instr per kernel
10.1.4	SALU	917504.00	917504.00	917504.00	Instr per kernel
10.1.5	SMEM	65536.00	65536.00	65536.00	Instr per kernel
10.1.6	Branch	131072.00	131072.00	131072.00	Instr per kernel

10.2 VALU Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.2.0 kernel	INT32	393216.00	393216.00	393216.00	Instr per
10.2.1	INT64	65536.00	65536.00	65536.00	Instr per

kernel

10.2.2	F16-ADD	0.00	0.00	0.00	Instr per
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kernel

10.2.3	F16-MUL	0.00	0.00	0.00	Instr per
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kernel

10.2.4	F16-FMA	0.00	0.00	0.00	Instr per
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kernel

10.2.5	F16-Trans	0.00	0.00	0.00	Instr per
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kernel

10.2.6	F32-ADD	0.00	0.00	0.00	Instr per
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kernel

10.2.7	F32-MUL	0.00	0.00	0.00	Instr per
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kernel

10.2.8	F32-FMA	0.00	0.00	0.00	Instr per
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kernel

10.2.9	F32-Trans	0.00	0.00	0.00	Instr per
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kernel

10.2.10	F64-ADD	0.00	0.00	0.00	Instr per
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kernel

10.2.11	F64-MUL	0.00	0.00	0.00	Instr per
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kernel

10.2.12	F64-FMA	0.00	0.00	0.00	Instr per
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kernel

10.2.13	F64-Trans	0.00	0.00	0.00	Instr per
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kernel

10.2.14	Conversion	0.00	0.00	0.00	Instr per
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kernel

10.3 VMEM Instr Mix

Metric_ID	Metric	Avg	Min	Max
Unit				
10.3.0	Global/Generic Instr	262144.0	262144.0	262144.0
Instr per kernel				
10.3.1	Global/Generic Read	0.0	0.0	0.0
Instr per kernel				
10.3.2	Global/Generic Write	262144.0	262144.0	262144.0
Instr per kernel				
10.3.3	Global/Generic Atomic	0.0	0.0	0.0
Instr per kernel				
10.3.4	Spill/Stack Instr	0.0	0.0	0.0
Instr per kernel				
10.3.5	Spill/Stack Coalesceable Instr			
Instr per kernel				
10.3.6	Spill/Stack Read	0.0	0.0	0.0
Instr per kernel				
10.3.7	Spill/Stack Write	0.0	0.0	0.0
Instr per kernel				
10.3.8	Spill/Stack Atomic	0.0	0.0	0.0
Instr per kernel				

10.4 MFMA Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.4.0	MFMA-I8	0.0	0.0	0.0	Instr per kernel
10.4.1	MFMA-F8	0.0	0.0	0.0	Instr per kernel
10.4.2	MFMA-F16	0.0	0.0	0.0	Instr per kernel
10.4.3	MFMA-BF16	0.0	0.0	0.0	Instr per kernel
10.4.4	MFMA-F32	0.0	0.0	0.0	Instr per kernel
10.4.5	MFMA-F64	0.0	0.0	0.0	Instr per kernel
10.4.6	MFMA-F6F4				Instr per kernel

11. Compute Units - Compute Pipeline

11.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit	Peak	Pct of Peak
11.1.0	VALU FLOPs	0.0	Gflop	81715.2	0.0
11.1.1	VALU IOPs	677.2	Giop	81715.2	0.83
11.1.2	MFMA FLOPs (F8)	0.0	Gflop	2614886.4	0.0
11.1.3	MFMA FLOPs (BF16)	0.0	Gflop	1307443.2	0.0
11.1.4	MFMA FLOPs (F16)	0.0	Gflop	1307443.2	0.0
11.1.5	MFMA FLOPs (F32)	0.0	Gflop	163430.4	0.0
11.1.6	MFMA FLOPs (F64)	0.0	Gflop	163430.4	0.0
11.1.7	MFMA FLOPs (F6F4)		Gflop		
11.1.8	MFMA IOPs (INT8)	0.0	Giop	2614886.4	0.0

11.2 Pipeline Stats

Metric_ID	Metric	Avg	Min	Max	Unit
11.2.0	IPC	0.1	0.02	0.1	Instr/cycle
11.2.1	IPC (Issued)	1.0	1.0	1.0	Instr/cycle
11.2.2	SALU Utilization	3.09	0.95	3.21	Pct
11.2.3	VALU Utilization	2.27	0.7	2.35	Pct
11.2.4	VALU Co-Issue Efficiency				Pct
11.2.5	VMEM Utilization	0.82	0.25	0.85	Pct
11.2.6	Branch Utilization	0.62	0.19	0.64	Pct
11.2.7	VALU Active Threads	64.0	64.0	64.0	Threads
11.2.8	MFMA Utilization	0.0	0.0	0.0	Pct
11.2.9	MFMA Instr Cycles				Cycles/instr
11.2.10	VMEM Latency				Cycles

11.2.11	SMEM Latency	Cycles
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11.3 Arithmetic Operations

Metric_ID	Metric	Avg	Min	Max	Unit
11.3.0 kernel	FLOPs (Total)	0.0	0.0	0.0	Ops per
11.3.1 kernel	IOPs (Total)	29360128.0	29360128.0	29360128.0	Ops per
11.3.2 kernel	F8 OPs	0.0	0.0	0.0	Ops per
11.3.3 kernel	F16 OPs	0.0	0.0	0.0	Ops per
11.3.4 kernel	BF16 OPs	0.0	0.0	0.0	Ops per
11.3.5 kernel	F32 OPs	0.0	0.0	0.0	Ops per
11.3.6 kernel	F64 OPs	0.0	0.0	0.0	Ops per
11.3.7 kernel	F6F4 OPs				Ops per
11.3.8 kernel	INT8 OPs	0.0	0.0	0.0	Ops per

12. Local Data Share (LDS)

12.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
12.1.0	Utilization	0.0	Pct of peak
12.1.1	Access Rate	0.0	Pct of peak
12.1.2	Theoretical Bandwidth (% of Peak)	0.0	Pct of peak
12.1.3	Bank Conflict Rate		Pct of peak

12.2 LDS Stats

Metric_ID	Metric	Avg	Min	Max	Unit
12.2.0 kernel	LDS Instrs	0.0	0.0	0.0	Instr per
12.2.1 kernel	LDS LOAD				Instr per
12.2.2 kernel	LDS STORE				Instr per
12.2.3 kernel	LDS ATOMIC				Instr per
12.2.4	LDS LOAD Bandwidth				Gbps
12.2.5	LDS STORE Bandwidth				Gbps
12.2.6	LDS ATOMIC Bandwidth				Gbps

12.2.7 kernel	Theoretical Bandwidth	0.0	0.0	0.0	Bytes per
12.2.8	LDS Latency				Cycles
12.2.9 Conflicts/access	Bank Conflicts/Access				
12.2.10 kernel	Index Accesses	0.0	0.0	0.0	Cycles per
12.2.11 kernel	Atomic Return Cycles	0.0	0.0	0.0	Cycles per
12.2.12 kernel	Bank Conflict	0.0	0.0	0.0	Cycles per
12.2.13 kernel	Addr Conflict	0.0	0.0	0.0	Cycles per
12.2.14 kernel	Unaligned Stall	0.0	0.0	0.0	Cycles per
12.2.15 per kernel	Mem Violations	0.0	0.0	0.0	Accesses
12.2.16 kernel	LDS Command FIFO Full Rate				Cycles per
12.2.17 kernel	LDS Data FIFO Full Rate				Cycles per

13. Instruction Cache

13.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
13.1.0	Bandwidth	3.15	Pct of peak
13.1.1	Cache Hit Rate	99.33	Pct of peak
13.1.2	L1I-L2 Bandwidth	0.01	Pct of peak

13.2 Instruction Cache Accesses

Metric_ID	Metric	Avg	Min	Max
Unit				
13.2.0	Req	458752.0	458752.0	458752.0
Req per kernel				
13.2.1	Hits	455386.25	454183.0	455634.0
Hits per kernel				
13.2.2	Misses - Non Duplicated	160.0	160.0	160.0
Misses per kernel				
13.2.3	Misses - Duplicated	2924.5	2800.0	4142.0
Misses per kernel				
13.2.4	Cache Hit Rate	99.33	99.06	99.35
Pct				
13.2.5	Instruction Fetch Latency			
Cycles				

13.3 Instruction Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
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13.3.0	L1I-L2 Bandwidth	245760.00	245760.00	245760.00	Bytes per kernel
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14. Scalar L1 Data Cache

14.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
14.1.0	Bandwidth	0.45	Pct of peak
14.1.1	Cache Hit Rate	92.77	Pct of peak
14.1.2	sL1D-L2 BW	0.00	Pct of peak

14.2 Scalar L1D Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
14.2.0 per kernel	Req	65536.00	65536.00	65536.00	Req
14.2.1 per kernel	Hits	60755.68	60640.00	60886.00	Req
14.2.2 per kernel	Misses - Non Duplicated	160.00	160.00	160.00	Req
14.2.3 per kernel	Misses- Duplicated	4577.23	4292.00	4711.00	Req
14.2.4	Cache Hit Rate	92.77	92.58	93.18	Pct

14.2.5 per kernel	Read Req (Total)	65536.00	65536.00	65536.00	Req
14.2.6 per kernel	Atomic Req	0.00	0.00	0.00	Req
14.2.7 per kernel	Read Req (1 DWord)	0.00	0.00	0.00	Req
14.2.8 per kernel	Read Req (2 DWord)	0.00	0.00	0.00	Req
14.2.9 per kernel	Read Req (4 DWord)	65536.00	65536.00	65536.00	Req
14.2.10 per kernel	Read Req (8 DWord)	0.00	0.00	0.00	Req
14.2.11 per kernel	Read Req (16 DWord)	0.00	0.00	0.00	Req

14.3 Scalar L1D Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
14.3.0	sL1D-L2 BW	10240.00	10240.00	10240.00	Bytes per kernel
14.3.1	Read Req	160.00	160.00	160.00	Req per kernel
14.3.2	Write Req	0.00	0.00	0.00	Req per kernel
14.3.3	Atomic Req	0.00	0.00	0.00	Req per kernel

14.3.4	Stall Cycles	0.00	0.00	0.00	Cycles per kernel
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15. Address Processing Unit and Data Return Path (TA/TD)
 15.1 Address Processing Unit

Metric_ID Max	Metric Unit	Avg	Min
15.1.0 78.34	Address Processing Unit Busy Pct	73.95	31.06
15.1.1 0.0	Address Stall Pct	0.0	0.0
15.1.2 68.16	Data Stall Pct	60.73	21.94
15.1.3 0.0	Data-Processor → Address Stall Pct	0.0	0.0
15.1.4	Sequencer → TA Address Stall Cycles per kernel		
15.1.5	Sequencer → TA Command Stall Cycles per kernel		
15.1.6	Sequencer → TA Data Stall Cycles per kernel		
15.1.7 262144.0	Total Instructions Instructions per kernel	262144.0	262144.0

15.1.8	Global/Generic Instructions	262144.0	262144.0
262144.0	Instructions per kernel		
15.1.9	Global/Generic Read Instructions	0.0	0.0
0.0	Instructions per kernel		
15.1.10	Global/Generic Read Instructions for LDS		
	Instructions per kernel		
15.1.11	Global/Generic Write Instructions	262144.0	262144.0
262144.0	Instructions per kernel		
15.1.12	Global/Generic Atomic Instructions	0.0	0.0
0.0	Instructions per kernel		
15.1.13	Spill/Stack Instructions	0.0	0.0
0.0	Instructions per kernel		
15.1.14	Spill/Stack Read Instructions	0.0	0.0
0.0	Instructions per kernel		
15.1.15	Spill/Stack Read Instructions for LDS		
	Instructions per kernel		
15.1.16	Spill/Stack Write Instructions	0.0	0.0
0.0	Instructions per kernel		
15.1.17	Spill/Stack Atomic Instructions	0.0	0.0
0.0	Instructions per kernel		
15.1.18	Spill/Stack Total Cycles	0.0	0.0
0.0	Cycles per kernel		
15.1.19	Spill/Stack Coalesced Read	0.0	0.0
0.0	Cycles per kernel		

15.1.20	Spill/Stack Coalesced Write	0.0	0.0
0.0	Cycles per kernel		

15.2 Data-Return Path

Metric_ID Max	Metric Unit	Avg	Min
15.2.0 79.52	Data-Return Busy Pct	74.96	26.35
15.2.1 78.71	Cache RAM → Data-Return Stall Pct	74.12	31.21
15.2.2 0.0	Workgroup manager → Data-Return Stall Pct	0.0	0.0
15.2.3 0.0	Coalescable Instructions Instructions per kernel	0.0	0.0
15.2.4 0.0	Read Instructions Instructions per kernel	0.0	0.0
15.2.5 262144.0	Write Instructions Instructions per kernel	262144.0	262144.0
15.2.6 0.0	Atomic Instructions Instructions per kernel	0.0	0.0
15.2.7	Write Ack Instructions Instructions per kernel		

INFO Not showing table with empty column(s): 16.6 L1D Addr Translation Stalls

16. Vector L1 Data Cache

16.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
16.1.0	Hit rate	0.00	Pct of peak
16.1.1	Bandwidth	15.15	Pct of peak
16.1.2	Utilization	95.95	Pct of peak
16.1.3	Coalescing	25.00	Pct of peak

16.2 L1D Cache Stalls (%)

Metric_ID	Metric	Min	Q1	Median	Q3	Max
16.2.0 94.98	Stalled on L2 Data	91.77	92.7	93.05	93.34	
16.2.1 82.6	Stalled on L2 Req	75.84	76.45	76.74	77.04	
16.2.2	Stalled on Address					
16.2.3	Stalled on Data					
16.2.4	Stalled on Latency FIFO					
16.2.5	Stalled on Request FIFO					
16.2.6	Stalled on Read Return					

16.2.7	Tag RAM Stall (Read)	0.0	0.0	0.0	0.0	0.0
16.2.8	Tag RAM Stall (Write)	0.0	0.0	0.0	0.0	0.0
16.2.9	Tag RAM Stall (Atomic)	0.0	0.0	0.0	0.0	0.0

16.3 L1D Cache Accesses

Metric_ID Unit	Metric	Avg	Min	Max
16.3.0 Req per kernel	Total Req	16777216.0	16777216.0	16777216.0
16.3.1 Req per kernel	Read Req	0.0	0.0	0.0
16.3.2 Req per kernel	Write Req	16777216.0	16777216.0	16777216.0
16.3.3 Req per kernel	Atomic Req	0.0	0.0	0.0
16.3.4 Bytes per kernel	Cache BW	536870912.0	536870912.0	536870912.0
16.3.5 Pct	Cache Hit Rate	0.0	0.0	0.0
16.3.6 Req per kernel	Cache Accesses	4194304.0	4194304.0	4194304.0

16.3.7	Cache Hits	0.0	0.0	0.0
Req per kernel				
16.3.8	Invalidations	304.0	304.0	304.0
Req per kernel				
16.3.9	L1-L2 BW	268435456.0	268435456.0	268435456.0
Bytes per kernel				
16.3.10	Tag RAM 0 Req			
Req per kernel				
16.3.11	Tag RAM 1 Req			
Req per kernel				
16.3.12	Tag RAM 2 Req			
Req per kernel				
16.3.13	Tag RAM 3 Req			
Req per kernel				
16.3.14	L1-L2 Read	0.0	0.0	0.0
Req per kernel				
16.3.15	L1-L2 Write	4194304.0	4194304.0	4194304.0
Req per kernel				
16.3.16	L1-L2 Atomic	0.0	0.0	0.0
Req per kernel				
16.3.17	L1 Access Latency			
Cycles				
16.3.18	L1-L2 Read Latency			
Cycles				

16.3.19 L1-L2 Write Latency
Cycles

16.4 L1D - L2 Transactions

Metric_ID Max Unit	Metric	Xfer	Coherency	Avg	Min
16.4.0 0.00	NC - Read Req per kernel	Read	NC	0.00	0.00
16.4.1 0.00	UC - Read Req per kernel	Read	UC	0.00	0.00
16.4.2 0.00	CC - Read Req per kernel	Read	CC	0.00	0.00
16.4.3 0.00	RW - Read Req per kernel	Read	RW	0.00	0.00
16.4.4 4194304.00	RW - Write Req per kernel	Write	RW	4194304.00	4194304.00
16.4.5 0.00	NC - Write Req per kernel	Write	NC	0.00	0.00
16.4.6 0.00	UC - Write Req per kernel	Write	UC	0.00	0.00
16.4.7 0.00	CC - Write Req per kernel	Write	CC	0.00	0.00
16.4.8 0.00	NC - Atomic Req per kernel	Atomic	NC	0.00	0.00

16.4.9	UC - Atomic	Atomic	UC	0.00	0.00
0.00	Req per kernel				

16.4.10	CC - Atomic	Atomic	CC	0.00	0.00
0.00	Req per kernel				

16.4.11	RW - Atomic	Atomic	RW	0.00	0.00
0.00	Req per kernel				

16.5 L1D Addr Translation

Metric_ID	Metric	Avg	Min	Max
Units				
16.5.0	Req	4194304.0	4194304.0	
4194304.0	Req per kernel			
16.5.1	Inflight Req			
Req per kernel				
16.5.2	Hit Ratio	91.77	91.71	92.12
Pct				
16.5.3	Hits	3849215.8	3846656.0	
3863764.0	Req per kernel			
16.5.4	Translation Misses	16377.24	16363.0	16384.0
Req per kernel				
16.5.5	Misses under Translation Miss			
Req per kernel				
16.5.6	Permission Misses	0.0	0.0	0.0
Req per kernel				

INFO Not showing table with empty column(s): 17.4 L2 Cache Stalls

17. L2 Cache

17.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
17.1.0	Utilization	84.03	Pct
17.1.1	Bandwidth	36.03	Pct
17.1.2	Hit Rate	50.05	Pct
17.1.3	L2-Fabric Read BW	0.31	Gb/s
17.1.4	L2-Fabric Write and Atomic BW	6191.55	Gb/s
17.1.5	HBM Bandwidth	5324.80	Gb/s

17.2 L2 - Fabric Transactions

Metric_ID	Metric	Avg	Min
Max	Unit		
17.2.0	Read BW	13669.29	12800.0
30720.0	Bytes per kernel		
17.2.1	HBM Read Traffic	100.0	100.0
100.0	Pct		
17.2.2	Remote Read Traffic	0.0	0.0
0.0	Pct		
17.2.3	Uncached Read Traffic	4.95	0.0
235.0	Pct		
17.2.4	Write and Atomic BW	268435456.0	268435456.0
268435456.0	Bytes per kernel		
17.2.5	HBM Write and Atomic Traffic	100.0	100.0

100.0	Pct		
17.2.6	Remote Write and Atomic Traffic	0.0	0.0
0.0	Pct		
17.2.7	Atomic Traffic	0.0	0.0
0.0	Pct		
17.2.8	Uncached Write and Atomic Traffic	0.0	0.0
0.0	Pct		
17.2.9	Read Latency	317.43	270.71
613.19	Cycles		
17.2.10	Write and Atomic Latency	404.7	391.46
555.68	Cycles		
17.2.11	Atomic Latency		
	Cycles		
17.2.12	Read Stall		
	Pct		
17.2.13	Write Stall		
	Pct		

17.3 L2 Cache Accesses

Metric_ID	Metric	Avg	Min	Max
Unit				
17.3.0	Bandwidth	537384860.55	537382912.0	537416192.0
	Bytes per kernel			
17.3.1	Read Bandwidth			
	Bytes per kernel			

17.3.2 Write Bandwidth
Bytes per kernel

17.3.3 Atomic Bandwidth
Bytes per kernel

17.3.4 Req	4198319.22	4198304.0	4198564.0
Req per kernel			

17.3.5 Read Req	4004.69	4000.0	4188.0
Req per kernel			

17.3.6 Write Req	4194304.0	4194304.0	4194304.0
Req per kernel			

17.3.7 Atomic Req	0.0	0.0	0.0
Req per kernel			

17.3.8 Streaming Req	0.0	0.0	0.0
Req per kernel			

17.3.9 Bypassss Req			
Req per kernel			

17.3.10 Probe Req	0.0	0.0	0.0
Req per kernel			

17.3.11 Input Buffer Req			
Req per kernel			

17.3.12 Cache Hit	50.05	50.04	50.05
Pct			

17.3.13 Hits	2101048.0	2101048.0	2101048.0
Hits per kernel			

17.3.14	Misses	2097257.76	2097256.0	2097437.0
Misses per kernel				
17.3.15	Writeback	2097204.6	2097200.0	2097208.0
Cachelines per kernel				
17.3.16	Writeback (Internal)	1966080.02	1966080.0	1966083.0
Cachelines per kernel				
17.3.17	Writeback (vL1D Req)	131071.96	131069.0	131072.0
Cachelines per kernel				
17.3.18	Evict (Internal)	1835117.83	1835103.0	1835120.0
Cachelines per kernel				
17.3.19	Evict (vL1D Req)	0.0	0.0	0.0
Cachelines per kernel				
17.3.20	NC Req	0.0	0.0	0.0
Req per kernel				
17.3.21	UC Req	4.69	0.0	188.0
Req per kernel				
17.3.22	CC Req	0.0	0.0	0.0
Req per kernel				
17.3.23	RW Req	4198304.0	4198304.0	4198304.0
Req per kernel				

17.5 L2 - Fabric Interface Stalls

Metric_ID	Metric				Type
Transaction	Avg	Min	Max	Unit	

17.5.0	Read - PCIe Stall Pct	PCIe Stall	Read
17.5.1	Read - Infinity Fabric Stall Pct	Infinity Fabric Stall	Read
17.5.2	Read - HBM Stall Pct	HBM Stall	Read
17.5.3	Write - PCIe Stall Pct	PCIe Stall	Write
17.5.4	Write - Infinity Fabric Stall Pct	Infinity Fabric Stall	Write
17.5.5	Write - HBM Stall Pct	HBM Stall	Write
17.5.6	Write - Credit Starvation Pct	Credit Starvation	Write
0.0	0.0	0.0	

17.6 L2 - Fabric Detailed Transaction Breakdown

Metric_ID Max	Metric Unit	Avg	Min
17.6.0 0.0	Read (32B) Req per kernel	0.0	0.0
17.6.1 480.0	Read (64B) Req per kernel	213.58	200.0
17.6.2 0.0	Read (128B) Req per kernel	0.0	0.0

17.6.3	Read (Uncached)	15.05	0.0
1128.0	Req per kernel		
17.6.4	HBM Read	213.58	200.0
480.0	Req per kernel		
17.6.5	Remote Read	0.0	0.0
0.0	Req per kernel		
17.6.6	Read Bandwidth - PCIe		
	Bytes per kernel		
17.6.7	Read Bandwidth - Infinity Fabric		
	Bytes per kernel		
17.6.8	Read Bandwidth - HBM		
	Bytes per kernel		
17.6.9	Write and Atomic (32B)	0.0	0.0
0.0	Req per kernel		
17.6.10	Write and Atomic (Uncached)	0.0	0.0
0.0	Req per kernel		
17.6.11	Write and Atomic (64B)	4194304.0	4194304.0
4194304.0	Req per kernel		
17.6.12	HBM Write and Atomic	4194304.0	4194304.0
4194304.0	Req per kernel		
17.6.13	Remote Write and Atomic	0.0	0.0
0.0	Req per kernel		
17.6.14	Write Bandwidth - PCIe		
	Bytes per kernel		

17.6.15 Write Bandwidth - Infinity Fabric
Bytes per kernel

17.6.16 Write Bandwidth - HBM
Bytes per kernel

17.6.17 Atomic 0.0 0.0
0.0 Req per kernel

17.6.18 Atomic - HBM
Req per kernel

17.6.19 Atomic Bandwidth - PCIe
Bytes per kernel

17.6.20 Atomic Bandwidth - Infinity Fabric
Bytes per kernel

17.6.21 Atomic Bandwidth - HBM
Bytes per kernel

INFO Not showing table with empty column(s): 18.9 L2-Fabric Read
Stall (Cycles per normUnit)

INFO Not showing table with empty column(s): 18.10 L2-Fabric Write
and Atomic Stall (Cycles per normUnit)

18. L2 Cache (per Channel)

18.1 Aggregate Stats (All channels)

Metric_ID	Metric	Avg	Std Dev	Min	Max	Unit
18.1.0	L2 Cache Hit Rate	50.05	0.00	50.04	50.05	Pct

18.2 L2 Cache Hit Rate (pct)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.2.0	0.00	50.10	50.12	50.12	50.12	50.14

18.2.1	1.00	50.10	50.12	50.12	50.12	50.14
18.2.2	2.00	50.10	50.12	50.12	50.12	50.14
18.2.3	3.00	50.11	50.12	50.12	50.12	50.14
18.2.4	4.00	50.11	50.12	50.12	50.12	50.14
18.2.5	5.00	50.00	50.00	50.00	50.00	50.03
18.2.6	6.00	49.95	50.00	50.00	50.00	50.03
18.2.7	7.00	49.99	50.00	50.00	50.00	50.03
18.2.8	8.00	50.00	50.00	50.00	50.00	50.03
18.2.9	9.00	50.00	50.00	50.00	50.00	50.03
18.2.10	10.00	50.00	50.00	50.00	50.00	50.03
18.2.11	11.00	50.11	50.12	50.12	50.12	50.14
18.2.12	12.00	50.00	50.00	50.00	50.00	50.03
18.2.13	13.00	50.00	50.00	50.00	50.00	50.03
18.2.14	14.00	50.00	50.00	50.00	50.00	50.03
18.2.15	15.00	50.00	50.00	50.00	50.00	50.03
18.2.16	16.00	50.00	50.00	50.00	50.00	50.03
18.2.17	17.00	50.00	50.00	50.00	50.00	50.03
18.2.18	18.00	50.00	50.00	50.00	50.00	50.03
18.2.19	19.00	50.11	50.12	50.12	50.12	50.14
18.2.20	20.00	50.00	50.00	50.00	50.00	50.03
18.2.21	21.00	50.00	50.00	50.00	50.00	50.03
18.2.22	22.00	50.00	50.00	50.00	50.00	50.03
18.2.23	23.00	50.00	50.00	50.00	50.00	50.03
18.2.24	24.00	50.10	50.12	50.12	50.12	50.14

18.2.25	25.00	50.10	50.12	50.12	50.12	50.14
18.2.26	26.00	50.10	50.12	50.12	50.12	50.14
18.2.27	27.00	50.11	50.12	50.12	50.12	50.14
18.2.28	28.00	50.11	50.12	50.12	50.12	50.14
18.2.29	29.00	50.00	50.00	50.00	50.00	50.03
18.2.30	30.00	49.95	50.00	50.00	50.00	50.03
18.2.31	31.00	49.99	50.00	50.00	50.00	50.03
18.2.32	32.00	50.10	50.12	50.12	50.12	50.14
18.2.33	33.00	50.10	50.12	50.12	50.12	50.14
18.2.34	34.00	50.10	50.12	50.12	50.12	50.14
18.2.35	35.00	50.11	50.12	50.12	50.12	50.14
18.2.36	36.00	50.11	50.12	50.12	50.12	50.14
18.2.37	37.00	50.00	50.00	50.00	50.00	50.03
18.2.38	38.00	49.95	50.00	50.00	50.00	50.03
18.2.39	39.00	49.99	50.00	50.00	50.00	50.03
18.2.40	40.00	50.00	50.00	50.00	50.00	50.03
18.2.41	41.00	50.00	50.00	50.00	50.00	50.03
18.2.42	42.00	50.00	50.00	50.00	50.00	50.03
18.2.43	43.00	50.11	50.12	50.12	50.12	50.14
18.2.44	44.00	50.00	50.00	50.00	50.00	50.03
18.2.45	45.00	50.00	50.00	50.00	50.00	50.03
18.2.46	46.00	50.00	50.00	50.00	50.00	50.03
18.2.47	47.00	50.00	50.00	50.00	50.00	50.03
18.2.48	48.00	50.00	50.00	50.00	50.00	50.03

18.2.49	49.00	50.00	50.00	50.00	50.00	50.03
18.2.50	50.00	50.00	50.00	50.00	50.00	50.03
18.2.51	51.00	50.11	50.12	50.12	50.12	50.14
18.2.52	52.00	50.00	50.00	50.00	50.00	50.03
18.2.53	53.00	50.00	50.00	50.00	50.00	50.03
18.2.54	54.00	50.00	50.00	50.00	50.00	50.03
18.2.55	55.00	50.00	50.00	50.00	50.00	50.03
18.2.56	56.00	50.10	50.12	50.12	50.12	50.14
18.2.57	57.00	50.10	50.12	50.12	50.12	50.14
18.2.58	58.00	50.10	50.12	50.12	50.12	50.14
18.2.59	59.00	50.11	50.12	50.12	50.12	50.14
18.2.60	60.00	50.11	50.12	50.12	50.12	50.14
18.2.61	61.00	50.00	50.00	50.00	50.00	50.03
18.2.62	62.00	49.95	50.00	50.00	50.00	50.03
18.2.63	63.00	49.99	50.00	50.00	50.00	50.03
18.2.64	64.00	50.00	50.00	50.00	50.00	50.03
18.2.65	65.00	50.00	50.00	50.00	50.00	50.03
18.2.66	66.00	50.00	50.00	50.00	50.00	50.03
18.2.67	67.00	50.11	50.12	50.12	50.12	50.14
18.2.68	68.00	50.00	50.00	50.00	50.00	50.03
18.2.69	69.00	50.00	50.00	50.00	50.00	50.03
18.2.70	70.00	50.00	50.00	50.00	50.00	50.03
18.2.71	71.00	50.00	50.00	50.00	50.00	50.03
18.2.72	72.00	50.10	50.12	50.12	50.12	50.14

18.2.73	73.00	50.11	50.12	50.12	50.12	50.14
18.2.74	74.00	50.11	50.12	50.12	50.12	50.14
18.2.75	75.00	50.11	50.12	50.12	50.12	50.14
18.2.76	76.00	50.11	50.12	50.12	50.12	50.14
18.2.77	77.00	50.00	50.00	50.00	50.00	50.03
18.2.78	78.00	49.93	50.00	50.00	50.00	50.03
18.2.79	79.00	49.99	50.00	50.00	50.00	50.03
18.2.80	80.00	50.10	50.12	50.12	50.12	50.14
18.2.81	81.00	50.11	50.12	50.12	50.12	50.14
18.2.82	82.00	50.11	50.12	50.12	50.12	50.14
18.2.83	83.00	50.11	50.12	50.12	50.12	50.14
18.2.84	84.00	50.11	50.12	50.12	50.12	50.14
18.2.85	85.00	50.00	50.00	50.00	50.00	50.03
18.2.86	86.00	49.93	50.00	50.00	50.00	50.03
18.2.87	87.00	49.99	50.00	50.00	50.00	50.03
18.2.88	88.00	50.00	50.00	50.00	50.00	50.03
18.2.89	89.00	50.00	50.00	50.00	50.00	50.03
18.2.90	90.00	50.00	50.00	50.00	50.00	50.03
18.2.91	91.00	50.11	50.12	50.12	50.12	50.14
18.2.92	92.00	50.00	50.00	50.00	50.00	50.03
18.2.93	93.00	50.00	50.00	50.00	50.00	50.03
18.2.94	94.00	50.00	50.00	50.00	50.00	50.03
18.2.95	95.00	50.00	50.00	50.00	50.00	50.03
18.2.96	96.00	50.00	50.00	50.00	50.00	50.03

18.2.97	97.00	50.00	50.00	50.00	50.00	50.03
18.2.98	98.00	50.00	50.00	50.00	50.00	50.03
18.2.99	99.00	50.11	50.12	50.12	50.12	50.14
18.2.100	100.00	50.00	50.00	50.00	50.00	50.03
18.2.101	101.00	50.00	50.00	50.00	50.00	50.03
18.2.102	102.00	50.00	50.00	50.00	50.00	50.03
18.2.103	103.00	50.00	50.00	50.00	50.00	50.03
18.2.104	104.00	50.10	50.12	50.12	50.12	50.14
18.2.105	105.00	50.10	50.12	50.12	50.12	50.14
18.2.106	106.00	50.10	50.12	50.12	50.12	50.14
18.2.107	107.00	50.11	50.12	50.12	50.12	50.14
18.2.108	108.00	50.11	50.12	50.12	50.12	50.14
18.2.109	109.00	50.00	50.00	50.00	50.00	50.03
18.2.110	110.00	49.95	50.00	50.00	50.00	50.03
18.2.111	111.00	49.99	50.00	50.00	50.00	50.03
18.2.112	112.00	50.10	50.12	50.12	50.12	50.14
18.2.113	113.00	50.10	50.12	50.12	50.12	50.14
18.2.114	114.00	50.11	50.12	50.12	50.12	50.14
18.2.115	115.00	50.11	50.12	50.12	50.12	50.14
18.2.116	116.00	50.11	50.12	50.12	50.12	50.14
18.2.117	117.00	50.00	50.00	50.00	50.00	50.03
18.2.118	118.00	49.95	50.00	50.00	50.00	50.03
18.2.119	119.00	49.99	50.00	50.00	50.00	50.03
18.2.120	120.00	50.00	50.00	50.00	50.00	50.03

18.2.121	121.00	50.00	50.00	50.00	50.00	50.03
18.2.122	122.00	50.00	50.00	50.00	50.00	50.03
18.2.123	123.00	50.11	50.12	50.12	50.12	50.14
18.2.124	124.00	50.00	50.00	50.00	50.00	50.03
18.2.125	125.00	50.00	50.00	50.00	50.00	50.03
18.2.126	126.00	50.00	50.00	50.00	50.00	50.03
18.2.127	127.00	50.00	50.00	50.00	50.00	50.03

18.3 L2 Requests (per normUnit)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.3.0	0.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.1	1.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.2	2.00	32848.00	32848.00	32848.00	32849.00	32869.00
18.3.3	3.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.4	4.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.5	5.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.6	6.00	32848.00	32848.00	32848.00	32848.00	32868.00

18.3.7	7.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.8	8.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.9	9.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.10	10.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.11	11.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.12	12.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.13	13.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.14	14.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.15	15.00	32768.00	32768.00	32768.00	32769.00	32789.00
18.3.16	16.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.17	17.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.18	18.00	32768.00	32768.00	32768.00	32768.00	32788.00

18.3.19	19.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.20	20.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.21	21.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.22	22.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.23	23.00	32768.00	32768.00	32768.00	32769.00	32789.00
18.3.24	24.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.25	25.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.26	26.00	32848.00	32848.00	32848.00	32849.00	32869.00
18.3.27	27.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.28	28.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.29	29.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.30	30.00	32848.00	32848.00	32848.00	32848.00	32868.00

18.3.31	31.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.32	32.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.33	33.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.34	34.00	32848.00	32848.00	32848.00	32849.00	32869.00
18.3.35	35.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.36	36.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.37	37.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.38	38.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.39	39.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.40	40.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.41	41.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.42	42.00	32768.00	32768.00	32768.00	32768.00	32788.00

18.3.43	43.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.44	44.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.45	45.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.46	46.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.47	47.00	32768.00	32768.00	32768.00	32769.00	32789.00
18.3.48	48.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.49	49.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.50	50.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.51	51.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.52	52.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.53	53.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.54	54.00	32768.00	32768.00	32768.00	32768.00	32788.00

18.3.55	55.00	32768.00	32768.00	32768.00	32769.00	32789.00
18.3.56	56.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.57	57.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.58	58.00	32848.00	32848.00	32848.00	32849.00	32869.00
18.3.59	59.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.60	60.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.61	61.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.62	62.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.63	63.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.64	64.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.65	65.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.66	66.00	32768.00	32768.00	32768.00	32768.00	32788.00

18.3.67	67.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.68	68.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.69	69.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.70	70.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.71	71.00	32768.00	32768.00	32768.00	32769.00	32789.00
18.3.72	72.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.73	73.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.74	74.00	32848.00	32848.00	32848.00	32849.00	32869.00
18.3.75	75.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.76	76.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.77	77.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.78	78.00	32848.00	32848.00	32848.00	32848.00	32868.00

18.3.79	79.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.80	80.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.81	81.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.82	82.00	32848.00	32848.00	32848.00	32849.00	32869.00
18.3.83	83.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.84	84.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.85	85.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.86	86.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.87	87.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.88	88.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.89	89.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.90	90.00	32768.00	32768.00	32768.00	32768.00	32788.00

18.3.91	91.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.92	92.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.93	93.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.94	94.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.95	95.00	32768.00	32768.00	32768.00	32769.00	32789.00
18.3.96	96.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.97	97.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.98	98.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.99	99.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.100	100.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.101	101.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.102	102.00	32768.00	32768.00	32768.00	32768.00	32788.00

18.3.103	103.00	32768.00	32768.00	32768.00	32769.00	32789.00
18.3.104	104.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.105	105.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.106	106.00	32848.00	32848.00	32848.00	32849.00	32869.00
18.3.107	107.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.108	108.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.109	109.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.110	110.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.111	111.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.112	112.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.113	113.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.114	114.00	32848.00	32848.00	32848.00	32849.00	32869.00

18.3.115	115.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.116	116.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.117	117.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.118	118.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.119	119.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.120	120.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.121	121.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.122	122.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.123	123.00	32768.00	32768.00	32768.00	32768.00	32789.00
18.3.124	124.00	32768.00	32768.00	32768.00	32768.00	32788.00
18.3.125	125.00	32848.00	32848.00	32848.00	32848.00	32868.00
18.3.126	126.00	32768.00	32768.00	32768.00	32768.00	32788.00

18.3.127	127.00	32768.00	32768.00	32768.00	32769.00	32789.00
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18.4 L2 Requests (per normUnit)

Metric_ID	Channel	L2 Read	L2 Write	L2 Atomic
18.4.0	0.00	81.18	32768.00	0.00
18.4.1	1.00	81.38	32768.00	0.00
18.4.2	2.00	81.18	32768.00	0.00
18.4.3	3.00	81.22	32768.00	0.00
18.4.4	4.00	81.20	32768.00	0.00
18.4.5	5.00	1.47	32768.00	0.00
18.4.6	6.00	1.22	32768.00	0.00
18.4.7	7.00	1.24	32768.00	0.00
18.4.8	8.00	1.63	32768.00	0.00
18.4.9	9.00	1.63	32768.00	0.00
18.4.10	10.00	1.48	32768.00	0.00
18.4.11	11.00	81.65	32768.00	0.00
18.4.12	12.00	1.11	32768.00	0.00
18.4.13	13.00	1.11	32768.00	0.00
18.4.14	14.00	1.50	32768.00	0.00
18.4.15	15.00	1.53	32768.00	0.00
18.4.16	16.00	1.63	32768.00	0.00
18.4.17	17.00	1.63	32768.00	0.00
18.4.18	18.00	1.48	32768.00	0.00
18.4.19	19.00	81.65	32768.00	0.00

18.4.20	20.00	1.10	32768.00	0.00
18.4.21	21.00	1.12	32768.00	0.00
18.4.22	22.00	1.50	32768.00	0.00
18.4.23	23.00	1.53	32768.00	0.00
18.4.24	24.00	81.18	32768.00	0.00
18.4.25	25.00	81.38	32768.00	0.00
18.4.26	26.00	81.18	32768.00	0.00
18.4.27	27.00	81.22	32768.00	0.00
18.4.28	28.00	81.20	32768.00	0.00
18.4.29	29.00	1.47	32768.00	0.00
18.4.30	30.00	1.24	32768.00	0.00
18.4.31	31.00	1.25	32768.00	0.00
18.4.32	32.00	81.18	32768.00	0.00
18.4.33	33.00	81.38	32768.00	0.00
18.4.34	34.00	81.18	32768.00	0.00
18.4.35	35.00	81.22	32768.00	0.00
18.4.36	36.00	81.20	32768.00	0.00
18.4.37	37.00	1.47	32768.00	0.00
18.4.38	38.00	1.22	32768.00	0.00
18.4.39	39.00	1.26	32768.00	0.00
18.4.40	40.00	1.63	32768.00	0.00
18.4.41	41.00	1.63	32768.00	0.00
18.4.42	42.00	1.48	32768.00	0.00
18.4.43	43.00	81.65	32768.00	0.00

18.4.44	44.00	1.10	32768.00	0.00
18.4.45	45.00	1.11	32768.00	0.00
18.4.46	46.00	1.50	32768.00	0.00
18.4.47	47.00	1.53	32768.00	0.00
18.4.48	48.00	1.63	32768.00	0.00
18.4.49	49.00	1.63	32768.00	0.00
18.4.50	50.00	1.48	32768.00	0.00
18.4.51	51.00	81.65	32768.00	0.00
18.4.52	52.00	1.09	32768.00	0.00
18.4.53	53.00	1.12	32768.00	0.00
18.4.54	54.00	1.50	32768.00	0.00
18.4.55	55.00	1.53	32768.00	0.00
18.4.56	56.00	81.18	32768.00	0.00
18.4.57	57.00	81.38	32768.00	0.00
18.4.58	58.00	81.18	32768.00	0.00
18.4.59	59.00	81.22	32768.00	0.00
18.4.60	60.00	81.20	32768.00	0.00
18.4.61	61.00	1.47	32768.00	0.00
18.4.62	62.00	1.23	32768.00	0.00
18.4.63	63.00	1.26	32768.00	0.00
18.4.64	64.00	1.63	32768.00	0.00
18.4.65	65.00	1.63	32768.00	0.00
18.4.66	66.00	1.48	32768.00	0.00
18.4.67	67.00	81.65	32768.00	0.00

18.4.68	68.00	1.09	32768.00	0.00
18.4.69	69.00	1.12	32768.00	0.00
18.4.70	70.00	1.50	32768.00	0.00
18.4.71	71.00	1.53	32768.00	0.00
18.4.72	72.00	81.18	32768.00	0.00
18.4.73	73.00	81.38	32768.00	0.00
18.4.74	74.00	81.18	32768.00	0.00
18.4.75	75.00	81.22	32768.00	0.00
18.4.76	76.00	81.20	32768.00	0.00
18.4.77	77.00	1.47	32768.00	0.00
18.4.78	78.00	1.23	32768.00	0.00
18.4.79	79.00	1.25	32768.00	0.00
18.4.80	80.00	81.18	32768.00	0.00
18.4.81	81.00	81.38	32768.00	0.00
18.4.82	82.00	81.18	32768.00	0.00
18.4.83	83.00	81.22	32768.00	0.00
18.4.84	84.00	81.20	32768.00	0.00
18.4.85	85.00	1.47	32768.00	0.00
18.4.86	86.00	1.24	32768.00	0.00
18.4.87	87.00	1.25	32768.00	0.00
18.4.88	88.00	1.63	32768.00	0.00
18.4.89	89.00	1.63	32768.00	0.00
18.4.90	90.00	1.48	32768.00	0.00
18.4.91	91.00	81.65	32768.00	0.00

18.4.92	92.00	1.10	32768.00	0.00
18.4.93	93.00	1.11	32768.00	0.00
18.4.94	94.00	1.50	32768.00	0.00
18.4.95	95.00	1.53	32768.00	0.00
18.4.96	96.00	1.63	32768.00	0.00
18.4.97	97.00	1.63	32768.00	0.00
18.4.98	98.00	1.48	32768.00	0.00
18.4.99	99.00	81.65	32768.00	0.00
18.4.100	100.00	1.09	32768.00	0.00
18.4.101	101.00	1.12	32768.00	0.00
18.4.102	102.00	1.50	32768.00	0.00
18.4.103	103.00	1.53	32768.00	0.00
18.4.104	104.00	81.18	32768.00	0.00
18.4.105	105.00	81.38	32768.00	0.00
18.4.106	106.00	81.18	32768.00	0.00
18.4.107	107.00	81.22	32768.00	0.00
18.4.108	108.00	81.20	32768.00	0.00
18.4.109	109.00	1.47	32768.00	0.00
18.4.110	110.00	1.22	32768.00	0.00
18.4.111	111.00	1.25	32768.00	0.00
18.4.112	112.00	81.18	32768.00	0.00
18.4.113	113.00	81.38	32768.00	0.00
18.4.114	114.00	81.18	32768.00	0.00
18.4.115	115.00	81.22	32768.00	0.00

18.4.116	116.00	81.20	32768.00	0.00
18.4.117	117.00	1.47	32768.00	0.00
18.4.118	118.00	1.24	32768.00	0.00
18.4.119	119.00	1.27	32768.00	0.00
18.4.120	120.00	1.63	32768.00	0.00
18.4.121	121.00	1.63	32768.00	0.00
18.4.122	122.00	1.48	32768.00	0.00
18.4.123	123.00	81.65	32768.00	0.00
18.4.124	124.00	1.10	32768.00	0.00
18.4.125	125.00	1.12	32768.00	0.00
18.4.126	126.00	1.50	32768.00	0.00
18.4.127	127.00	1.53	32768.00	0.00

18.5 L2-Fabric Requests (per normUnit)

Metric_ID	Channel	L2-Fabric Read	L2-Fabric Write and Atomic
L2-Fabric Atomic			
18.5.0 0.00	0.00	0.14	32768.00
18.5.1 0.00	1.00	0.18	32768.00
18.5.2 0.00	2.00	4.27	32768.00
18.5.3 0.00	3.00	0.10	32768.00

18.5.4 0.00	4.00	4.50	32768.00
18.5.5 0.00	5.00	4.10	32768.00
18.5.6 0.00	6.00	4.12	32768.00
18.5.7 0.00	7.00	4.12	32768.00
18.5.8 0.00	8.00	0.11	32768.00
18.5.9 0.00	9.00	0.14	32768.00
18.5.10 0.00	10.00	0.12	32768.00
18.5.11 0.00	11.00	0.14	32768.00
18.5.12 0.00	12.00	0.10	32768.00
18.5.13 0.00	13.00	4.13	32768.00
18.5.14 0.00	14.00	0.08	32768.00
18.5.15 0.00	15.00	0.12	32768.00

18.5.16 0.00	16.00	0.11	32768.00
18.5.17 0.00	17.00	0.14	32768.00
18.5.18 0.00	18.00	0.12	32768.00
18.5.19 0.00	19.00	0.14	32768.00
18.5.20 0.00	20.00	0.10	32768.00
18.5.21 0.00	21.00	4.13	32768.00
18.5.22 0.00	22.00	0.08	32768.00
18.5.23 0.00	23.00	0.12	32768.00
18.5.24 0.00	24.00	0.15	32768.00
18.5.25 0.00	25.00	0.17	32768.00
18.5.26 0.00	26.00	4.27	32768.00
18.5.27 0.00	27.00	0.10	32768.00

18.5.28 0.00	28.00	4.50	32768.00
18.5.29 0.00	29.00	4.10	32768.00
18.5.30 0.00	30.00	4.12	32768.00
18.5.31 0.00	31.00	4.12	32768.00
18.5.32 0.00	32.00	0.14	32768.00
18.5.33 0.00	33.00	0.19	32768.00
18.5.34 0.00	34.00	4.27	32768.00
18.5.35 0.00	35.00	0.10	32768.00
18.5.36 0.00	36.00	4.50	32768.00
18.5.37 0.00	37.00	4.10	32768.00
18.5.38 0.00	38.00	4.12	32768.00
18.5.39 0.00	39.00	4.13	32768.00

18.5.40 0.00	40.00	0.11	32768.00
18.5.41 0.00	41.00	0.14	32768.00
18.5.42 0.00	42.00	0.12	32768.00
18.5.43 0.00	43.00	0.14	32768.00
18.5.44 0.00	44.00	0.10	32768.00
18.5.45 0.00	45.00	4.13	32768.00
18.5.46 0.00	46.00	0.08	32768.00
18.5.47 0.00	47.00	0.12	32768.00
18.5.48 0.00	48.00	0.11	32768.00
18.5.49 0.00	49.00	0.14	32768.00
18.5.50 0.00	50.00	0.12	32768.00
18.5.51 0.00	51.00	0.14	32768.00

18.5.52 0.00	52.00	0.10	32768.00
18.5.53 0.00	53.00	4.13	32768.00
18.5.54 0.00	54.00	0.08	32768.00
18.5.55 0.00	55.00	0.12	32768.00
18.5.56 0.00	56.00	0.15	32768.00
18.5.57 0.00	57.00	0.19	32768.00
18.5.58 0.00	58.00	4.27	32768.00
18.5.59 0.00	59.00	0.10	32768.00
18.5.60 0.00	60.00	4.50	32768.00
18.5.61 0.00	61.00	4.10	32768.00
18.5.62 0.00	62.00	4.12	32768.00
18.5.63 0.00	63.00	4.13	32768.00

18.5.64 0.00	64.00	0.11	32768.00
18.5.65 0.00	65.00	0.14	32768.00
18.5.66 0.00	66.00	0.12	32768.00
18.5.67 0.00	67.00	0.14	32768.00
18.5.68 0.00	68.00	0.10	32768.00
18.5.69 0.00	69.00	4.13	32768.00
18.5.70 0.00	70.00	0.08	32768.00
18.5.71 0.00	71.00	0.12	32768.00
18.5.72 0.00	72.00	0.15	32768.00
18.5.73 0.00	73.00	0.18	32768.00
18.5.74 0.00	74.00	4.27	32768.00
18.5.75 0.00	75.00	0.10	32768.00

18.5.76 0.00	76.00	4.50	32768.00
18.5.77 0.00	77.00	4.10	32768.00
18.5.78 0.00	78.00	4.12	32768.00
18.5.79 0.00	79.00	4.13	32768.00
18.5.80 0.00	80.00	0.15	32768.00
18.5.81 0.00	81.00	0.20	32768.00
18.5.82 0.00	82.00	4.27	32768.00
18.5.83 0.00	83.00	0.10	32768.00
18.5.84 0.00	84.00	4.50	32768.00
18.5.85 0.00	85.00	4.10	32768.00
18.5.86 0.00	86.00	4.14	32768.00
18.5.87 0.00	87.00	4.14	32768.00

18.5.88 0.00	88.00	0.11	32768.00
18.5.89 0.00	89.00	0.14	32768.00
18.5.90 0.00	90.00	0.12	32768.00
18.5.91 0.00	91.00	0.14	32768.00
18.5.92 0.00	92.00	0.10	32768.00
18.5.93 0.00	93.00	4.13	32768.00
18.5.94 0.00	94.00	0.08	32768.00
18.5.95 0.00	95.00	0.12	32768.00
18.5.96 0.00	96.00	0.11	32768.00
18.5.97 0.00	97.00	0.14	32768.00
18.5.98 0.00	98.00	0.12	32768.00
18.5.99 0.00	99.00	0.14	32768.00

18.5.100 0.00	100.00	0.10	32768.00
18.5.101 0.00	101.00	4.13	32768.00
18.5.102 0.00	102.00	0.08	32768.00
18.5.103 0.00	103.00	0.12	32768.00
18.5.104 0.00	104.00	0.14	32768.00
18.5.105 0.00	105.00	0.19	32768.00
18.5.106 0.00	106.00	4.27	32768.00
18.5.107 0.00	107.00	0.10	32768.00
18.5.108 0.00	108.00	4.50	32768.00
18.5.109 0.00	109.00	4.10	32768.00
18.5.110 0.00	110.00	4.12	32768.00
18.5.111 0.00	111.00	4.14	32768.00

18.5.112 0.00	112.00	0.15	32768.00
18.5.113 0.00	113.00	0.19	32768.00
18.5.114 0.00	114.00	4.27	32768.00
18.5.115 0.00	115.00	0.10	32768.00
18.5.116 0.00	116.00	4.50	32768.00
18.5.117 0.00	117.00	4.10	32768.00
18.5.118 0.00	118.00	4.14	32768.00
18.5.119 0.00	119.00	4.12	32768.00
18.5.120 0.00	120.00	0.11	32768.00
18.5.121 0.00	121.00	0.14	32768.00
18.5.122 0.00	122.00	0.12	32768.00
18.5.123 0.00	123.00	0.14	32768.00

18.5.124 0.00	124.00	0.10	32768.00
18.5.125 0.00	125.00	4.13	32768.00
18.5.126 0.00	126.00	0.08	32768.00
18.5.127 0.00	127.00	0.12	32768.00

18.6 L2-Fabric Read Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.6.0	0.00	465.75	1387.00	1416.00	1460.00	1818.00
18.6.1	1.00	401.60	1494.00	1499.00	1578.00	1976.00
18.6.2	2.00	251.17	377.50	379.00	399.07	658.50
18.6.3	3.00	1511.00	1516.00	1519.50	1700.25	2406.00
18.6.4	4.00	247.67	301.00	374.00	404.75	716.75
18.6.5	5.00	0.00	0.00	0.00	33.25	168.25
18.6.6	6.00	0.00	0.00	0.00	0.00	161.00
18.6.7	7.00	0.00	0.00	0.00	0.00	180.50
18.6.8	8.00	0.00	0.00	0.00	0.00	148.00
18.6.9	9.00	0.00	0.00	0.00	0.00	324.00
18.6.10	10.00	0.00	0.00	0.00	0.00	0.00
18.6.11	11.00	1453.00	1476.75	1531.00	1660.25	2326.00
18.6.12	12.00	0.00	0.00	0.00	0.00	0.00
18.6.13	13.00	0.00	0.00	0.00	0.00	360.75

18.6.14	14.00	0.00	0.00	0.00	0.00	0.00
18.6.15	15.00	0.00	0.00	0.00	0.00	358.00
18.6.16	16.00	0.00	0.00	0.00	0.00	150.00
18.6.17	17.00	0.00	0.00	0.00	0.00	492.00
18.6.18	18.00	0.00	0.00	0.00	0.00	0.00
18.6.19	19.00	1450.00	1477.75	1572.00	1785.50	2339.00
18.6.20	20.00	0.00	0.00	0.00	0.00	0.00
18.6.21	21.00	0.00	0.00	0.00	0.00	445.00
18.6.22	22.00	0.00	0.00	0.00	0.00	0.00
18.6.23	23.00	0.00	0.00	0.00	0.00	632.00
18.6.24	24.00	368.40	1366.00	1383.00	1438.00	1934.00
18.6.25	25.00	494.75	1483.50	1509.00	1648.50	2122.00
18.6.26	26.00	253.00	376.00	378.75	419.38	671.50
18.6.27	27.00	1504.00	1512.00	1519.50	1657.75	2883.00
18.6.28	28.00	249.17	321.70	375.00	436.50	638.00
18.6.29	29.00	0.00	0.00	0.00	48.38	245.75
18.6.30	30.00	0.00	0.00	0.00	0.00	256.00
18.6.31	31.00	0.00	0.00	0.00	0.00	181.25
18.6.32	32.00	384.50	1031.00	1078.00	1106.00	1569.00
18.6.33	33.00	370.50	1109.50	1364.00	1478.50	2169.00
18.6.34	34.00	217.80	293.80	355.50	392.38	608.00
18.6.35	35.00	1107.00	1442.25	1578.50	1641.75	2477.00
18.6.36	36.00	175.17	264.25	321.50	391.88	565.75
18.6.37	37.00	0.00	0.00	0.00	72.50	233.75

18.6.38	38.00	0.00	0.00	0.00	0.00	314.00
18.6.39	39.00	0.00	0.00	0.00	0.00	262.00
18.6.40	40.00	0.00	0.00	0.00	0.00	291.00
18.6.41	41.00	0.00	0.00	0.00	0.00	411.00
18.6.42	42.00	0.00	0.00	0.00	0.00	0.00
18.6.43	43.00	1058.00	1065.75	1118.00	1268.50	1961.00
18.6.44	44.00	0.00	0.00	0.00	0.00	0.00
18.6.45	45.00	0.00	0.00	0.00	0.00	616.50
18.6.46	46.00	0.00	0.00	0.00	0.00	0.00
18.6.47	47.00	0.00	0.00	0.00	0.00	706.00
18.6.48	48.00	0.00	0.00	0.00	0.00	290.00
18.6.49	49.00	0.00	0.00	0.00	0.00	385.00
18.6.50	50.00	0.00	0.00	0.00	0.00	0.00
18.6.51	51.00	1040.00	1068.25	1136.00	1252.75	1868.00
18.6.52	52.00	0.00	0.00	0.00	0.00	0.00
18.6.53	53.00	0.00	0.00	0.00	0.00	690.50
18.6.54	54.00	0.00	0.00	0.00	0.00	0.00
18.6.55	55.00	0.00	0.00	0.00	0.00	626.00
18.6.56	56.00	260.80	1010.00	1048.00	1142.00	1406.00
18.6.57	57.00	273.83	1148.50	1398.00	1531.50	2223.00
18.6.58	58.00	207.50	278.48	342.75	396.62	556.00
18.6.59	59.00	1095.00	1197.75	1495.00	1628.75	2539.00
18.6.60	60.00	175.50	263.00	296.20	368.07	568.75
18.6.61	61.00	0.00	0.00	0.00	72.25	243.25

18.6.62	62.00	0.00	0.00	0.00	0.00	338.67
18.6.63	63.00	0.00	0.00	0.00	0.00	291.40
18.6.64	64.00	0.00	0.00	0.00	0.00	465.00
18.6.65	65.00	0.00	0.00	0.00	0.00	283.00
18.6.66	66.00	0.00	0.00	0.00	0.00	0.00
18.6.67	67.00	509.00	526.75	586.00	678.50	1594.00
18.6.68	68.00	0.00	0.00	0.00	0.00	0.00
18.6.69	69.00	0.00	0.00	0.00	0.00	1183.75
18.6.70	70.00	0.00	0.00	0.00	0.00	0.00
18.6.71	71.00	0.00	0.00	0.00	0.00	894.00
18.6.72	72.00	138.00	503.00	552.00	626.00	1119.00
18.6.73	73.00	220.00	641.00	798.00	1032.50	1783.00
18.6.74	74.00	87.83	165.27	206.25	262.50	458.00
18.6.75	75.00	583.00	830.75	976.00	1079.75	2153.00
18.6.76	76.00	85.00	161.88	206.20	268.50	452.00
18.6.77	77.00	0.00	0.00	0.00	57.00	248.50
18.6.78	78.00	0.00	0.00	0.00	0.00	593.00
18.6.79	79.00	0.00	0.00	0.00	0.00	358.70
18.6.80	80.00	138.40	526.00	574.00	626.00	1116.00
18.6.81	81.00	132.62	736.50	962.00	1076.00	1545.00
18.6.82	82.00	87.83	189.62	231.00	285.25	489.75
18.6.83	83.00	664.00	789.25	974.50	1142.75	1833.00
18.6.84	84.00	115.20	176.05	220.80	285.12	464.75
18.6.85	85.00	0.00	0.00	0.00	55.00	250.00

18.6.86	86.00	0.00	0.00	0.00	0.00	483.27
18.6.87	87.00	0.00	0.00	0.00	0.00	342.36
18.6.88	88.00	0.00	0.00	0.00	0.00	477.00
18.6.89	89.00	0.00	0.00	0.00	0.00	305.00
18.6.90	90.00	0.00	0.00	0.00	0.00	0.00
18.6.91	91.00	498.00	532.50	591.50	788.50	1491.00
18.6.92	92.00	0.00	0.00	0.00	0.00	0.00
18.6.93	93.00	0.00	0.00	0.00	0.00	897.75
18.6.94	94.00	0.00	0.00	0.00	0.00	0.00
18.6.95	95.00	0.00	0.00	0.00	0.00	880.00
18.6.96	96.00	0.00	0.00	0.00	0.00	329.00
18.6.97	97.00	0.00	0.00	0.00	0.00	391.00
18.6.98	98.00	0.00	0.00	0.00	0.00	0.00
18.6.99	99.00	928.00	977.00	1049.00	1290.50	1801.00
18.6.100	100.00	0.00	0.00	0.00	0.00	0.00
18.6.101	101.00	0.00	0.00	0.00	0.00	847.25
18.6.102	102.00	0.00	0.00	0.00	0.00	0.00
18.6.103	103.00	0.00	0.00	0.00	0.00	849.00
18.6.104	104.00	342.75	936.00	1001.00	1054.00	1616.00
18.6.105	105.00	297.67	1165.00	1342.00	1516.00	1883.00
18.6.106	106.00	210.80	291.82	329.25	382.62	567.25
18.6.107	107.00	1083.00	1290.75	1530.00	1638.00	2614.00
18.6.108	108.00	203.80	295.95	325.00	397.62	608.50
18.6.109	109.00	0.00	0.00	0.00	60.50	256.00

18.6.110	110.00	0.00	0.00	0.00	0.00	230.33
18.6.111	111.00	0.00	0.00	0.00	0.00	210.00
18.6.112	112.00	245.40	924.00	959.00	1022.00	1574.00
18.6.113	113.00	220.17	983.00	1332.00	1481.50	1982.00
18.6.114	114.00	189.80	275.00	331.50	374.25	617.75
18.6.115	115.00	1093.00	1249.75	1338.00	1552.50	2273.00
18.6.116	116.00	193.00	289.18	334.75	403.18	526.00
18.6.117	117.00	0.00	0.00	0.00	57.08	235.75
18.6.118	118.00	0.00	0.00	0.00	0.00	280.75
18.6.119	119.00	0.00	0.00	0.00	0.00	368.56
18.6.120	120.00	0.00	0.00	0.00	0.00	329.00
18.6.121	121.00	0.00	0.00	0.00	0.00	454.00
18.6.122	122.00	0.00	0.00	0.00	0.00	0.00
18.6.123	123.00	895.00	947.25	1014.50	1213.75	1887.00
18.6.124	124.00	0.00	0.00	0.00	0.00	0.00
18.6.125	125.00	0.00	0.00	0.00	0.00	719.00
18.6.126	126.00	0.00	0.00	0.00	0.00	0.00
18.6.127	127.00	0.00	0.00	0.00	0.00	796.00

18.7 L2-Fabric Write and Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.7.0	0.00	331.50	345.64	351.10	357.71	501.64
18.7.1	1.00	324.41	334.75	339.43	344.80	477.93
18.7.2	2.00	342.63	353.57	358.87	363.55	508.43
18.7.3	3.00	340.07	350.62	356.13	361.84	503.25

18.7.4	4.00	357.40	376.93	383.71	390.41	546.35
18.7.5	5.00	346.35	365.38	372.35	378.99	529.99
18.7.6	6.00	372.82	387.84	395.48	401.16	556.02
18.7.7	7.00	360.55	372.81	378.81	384.31	535.95
18.7.8	8.00	430.67	458.08	466.37	471.91	666.69
18.7.9	9.00	462.02	466.88	468.58	471.48	662.86
18.7.10	10.00	418.02	449.95	460.46	469.09	669.36
18.7.11	11.00	404.26	438.01	453.20	467.58	660.86
18.7.12	12.00	392.72	410.22	421.15	430.12	623.53
18.7.13	13.00	381.37	406.94	414.94	421.69	607.71
18.7.14	14.00	400.11	427.68	440.54	454.76	666.91
18.7.15	15.00	407.08	450.36	463.16	472.69	667.07
18.7.16	16.00	425.38	451.32	459.30	466.08	659.45
18.7.17	17.00	457.45	462.05	463.93	466.61	657.31
18.7.18	18.00	410.45	441.79	455.55	463.70	661.01
18.7.19	19.00	402.73	432.76	445.66	461.12	656.36
18.7.20	20.00	381.50	408.21	416.20	426.25	612.60
18.7.21	21.00	371.96	400.51	410.05	417.27	605.26
18.7.22	22.00	392.60	423.23	436.80	449.59	653.20
18.7.23	23.00	400.26	444.07	457.95	466.78	660.80
18.7.24	24.00	323.72	341.76	346.95	352.97	499.14
18.7.25	25.00	318.62	329.86	335.13	340.38	465.40
18.7.26	26.00	333.08	350.25	353.89	359.70	503.23
18.7.27	27.00	332.36	346.46	351.12	357.09	497.77

18.7.28	28.00	350.35	372.17	378.53	385.95	546.39
18.7.29	29.00	342.41	361.41	367.23	373.54	527.79
18.7.30	30.00	368.27	384.96	389.61	396.56	550.72
18.7.31	31.00	354.80	368.98	372.81	379.54	532.12
18.7.32	32.00	406.70	446.18	457.37	468.47	655.39
18.7.33	33.00	459.68	466.96	469.15	471.01	654.63
18.7.34	34.00	415.37	444.01	458.54	468.31	649.05
18.7.35	35.00	401.60	438.63	453.29	466.67	646.92
18.7.36	36.00	380.41	407.31	416.09	427.32	608.11
18.7.37	37.00	378.03	400.64	408.52	420.58	585.51
18.7.38	38.00	394.61	416.51	426.27	437.05	602.62
18.7.39	39.00	401.45	432.64	444.91	460.47	627.57
18.7.40	40.00	324.47	339.41	346.67	351.05	491.18
18.7.41	41.00	318.41	330.46	334.52	339.33	472.52
18.7.42	42.00	334.88	347.05	352.50	356.11	497.73
18.7.43	43.00	331.68	343.92	350.12	354.01	493.54
18.7.44	44.00	354.86	369.92	375.50	381.74	529.50
18.7.45	45.00	348.79	360.75	365.29	370.92	514.17
18.7.46	46.00	372.92	387.10	391.22	395.61	557.83
18.7.47	47.00	351.82	363.02	367.58	371.42	521.69
18.7.48	48.00	325.64	337.19	342.52	348.21	486.57
18.7.49	49.00	318.49	326.70	331.15	336.70	466.77
18.7.50	50.00	334.32	343.27	347.95	353.53	491.93
18.7.51	51.00	333.00	341.73	346.44	352.26	489.05

18.7.52	52.00	353.56	366.18	372.21	378.86	519.78
18.7.53	53.00	345.26	356.80	362.78	368.55	502.99
18.7.54	54.00	369.89	382.45	387.85	393.32	548.49
18.7.55	55.00	350.34	360.00	364.24	369.03	512.43
18.7.56	56.00	407.63	440.89	454.18	462.21	650.32
18.7.57	57.00	453.29	462.17	464.12	466.38	650.24
18.7.58	58.00	410.32	438.18	451.99	462.10	643.33
18.7.59	59.00	394.44	433.34	446.71	460.11	647.15
18.7.60	60.00	376.22	402.58	411.51	424.72	601.49
18.7.61	61.00	367.33	393.29	404.60	416.20	580.68
18.7.62	62.00	388.24	410.46	423.67	435.07	603.96
18.7.63	63.00	392.60	429.61	442.11	457.39	622.85
18.7.64	64.00	320.33	333.52	338.29	343.12	472.49
18.7.65	65.00	315.16	324.24	327.53	330.81	451.53
18.7.66	66.00	329.24	340.45	344.60	348.24	478.99
18.7.67	67.00	327.21	339.58	343.34	346.70	475.53
18.7.68	68.00	349.09	366.20	370.74	378.57	537.27
18.7.69	69.00	342.15	356.65	360.93	366.06	512.31
18.7.70	70.00	367.86	381.08	385.72	392.99	552.88
18.7.71	71.00	347.29	357.93	361.85	366.67	518.86
18.7.72	72.00	408.59	449.20	459.67	465.58	658.77
18.7.73	73.00	456.98	462.54	464.56	467.21	656.20
18.7.74	74.00	395.02	437.83	454.63	462.59	661.28
18.7.75	75.00	392.49	427.18	446.07	458.02	662.65

18.7.76	76.00	383.80	407.52	418.51	427.60	621.84
18.7.77	77.00	382.00	400.16	408.04	416.42	598.06
18.7.78	78.00	380.10	411.34	420.42	429.92	600.31
18.7.79	79.00	393.02	431.57	445.08	454.27	649.42
18.7.80	80.00	419.17	456.28	465.08	468.63	661.16
18.7.81	81.00	459.62	464.32	466.50	468.44	655.97
18.7.82	82.00	398.60	444.25	458.10	465.50	661.75
18.7.83	83.00	393.95	431.32	449.92	462.89	664.60
18.7.84	84.00	385.90	410.42	421.51	430.21	609.75
18.7.85	85.00	383.08	403.42	411.27	420.12	591.09
18.7.86	86.00	382.05	414.44	423.71	433.10	604.81
18.7.87	87.00	395.74	435.88	446.37	457.32	645.38
18.7.88	88.00	323.03	336.65	342.26	347.67	478.16
18.7.89	89.00	314.55	326.45	330.99	335.42	456.79
18.7.90	90.00	328.86	343.14	347.63	352.15	485.04
18.7.91	91.00	328.14	341.31	346.31	351.38	481.38
18.7.92	92.00	347.64	367.83	373.91	380.94	549.43
18.7.93	93.00	342.94	358.91	363.01	369.65	521.91
18.7.94	94.00	367.83	382.80	389.05	395.10	558.95
18.7.95	95.00	346.49	360.03	364.44	368.97	520.27
18.7.96	96.00	403.29	442.18	450.92	460.09	664.52
18.7.97	97.00	453.36	460.64	462.35	464.62	659.48
18.7.98	98.00	390.80	425.21	440.34	456.48	656.69
18.7.99	99.00	385.11	413.45	432.18	444.10	622.07

18.7.100	100.00	371.73	399.72	407.29	415.31	608.69
18.7.101	101.00	368.06	393.81	401.84	409.59	582.70
18.7.102	102.00	383.99	416.78	428.35	436.21	619.85
18.7.103	103.00	402.62	438.11	449.46	459.27	656.81
18.7.104	104.00	319.14	333.06	338.45	343.74	471.82
18.7.105	105.00	313.95	324.71	328.07	332.63	452.03
18.7.106	106.00	329.94	342.31	346.31	351.26	480.66
18.7.107	107.00	326.68	339.90	343.05	347.22	474.36
18.7.108	108.00	348.85	362.69	367.41	373.12	514.08
18.7.109	109.00	341.88	353.32	356.93	362.22	495.39
18.7.110	110.00	366.88	378.69	383.33	387.40	527.11
18.7.111	111.00	346.12	358.34	362.24	366.44	500.63
18.7.112	112.00	321.36	335.16	340.11	345.75	488.53
18.7.113	113.00	316.79	326.87	330.14	334.10	468.07
18.7.114	114.00	332.60	343.72	348.57	352.84	498.91
18.7.115	115.00	330.93	341.84	345.79	350.65	490.76
18.7.116	116.00	349.85	364.58	370.18	375.81	517.20
18.7.117	117.00	341.22	355.00	359.82	364.25	505.75
18.7.118	118.00	365.70	381.43	385.91	391.11	541.14
18.7.119	119.00	349.10	360.95	365.16	369.69	506.50
18.7.120	120.00	408.70	445.76	455.13	461.73	655.39
18.7.121	121.00	454.31	462.28	463.59	465.69	657.10
18.7.122	122.00	397.39	430.44	446.29	458.73	653.68
18.7.123	123.00	395.05	420.60	436.68	449.32	618.74

18.7.124	124.00	373.55	402.22	410.46	419.06	616.11
18.7.125	125.00	370.23	396.62	404.69	413.33	596.10
18.7.126	126.00	387.92	418.93	430.40	439.31	635.21
18.7.127	127.00	402.73	441.32	450.74	461.27	660.50

18.8 L2-Fabric Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.8.0	0.00	0.00	0.00	0.00	0.00	0.00
18.8.1	1.00	0.00	0.00	0.00	0.00	0.00
18.8.2	2.00	0.00	0.00	0.00	0.00	0.00
18.8.3	3.00	0.00	0.00	0.00	0.00	0.00
18.8.4	4.00	0.00	0.00	0.00	0.00	0.00
18.8.5	5.00	0.00	0.00	0.00	0.00	0.00
18.8.6	6.00	0.00	0.00	0.00	0.00	0.00
18.8.7	7.00	0.00	0.00	0.00	0.00	0.00
18.8.8	8.00	0.00	0.00	0.00	0.00	0.00
18.8.9	9.00	0.00	0.00	0.00	0.00	0.00
18.8.10	10.00	0.00	0.00	0.00	0.00	0.00
18.8.11	11.00	0.00	0.00	0.00	0.00	0.00
18.8.12	12.00	0.00	0.00	0.00	0.00	0.00
18.8.13	13.00	0.00	0.00	0.00	0.00	0.00
18.8.14	14.00	0.00	0.00	0.00	0.00	0.00
18.8.15	15.00	0.00	0.00	0.00	0.00	0.00
18.8.16	16.00	0.00	0.00	0.00	0.00	0.00
18.8.17	17.00	0.00	0.00	0.00	0.00	0.00

18.8.18	18.00	0.00	0.00	0.00	0.00	0.00
18.8.19	19.00	0.00	0.00	0.00	0.00	0.00
18.8.20	20.00	0.00	0.00	0.00	0.00	0.00
18.8.21	21.00	0.00	0.00	0.00	0.00	0.00
18.8.22	22.00	0.00	0.00	0.00	0.00	0.00
18.8.23	23.00	0.00	0.00	0.00	0.00	0.00
18.8.24	24.00	0.00	0.00	0.00	0.00	0.00
18.8.25	25.00	0.00	0.00	0.00	0.00	0.00
18.8.26	26.00	0.00	0.00	0.00	0.00	0.00
18.8.27	27.00	0.00	0.00	0.00	0.00	0.00
18.8.28	28.00	0.00	0.00	0.00	0.00	0.00
18.8.29	29.00	0.00	0.00	0.00	0.00	0.00
18.8.30	30.00	0.00	0.00	0.00	0.00	0.00
18.8.31	31.00	0.00	0.00	0.00	0.00	0.00
18.8.32	32.00	0.00	0.00	0.00	0.00	0.00
18.8.33	33.00	0.00	0.00	0.00	0.00	0.00
18.8.34	34.00	0.00	0.00	0.00	0.00	0.00
18.8.35	35.00	0.00	0.00	0.00	0.00	0.00
18.8.36	36.00	0.00	0.00	0.00	0.00	0.00
18.8.37	37.00	0.00	0.00	0.00	0.00	0.00
18.8.38	38.00	0.00	0.00	0.00	0.00	0.00
18.8.39	39.00	0.00	0.00	0.00	0.00	0.00
18.8.40	40.00	0.00	0.00	0.00	0.00	0.00
18.8.41	41.00	0.00	0.00	0.00	0.00	0.00

18.8.42	42.00	0.00	0.00	0.00	0.00	0.00
18.8.43	43.00	0.00	0.00	0.00	0.00	0.00
18.8.44	44.00	0.00	0.00	0.00	0.00	0.00
18.8.45	45.00	0.00	0.00	0.00	0.00	0.00
18.8.46	46.00	0.00	0.00	0.00	0.00	0.00
18.8.47	47.00	0.00	0.00	0.00	0.00	0.00
18.8.48	48.00	0.00	0.00	0.00	0.00	0.00
18.8.49	49.00	0.00	0.00	0.00	0.00	0.00
18.8.50	50.00	0.00	0.00	0.00	0.00	0.00
18.8.51	51.00	0.00	0.00	0.00	0.00	0.00
18.8.52	52.00	0.00	0.00	0.00	0.00	0.00
18.8.53	53.00	0.00	0.00	0.00	0.00	0.00
18.8.54	54.00	0.00	0.00	0.00	0.00	0.00
18.8.55	55.00	0.00	0.00	0.00	0.00	0.00
18.8.56	56.00	0.00	0.00	0.00	0.00	0.00
18.8.57	57.00	0.00	0.00	0.00	0.00	0.00
18.8.58	58.00	0.00	0.00	0.00	0.00	0.00
18.8.59	59.00	0.00	0.00	0.00	0.00	0.00
18.8.60	60.00	0.00	0.00	0.00	0.00	0.00
18.8.61	61.00	0.00	0.00	0.00	0.00	0.00
18.8.62	62.00	0.00	0.00	0.00	0.00	0.00
18.8.63	63.00	0.00	0.00	0.00	0.00	0.00
18.8.64	64.00	0.00	0.00	0.00	0.00	0.00
18.8.65	65.00	0.00	0.00	0.00	0.00	0.00

18.8.66	66.00	0.00	0.00	0.00	0.00	0.00
18.8.67	67.00	0.00	0.00	0.00	0.00	0.00
18.8.68	68.00	0.00	0.00	0.00	0.00	0.00
18.8.69	69.00	0.00	0.00	0.00	0.00	0.00
18.8.70	70.00	0.00	0.00	0.00	0.00	0.00
18.8.71	71.00	0.00	0.00	0.00	0.00	0.00
18.8.72	72.00	0.00	0.00	0.00	0.00	0.00
18.8.73	73.00	0.00	0.00	0.00	0.00	0.00
18.8.74	74.00	0.00	0.00	0.00	0.00	0.00
18.8.75	75.00	0.00	0.00	0.00	0.00	0.00
18.8.76	76.00	0.00	0.00	0.00	0.00	0.00
18.8.77	77.00	0.00	0.00	0.00	0.00	0.00
18.8.78	78.00	0.00	0.00	0.00	0.00	0.00
18.8.79	79.00	0.00	0.00	0.00	0.00	0.00
18.8.80	80.00	0.00	0.00	0.00	0.00	0.00
18.8.81	81.00	0.00	0.00	0.00	0.00	0.00
18.8.82	82.00	0.00	0.00	0.00	0.00	0.00
18.8.83	83.00	0.00	0.00	0.00	0.00	0.00
18.8.84	84.00	0.00	0.00	0.00	0.00	0.00
18.8.85	85.00	0.00	0.00	0.00	0.00	0.00
18.8.86	86.00	0.00	0.00	0.00	0.00	0.00
18.8.87	87.00	0.00	0.00	0.00	0.00	0.00
18.8.88	88.00	0.00	0.00	0.00	0.00	0.00
18.8.89	89.00	0.00	0.00	0.00	0.00	0.00

18.8.90	90.00	0.00	0.00	0.00	0.00	0.00
18.8.91	91.00	0.00	0.00	0.00	0.00	0.00
18.8.92	92.00	0.00	0.00	0.00	0.00	0.00
18.8.93	93.00	0.00	0.00	0.00	0.00	0.00
18.8.94	94.00	0.00	0.00	0.00	0.00	0.00
18.8.95	95.00	0.00	0.00	0.00	0.00	0.00
18.8.96	96.00	0.00	0.00	0.00	0.00	0.00
18.8.97	97.00	0.00	0.00	0.00	0.00	0.00
18.8.98	98.00	0.00	0.00	0.00	0.00	0.00
18.8.99	99.00	0.00	0.00	0.00	0.00	0.00
18.8.100	100.00	0.00	0.00	0.00	0.00	0.00
18.8.101	101.00	0.00	0.00	0.00	0.00	0.00
18.8.102	102.00	0.00	0.00	0.00	0.00	0.00
18.8.103	103.00	0.00	0.00	0.00	0.00	0.00
18.8.104	104.00	0.00	0.00	0.00	0.00	0.00
18.8.105	105.00	0.00	0.00	0.00	0.00	0.00
18.8.106	106.00	0.00	0.00	0.00	0.00	0.00
18.8.107	107.00	0.00	0.00	0.00	0.00	0.00
18.8.108	108.00	0.00	0.00	0.00	0.00	0.00
18.8.109	109.00	0.00	0.00	0.00	0.00	0.00
18.8.110	110.00	0.00	0.00	0.00	0.00	0.00
18.8.111	111.00	0.00	0.00	0.00	0.00	0.00
18.8.112	112.00	0.00	0.00	0.00	0.00	0.00
18.8.113	113.00	0.00	0.00	0.00	0.00	0.00

18.8.114	114.00	0.00	0.00	0.00	0.00	0.00
18.8.115	115.00	0.00	0.00	0.00	0.00	0.00
18.8.116	116.00	0.00	0.00	0.00	0.00	0.00
18.8.117	117.00	0.00	0.00	0.00	0.00	0.00
18.8.118	118.00	0.00	0.00	0.00	0.00	0.00
18.8.119	119.00	0.00	0.00	0.00	0.00	0.00
18.8.120	120.00	0.00	0.00	0.00	0.00	0.00
18.8.121	121.00	0.00	0.00	0.00	0.00	0.00
18.8.122	122.00	0.00	0.00	0.00	0.00	0.00
18.8.123	123.00	0.00	0.00	0.00	0.00	0.00
18.8.124	124.00	0.00	0.00	0.00	0.00	0.00
18.8.125	125.00	0.00	0.00	0.00	0.00	0.00
18.8.126	126.00	0.00	0.00	0.00	0.00	0.00
18.8.127	127.00	0.00	0.00	0.00	0.00	0.00

18.12 L2-Fabric (128B read requests per normUnit)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.12.0	0.00	0.00	0.00	0.00	0.00	0.00
18.12.1	1.00	0.00	0.00	0.00	0.00	0.00
18.12.2	2.00	0.00	0.00	0.00	0.00	0.00
18.12.3	3.00	0.00	0.00	0.00	0.00	0.00
18.12.4	4.00	0.00	0.00	0.00	0.00	0.00
18.12.5	5.00	0.00	0.00	0.00	0.00	0.00
18.12.6	6.00	0.00	0.00	0.00	0.00	0.00
18.12.7	7.00	0.00	0.00	0.00	0.00	0.00

18.12.8	8.00	0.00	0.00	0.00	0.00	0.00
18.12.9	9.00	0.00	0.00	0.00	0.00	0.00
18.12.10	10.00	0.00	0.00	0.00	0.00	0.00
18.12.11	11.00	0.00	0.00	0.00	0.00	0.00
18.12.12	12.00	0.00	0.00	0.00	0.00	0.00
18.12.13	13.00	0.00	0.00	0.00	0.00	0.00
18.12.14	14.00	0.00	0.00	0.00	0.00	0.00
18.12.15	15.00	0.00	0.00	0.00	0.00	0.00
18.12.16	16.00	0.00	0.00	0.00	0.00	0.00
18.12.17	17.00	0.00	0.00	0.00	0.00	0.00
18.12.18	18.00	0.00	0.00	0.00	0.00	0.00
18.12.19	19.00	0.00	0.00	0.00	0.00	0.00
18.12.20	20.00	0.00	0.00	0.00	0.00	0.00
18.12.21	21.00	0.00	0.00	0.00	0.00	0.00
18.12.22	22.00	0.00	0.00	0.00	0.00	0.00
18.12.23	23.00	0.00	0.00	0.00	0.00	0.00
18.12.24	24.00	0.00	0.00	0.00	0.00	0.00
18.12.25	25.00	0.00	0.00	0.00	0.00	0.00
18.12.26	26.00	0.00	0.00	0.00	0.00	0.00
18.12.27	27.00	0.00	0.00	0.00	0.00	0.00
18.12.28	28.00	0.00	0.00	0.00	0.00	0.00
18.12.29	29.00	0.00	0.00	0.00	0.00	0.00
18.12.30	30.00	0.00	0.00	0.00	0.00	0.00
18.12.31	31.00	0.00	0.00	0.00	0.00	0.00

18.12.32	32.00	0.00	0.00	0.00	0.00	0.00
18.12.33	33.00	0.00	0.00	0.00	0.00	0.00
18.12.34	34.00	0.00	0.00	0.00	0.00	0.00
18.12.35	35.00	0.00	0.00	0.00	0.00	0.00
18.12.36	36.00	0.00	0.00	0.00	0.00	0.00
18.12.37	37.00	0.00	0.00	0.00	0.00	0.00
18.12.38	38.00	0.00	0.00	0.00	0.00	0.00
18.12.39	39.00	0.00	0.00	0.00	0.00	0.00
18.12.40	40.00	0.00	0.00	0.00	0.00	0.00
18.12.41	41.00	0.00	0.00	0.00	0.00	0.00
18.12.42	42.00	0.00	0.00	0.00	0.00	0.00
18.12.43	43.00	0.00	0.00	0.00	0.00	0.00
18.12.44	44.00	0.00	0.00	0.00	0.00	0.00
18.12.45	45.00	0.00	0.00	0.00	0.00	0.00
18.12.46	46.00	0.00	0.00	0.00	0.00	0.00
18.12.47	47.00	0.00	0.00	0.00	0.00	0.00
18.12.48	48.00	0.00	0.00	0.00	0.00	0.00
18.12.49	49.00	0.00	0.00	0.00	0.00	0.00
18.12.50	50.00	0.00	0.00	0.00	0.00	0.00
18.12.51	51.00	0.00	0.00	0.00	0.00	0.00
18.12.52	52.00	0.00	0.00	0.00	0.00	0.00
18.12.53	53.00	0.00	0.00	0.00	0.00	0.00
18.12.54	54.00	0.00	0.00	0.00	0.00	0.00
18.12.55	55.00	0.00	0.00	0.00	0.00	0.00

18.12.56	56.00	0.00	0.00	0.00	0.00	0.00
18.12.57	57.00	0.00	0.00	0.00	0.00	0.00
18.12.58	58.00	0.00	0.00	0.00	0.00	0.00
18.12.59	59.00	0.00	0.00	0.00	0.00	0.00
18.12.60	60.00	0.00	0.00	0.00	0.00	0.00
18.12.61	61.00	0.00	0.00	0.00	0.00	0.00
18.12.62	62.00	0.00	0.00	0.00	0.00	0.00
18.12.63	63.00	0.00	0.00	0.00	0.00	0.00
18.12.64	64.00	0.00	0.00	0.00	0.00	0.00
18.12.65	65.00	0.00	0.00	0.00	0.00	0.00
18.12.66	66.00	0.00	0.00	0.00	0.00	0.00
18.12.67	67.00	0.00	0.00	0.00	0.00	0.00
18.12.68	68.00	0.00	0.00	0.00	0.00	0.00
18.12.69	69.00	0.00	0.00	0.00	0.00	0.00
18.12.70	70.00	0.00	0.00	0.00	0.00	0.00
18.12.71	71.00	0.00	0.00	0.00	0.00	0.00
18.12.72	72.00	0.00	0.00	0.00	0.00	0.00
18.12.73	73.00	0.00	0.00	0.00	0.00	0.00
18.12.74	74.00	0.00	0.00	0.00	0.00	0.00
18.12.75	75.00	0.00	0.00	0.00	0.00	0.00
18.12.76	76.00	0.00	0.00	0.00	0.00	0.00
18.12.77	77.00	0.00	0.00	0.00	0.00	0.00
18.12.78	78.00	0.00	0.00	0.00	0.00	0.00
18.12.79	79.00	0.00	0.00	0.00	0.00	0.00

18.12.80	80.00	0.00	0.00	0.00	0.00	0.00
18.12.81	81.00	0.00	0.00	0.00	0.00	0.00
18.12.82	82.00	0.00	0.00	0.00	0.00	0.00
18.12.83	83.00	0.00	0.00	0.00	0.00	0.00
18.12.84	84.00	0.00	0.00	0.00	0.00	0.00
18.12.85	85.00	0.00	0.00	0.00	0.00	0.00
18.12.86	86.00	0.00	0.00	0.00	0.00	0.00
18.12.87	87.00	0.00	0.00	0.00	0.00	0.00
18.12.88	88.00	0.00	0.00	0.00	0.00	0.00
18.12.89	89.00	0.00	0.00	0.00	0.00	0.00
18.12.90	90.00	0.00	0.00	0.00	0.00	0.00
18.12.91	91.00	0.00	0.00	0.00	0.00	0.00
18.12.92	92.00	0.00	0.00	0.00	0.00	0.00
18.12.93	93.00	0.00	0.00	0.00	0.00	0.00
18.12.94	94.00	0.00	0.00	0.00	0.00	0.00
18.12.95	95.00	0.00	0.00	0.00	0.00	0.00
18.12.96	96.00	0.00	0.00	0.00	0.00	0.00
18.12.97	97.00	0.00	0.00	0.00	0.00	0.00
18.12.98	98.00	0.00	0.00	0.00	0.00	0.00
18.12.99	99.00	0.00	0.00	0.00	0.00	0.00
18.12.100	100.00	0.00	0.00	0.00	0.00	0.00
18.12.101	101.00	0.00	0.00	0.00	0.00	0.00
18.12.102	102.00	0.00	0.00	0.00	0.00	0.00
18.12.103	103.00	0.00	0.00	0.00	0.00	0.00

18.12.104	104.00	0.00	0.00	0.00	0.00	0.00
18.12.105	105.00	0.00	0.00	0.00	0.00	0.00
18.12.106	106.00	0.00	0.00	0.00	0.00	0.00
18.12.107	107.00	0.00	0.00	0.00	0.00	0.00
18.12.108	108.00	0.00	0.00	0.00	0.00	0.00
18.12.109	109.00	0.00	0.00	0.00	0.00	0.00
18.12.110	110.00	0.00	0.00	0.00	0.00	0.00
18.12.111	111.00	0.00	0.00	0.00	0.00	0.00
18.12.112	112.00	0.00	0.00	0.00	0.00	0.00
18.12.113	113.00	0.00	0.00	0.00	0.00	0.00
18.12.114	114.00	0.00	0.00	0.00	0.00	0.00
18.12.115	115.00	0.00	0.00	0.00	0.00	0.00
18.12.116	116.00	0.00	0.00	0.00	0.00	0.00
18.12.117	117.00	0.00	0.00	0.00	0.00	0.00
18.12.118	118.00	0.00	0.00	0.00	0.00	0.00
18.12.119	119.00	0.00	0.00	0.00	0.00	0.00
18.12.120	120.00	0.00	0.00	0.00	0.00	0.00
18.12.121	121.00	0.00	0.00	0.00	0.00	0.00
18.12.122	122.00	0.00	0.00	0.00	0.00	0.00
18.12.123	123.00	0.00	0.00	0.00	0.00	0.00
18.12.124	124.00	0.00	0.00	0.00	0.00	0.00
18.12.125	125.00	0.00	0.00	0.00	0.00	0.00
18.12.126	126.00	0.00	0.00	0.00	0.00	0.00
18.12.127	127.00	0.00	0.00	0.00	0.00	0.00

Large Input

```
[27]: !rocprof-compute analyze -p workloads/fusion_matmulbias_32768x32768/MI300X_A1_
      ↪ -k 0
```

[illegible]

```
INFO Analysis mode = cli
```

```
INFO [analysis] deriving rocprofiler-compute metrics...
```

```
WARNING PC sampling: can not detect pc sampling method without /workspa
ce/user/workloads/fusion_matmulbias_32768x32768/MI300X_A1/ps_file_pc_sampling_ho
st_trap.csv
```

0. Top Stats

0.1 Top Kernels

Kernel_Name				Count	Sum(ns)
Mean(ns)	Median(ns)	Pct	S		
0	matmulbias_kernel			16.00	4020884164.83
251305260.30	252557078.75	99.89	*		
1	void at::native::(anonymous namespace)::			3.00	3329975.00
1109991.67	1565824.92	0.08			
	distribution_elementwise_grid_stride_ker				
	nel<float, 4, at::native::templates::...				
2	void at::native::vectorized_elementwise_			12.00	640889.25
53407.44	50715.25	0.02			
	kernel<4, at::native::FillFunctor<int>,				

	std::array<char*, 1ul> >(int, at::nat...		
3	__amd_rocclr_copyBuffer	89.00	264123.84
2967.68	2936.75 0.01		
4	void at::native::(anonymous namespace)::	7.00	18749.00
2678.43	2582.58 0.00		
	CatArrayBatchedCopy_contig<at::native::(		
	anonymous namespace)::OpaqueType<2u>,...		
5	void at::native::vectorized_elementwise_	3.00	10400.50
3466.83	3454.67 0.00		
	kernel<4, at::native::compare_scalar_ker		
	nel<double>(at::TensorIteratorBase&, ...		
6	void at::native::vectorized_elementwise_	2.00	7082.92
3541.46	3541.46 0.00		
	kernel<8, at::native::UnaryFunctor<c10:		
	:Half, c10::Half, bool, at::native::(...		
7	void at::native::vectorized_elementwise_	2.00	5502.67
2751.33	2751.33 0.00		
	kernel<8, at::native::AbsFunctor<c10::Ha		
	lf>, std::array<char*, 2ul> >(int, at...		
8	void at::native::vectorized_elementwise_	2.00	5455.50
2727.75	2727.75 0.00		
	kernel<8, at::native::BinaryFunctor<c10:		
	:Half, c10::Half, bool, at::native::(...		

```

9 void rocprim::ROCPRIM_400001_NS::detail:      1.00      4496.83
4496.83      4496.83      0.00
    :partition_kernel<(rocprim::ROCPRIM_4000
    01_NS::detail::select_method)0, true,...

```

0.2 Dispatch List

Dispatch_ID	Kernel_Name	GPU_ID
0	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4, at::native::templates::cuda::normal_and_transform<c10::Half, float, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel<at::CUDAGeneratorImpl*>(at::TensorBase const&, double, double, at::CUDAGeneratorImpl*)::...	0
1	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4, at::native::templates::cuda::normal_and_transform<c10::Half, float, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel<at::CUDAGeneratorImpl*>(at::TensorBase const&, double, double, at::CUDAGeneratorImpl*)::...	0
2	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4, at::native::templates::cuda::normal_and_transform<c10::Half, float, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel<at::CUDAGeneratorImpl*>(at::TensorBase const&, double, double, at::CUDAGeneratorImpl*)::...	0
3	matmulbias_kernel	0

```

4          4 void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
          std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
          1>)

```

```

5          5 matmulbias_kernel
0

```

```

6          6 void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
          std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
          1>)

```

```

7          7 matmulbias_kernel
0

```

```

8          8 void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
          std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
          1>)

```

```

9          9 matmulbias_kernel
0

```

1. System Info

Info

workload_name	fusion_matmulbias_32768x32768
command fusion	python triton/matmulbias.py --M 32768 --N 32768
ip_blocks	SQ LDS SQC TA TD TCP TCC SPI CPC CPF roofline
timestamp	Fri Jan 23 18:48:18 2026 (UTC)
version	3
hostname	6b82f9811751
cpu_model	Intel(R) Xeon(R) Platinum 8470
sbios	SeaBIOS1.15.0-1
linux_distro	CentOS Stream 10 (Coughlan)
linux_kernel_version	5.15.0-164-generic
amd_gpu_kernel_version	nan
cpu_memory	230893524

gpu_memory	nan
rocm_version	7.0.3
vbios	113-M3000108-103
compute_partition	SPX
memory_partition	nan
gpu_series	MI300
gpu_model	MI300X_A1
gpu_arch	gfx942
gpu_chip_id	29877
gpu_l1	32
gpu_l2	4096
cu_per_gpu	304

<code>simd_per_cu</code>	4
<code>se_per_gpu</code>	32
<code>wave_size</code>	64
<code>workgroup_max_size</code>	1024
<code>max_waves_per_cu</code>	32
<code>max_sclk</code>	2100
<code>max_mclk</code>	1300
<code>cur_sclk</code>	2100
<code>cur_mclk</code>	1300
<code>total_l2_chan</code>	128
<code>lds_banks_per_cu</code>	32
<code>sqc_per_gpu</code>	160

pipes_per_gpu	4
num_xcd	8
num_hbm_channels	128

2. System Speed-of-Light

2.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit	Peak
Pct of Peak				
2.1.0	VALU FLOPs	4.0	Gflop/s	
81715.2	0.0			
2.1.1	VALU IOPs	1710.82	Giop/s	
81715.2	2.09			
2.1.2	MFMA FLOPs (F8)	0.0	Gflop/s	
2614886.4	0.0			
2.1.3	MFMA FLOPs (BF16)	0.0	Gflop/s	
1307443.2	0.0			
2.1.4	MFMA FLOPs (F16)	280163.54	Gflop/s	
1307443.2	21.43			
2.1.5	MFMA FLOPs (F32)	0.0	Gflop/s	
163430.4	0.0			
2.1.6	MFMA FLOPs (F64)	0.0	Gflop/s	

163430.4 0.0

2.1.7 MFMA FLOPs (F6F4) Gflop/s

2.1.8 MFMA IOPs (Int8) 0.0 Giop/s
2614886.4 0.0

2.1.9 Active CUs 298.0 Cus 304.0
98.03

2.1.10 SALU Utilization 6.04 Pct 100.0
6.04

2.1.11 VALU Utilization 20.31 Pct 100.0
20.31

2.1.12 MFMA Utilization 39.14 Pct 100.0
39.14

2.1.13 VMEM Utilization 1.23 Pct 100.0
1.23

2.1.14 Branch Utilization 2.17 Pct 100.0
2.17

2.1.15 VALU Active Threads 64.0 Threads 64.0
100.0

2.1.16 IPC 0.39 Instr/cycle 5.0
7.72

2.1.17 Wavefront Occupancy 1777.51 Wavefronts
9728.0 18.27

2.1.18 Theoretical LDS Bandwidth 15275.08 Gb/s

81715.2	18.69				
2.1.19 0.0	LDS Bank Conflicts/Access	0.0	Conflicts/access	32.0	
2.1.20 50.09	vL1D Cache Hit Rate	50.09	Pct	100.0	
2.1.21 81715.2	vL1D Cache BW	8909.12	Gb/s		
10.9					
2.1.22 47.4	L2 Cache Hit Rate	47.4	Pct	100.0	
2.1.23 34406.4	L2 Cache BW	4446.19	Gb/s		
12.92					
2.1.24 5324.8	L2-Fabric Read BW	2334.46	Gb/s		
43.84					
2.1.25 5324.8	L2-Fabric Write BW	8.58	Gb/s		
0.16					
2.1.26	L2-Fabric Read Latency	613.3	Cycles		
2.1.27	L2-Fabric Write Latency	230.09	Cycles		
2.1.28 0.0	sL1D Cache Hit Rate	0.0	Pct	100.0	
2.1.29 21504.0	sL1D Cache BW	0.0	Gb/s		
0.0					
2.1.30	L1I Hit Rate	100.0	Pct	100.0	

100.0

2.1.31	L1I BW	1562.78	Gb/s
21504.0	7.27		

2.1.32	L1I Fetch Latency	Cycles
--------	-------------------	--------

3. Memory Chart

3.1 Memory Chart

(Normalization: per_kernel)

Instr Buff	Instr Dispatch	Exec
LDS	L2 Cache	

SALU: 5205674428

CUs	Req	: 53705465	Util	: 35 %	Active
			xGMI/PCIE		

Wave	0	Instr Buf	29
8: 304		<-----	

	SMEM:	
3116	Lat	:
N/A	Hit	: 47

```

%
-----
>
~
|
RVGPRseq: 30.0

| |
Wave N-1 Instr Buf      VALU:
17646000812            Vector L1 Cache
Request                  |
|
-----
>
|
v
SGPRs : 48

MFMA:
4294967296      Rd : 68854501Hit : 50 %      Rd
: 8590163574 Rd : 8590192357

-----
> <----- <-----
-
Fabric

Wt : 26936856Lat : N/A      Wr : 134397088 Wr :
134413798 Rd :
4599351361

VMEM: 1080124024 LDS Alloc: -
-----> -----
> <----- Latency (cycles) Rd:
4462405226

----->
0 Atomic: 0 Coales: 25 % Atomic: 0 Atomic:
0 Wr : 33698328 <----- HBM
<
-----> <-----
> -----> Rd : 613 Wr:
33753553

LDS:
5370546560 Stall : 0
cycles Atomic: 0 Wr :
230 ----->
-----> Scratch Alloc:

```



```

-----> Atomic:      0
                                0

Wave Occupancy      GWS: 0
Scalar L1D
Cache      Latency (cycles)

      6      -----
--->                                Rd      : 35
84

Wave Life
      <-----
      ^

|
      1320185
      BRANCH: 1477754229 Wavefronts:      Rd      :
5599 Hit      :      0.0 %      Wr      : 5603      Rd      :
N/A      |
|
      ----->                                0 <
-----
>      |
|

Lat      :      N/A      Atomic:      0      Wr      :
N/A      |
v
      ^
      <-----

>

|      Workgroups:

|

0      Instr L1 Cache
      GMI

|

|

|      Fetch :
6151483738      Hit      :      100 %      Req
: 27283

```

-----<-----

Lat : 1.0 cycles

4. Roofline

4.1 Roofline

280017.0

139812GFLOP/s

64540.4

14875.8

3428.7

790.3

63613 GB/s

33436 GB/s

182.1 25499 GB/s

42.0 4198 GB/s

0.01

0.18

3.16

56.23

1000.00

Performance (GFLOP/sec) Arithmetic Intensity (FLOP)s/Byte)

5. Command Processor (CPC/CPF)

5.1 Command Processor Fetcher

Metric_ID	Metric	Avg	Min	Max	Unit
5.1.0	CPF Utilization	100.00	100.00	100.00	Pct
5.1.1	CPF Stall	0.00	0.00	0.00	Pct
5.1.2	CPF-L2 Utilization	0.03	0.00	0.23	Pct
5.1.3	CPF-L2 Stall	0.00	0.00	0.00	Pct
5.1.4	CPF-UTCL1 Stall	0.00	0.00	0.00	Pct

5.2 Packet Processor

Metric_ID	Metric	Avg	Min	Max	Unit
-----------	--------	-----	-----	-----	------

5.2.0	CPC SYNC FIFO Full Rate				Pct
5.2.1	CPC CANE Stall Rate				Pct
5.2.2	CPC ADC Utilization				Pct
5.2.3	CPC Utilization	99.9	98.41	100.0	Pct
5.2.4	CPC Stall Rate	0.42	0.1	0.56	Pct
5.2.5	CPC Packet Decoding Utilization	2.3	0.44	3.45	Pct
5.2.6	CPC-Workgroup Manager Utilization	97.65	91.83	102.0	Pct
5.2.7	CPC-L2 Utilization	0.0	0.0	0.0	Pct
5.2.8	CPC-UTCL1 Stall	0.0	0.0	0.0	Pct
5.2.9	CPC-UTCL2 Utilization	0.0	0.0	0.0	Pct

6. Workgroup Manager (SPI)

6.1 Workgroup Manager Utilizations

Metric_ID	Metric	Avg	Min	Max	Unit
6.1.0	Schedule-Pipe Wave Occupancy				Wave
6.1.1	Accelerator Utilization	100.17	97.52	102.83	Pct
6.1.2	Scheduler-Pipe Utilization	0.0	0.0	0.0	Pct
6.1.3	Scheduler-Pipe Wave Utilization				Pct
6.1.4	Workgroup Manager Utilization	97.93	95.6	99.99	Pct
6.1.5	Shader Engine Utilization	97.94	95.43	104.19	Pct
6.1.6	SIMD Utilization	97.53	94.59	99.9	Pct
6.1.7 Workgroups	Dispatched Workgroups	0.0	0.0	0.0	
6.1.8 Wavefronts	Dispatched Wavefronts	0.0	0.0	0.0	
6.1.9 Cycles/wave	VGPR Writes				
6.1.10 Cycles/wave	SGPR Writes				

6.2 Workgroup Manager - Resource Allocation

Metric_ID Unit	Metric	Avg	Min	Max
6.2.0 Pct	Not-scheduled Rate (Workgroup Manager)	0.0	0.0	0.0
6.2.1 Pct	Not-scheduled Rate (Scheduler-Pipe)	0.0	0.0	0.0
6.2.2 Pct	Scheduler-Pipe FIFO Full Rate			
6.2.3 Pct	Scheduler-Pipe Stall Rate	0.0	0.0	0.0
6.2.4 Pct	Scratch Stall Rate	0.0	0.0	0.0
6.2.5 Pct	Insufficient SIMD Waveslots	0.0	0.0	0.0
6.2.6 Pct	Insufficient SIMD VGPRs	0.0	0.0	0.0
6.2.7 Pct	Insufficient SIMD SGPRs	0.0	0.0	0.0
6.2.8 Pct	Insufficient CU LDS	0.0	0.0	0.0
6.2.9 Pct	Insufficient CU Barriers	0.0	0.0	0.0

6.2.10	Reached CU Workgroup Limit	0.0	0.0	0.0
Pct				

6.2.11	Reached CU Wavefront Limit	0.0	0.0	0.0
Pct				

7. Wavefront

7.1 Wavefront Launch Stats

Metric_ID Unit	Metric	Avg	Min	Max
7.1.0 Work items	Grid Size	25165824.00	16777216.00	33554432.00
7.1.1 Work items	Workgroup Size	384.00	256.00	512.00
7.1.2 Wavefronts	Total Wavefronts	0.00	0.00	0.00
7.1.3 Wavefronts	Saved Wavefronts	228.00	0.00	2432.00
7.1.4 Wavefronts	Restored Wavefronts	218.62	0.00	2282.00
7.1.5 Registers	VGPRs	30.00	20.00	40.00
7.1.6 Registers	AGPRs	162.00	132.00	192.00

7.1.7	SGPRs	48.00	48.00	48.00
Registers				

7.1.8	LDS Allocation	0.00	0.00	0.00
Bytes				

7.1.9	Scratch Allocation	0.00	0.00	0.00
Bytes/workitem				

7.2 Wavefront Runtime Stats

Metric_ID	Metric	Avg	Min
Max Unit			

7.2.0	Kernel Time (Nanosec)	251305260.30	241297375.75
258926448.25	Ns		

7.2.1	Kernel Time (Cycles)	292479100.07	261682905.62
392016010.88	Cycle		

7.2.2	Instructions per wavefront	89036.27	70301.22
107540.81	Instr/wavefront		

7.2.3	Wave Cycles	501487515003.25	365763059808.00
622468643880.00	Cycles per kernel		

7.2.4	Dependency Wait Cycles	141746631770.75	89130480040.00
194457533040.00	Cycles per kernel		

7.2.5	Issue Wait Cycles	218745137356.75	152184059316.00
285530073532.00	Cycles per kernel		

7.2.6	Active Cycles	142610345020.25	123753912032.00
161419120480.00	Cycles per kernel		

7.2.7	Wavefront Occupancy	1777.51	1184.23
-------	---------------------	---------	---------

2371.38 Wavefronts

10. Compute Units - Instruction Mix

10.1 Overall Instruction Mix

Metric_ID Unit	Metric	Avg	Min	Max
10.1.0 Instr per kernel	VALU	13351033516.00	11501830144.00	15200157696.00
10.1.1 Instr per kernel	VMEM	1080124024.00	1080033280.00	1080889344.00
10.1.2 Instr per kernel	LDS	5370546560.00	3761766400.00	6979321856.00
10.1.3 Instr per kernel	MFMA	4294967296.00	4294967296.00	4294967296.00
10.1.4 Instr per kernel	SALU	5205674428.00	4188798976.00	6223709184.00
10.1.5 Instr per kernel	SMEM	3116.00	0.00	49856.00
10.1.6 Instr per kernel	Branch	1477754228.94	1343750144.00	1612437327.00

10.2 VALU Arithmetic Instr Mix

Metric_ID Unit	Metric	Avg	Min	Max
-------------------	--------	-----	-----	-----

10.2.0	INT32	5674111801.00	3795320832.00	7552971664.00
Instr per kernel				
10.2.1	INT64	1079902208.00	1079246848.00	1080557568.00
Instr per kernel				
10.2.2	F16-ADD	12582912.00	12582912.00	12582912.00
Instr per kernel				
10.2.3	F16-MUL	0.00	0.00	0.00
Instr per kernel				
10.2.4	F16-FMA	0.00	0.00	0.00
Instr per kernel				
10.2.5	F16-Trans	0.00	0.00	0.00
Instr per kernel				
10.2.6	F32-ADD	0.00	0.00	0.00
Instr per kernel				
10.2.7	F32-MUL	1572864.00	1048576.00	2097152.00
Instr per kernel				
10.2.8	F32-FMA	0.00	0.00	0.00
Instr per kernel				
10.2.9	F32-Trans	1572864.00	1048576.00	2097152.00
Instr per kernel				
10.2.10	F64-ADD	0.00	0.00	0.00
Instr per kernel				
10.2.11	F64-MUL	0.00	0.00	0.00
Instr per kernel				

10.2.12	F64-FMA	0.00	0.00	0.00
Instr per kernel				

10.2.13	F64-Trans	0.00	0.00	0.00
Instr per kernel				

10.2.14	Conversion	19922944.00	18874368.00	20971520.00
Instr per kernel				

10.3 VMEM Instr Mix

Metric_ID Max	Metric Unit	Avg	Min
10.3.0 1077936128.0	Global/Generic Instr Instr per kernel	1077936128.0	1077936128.0
10.3.1 1073741824.0	Global/Generic Read Instr per kernel	1073741824.0	1073741824.0
10.3.2 4194304.0	Global/Generic Write Instr per kernel	4194304.0	4194304.0
10.3.3 0.0	Global/Generic Atomic Instr per kernel	0.0	0.0
10.3.4 2931680.0	Spill/Stack Instr Instr per kernel	2191870.0	2097152.0
10.3.5	Spill/Stack Coalesceable Instr Instr per kernel		
10.3.6 2533424.0	Spill/Stack Read Instr per kernel	2146915.0	2097152.0

10.3.7	Spill/Stack Write	20064.0	0.0
321024.0	Instr per kernel		

10.3.8	Spill/Stack Atomic	0.0	0.0
0.0	Instr per kernel		

10.4 MFMA Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.4.0 per kernel	MFMA-I8	0.0	0.0	0.0	Instr
10.4.1 per kernel	MFMA-F8	0.0	0.0	0.0	Instr
10.4.2 per kernel	MFMA-F16	4294967296.0	4294967296.0	4294967296.0	Instr
10.4.3 per kernel	MFMA-BF16	0.0	0.0	0.0	Instr
10.4.4 per kernel	MFMA-F32	0.0	0.0	0.0	Instr
10.4.5 per kernel	MFMA-F64	0.0	0.0	0.0	Instr
10.4.6 per kernel	MFMA-F6F4				Instr

11. Compute Units - Compute Pipeline

11.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit	Peak	Pct of Peak
11.1.0	VALU FLOPs	4.0	Gflop	81715.2	0.0
11.1.1	VALU IOPs	1710.82	Giop	81715.2	2.09
11.1.2	MFMA FLOPs (F8)	0.0	Gflop	2614886.4	0.0
11.1.3	MFMA FLOPs (BF16)	0.0	Gflop	1307443.2	0.0
11.1.4	MFMA FLOPs (F16)	280163.54	Gflop	1307443.2	21.43
11.1.5	MFMA FLOPs (F32)	0.0	Gflop	163430.4	0.0
11.1.6	MFMA FLOPs (F64)	0.0	Gflop	163430.4	0.0
11.1.7	MFMA FLOPs (F6F4)		Gflop		
11.1.8	MFMA IOPs (INT8)	0.0	Giop	2614886.4	0.0

11.2 Pipeline Stats

Metric_ID	Metric	Avg	Min	Max	Unit
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11.2.0	IPC	0.39	0.24	0.48	Instr/cycle
11.2.1	IPC (Issued)	0.86	0.84	0.88	Instr/cycle
11.2.2	SALU Utilization	6.04	3.51	7.82	Pct
11.2.3	VALU Utilization	20.31	13.26	24.51	Pct
11.2.4	VALU Co-Issue Efficiency				Pct
11.2.5	VMEM Utilization	1.23	0.91	1.36	Pct
11.2.6	Branch Utilization	2.17	1.35	2.7	Pct
11.2.7	VALU Active Threads	64.0	63.99	64.0	Threads
11.2.8	MFMA Utilization	39.14	28.83	43.19	Pct
11.2.9	MFMA Instr Cycles	32.0	32.0	32.0	Cycles/instr
11.2.10	VMEM Latency				Cycles
11.2.11	SMEM Latency				Cycles

11.3 Arithmetic Operations

Metric_ID Unit	Metric	Avg	Min	Max
11.3.0	FLOPs (Total)	70369750810624.0	70369683701760.0	
70369817919488.0	Ops per kernel			
11.3.1	IOPs (Total)	432256896576.0	311972331520.0	
552545870848.0	Ops per kernel			
11.3.2	F8 OPs	0.0	0.0	0.0
Ops per kernel				
11.3.3	F16 OPs	70369549484032.0	70369549484032.0	
70369549484032.0	Ops per kernel			
11.3.4	BF16 OPs	0.0	0.0	0.0
Ops per kernel				
11.3.5	F32 OPs	201326592.0	134217728.0	
268435456.0	Ops per kernel			
11.3.6	F64 OPs	0.0	0.0	0.0
Ops per kernel				
11.3.7	F6F4 OPs			
Ops per kernel				
11.3.8	INT8 OPs	0.0	0.0	0.0
Ops per kernel				

12. Local Data Share (LDS)

12.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
12.1.0	Utilization	34.72	Pct of peak
12.1.1	Access Rate	22.73	Pct of peak
12.1.2	Theoretical Bandwidth (% of Peak)	18.69	Pct of peak
12.1.3	Bank Conflict Rate	0.00	Pct of peak

12.2 LDS Stats

Metric_ID	Metric	Avg	Min
Max	Unit		
12.2.0	LDS Instrs	5370546560.0	3761766400.0
6979321856.0	Instr per kernel		
12.2.1	LDS LOAD		
Instr per kernel			
12.2.2	LDS STORE		
Instr per kernel			
12.2.3	LDS ATOMIC		
Instr per kernel			
12.2.4	LDS LOAD Bandwidth		
Gbps			
12.2.5	LDS STORE Bandwidth		
Gbps			
12.2.6	LDS ATOMIC Bandwidth		
Gbps			
12.2.7	Theoretical Bandwidth	3848293187584.0	3298534883328.0

4398066434048.0 Bytes per kernel

12.2.8 LDS Latency
Cycles

12.2.9	Bank Conflicts/Access	0.0	0.0
0.0	Conflicts/access		

12.2.10	Index Accesses	30064790528.0	25769803776.0
34359894016.0	Cycles per kernel		

12.2.11	Atomic Return Cycles	0.0	0.0
0.0	Cycles per kernel		

12.2.12	Bank Conflict	0.0	0.0
0.0	Cycles per kernel		

12.2.13	Addr Conflict	0.0	0.0
0.0	Cycles per kernel		

12.2.14	Unaligned Stall	0.0	0.0
0.0	Cycles per kernel		

12.2.15	Mem Violations	0.0	0.0
0.0	Accesses per kernel		

12.2.16 LDS Command FIFO Full Rate
Cycles per kernel

12.2.17 LDS Data FIFO Full Rate
Cycles per kernel

13. Instruction Cache
13.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
13.1.0	Bandwidth	7.27	Pct of peak
13.1.1	Cache Hit Rate	100.00	Pct of peak
13.1.2	L1I-L2 Bandwidth	0.00	Pct of peak

13.2 Instruction Cache Accesses

Metric_ID Unit	Metric	Avg	Min	Max
13.2.0 7016808202.0	Req Req per kernel	6151483737.56	5287182336.0	
13.2.1 7016746468.0	Hits Hits per kernel	6151415008.69	5287178797.0	
13.2.2 2530.0	Misses - Non Duplicated Misses per kernel	710.88	320.0	
13.2.3 17094.0	Misses - Duplicated Misses per kernel	4275.12	2677.0	
13.2.4 Pct	Cache Hit Rate	100.0	100.0	100.0
13.2.5 Cycles	Instruction Fetch Latency			

13.3 Instruction Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
13.3.0	L1I-L2 Bandwidth	1746092.00	921600.00	5099520.00	Bytes

per kernel

14. Scalar L1 Data Cache

14.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
14.1.0	Bandwidth	0.00	Pct of peak
14.1.1	Cache Hit Rate	0.00	Pct of peak
14.1.2	sL1D-L2 BW	0.00	Pct of peak

14.2 Scalar L1D Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
14.2.0 kernel	Req	5599.06	0.00	89585.00	Req per
14.2.1 kernel	Hits	0.00	0.00	0.00	Req per
14.2.2 kernel	Misses - Non Duplicated	3116.00	0.00	49856.00	Req per
14.2.3 kernel	Misses- Duplicated	0.00	0.00	0.00	Req per
14.2.4	Cache Hit Rate	0.00	0.00	0.00	Pct
14.2.5 kernel	Read Req (Total)	3611.88	0.00	31488.00	Req per

14.2.6 kernel	Atomic Req	0.00	0.00	0.00	Req per
14.2.7 kernel	Read Req (1 DWord)	2508.00	0.00	26752.00	Req per
14.2.8 kernel	Read Req (2 DWord)	0.00	0.00	0.00	Req per
14.2.9 kernel	Read Req (4 DWord)	448.00	0.00	4736.00	Req per
14.2.10 kernel	Read Req (8 DWord)	0.00	0.00	0.00	Req per
14.2.11 kernel	Read Req (16 DWord)	655.88	0.00	6846.00	Req per

14.3 Scalar L1D Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
14.3.0	sL1D-L2 BW	587976.00	0.00	3792000.00	Bytes per kernel
14.3.1	Read Req	3584.00	0.00	37888.00	Req per kernel
14.3.2	Write Req	5603.12	0.00	59250.00	Req per kernel
14.3.3	Atomic Req	0.00	0.00	0.00	Req per kernel
14.3.4	Stall Cycles	43.38	0.00	594.00	Cycles per

kernel

15. Address Processing Unit and Data Return Path (TA/TD)

15.1 Address Processing Unit

Metric_ID Max	Metric Unit	Avg	Min
15.1.0 41.55	Address Processing Unit Busy Pct	38.16	34.82
15.1.1 14.66	Address Stall Pct	11.79	9.32
15.1.2 0.77	Data Stall Pct	0.73	0.6
15.1.3 0.0	Data-Processor → Address Stall Pct	0.0	0.0
15.1.4	Sequencer → TA Address Stall Cycles per kernel		
15.1.5	Sequencer → TA Command Stall Cycles per kernel		
15.1.6	Sequencer → TA Data Stall Cycles per kernel		
15.1.7 1080033280.0	Total Instructions 1080802736.0 Instructions per kernel	1080081371.0	
15.1.8 1077936128.0	Global/Generic Instructions 1077936128.0 Instructions per kernel	1077936128.0	

15.1.9	Global/Generic Read Instructions	1073741824.0	
1073741824.0	1073741824.0 Instructions per kernel		
15.1.10	Global/Generic Read Instructions for LDS		
	Instructions per kernel		
15.1.11	Global/Generic Write Instructions	4194304.0	
4194304.0	4194304.0 Instructions per kernel		
15.1.12	Global/Generic Atomic Instructions	0.0	0.0
0.0	Instructions per kernel		
15.1.13	Spill/Stack Instructions	2191870.0	
2097152.0	2931680.0 Instructions per kernel		
15.1.14	Spill/Stack Read Instructions	2146915.0	
2097152.0	2533424.0 Instructions per kernel		
15.1.15	Spill/Stack Read Instructions for LDS		
	Instructions per kernel		
15.1.16	Spill/Stack Write Instructions	20064.0	0.0
321024.0	Instructions per kernel		
15.1.17	Spill/Stack Atomic Instructions	0.0	0.0
0.0	Instructions per kernel		
15.1.18	Spill/Stack Total Cycles	33997056.0	
33554432.0	37445632.0 Cycles per kernel		
15.1.19	Spill/Stack Coalesced Read	195260.0	0.0
1684416.0	Cycles per kernel		
15.1.20	Spill/Stack Coalesced Write	83030.0	0.0
1328480.0	Cycles per kernel		

15.2 Data-Return Path

Metric_ID Max	Metric Unit	Avg	Min
15.2.0 78.38	Data-Return Busy Pct	73.21	68.73
15.2.1 62.98	Cache RAM → Data-Return Stall Pct	50.77	46.21
15.2.2 0.0	Workgroup manager → Data-Return Stall Pct	0.0	0.0
15.2.3 767968.0	Coalescable Instructions Instructions per kernel	88126.0	0.0
15.2.4 1075440720.0	Read Instructions 1076608432.0 Instructions per kernel	1075842112.0	
15.2.5 4592560.0	Write Instructions Instructions per kernel	4239259.0	4194304.0
15.2.6 0.0	Atomic Instructions Instructions per kernel	0.0	0.0
15.2.7	Write Ack Instructions Instructions per kernel		

INFO Not showing table with empty column(s): 16.6 L1D Addr Translation Stalls

16. Vector L1 Data Cache

16.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
16.1.0	Hit rate	50.09	Pct of peak
16.1.1	Bandwidth	10.90	Pct of peak
16.1.2	Utilization	80.11	Pct of peak
16.1.3	Coalescing	25.00	Pct of peak

16.2 L1D Cache Stalls (%)

Metric_ID	Metric	Min	Q1	Median	Q3	Max
16.2.0 61.33	Stalled on L2 Data	43.94	46.13	48.22	49.8	
16.2.1	Stalled on L2 Req	0.0	0.01	0.02	0.02	0.11
16.2.2	Stalled on Address					
16.2.3	Stalled on Data					
16.2.4	Stalled on Latency FIFO					
16.2.5	Stalled on Request FIFO					
16.2.6	Stalled on Read Return					
16.2.7	Tag RAM Stall (Read)	0.0	0.0	0.0	0.0	0.0

16.2.8	Tag RAM Stall (Write)	0.17	0.22	0.24	0.26	0.26
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16.2.9	Tag RAM Stall (Atomic)	0.0	0.0	0.0	0.0	0.0
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16.3 L1D Cache Accesses

Metric_ID Unit	Metric	Avg	Min	Max
16.3.0 69137285120.0	Total Req Req per kernel	69123874816.0	69122129920.0	
16.3.1 68860854272.0	Read Req Req per kernel	68854501888.0	68853694464.0	
16.3.2 276360960.0	Write Req Req per kernel	269368560.0	268435456.0	
16.3.3 Req per kernel	Atomic Req	0.0	0.0	0.0
16.3.4 2238125793280.0	Cache BW Bytes per kernel	2237705950720.0	2237677961216.0	
16.3.5 Pct	Cache Hit Rate	50.09	50.08	50.11
16.3.6 17485357760.0	Cache Accesses Req per kernel	17482077740.0	17481859072.0	
16.3.7 8761073810.0	Cache Hits Req per kernel	8757517077.62	8755785413.0	

16.3.8	Invalidations	577.5	304.0	2704.0
	Req per kernel			
16.3.9	L1-L2 BW	1108142351152.0	1108118339584.0	
1108232936832.0	Bytes per kernel			
16.3.10	Tag RAM 0 Req			
	Req per kernel			
16.3.11	Tag RAM 1 Req			
	Req per kernel			
16.3.12	Tag RAM 2 Req			
	Req per kernel			
16.3.13	Tag RAM 3 Req			
	Req per kernel			
16.3.14	L1-L2 Read	8590163574.38	8590065664.0	
8590908662.0	Req per kernel			
16.3.15	L1-L2 Write	134397088.0	134217728.0	
136007680.0	Req per kernel			
16.3.16	L1-L2 Atomic	0.0	0.0	0.0
	Req per kernel			
16.3.17	L1 Access Latency			
	Cycles			
16.3.18	L1-L2 Read Latency			
	Cycles			
16.3.19	L1-L2 Write Latency			
	Cycles			

16.4 L1D - L2 Transactions

Metric_ID Min	Metric Max	Unit	Xfer	Coherency	Avg
16.4.0 0.00	NC - Read 0.00	Read Req per kernel	NC		0.00
16.4.1 0.00	UC - Read 0.00	Read Req per kernel	UC		0.00
16.4.2 0.00	CC - Read 0.00	Read Req per kernel	CC		0.00
16.4.3 8590065664.00	RW - Read 8590961589.00	Read Req per kernel	RW		8590166837.25
16.4.4 134217728.00	RW - Write 135902144.00	Write Req per kernel	RW		134412988.00
16.4.5 0.00	NC - Write 0.00	Write Req per kernel	NC		0.00
16.4.6 0.00	UC - Write 0.00	Write Req per kernel	UC		0.00
16.4.7 0.00	CC - Write 0.00	Write Req per kernel	CC		0.00
16.4.8 0.00	NC - Atomic 0.00	Atomic Req per kernel	NC		0.00
16.4.9 0.00	UC - Atomic 0.00	Atomic Req per kernel	UC		0.00

16.4.10	CC - Atomic	Atomic	CC	0.00
0.00	0.00	Req per kernel		

16.4.11	RW - Atomic	Atomic	RW	0.00
0.00	0.00	Req per kernel		

16.5 L1D Addr Translation

Metric_ID	Metric	Avg	Min
Max	Units		
16.5.0	Req	17482077740.0	17481859072.0
17485357760.0	Req per kernel		
16.5.1	Inflight Req		
Req per kernel			
16.5.2	Hit Ratio	99.59	99.57
99.61	Pct		
16.5.3	Hits	17410821107.75	17410385920.0
17414169111.0	Req per kernel		
16.5.4	Translation Misses	67475702.56	67475456.0
67477472.0	Req per kernel		
16.5.5	Misses under Translation Miss		
Req per kernel			
16.5.6	Permission Misses	0.0	0.0
0.0	Req per kernel		

INFO Not showing table with empty column(s): 17.4 L2 Cache Stalls

17. L2 Cache

17.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
17.1.0	Utilization	81.95	Pct
17.1.1	Bandwidth	12.92	Pct
17.1.2	Hit Rate	47.40	Pct
17.1.3	L2-Fabric Read BW	2334.46	Gb/s
17.1.4	L2-Fabric Write and Atomic BW	8.58	Gb/s
17.1.5	HBM Bandwidth	5324.80	Gb/s

17.2 L2 - Fabric Transactions

Metric_ID	Metric	Avg	Min
Max	Unit		
17.2.0	Read BW	586771569868.0	
566283176576.0	625857132288.0 Bytes per kernel		
17.2.1	HBM Read Traffic	97.17	90.2
100.31	Pct		
17.2.2	Remote Read Traffic	2.85	0.0
9.8	Pct		
17.2.3	Uncached Read Traffic	0.0	0.0
0.0	Pct		
17.2.4	Write and Atomic BW	2155561480.0	
2147483648.0	2203521376.0 Bytes per kernel		
17.2.5	HBM Write and Atomic Traffic	100.18	95.15
105.53	Pct		
17.2.6	Remote Write and Atomic Traffic	0.41	0.0

4.85	Pct			
17.2.7	Atomic Traffic	0.0	0.0	
0.0	Pct			
17.2.8	Uncached Write and Atomic Traffic	0.01	0.0	
0.15	Pct			
17.2.9	Read Latency	613.3	479.13	
1097.34	Cycles			
17.2.10	Write and Atomic Latency	230.09	156.83	
508.2	Cycles			
17.2.11	Atomic Latency			
	Cycles			
17.2.12	Read Stall			
	Pct			
17.2.13	Write Stall			
	Pct			
17.3 L2 Cache Accesses				
Metric_ID	Metric	Avg	Min	Max
Unit				
17.3.0	Bandwidth	1116751209112.0	1116710137472.0	
1117061963392.0	Bytes per kernel			
17.3.1	Read Bandwidth			
Bytes per kernel				
17.3.2	Write Bandwidth			
Bytes per kernel				

17.3.3	Atomic Bandwidth			
Bytes per kernel				
17.3.4	Req	8724618821.19	8724297949.0	
8727046589.0	Req per kernel			
17.3.5	Read Req	8590192356.88	8590080251.0	
8591051325.0	Req per kernel			
17.3.6	Write Req	134413798.5	134217728.0	
136068480.0	Req per kernel			
17.3.7	Atomic Req	0.0	0.0	0.0
Req per kernel				
17.3.8	Streaming Req	279168.0	0.0	
2462720.0	Req per kernel			
17.3.9	Bypassss Req			
Req per kernel				
17.3.10	Probe Req	3625.0	0.0	
58000.0	Req per kernel			
17.3.11	Input Buffer Req			
Req per kernel				
17.3.12	Cache Hit	47.4	44.36	49.1
Pct				
17.3.13	Hits	4140062296.12	3812029704.0	
4284023540.0	Hits per kernel			
17.3.14	Misses	4594228349.94	4440870189.0	
4939754585.0	Misses per kernel			

17.3.15	Writeback	16833399.38	16777328.0	
17554745.0	Cachelines per kernel			
17.3.16	Writeback (Internal)	16826535.88	16732160.0	
17612709.0	Cachelines per kernel			
17.3.17	Writeback (vL1D Req)	43948.88	27774.0	
73591.0	Cachelines per kernel			
17.3.18	Evict (Internal)	4615895192.25	4440592380.0	
4964886116.0	Cachelines per kernel			
17.3.19	Evict (vL1D Req)	0.0	0.0	0.0
	Cachelines per kernel			
17.3.20	NC Req	0.0	0.0	0.0
	Req per kernel			
17.3.21	UC Req	1308.5	0.0	
17855.0	Req per kernel			
17.3.22	CC Req	0.0	0.0	0.0
	Req per kernel			
17.3.23	RW Req	8724616509.0	8724297933.0	
8727029049.0	Req per kernel			

17.5 L2 - Fabric Interface Stalls

Metric_ID	Metric	Min	Max	Unit	Type
Transaction	Avg				
17.5.0	Read - PCIe Stall				PCIe Stall
	Pct				Read

17.5.1	Read - Infinity Fabric Stall Pct	Infinity Fabric Stall	Read
17.5.2	Read - HBM Stall Pct	HBM Stall	Read
17.5.3	Write - PCIe Stall Pct	PCIe Stall	Write
17.5.4	Write - Infinity Fabric Stall Pct	Infinity Fabric Stall	Write
17.5.5	Write - HBM Stall Pct	HBM Stall	Write
17.5.6	Write - Credit Starvation 0.0 0.0 0.0 Pct	Credit Starvation	Write

17.6 L2 - Fabric Detailed Transaction Breakdown

Metric_ID Max	Metric Unit	Avg	Min
17.6.0 0.0	Read (32B) Req per kernel	0.0	0.0
17.6.1 -268773176.0	Read (64B) 391814215.0 Req per kernel	30396942.31	
17.6.2 4424096001.0	Read (128B) 4865144388.0 Req per kernel	4568954418.44	
17.6.3 35936.0	Read (Uncached) Req per kernel	4618.88	0.0

17.6.4	HBM Read	4462405225.75	
4424077666.0	4544793344.0 Req per kernel		
17.6.5	Remote Read	137826091.38	0.0
485047654.0	Req per kernel		
17.6.6	Read Bandwidth - PCIe Bytes per kernel		
17.6.7	Read Bandwidth - Infinity Fabric Bytes per kernel		
17.6.8	Read Bandwidth - HBM Bytes per kernel		
17.6.9	Write and Atomic (32B)	35358.88	-1721987.0
1711205.0	Req per kernel		
17.6.10	Write and Atomic (Uncached)	3875.0	0.0
54400.0	Req per kernel		
17.6.11	Write and Atomic (64B)	33662968.69	33554432.0
35291015.0	Req per kernel		
17.6.12	HBM Write and Atomic	33753553.44	33554432.0
35425454.0	Req per kernel		
17.6.13	Remote Write and Atomic	142533.0	0.0
1711141.0	Req per kernel		
17.6.14	Write Bandwidth - PCIe Bytes per kernel		
17.6.15	Write Bandwidth - Infinity Fabric Bytes per kernel		

17.6.16	Write Bandwidth - HBM Bytes per kernel		
17.6.17	Atomic Req per kernel	0.0	0.0
17.6.18	Atomic - HBM Req per kernel		
17.6.19	Atomic Bandwidth - PCIe Bytes per kernel		
17.6.20	Atomic Bandwidth - Infinity Fabric Bytes per kernel		
17.6.21	Atomic Bandwidth - HBM Bytes per kernel		

INFO Not showing table with empty column(s): 18.9 L2-Fabric Read Stall (Cycles per normUnit)

INFO Not showing table with empty column(s): 18.10 L2-Fabric Write and Atomic Stall (Cycles per normUnit)

18. L2 Cache (per Channel)

18.1 Aggregate Stats (All channels)

Metric_ID	Metric	Avg	Std Dev	Min	Max	Unit
18.1.0	L2 Cache Hit Rate	47.43	2.09	43.38	49.10	Pct

18.2 L2 Cache Hit Rate (pct)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.2.0	0.00	43.38	45.82	48.46	49.10	49.10
18.2.1	1.00	43.38	45.82	48.46	49.10	49.10
18.2.2	2.00	43.38	45.82	48.46	49.10	49.10

18.2.3	3.00	43.38	45.82	48.46	49.10	49.10
18.2.4	4.00	43.38	45.82	48.46	49.10	49.10
18.2.5	5.00	43.38	45.82	48.46	49.10	49.10
18.2.6	6.00	43.38	45.82	48.46	49.10	49.10
18.2.7	7.00	43.38	45.82	48.46	49.10	49.10
18.2.8	8.00	43.38	45.82	48.46	49.10	49.10
18.2.9	9.00	43.38	45.82	48.46	49.10	49.10
18.2.10	10.00	43.38	45.82	48.46	49.10	49.10
18.2.11	11.00	43.38	45.82	48.46	49.10	49.10
18.2.12	12.00	43.38	45.82	48.46	49.10	49.10
18.2.13	13.00	43.38	45.82	48.46	49.10	49.10
18.2.14	14.00	43.38	45.82	48.46	49.10	49.10
18.2.15	15.00	43.38	45.82	48.46	49.10	49.10
18.2.16	16.00	43.64	44.87	48.39	49.09	49.10
18.2.17	17.00	43.65	44.87	48.39	49.09	49.10
18.2.18	18.00	43.64	44.87	48.39	49.09	49.10
18.2.19	19.00	43.65	44.87	48.39	49.09	49.10
18.2.20	20.00	43.64	44.87	48.39	49.09	49.10
18.2.21	21.00	43.65	44.87	48.39	49.09	49.10
18.2.22	22.00	43.64	44.87	48.39	49.09	49.10
18.2.23	23.00	43.65	44.87	48.39	49.09	49.10
18.2.24	24.00	43.64	44.87	48.39	49.09	49.10
18.2.25	25.00	43.65	44.87	48.39	49.09	49.10
18.2.26	26.00	43.65	44.87	48.39	49.09	49.10

18.2.27	27.00	43.65	44.87	48.39	49.09	49.10
18.2.28	28.00	43.64	44.87	48.39	49.09	49.10
18.2.29	29.00	43.65	44.87	48.39	49.09	49.10
18.2.30	30.00	43.65	44.87	48.39	49.09	49.10
18.2.31	31.00	43.65	44.87	48.39	49.09	49.10
18.2.32	32.00	43.04	43.68	47.68	49.10	49.10
18.2.33	33.00	43.04	43.68	47.68	49.10	49.10
18.2.34	34.00	43.04	43.68	47.68	49.10	49.10
18.2.35	35.00	43.04	43.67	47.68	49.10	49.10
18.2.36	36.00	43.04	43.68	47.68	49.10	49.10
18.2.37	37.00	43.04	43.68	47.68	49.10	49.10
18.2.38	38.00	43.04	43.68	47.68	49.10	49.10
18.2.39	39.00	43.04	43.67	47.68	49.10	49.10
18.2.40	40.00	43.04	43.68	47.68	49.10	49.10
18.2.41	41.00	43.04	43.68	47.68	49.10	49.10
18.2.42	42.00	43.04	43.68	47.68	49.10	49.10
18.2.43	43.00	43.04	43.68	47.68	49.10	49.10
18.2.44	44.00	43.04	43.68	47.68	49.10	49.10
18.2.45	45.00	43.04	43.68	47.68	49.10	49.10
18.2.46	46.00	43.04	43.68	47.68	49.10	49.10
18.2.47	47.00	43.04	43.68	47.68	49.10	49.10
18.2.48	48.00	43.86	46.87	48.61	49.09	49.10
18.2.49	49.00	43.86	46.87	48.61	49.09	49.10
18.2.50	50.00	43.86	46.87	48.61	49.09	49.10

18.2.51	51.00	43.86	46.87	48.61	49.09	49.10
18.2.52	52.00	43.86	46.87	48.61	49.09	49.10
18.2.53	53.00	43.86	46.87	48.61	49.09	49.10
18.2.54	54.00	43.86	46.87	48.61	49.09	49.10
18.2.55	55.00	43.86	46.87	48.61	49.09	49.10
18.2.56	56.00	43.86	46.87	48.61	49.09	49.10
18.2.57	57.00	43.86	46.87	48.61	49.09	49.10
18.2.58	58.00	43.86	46.87	48.61	49.09	49.10
18.2.59	59.00	43.86	46.87	48.61	49.09	49.10
18.2.60	60.00	43.86	46.87	48.61	49.09	49.10
18.2.61	61.00	43.86	46.87	48.61	49.09	49.10
18.2.62	62.00	43.86	46.87	48.61	49.09	49.10
18.2.63	63.00	43.86	46.87	48.61	49.09	49.10
18.2.64	64.00	43.93	45.64	48.26	49.10	49.10
18.2.65	65.00	43.93	45.64	48.26	49.10	49.10
18.2.66	66.00	43.93	45.64	48.26	49.10	49.10
18.2.67	67.00	43.93	45.64	48.26	49.10	49.10
18.2.68	68.00	43.93	45.64	48.26	49.10	49.10
18.2.69	69.00	43.93	45.64	48.26	49.10	49.10
18.2.70	70.00	43.93	45.64	48.26	49.10	49.10
18.2.71	71.00	43.93	45.64	48.26	49.10	49.10
18.2.72	72.00	43.93	45.64	48.26	49.10	49.10
18.2.73	73.00	43.93	45.64	48.26	49.10	49.10
18.2.74	74.00	43.93	45.64	48.26	49.10	49.10

18.2.75	75.00	43.93	45.64	48.26	49.10	49.10
18.2.76	76.00	43.93	45.64	48.26	49.10	49.10
18.2.77	77.00	43.93	45.64	48.26	49.10	49.10
18.2.78	78.00	43.93	45.64	48.26	49.10	49.10
18.2.79	79.00	43.93	45.64	48.26	49.10	49.10
18.2.80	80.00	43.46	45.23	48.19	49.09	49.10
18.2.81	81.00	43.46	45.23	48.19	49.09	49.10
18.2.82	82.00	43.46	45.23	48.19	49.09	49.10
18.2.83	83.00	43.46	45.23	48.19	49.09	49.10
18.2.84	84.00	43.46	45.23	48.19	49.09	49.10
18.2.85	85.00	43.46	45.23	48.19	49.09	49.10
18.2.86	86.00	43.46	45.23	48.19	49.09	49.10
18.2.87	87.00	43.46	45.23	48.19	49.09	49.10
18.2.88	88.00	43.46	45.23	48.19	49.09	49.10
18.2.89	89.00	43.46	45.23	48.19	49.09	49.10
18.2.90	90.00	43.46	45.23	48.19	49.09	49.10
18.2.91	91.00	43.46	45.23	48.19	49.09	49.10
18.2.92	92.00	43.46	45.23	48.19	49.09	49.10
18.2.93	93.00	43.46	45.23	48.19	49.09	49.10
18.2.94	94.00	43.46	45.23	48.19	49.09	49.10
18.2.95	95.00	43.46	45.23	48.19	49.09	49.10
18.2.96	96.00	44.39	46.03	48.24	49.10	49.10
18.2.97	97.00	44.39	46.03	48.24	49.10	49.10
18.2.98	98.00	44.39	46.03	48.24	49.10	49.10

18.2.99	99.00	44.39	46.03	48.24	49.10	49.10
18.2.100	100.00	44.39	46.03	48.24	49.10	49.10
18.2.101	101.00	44.39	46.03	48.24	49.10	49.10
18.2.102	102.00	44.39	46.03	48.24	49.10	49.10
18.2.103	103.00	44.39	46.03	48.24	49.10	49.10
18.2.104	104.00	44.39	46.03	48.24	49.10	49.10
18.2.105	105.00	44.39	46.03	48.24	49.10	49.10
18.2.106	106.00	44.39	46.03	48.24	49.10	49.10
18.2.107	107.00	44.39	46.03	48.24	49.10	49.10
18.2.108	108.00	44.39	46.03	48.24	49.10	49.10
18.2.109	109.00	44.39	46.03	48.24	49.10	49.10
18.2.110	110.00	44.39	46.03	48.24	49.10	49.10
18.2.111	111.00	44.39	46.03	48.24	49.10	49.10
18.2.112	112.00	44.14	45.27	48.55	49.09	49.10
18.2.113	113.00	44.14	45.27	48.55	49.09	49.10
18.2.114	114.00	44.15	45.27	48.55	49.09	49.10
18.2.115	115.00	44.14	45.27	48.55	49.09	49.10
18.2.116	116.00	44.14	45.27	48.55	49.09	49.10
18.2.117	117.00	44.14	45.27	48.55	49.09	49.10
18.2.118	118.00	44.14	45.27	48.55	49.09	49.10
18.2.119	119.00	44.14	45.27	48.55	49.09	49.10
18.2.120	120.00	44.14	45.27	48.55	49.09	49.10
18.2.121	121.00	44.14	45.27	48.55	49.09	49.10
18.2.122	122.00	44.14	45.27	48.55	49.09	49.10

18.2.123	123.00	44.14	45.27	48.55	49.09	49.10
18.2.124	124.00	44.14	45.27	48.55	49.09	49.10
18.2.125	125.00	44.14	45.27	48.55	49.09	49.10
18.2.126	126.00	44.14	45.27	48.55	49.09	49.10
18.2.127	127.00	44.14	45.27	48.55	49.09	49.10

18.3 L2 Requests (per normUnit)

Metric_ID Q3	Channel Max	Min	Q1	Median
18.3.0 68158624.00	0.00 68180548.00	68158544.00	68158546.00	68158624.00
18.3.1 68158704.00	1.00 68180580.00	68158545.00	68158547.00	68158704.00
18.3.2 68158704.00	2.00 68180584.00	68158544.00	68158546.00	68158704.00
18.3.3 68158704.00	3.00 68180504.00	68158504.00	68158505.75	68158704.00
18.3.4 68158704.00	4.00 68180535.00	68158544.00	68158546.75	68158704.00
18.3.5 68158704.00	5.00 68180669.00	68158544.00	68158545.75	68158704.00
18.3.6 68158705.00	6.00 68180753.00	68158544.00	68158545.75	68158704.00
18.3.7 68158704.00	7.00 68180525.00	68158544.00	68158547.00	68158704.00

18.3.8	8.00	68158624.00	68158626.00	68158704.00
68158704.00	68180744.00			
18.3.9	9.00	68158624.00	68158626.75	68158704.00
68158704.00	68180827.00			
18.3.10	10.00	68158624.00	68158626.00	68158644.00
68158644.00	68180723.00			
18.3.11	11.00	68158624.00	68158627.75	68158704.00
68158704.00	68180718.00			
18.3.12	12.00	68158624.00	68158626.75	68158704.00
68158704.00	68180550.00			
18.3.13	13.00	68158624.00	68158627.00	68158704.00
68158704.00	68180415.00			
18.3.14	14.00	68158624.00	68158626.75	68158704.00
68158704.00	68180801.00			
18.3.15	15.00	68158544.00	68158548.75	68158704.00
68158704.25	68180693.00			
18.3.16	16.00	68158624.00	68158626.00	68158704.00
68158704.00	68180679.00			
18.3.17	17.00	68158624.00	68158625.75	68158704.00
68158704.00	68180705.00			
18.3.18	18.00	68158624.00	68158624.75	68158644.00
68158644.00	68180574.00			
18.3.19	19.00	68158624.00	68158626.50	68158704.00
68158704.00	68180499.00			

18.3.20	20.00	68158624.00	68158625.75	68158704.00
68158704.00	68180574.00			
18.3.21	21.00	68158624.00	68158625.75	68158704.00
68158704.00	68180598.00			
18.3.22	22.00	68158624.00	68158625.00	68158704.00
68158704.00	68180789.00			
18.3.23	23.00	68158544.00	68158546.00	68158704.00
68158704.25	68180787.00			
18.3.24	24.00	68158544.00	68158546.00	68158624.00
68158624.00	68180584.00			
18.3.25	25.00	68158544.00	68158546.75	68158704.00
68158704.00	68180618.00			
18.3.26	26.00	68158544.00	68158545.00	68158704.00
68158704.00	68180737.00			
18.3.27	27.00	68158504.00	68158505.50	68158704.00
68158704.00	68180700.00			
18.3.28	28.00	68158544.00	68158546.75	68158704.00
68158704.00	68180558.00			
18.3.29	29.00	68158544.00	68158544.75	68158704.00
68158704.00	68180596.00			
18.3.30	30.00	68158544.00	68158544.75	68158704.00
68158705.00	68180563.00			
18.3.31	31.00	68158544.00	68158545.75	68158704.00
68158704.00	68180413.00			

18.3.32	32.00	68158544.00	68158544.75	68158624.00
68158624.00	68180585.00			
18.3.33	33.00	68158544.00	68158544.75	68158704.00
68158704.00	68180540.00			
18.3.34	34.00	68158544.00	68158544.00	68158704.00
68158704.00	68180712.00			
18.3.35	35.00	68158504.00	68158504.00	68158704.00
68158704.00	68180496.00			
18.3.36	36.00	68158544.00	68158545.75	68158704.00
68158704.00	68180511.00			
18.3.37	37.00	68158544.00	68158545.50	68158704.00
68158704.00	68180495.00			
18.3.38	38.00	68158544.00	68158544.00	68158704.00
68158705.00	68180592.00			
18.3.39	39.00	68158544.00	68158545.00	68158704.00
68158704.00	68180574.00			
18.3.40	40.00	68158624.00	68158624.00	68158704.00
68158704.00	68180670.00			
18.3.41	41.00	68158624.00	68158625.00	68158704.00
68158704.00	68180702.00			
18.3.42	42.00	68158624.00	68158624.00	68158644.00
68158644.00	68180630.00			
18.3.43	43.00	68158624.00	68158624.75	68158704.00
68158704.00	68180794.00			

18.3.44	44.00	68158624.00	68158624.00	68158704.00
68158704.00	68180701.00			
18.3.45	45.00	68158624.00	68158624.00	68158704.00
68158704.00	68180611.00			
18.3.46	46.00	68158624.00	68158624.00	68158704.00
68158704.00	68180710.00			
18.3.47	47.00	68158544.00	68158545.00	68158704.00
68158704.25	68180646.00			
18.3.48	48.00	68158624.00	68158628.00	68158704.00
68158704.00	68180748.00			
18.3.49	49.00	68158624.00	68158628.00	68158704.00
68158704.00	68180620.00			
18.3.50	50.00	68158624.00	68158628.75	68158644.00
68158644.00	68180667.00			
18.3.51	51.00	68158624.00	68158626.75	68158704.00
68158704.00	68180742.00			
18.3.52	52.00	68158625.00	68158628.50	68158704.00
68158704.00	68180676.00			
18.3.53	53.00	68158624.00	68158627.25	68158704.00
68158704.00	68180593.00			
18.3.54	54.00	68158624.00	68158627.75	68158704.00
68158704.00	68180761.00			
18.3.55	55.00	68158544.00	68158549.75	68158704.00
68158704.25	68180604.00			

18.3.56	56.00	68158545.00	68158551.25	68158624.00
68158624.00	68180555.00			
18.3.57	57.00	68158544.00	68158546.00	68158704.00
68158704.00	68180605.00			
18.3.58	58.00	68158545.00	68158547.75	68158704.00
68158704.00	68180637.00			
18.3.59	59.00	68158506.00	68158509.75	68158704.00
68158704.00	68180652.00			
18.3.60	60.00	68158544.00	68158548.75	68158704.00
68158704.00	68180568.00			
18.3.61	61.00	68158544.00	68158548.75	68158704.00
68158704.00	68180577.00			
18.3.62	62.00	68158544.00	68158547.75	68158704.00
68158705.00	68180589.00			
18.3.63	63.00	68158546.00	68158550.75	68158704.00
68158704.00	68180518.00			
18.3.64	64.00	68158624.00	68158626.75	68158704.00
68158704.00	68168753.00			
18.3.65	65.00	68158624.00	68158627.00	68158704.00
68158704.00	68168765.00			
18.3.66	66.00	68158624.00	68158625.00	68158644.00
68158644.00	68168820.00			
18.3.67	67.00	68158624.00	68158625.00	68158704.00
68158704.00	68168842.00			

18.3.68	68.00	68158624.00	68158626.75	68158704.00
68158704.00	68168833.00			
18.3.69	69.00	68158624.00	68158628.50	68158704.00
68158704.00	68168701.00			
18.3.70	70.00	68158624.00	68158625.50	68158704.00
68158704.00	68168853.00			
18.3.71	71.00	68158545.00	68158547.50	68158704.00
68158704.25	68168677.00			
18.3.72	72.00	68158544.00	68158548.50	68158624.00
68158624.00	68168744.00			
18.3.73	73.00	68158544.00	68158548.00	68158704.00
68158704.00	68168670.00			
18.3.74	74.00	68158544.00	68158545.00	68158704.00
68158704.00	68168634.00			
18.3.75	75.00	68158504.00	68158505.00	68158704.00
68158704.00	68168530.00			
18.3.76	76.00	68158544.00	68158546.75	68158704.00
68158704.00	68168660.00			
18.3.77	77.00	68158544.00	68158546.50	68158704.00
68158704.00	68168719.00			
18.3.78	78.00	68158544.00	68158544.75	68158704.00
68158705.00	68168754.00			
18.3.79	79.00	68158545.00	68158545.00	68158704.00
68158704.00	68168676.00			

18.3.80	80.00	68158544.00	68158545.00	68158624.00
68158624.00	68180490.00			
18.3.81	81.00	68158544.00	68158545.00	68158704.00
68158704.00	68180438.00			
18.3.82	82.00	68158544.00	68158544.00	68158704.00
68158704.00	68180464.00			
18.3.83	83.00	68158504.00	68158504.00	68158704.00
68158704.00	68180498.00			
18.3.84	84.00	68158544.00	68158544.75	68158704.00
68158704.00	68180699.00			
18.3.85	85.00	68158544.00	68158546.00	68158704.00
68158704.00	68180688.00			
18.3.86	86.00	68158544.00	68158544.75	68158704.00
68158705.00	68180665.00			
18.3.87	87.00	68158544.00	68158545.00	68158704.00
68158704.00	68180532.00			
18.3.88	88.00	68158624.00	68158624.75	68158704.00
68158704.00	68181034.00			
18.3.89	89.00	68158624.00	68158624.75	68158704.00
68158704.00	68180891.00			
18.3.90	90.00	68158624.00	68158625.00	68158644.00
68158644.00	68180706.00			
18.3.91	91.00	68158624.00	68158624.00	68158704.00
68158704.00	68180663.00			

18.3.92	92.00	68158624.00	68158626.00	68158704.00
68158704.00	68180383.00			
18.3.93	93.00	68158624.00	68158625.00	68158704.00
68158704.00	68180484.00			
18.3.94	94.00	68158624.00	68158625.00	68158704.00
68158704.00	68180675.00			
18.3.95	95.00	68158544.00	68158545.75	68158704.00
68158704.25	68180618.00			
18.3.96	96.00	68158625.00	68158627.00	68158704.00
68158704.00	68174794.00			
18.3.97	97.00	68158624.00	68158626.50	68158704.00
68158704.00	68174896.00			
18.3.98	98.00	68158624.00	68158628.25	68158644.00
68158644.00	68174914.00			
18.3.99	99.00	68158626.00	68158626.00	68158704.00
68158704.00	68175064.00			
18.3.100	100.00	68158624.00	68158624.75	68158704.00
68158704.00	68175010.00			
18.3.101	101.00	68158624.00	68158627.00	68158704.00
68158704.00	68174867.00			
18.3.102	102.00	68158624.00	68158626.75	68158704.00
68158704.00	68175009.00			
18.3.103	103.00	68158545.00	68158546.75	68158704.00
68158704.25	68174816.00			

18.3.104	104.00	68158545.00	68158549.00	68158624.00
68158624.00	68175052.00			
18.3.105	105.00	68158544.00	68158545.00	68158704.00
68158704.00	68175009.00			
18.3.106	106.00	68158545.00	68158546.75	68158704.00
68158704.00	68174858.00			
18.3.107	107.00	68158505.00	68158507.50	68158704.00
68158704.00	68174653.00			
18.3.108	108.00	68158544.00	68158547.00	68158704.00
68158704.00	68174700.00			
18.3.109	109.00	68158544.00	68158546.75	68158704.00
68158704.00	68174789.00			
18.3.110	110.00	68158544.00	68158547.50	68158704.00
68158705.00	68174863.00			
18.3.111	111.00	68158546.00	68158549.75	68158704.00
68158704.00	68174809.00			
18.3.112	112.00	68158545.00	68158548.50	68158624.00
68158624.00	68180555.00			
18.3.113	113.00	68158545.00	68158548.75	68158704.00
68158704.00	68180624.00			
18.3.114	114.00	68158544.00	68158547.75	68158704.00
68158704.00	68180575.00			
18.3.115	115.00	68158505.00	68158508.50	68158704.00
68158704.00	68180507.00			

18.3.116	116.00	68158545.00	68158549.00	68158704.00
68158704.00	68180652.00			
18.3.117	117.00	68158545.00	68158546.75	68158704.00
68158704.00	68180414.00			
18.3.118	118.00	68158544.00	68158545.00	68158704.00
68158705.00	68180748.00			
18.3.119	119.00	68158545.00	68158548.75	68158704.00
68158704.00	68180570.00			
18.3.120	120.00	68158625.00	68158627.75	68158704.00
68158704.00	68180629.00			
18.3.121	121.00	68158625.00	68158629.50	68158704.00
68158704.00	68180697.00			
18.3.122	122.00	68158624.00	68158627.75	68158644.00
68158644.00	68180782.00			
18.3.123	123.00	68158624.00	68158628.75	68158704.00
68158704.00	68180791.00			
18.3.124	124.00	68158625.00	68158627.00	68158704.00
68158704.00	68180604.00			
18.3.125	125.00	68158624.00	68158627.00	68158704.00
68158704.00	68180640.00			
18.3.126	126.00	68158624.00	68158628.00	68158704.00
68158704.00	68180692.00			
18.3.127	127.00	68158544.00	68158549.50	68158704.00
68158704.25	68180520.00			

18.4 L2 Requests (per normUnit)

Metric_ID	Channel	L2 Read	L2 Write	L2 Atomic
18.4.0	0.00	67110908.06	1050195.50	0.00
18.4.1	1.00	67110949.50	1050192.69	0.00
18.4.2	2.00	67110946.81	1050194.94	0.00
18.4.3	3.00	67110927.75	1050200.00	0.00
18.4.4	4.00	67110967.81	1050202.62	0.00
18.4.5	5.00	67110927.12	1050193.38	0.00
18.4.6	6.00	67110963.12	1050196.06	0.00
18.4.7	7.00	67110964.94	1050201.25	0.00
18.4.8	8.00	67110932.00	1050203.38	0.00
18.4.9	9.00	67110900.25	1050198.50	0.00
18.4.10	10.00	67110900.44	1050194.62	0.00
18.4.11	11.00	67110896.50	1050197.19	0.00
18.4.12	12.00	67110904.88	1050189.75	0.00
18.4.13	13.00	67110910.25	1050197.62	0.00
18.4.14	14.00	67110913.62	1050202.88	0.00
18.4.15	15.00	67110872.44	1050200.12	0.00
18.4.16	16.00	67110922.69	1050195.38	0.00
18.4.17	17.00	67110900.69	1050199.62	0.00
18.4.18	18.00	67110892.38	1050187.44	0.00
18.4.19	19.00	67110896.38	1050193.44	0.00
18.4.20	20.00	67110906.88	1050196.19	0.00
18.4.21	21.00	67110908.38	1050196.94	0.00

18.4.22	22.00	67110915.75	1050204.31	0.00
18.4.23	23.00	67110869.38	1050202.75	0.00
18.4.24	24.00	67110907.44	1050198.12	0.00
18.4.25	25.00	67110952.38	1050196.12	0.00
18.4.26	26.00	67110956.06	1050207.56	0.00
18.4.27	27.00	67110928.81	1050205.00	0.00
18.4.28	28.00	67110963.25	1050196.38	0.00
18.4.29	29.00	67110928.81	1050196.50	0.00
18.4.30	30.00	67110961.12	1050192.50	0.00
18.4.31	31.00	67110958.81	1050192.25	0.00
18.4.32	32.00	67110735.00	1049863.56	0.00
18.4.33	33.00	67110778.56	1049860.50	0.00
18.4.34	34.00	67110774.94	1049860.50	0.00
18.4.35	35.00	67110759.19	1049869.88	0.00
18.4.36	36.00	67110793.19	1049864.56	0.00
18.4.37	37.00	67110760.00	1049867.06	0.00
18.4.38	38.00	67110793.50	1049865.94	0.00
18.4.39	39.00	67110790.88	1049865.88	0.00
18.4.40	40.00	67110752.50	1049862.88	0.00
18.4.41	41.00	67110729.00	1049859.75	0.00
18.4.42	42.00	67110737.06	1049869.25	0.00
18.4.43	43.00	67110728.00	1049862.75	0.00
18.4.44	44.00	67110740.19	1049865.25	0.00
18.4.45	45.00	67110742.62	1049870.25	0.00

18.4.46	46.00	67110739.62	1049864.00	0.00
18.4.47	47.00	67110694.38	1049861.50	0.00
18.4.48	48.00	67110921.25	1050193.31	0.00
18.4.49	49.00	67110899.31	1050198.31	0.00
18.4.50	50.00	67110903.31	1050200.00	0.00
18.4.51	51.00	67110897.62	1050195.44	0.00
18.4.52	52.00	67110910.69	1050197.75	0.00
18.4.53	53.00	67110909.38	1050197.19	0.00
18.4.54	54.00	67110913.38	1050197.81	0.00
18.4.55	55.00	67110872.31	1050201.81	0.00
18.4.56	56.00	67110910.88	1050195.44	0.00
18.4.57	57.00	67110954.44	1050194.75	0.00
18.4.58	58.00	67110955.25	1050205.12	0.00
18.4.59	59.00	67110930.75	1050201.38	0.00
18.4.60	60.00	67110968.06	1050195.69	0.00
18.4.61	61.00	67110930.62	1050198.19	0.00
18.4.62	62.00	67110962.94	1050196.56	0.00
18.4.63	63.00	67110960.31	1050191.75	0.00
18.4.64	64.00	67110923.00	1050198.38	0.00
18.4.65	65.00	67110900.31	1050198.94	0.00
18.4.66	66.00	67110903.69	1050202.12	0.00
18.4.67	67.00	67110899.44	1050198.94	0.00
18.4.68	68.00	67110905.94	1050191.88	0.00
18.4.69	69.00	67110911.31	1050198.81	0.00

18.4.70	70.00	67110910.75	1050198.50	0.00
18.4.71	71.00	67110869.88	1050198.06	0.00
18.4.72	72.00	67110907.88	1050196.31	0.00
18.4.73	73.00	67110953.88	1050198.25	0.00
18.4.74	74.00	67110945.50	1050191.94	0.00
18.4.75	75.00	67110921.88	1050192.25	0.00
18.4.76	76.00	67110965.81	1050200.25	0.00
18.4.77	77.00	67110931.81	1050200.06	0.00
18.4.78	78.00	67110964.75	1050200.00	0.00
18.4.79	79.00	67110962.19	1050195.81	0.00
18.4.80	80.00	67110901.38	1050189.00	0.00
18.4.81	81.00	67110950.44	1050193.06	0.00
18.4.82	82.00	67110950.94	1050202.50	0.00
18.4.83	83.00	67110923.88	1050196.06	0.00
18.4.84	84.00	67110966.81	1050201.75	0.00
18.4.85	85.00	67110933.00	1050205.19	0.00
18.4.86	86.00	67110961.38	1050197.25	0.00
18.4.87	87.00	67110963.81	1050199.94	0.00
18.4.88	88.00	67110927.81	1050208.50	0.00
18.4.89	89.00	67110903.69	1050204.56	0.00
18.4.90	90.00	67110898.38	1050196.06	0.00
18.4.91	91.00	67110895.75	1050195.50	0.00
18.4.92	92.00	67110902.38	1050190.12	0.00
18.4.93	93.00	67110901.31	1050186.50	0.00

18.4.94	94.00	67110910.06	1050197.81	0.00
18.4.95	95.00	67110866.00	1050196.69	0.00
18.4.96	96.00	67110927.50	1050199.75	0.00
18.4.97	97.00	67110902.38	1050200.56	0.00
18.4.98	98.00	67110895.81	1050187.69	0.00
18.4.99	99.00	67110894.94	1050188.31	0.00
18.4.100	100.00	67110908.88	1050198.25	0.00
18.4.101	101.00	67110909.44	1050196.50	0.00
18.4.102	102.00	67110914.12	1050204.88	0.00
18.4.103	103.00	67110872.62	1050198.81	0.00
18.4.104	104.00	67110909.69	1050198.81	0.00
18.4.105	105.00	67110951.12	1050193.81	0.00
18.4.106	106.00	67110953.12	1050203.50	0.00
18.4.107	107.00	67110936.81	1050210.06	0.00
18.4.108	108.00	67110964.56	1050196.94	0.00
18.4.109	109.00	67110931.00	1050199.25	0.00
18.4.110	110.00	67110959.19	1050189.62	0.00
18.4.111	111.00	67110965.12	1050193.75	0.00
18.4.112	112.00	67110864.12	1050109.19	0.00
18.4.113	113.00	67110907.44	1050108.19	0.00
18.4.114	114.00	67110899.50	1050099.06	0.00
18.4.115	115.00	67110880.06	1050104.62	0.00
18.4.116	116.00	67110916.94	1050102.81	0.00
18.4.117	117.00	67110882.00	1050103.25	0.00

18.4.118	118.00	67110921.31	1050108.56	0.00
18.4.119	119.00	67110913.69	1050100.44	0.00
18.4.120	120.00	67110877.38	1050100.62	0.00
18.4.121	121.00	67110847.50	1050098.38	0.00
18.4.122	122.00	67110853.19	1050095.56	0.00
18.4.123	123.00	67110855.56	1050106.44	0.00
18.4.124	124.00	67110863.81	1050107.38	0.00
18.4.125	125.00	67110865.31	1050104.94	0.00
18.4.126	126.00	67110864.12	1050102.06	0.00
18.4.127	127.00	67110823.12	1050105.62	0.00

18.5 L2-Fabric Requests (per normUnit)

Metric_ID L2-Fabric Atomic	Channel	L2-Fabric Read	L2-Fabric Write and Atomic
18.5.0 0.00	0.00	35586868.38	263666.75
18.5.1 0.00	1.00	35586781.94	263666.94
18.5.2 0.00	2.00	35586826.75	263676.88
18.5.3 0.00	3.00	35586847.50	263666.12
18.5.4 0.00	4.00	35586877.50	263680.62

18.5.5 0.00	5.00	35586792.31	263666.62
18.5.6 0.00	6.00	35586834.69	263673.38
18.5.7 0.00	7.00	35586858.81	263662.81
18.5.8 0.00	8.00	35587020.19	263678.38
18.5.9 0.00	9.00	35586889.12	263660.38
18.5.10 0.00	10.00	35586945.94	263676.00
18.5.11 0.00	11.00	35586935.31	263663.44
18.5.12 0.00	12.00	35587000.38	263672.44
18.5.13 0.00	13.00	35586899.25	263658.69
18.5.14 0.00	14.00	35586953.00	263671.31
18.5.15 0.00	15.00	35586951.06	263665.38
18.5.16 0.00	16.00	35544418.75	263844.50

18.5.17 0.00	17.00	35544299.19	263833.50
18.5.18 0.00	18.00	35544321.25	263840.44
18.5.19 0.00	19.00	35544350.94	263824.69
18.5.20 0.00	20.00	35544399.69	263849.06
18.5.21 0.00	21.00	35544308.75	263831.75
18.5.22 0.00	22.00	35544347.88	263857.19
18.5.23 0.00	23.00	35544357.94	263841.12
18.5.24 0.00	24.00	35544380.06	263845.38
18.5.25 0.00	25.00	35544285.81	263833.25
18.5.26 0.00	26.00	35544312.44	263855.81
18.5.27 0.00	27.00	35544336.38	263838.88
18.5.28 0.00	28.00	35544392.38	263845.94

18.5.29 0.00	29.00	35544290.12	263835.38
18.5.30 0.00	30.00	35544309.19	263838.38
18.5.31 0.00	31.00	35544334.62	263830.50
18.5.32 0.00	32.00	36044296.19	264065.31
18.5.33 0.00	33.00	36044242.88	264052.81
18.5.34 0.00	34.00	36044236.25	264066.44
18.5.35 0.00	35.00	36044288.81	264049.25
18.5.36 0.00	36.00	36044315.69	264067.12
18.5.37 0.00	37.00	36044257.81	264043.06
18.5.38 0.00	38.00	36044257.06	264066.19
18.5.39 0.00	39.00	36044297.50	264041.19
18.5.40 0.00	40.00	36044081.56	264063.25

18.5.41 0.00	41.00	36043999.62	264049.81
18.5.42 0.00	42.00	36044029.31	264062.12
18.5.43 0.00	43.00	36044048.31	264045.50
18.5.44 0.00	44.00	36044066.12	264063.81
18.5.45 0.00	45.00	36044018.25	264045.94
18.5.46 0.00	46.00	36044018.38	264063.94
18.5.47 0.00	47.00	36044053.19	264058.94
18.5.48 0.00	48.00	35272115.88	263812.00
18.5.49 0.00	49.00	35272118.19	263817.75
18.5.50 0.00	50.00	35272110.00	263812.94
18.5.51 0.00	51.00	35272066.25	263810.56
18.5.52 0.00	52.00	35272106.81	263812.50

18.5.53 0.00	53.00	35272132.81	263814.88
18.5.54 0.00	54.00	35272126.44	263821.94
18.5.55 0.00	55.00	35272081.12	263821.31
18.5.56 0.00	56.00	35272208.62	263814.12
18.5.57 0.00	57.00	35272240.31	263817.62
18.5.58 0.00	58.00	35272249.38	263816.44
18.5.59 0.00	59.00	35272203.44	263821.69
18.5.60 0.00	60.00	35272225.62	263809.56
18.5.61 0.00	61.00	35272251.62	263818.88
18.5.62 0.00	62.00	35272241.06	263814.94
18.5.63 0.00	63.00	35272201.56	263815.75
18.5.64 0.00	64.00	35709372.56	263884.12

18.5.65 0.00	65.00	35709399.50	263887.81
18.5.66 0.00	66.00	35709359.06	263881.00
18.5.67 0.00	67.00	35709435.06	263884.62
18.5.68 0.00	68.00	35709359.06	263881.00
18.5.69 0.00	69.00	35709404.19	263888.06
18.5.70 0.00	70.00	35709358.38	263874.12
18.5.71 0.00	71.00	35709438.00	263882.06
18.5.72 0.00	72.00	35709546.38	263875.19
18.5.73 0.00	73.00	35709611.38	263889.00
18.5.74 0.00	74.00	35709547.06	263881.44
18.5.75 0.00	75.00	35709621.62	263887.56
18.5.76 0.00	76.00	35709553.25	263881.75

18.5.77 0.00	77.00	35709625.12	263885.62
18.5.78 0.00	78.00	35709564.69	263885.12
18.5.79 0.00	79.00	35709630.62	263887.50
18.5.80 0.00	80.00	35812180.06	263843.00
18.5.81 0.00	81.00	35812505.56	263855.75
18.5.82 0.00	82.00	35812209.69	263849.06
18.5.83 0.00	83.00	35812518.31	263859.19
18.5.84 0.00	84.00	35812202.50	263860.94
18.5.85 0.00	85.00	35812519.38	263869.94
18.5.86 0.00	86.00	35812215.69	263848.81
18.5.87 0.00	87.00	35812531.94	263857.69
18.5.88 0.00	88.00	35812227.50	263853.38

18.5.89 0.00	89.00	35812533.00	263872.62
18.5.90 0.00	90.00	35812219.69	263849.50
18.5.91 0.00	91.00	35812521.75	263863.44
18.5.92 0.00	92.00	35812195.06	263841.12
18.5.93 0.00	93.00	35812528.88	263860.69
18.5.94 0.00	94.00	35812232.88	263854.69
18.5.95 0.00	95.00	35812534.62	263860.25
18.5.96 0.00	96.00	35509464.44	263934.38
18.5.97 0.00	97.00	35509476.12	263931.75
18.5.98 0.00	98.00	35509456.75	263925.56
18.5.99 0.00	99.00	35509478.88	263916.31
18.5.100 0.00	100.00	35509442.38	263936.00

18.5.101 0.00	101.00	35509478.69	263930.81
18.5.102 0.00	102.00	35509476.25	263941.81
18.5.103 0.00	103.00	35509493.38	263925.62
18.5.104 0.00	104.00	35509421.44	263935.38
18.5.105 0.00	105.00	35509460.69	263925.81
18.5.106 0.00	106.00	35509454.56	263939.06
18.5.107 0.00	107.00	35509471.00	263935.56
18.5.108 0.00	108.00	35509432.00	263933.12
18.5.109 0.00	109.00	35509475.25	263931.50
18.5.110 0.00	110.00	35509466.00	263926.00
18.5.111 0.00	111.00	35509479.19	263921.38
18.5.112 0.00	112.00	35550310.56	263854.81

18.5.113 0.00	113.00	35550213.88	263847.44
18.5.114 0.00	114.00	35550360.56	263852.44
18.5.115 0.00	115.00	35550202.19	263847.88
18.5.116 0.00	116.00	35550320.19	263857.19
18.5.117 0.00	117.00	35550222.44	263840.38
18.5.118 0.00	118.00	35550379.25	263857.62
18.5.119 0.00	119.00	35550212.44	263841.69
18.5.120 0.00	120.00	35550426.44	263850.44
18.5.121 0.00	121.00	35550297.25	263844.19
18.5.122 0.00	122.00	35550442.38	263847.12
18.5.123 0.00	123.00	35550306.62	263843.69
18.5.124 0.00	124.00	35550411.50	263854.75

18.5.125 0.00	125.00	35550307.69	263833.69
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18.5.126 0.00	126.00	35550456.44	263852.00
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18.5.127 0.00	127.00	35550306.31	263848.31
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18.6 L2-Fabric Read Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.6.0	0.00	436.83	529.74	571.42	611.76	1102.77
18.6.1	1.00	443.50	538.50	580.47	625.01	1128.46
18.6.2	2.00	447.39	542.65	586.33	630.58	1143.52
18.6.3	3.00	447.74	542.82	586.79	631.50	1145.48
18.6.4	4.00	455.94	550.05	591.99	629.94	1131.62
18.6.5	5.00	447.64	542.42	584.99	628.75	1136.52
18.6.6	6.00	473.99	571.97	613.68	654.83	1181.64
18.6.7	7.00	459.67	556.10	601.71	646.26	1183.11
18.6.8	8.00	436.45	529.37	571.10	608.74	1089.19
18.6.9	9.00	447.48	542.93	586.01	629.04	1133.08
18.6.10	10.00	441.81	536.80	578.53	621.34	1111.77
18.6.11	11.00	447.48	542.79	587.62	631.64	1143.00
18.6.12	12.00	456.35	550.28	592.67	628.85	1123.67
18.6.13	13.00	447.14	541.94	585.30	628.60	1131.18
18.6.14	14.00	467.25	564.91	603.86	643.59	1143.25
18.6.15	15.00	452.48	548.59	593.42	638.00	1154.12

18.6.16	16.00	516.40	533.66	557.85	592.38	1006.21
18.6.17	17.00	527.14	545.11	572.64	600.58	1048.70
18.6.18	18.00	521.87	538.00	565.84	594.67	1027.00
18.6.19	19.00	526.86	546.66	573.90	601.78	1057.96
18.6.20	20.00	535.47	550.68	576.52	611.77	1039.83
18.6.21	21.00	526.42	543.40	571.46	599.46	1046.74
18.6.22	22.00	548.04	562.82	590.49	620.97	1057.97
18.6.23	23.00	532.05	550.77	579.27	606.70	1068.69
18.6.24	24.00	513.74	531.33	555.88	587.54	1018.20
18.6.25	25.00	522.94	539.60	567.70	595.20	1043.68
18.6.26	26.00	526.87	545.66	572.68	601.78	1057.61
18.6.27	27.00	526.87	545.36	573.10	601.48	1059.91
18.6.28	28.00	535.42	551.71	576.64	610.99	1047.12
18.6.29	29.00	526.89	543.45	571.45	599.94	1051.47
18.6.30	30.00	554.09	570.63	599.18	629.84	1095.63
18.6.31	31.00	538.55	559.93	587.05	615.42	1096.79
18.6.32	32.00	530.27	578.41	646.72	695.57	1123.59
18.6.33	33.00	542.56	591.93	663.51	717.47	1159.37
18.6.34	34.00	537.93	586.57	656.64	708.29	1144.92
18.6.35	35.00	543.20	592.40	665.03	719.68	1166.08
18.6.36	36.00	548.49	597.70	666.39	716.00	1155.05
18.6.37	37.00	542.02	591.15	662.76	716.64	1161.72
18.6.38	38.00	563.98	613.36	684.50	736.14	1189.87
18.6.39	39.00	548.11	596.89	667.93	719.60	1168.43

18.6.40	40.00	528.92	576.56	648.80	698.92	1130.03
18.6.41	41.00	538.35	585.71	658.43	714.74	1154.04
18.6.42	42.00	540.67	589.70	662.68	719.21	1163.67
18.6.43	43.00	541.30	590.91	664.26	720.62	1165.90
18.6.44	44.00	548.98	597.08	667.53	720.56	1159.12
18.6.45	45.00	543.29	591.93	664.57	721.10	1164.93
18.6.46	46.00	568.62	619.94	691.89	746.11	1204.28
18.6.47	47.00	554.13	604.97	680.11	739.17	1202.30
18.6.48	48.00	437.50	494.31	548.71	607.02	952.68
18.6.49	49.00	438.79	501.67	555.89	614.82	974.26
18.6.50	50.00	442.41	504.99	559.91	619.79	983.14
18.6.51	51.00	443.24	506.29	561.09	620.69	984.66
18.6.52	52.00	450.86	513.30	565.26	625.42	979.46
18.6.53	53.00	444.75	507.60	561.89	621.51	984.46
18.6.54	54.00	473.47	535.20	587.47	650.45	1021.95
18.6.55	55.00	458.29	519.85	575.64	636.41	1019.56
18.6.56	56.00	427.79	492.11	545.06	604.14	945.64
18.6.57	57.00	443.77	507.51	561.19	621.30	980.17
18.6.58	58.00	437.35	501.88	555.10	615.18	964.23
18.6.59	59.00	444.13	507.59	562.23	622.30	985.68
18.6.60	60.00	451.00	513.83	564.46	625.58	974.77
18.6.61	61.00	443.51	507.20	560.25	620.36	980.70
18.6.62	62.00	466.51	529.10	580.51	642.86	1006.28
18.6.63	63.00	448.87	512.42	565.08	625.79	985.19

18.6.64	64.00	386.75	454.74	595.19	620.33	1144.70
18.6.65	65.00	396.28	464.15	606.61	635.37	1169.92
18.6.66	66.00	395.85	458.18	608.14	639.00	1179.41
18.6.67	67.00	397.44	459.79	609.62	640.23	1181.75
18.6.68	68.00	402.01	462.11	609.76	637.20	1167.34
18.6.69	69.00	396.14	456.64	606.27	636.90	1172.03
18.6.70	70.00	422.56	486.89	633.77	663.11	1212.99
18.6.71	71.00	407.51	470.87	621.85	656.80	1211.40
18.6.72	72.00	380.09	439.15	588.16	612.19	1130.62
18.6.73	73.00	395.39	456.01	606.30	636.56	1168.99
18.6.74	74.00	390.29	450.16	599.26	624.86	1152.78
18.6.75	75.00	396.07	456.82	607.60	638.52	1175.79
18.6.76	76.00	401.52	461.95	608.90	634.19	1162.30
18.6.77	77.00	395.13	455.76	604.97	634.64	1168.65
18.6.78	78.00	417.49	479.54	626.78	655.06	1198.72
18.6.79	79.00	401.20	461.79	609.96	636.17	1174.97
18.6.80	80.00	505.21	532.51	554.58	587.65	986.62
18.6.81	81.00	524.09	552.86	570.61	601.41	1026.46
18.6.82	82.00	515.13	542.35	563.44	595.91	1006.49
18.6.83	83.00	525.13	553.88	571.51	602.01	1031.75
18.6.84	84.00	522.90	552.36	573.18	606.88	1019.08
18.6.85	85.00	521.55	550.45	569.25	600.59	1025.42
18.6.86	86.00	540.78	571.21	589.57	623.38	1054.06
18.6.87	87.00	524.22	553.39	573.45	606.63	1028.85

18.6.88	88.00	507.96	537.15	558.30	593.75	1000.73
18.6.89	89.00	521.61	551.81	570.15	605.44	1025.59
18.6.90	90.00	525.47	554.05	571.53	603.12	1033.73
18.6.91	91.00	526.15	555.64	573.05	604.49	1035.40
18.6.92	92.00	525.13	555.61	573.97	606.51	1023.31
18.6.93	93.00	523.17	553.21	570.22	601.43	1027.84
18.6.94	94.00	547.84	578.93	596.28	630.27	1067.82
18.6.95	95.00	539.54	566.25	584.40	614.91	1066.71
18.6.96	96.00	368.17	536.57	609.81	664.74	1026.15
18.6.97	97.00	381.13	546.06	623.66	684.94	1068.34
18.6.98	98.00	374.46	537.37	614.69	677.22	1047.65
18.6.99	99.00	380.18	543.68	623.04	688.83	1078.55
18.6.100	100.00	386.60	555.63	630.04	683.98	1058.19
18.6.101	101.00	379.91	545.60	623.54	684.13	1065.61
18.6.102	102.00	401.91	565.17	639.93	698.91	1076.90
18.6.103	103.00	386.20	548.92	628.27	694.13	1087.33
18.6.104	104.00	367.05	534.52	610.58	667.99	1041.76
18.6.105	105.00	379.34	543.69	621.93	682.45	1067.15
18.6.106	106.00	381.22	544.78	623.70	688.67	1082.21
18.6.107	107.00	382.22	545.99	625.02	690.41	1084.38
18.6.108	108.00	386.79	555.30	630.90	685.45	1066.53
18.6.109	109.00	380.11	545.59	624.09	684.39	1070.37
18.6.110	110.00	408.71	572.56	649.69	711.51	1115.67
18.6.111	111.00	393.12	556.65	636.95	703.82	1116.76

18.6.112	112.00	490.16	515.85	551.01	661.57	1059.79
18.6.113	113.00	501.51	522.42	560.34	673.73	1085.19
18.6.114	114.00	506.41	525.30	562.52	681.52	1098.63
18.6.115	115.00	507.44	526.55	563.46	682.01	1102.12
18.6.116	116.00	505.69	534.06	568.34	676.86	1083.83
18.6.117	117.00	503.18	524.40	562.26	675.71	1088.67
18.6.118	118.00	528.13	551.26	586.75	702.77	1131.51
18.6.119	119.00	518.52	536.54	574.82	694.51	1131.49
18.6.120	120.00	486.66	513.42	548.18	658.17	1045.73
18.6.121	121.00	504.73	525.20	562.86	677.49	1086.52
18.6.122	122.00	497.14	518.70	554.20	670.42	1067.68
18.6.123	123.00	506.59	524.55	561.94	680.78	1096.32
18.6.124	124.00	504.80	534.17	567.70	675.44	1076.86
18.6.125	125.00	503.21	524.14	561.98	675.13	1084.85
18.6.126	126.00	517.08	544.47	577.70	690.25	1097.22
18.6.127	127.00	511.00	529.46	566.51	683.62	1105.73

18.7 L2-Fabric Write and Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.7.0	0.00	156.26	172.27	211.67	242.11	501.77
18.7.1	1.00	160.81	177.58	212.07	246.89	511.97
18.7.2	2.00	162.50	178.61	215.82	254.33	526.86
18.7.3	3.00	164.06	181.20	218.14	252.93	524.67
18.7.4	4.00	166.00	183.54	223.50	247.19	511.37
18.7.5	5.00	163.57	180.99	214.00	248.16	513.53

18.7.6	6.00	175.72	192.98	224.74	256.57	526.20
18.7.7	7.00	169.35	186.85	221.35	255.04	533.31
18.7.8	8.00	156.47	174.56	217.08	241.30	508.01
18.7.9	9.00	162.93	180.95	214.22	249.16	522.73
18.7.10	10.00	159.19	175.22	208.76	243.28	507.06
18.7.11	11.00	163.20	180.36	214.77	248.83	522.25
18.7.12	12.00	164.47	182.60	224.15	248.29	515.72
18.7.13	13.00	162.10	179.49	213.45	247.69	520.91
18.7.14	14.00	171.41	188.10	221.53	255.99	524.84
18.7.15	15.00	164.13	181.20	218.02	252.87	529.41
18.7.16	16.00	150.98	171.08	208.68	232.50	495.73
18.7.17	17.00	157.43	178.17	208.25	241.28	512.07
18.7.18	18.00	153.47	171.30	202.86	233.79	494.94
18.7.19	19.00	158.88	176.63	209.17	241.32	508.49
18.7.20	20.00	159.94	180.38	216.07	238.56	505.77
18.7.21	21.00	157.51	178.23	207.86	238.97	507.19
18.7.22	22.00	166.58	184.95	215.00	245.18	511.90
18.7.23	23.00	159.75	177.53	210.74	244.16	516.66
18.7.24	24.00	150.95	170.30	200.15	231.64	492.08
18.7.25	25.00	155.68	175.50	213.56	236.78	498.98
18.7.26	26.00	157.01	177.35	208.44	241.71	516.01
18.7.27	27.00	158.02	175.38	209.41	243.04	513.16
18.7.28	28.00	159.39	179.56	209.33	237.88	499.78
18.7.29	29.00	157.21	177.85	206.86	238.42	503.86

18.7.30	30.00	169.22	187.82	217.40	246.47	515.85
18.7.31	31.00	162.70	181.01	213.84	245.55	519.89
18.7.32	32.00	151.33	170.45	207.84	252.91	516.37
18.7.33	33.00	157.92	177.35	215.70	261.88	537.65
18.7.34	34.00	153.46	173.25	212.93	258.51	521.97
18.7.35	35.00	159.15	178.89	218.58	264.17	535.58
18.7.36	36.00	160.40	179.72	215.52	259.72	519.47
18.7.37	37.00	158.33	177.54	215.12	259.70	526.89
18.7.38	38.00	167.05	187.28	222.40	264.46	527.04
18.7.39	39.00	159.94	179.36	215.12	257.84	517.82
18.7.40	40.00	149.81	170.81	208.98	248.57	500.02
18.7.41	41.00	154.95	173.93	211.04	254.82	513.54
18.7.42	42.00	155.23	175.20	212.40	255.34	514.93
18.7.43	43.00	157.50	176.65	213.42	256.38	515.45
18.7.44	44.00	159.57	178.75	215.08	257.09	518.80
18.7.45	45.00	157.81	177.31	214.63	258.81	522.88
18.7.46	46.00	169.02	189.96	227.68	272.00	541.22
18.7.47	47.00	163.18	183.14	222.77	267.76	542.61
18.7.48	48.00	148.47	168.59	195.77	226.75	489.88
18.7.49	49.00	153.89	173.54	201.42	232.74	498.94
18.7.50	50.00	154.09	174.55	202.30	233.10	500.69
18.7.51	51.00	157.43	177.12	204.85	234.16	499.71
18.7.52	52.00	159.18	179.40	205.71	234.63	503.84
18.7.53	53.00	157.43	177.82	205.17	235.28	507.48

18.7.54	54.00	168.67	190.13	217.29	247.84	526.53
18.7.55	55.00	163.05	183.55	212.86	245.46	528.55
18.7.56	56.00	150.17	170.17	198.18	230.47	503.53
18.7.57	57.00	156.77	177.07	205.68	239.05	522.55
18.7.58	58.00	152.70	172.80	202.82	235.97	506.21
18.7.59	59.00	157.28	177.24	207.47	241.57	520.11
18.7.60	60.00	158.12	178.72	205.00	236.24	509.17
18.7.61	61.00	155.85	176.26	204.11	236.80	512.84
18.7.62	62.00	165.11	186.03	212.11	240.99	514.41
18.7.63	63.00	157.51	177.62	204.83	233.82	503.44
18.7.64	64.00	153.57	173.37	227.97	246.42	486.84
18.7.65	65.00	159.00	178.30	236.52	252.20	496.25
18.7.66	66.00	159.73	179.62	213.97	254.10	498.36
18.7.67	67.00	162.44	182.56	215.17	254.16	500.03
18.7.68	68.00	163.53	183.11	216.04	254.14	500.08
18.7.69	69.00	161.58	181.30	215.55	255.10	503.65
18.7.70	70.00	173.48	194.16	228.25	268.91	524.24
18.7.71	71.00	167.09	187.56	223.14	265.57	526.43
18.7.72	72.00	154.28	173.58	209.73	249.52	500.08
18.7.73	73.00	160.83	180.77	216.95	258.73	516.99
18.7.74	74.00	156.83	176.35	213.33	256.02	507.20
18.7.75	75.00	161.16	181.49	218.85	261.72	522.29
18.7.76	76.00	162.32	181.95	215.46	254.23	500.94
18.7.77	77.00	160.01	179.85	215.31	256.49	505.97

18.7.78	78.00	169.50	189.56	221.74	260.73	509.42
18.7.79	79.00	161.75	181.51	215.32	254.05	500.51
18.7.80	80.00	147.84	167.34	201.00	233.98	497.24
18.7.81	81.00	154.02	174.04	206.60	242.63	512.18
18.7.82	82.00	149.90	170.10	204.65	239.93	506.70
18.7.83	83.00	155.28	175.37	209.21	246.15	519.16
18.7.84	84.00	156.92	176.72	208.89	239.46	499.34
18.7.85	85.00	154.37	174.37	205.80	241.28	507.00
18.7.86	86.00	163.55	184.24	215.55	245.70	509.24
18.7.87	87.00	155.97	175.71	205.67	239.60	499.21
18.7.88	88.00	147.30	169.43	208.81	233.36	486.96
18.7.89	89.00	152.37	171.78	215.68	239.25	498.45
18.7.90	90.00	152.74	173.38	206.26	239.25	497.26
18.7.91	91.00	154.67	174.41	204.96	240.28	499.02
18.7.92	92.00	156.14	175.84	205.26	239.83	497.88
18.7.93	93.00	153.97	173.67	204.71	239.72	499.97
18.7.94	94.00	165.45	186.92	219.95	252.77	522.96
18.7.95	95.00	158.98	179.26	212.56	249.99	525.25
18.7.96	96.00	155.18	175.34	206.27	243.50	508.38
18.7.97	97.00	161.52	180.56	214.84	251.96	526.26
18.7.98	98.00	157.71	175.89	209.12	245.40	504.53
18.7.99	99.00	163.22	182.19	215.54	251.72	522.54
18.7.100	100.00	164.47	184.68	214.82	250.00	515.46
18.7.101	101.00	161.91	180.58	214.36	250.68	522.30

18.7.102	102.00	171.14	189.94	222.33	258.61	524.48
18.7.103	103.00	164.18	182.87	217.72	256.23	532.16
18.7.104	104.00	155.13	174.67	209.59	244.50	502.79
18.7.105	105.00	160.01	178.79	213.08	249.81	513.34
18.7.106	106.00	161.14	179.40	216.33	257.32	526.28
18.7.107	107.00	162.70	180.98	218.04	256.81	529.34
18.7.108	108.00	164.06	183.99	216.43	249.76	510.10
18.7.109	109.00	161.46	180.47	213.61	250.70	515.50
18.7.110	110.00	173.97	193.00	224.80	259.99	526.02
18.7.111	111.00	167.20	185.99	220.86	257.47	532.26
18.7.112	112.00	149.19	172.37	204.92	228.79	492.61
18.7.113	113.00	154.00	176.92	213.29	234.99	505.08
18.7.114	114.00	154.86	174.87	208.01	240.68	511.97
18.7.115	115.00	157.31	177.44	210.11	240.77	516.24
18.7.116	116.00	159.03	182.62	215.09	236.01	501.30
18.7.117	117.00	156.51	179.96	214.58	235.47	504.58
18.7.118	118.00	168.44	189.16	217.34	245.06	514.72
18.7.119	119.00	161.78	182.23	213.06	243.20	519.94
18.7.120	120.00	149.52	173.49	208.45	230.21	497.39
18.7.121	121.00	155.76	179.82	213.31	239.12	513.52
18.7.122	122.00	151.76	171.45	201.53	232.28	494.34
18.7.123	123.00	156.08	176.32	207.60	238.64	510.97
18.7.124	124.00	157.57	181.30	214.78	235.57	508.73
18.7.125	125.00	155.01	178.68	212.83	236.43	511.80

18.7.126	126.00	164.06	184.12	213.80	242.80	510.77
18.7.127	127.00	156.91	177.10	209.88	241.35	519.83

18.8 L2-Fabric Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.8.0	0.00	0.00	0.00	0.00	0.00	0.00
18.8.1	1.00	0.00	0.00	0.00	0.00	0.00
18.8.2	2.00	0.00	0.00	0.00	0.00	0.00
18.8.3	3.00	0.00	0.00	0.00	0.00	0.00
18.8.4	4.00	0.00	0.00	0.00	0.00	0.00
18.8.5	5.00	0.00	0.00	0.00	0.00	0.00
18.8.6	6.00	0.00	0.00	0.00	0.00	0.00
18.8.7	7.00	0.00	0.00	0.00	0.00	0.00
18.8.8	8.00	0.00	0.00	0.00	0.00	0.00
18.8.9	9.00	0.00	0.00	0.00	0.00	0.00
18.8.10	10.00	0.00	0.00	0.00	0.00	0.00
18.8.11	11.00	0.00	0.00	0.00	0.00	0.00
18.8.12	12.00	0.00	0.00	0.00	0.00	0.00
18.8.13	13.00	0.00	0.00	0.00	0.00	0.00
18.8.14	14.00	0.00	0.00	0.00	0.00	0.00
18.8.15	15.00	0.00	0.00	0.00	0.00	0.00
18.8.16	16.00	0.00	0.00	0.00	0.00	0.00
18.8.17	17.00	0.00	0.00	0.00	0.00	0.00
18.8.18	18.00	0.00	0.00	0.00	0.00	0.00
18.8.19	19.00	0.00	0.00	0.00	0.00	0.00

18.8.20	20.00	0.00	0.00	0.00	0.00	0.00
18.8.21	21.00	0.00	0.00	0.00	0.00	0.00
18.8.22	22.00	0.00	0.00	0.00	0.00	0.00
18.8.23	23.00	0.00	0.00	0.00	0.00	0.00
18.8.24	24.00	0.00	0.00	0.00	0.00	0.00
18.8.25	25.00	0.00	0.00	0.00	0.00	0.00
18.8.26	26.00	0.00	0.00	0.00	0.00	0.00
18.8.27	27.00	0.00	0.00	0.00	0.00	0.00
18.8.28	28.00	0.00	0.00	0.00	0.00	0.00
18.8.29	29.00	0.00	0.00	0.00	0.00	0.00
18.8.30	30.00	0.00	0.00	0.00	0.00	0.00
18.8.31	31.00	0.00	0.00	0.00	0.00	0.00
18.8.32	32.00	0.00	0.00	0.00	0.00	0.00
18.8.33	33.00	0.00	0.00	0.00	0.00	0.00
18.8.34	34.00	0.00	0.00	0.00	0.00	0.00
18.8.35	35.00	0.00	0.00	0.00	0.00	0.00
18.8.36	36.00	0.00	0.00	0.00	0.00	0.00
18.8.37	37.00	0.00	0.00	0.00	0.00	0.00
18.8.38	38.00	0.00	0.00	0.00	0.00	0.00
18.8.39	39.00	0.00	0.00	0.00	0.00	0.00
18.8.40	40.00	0.00	0.00	0.00	0.00	0.00
18.8.41	41.00	0.00	0.00	0.00	0.00	0.00
18.8.42	42.00	0.00	0.00	0.00	0.00	0.00
18.8.43	43.00	0.00	0.00	0.00	0.00	0.00

18.8.44	44.00	0.00	0.00	0.00	0.00	0.00
18.8.45	45.00	0.00	0.00	0.00	0.00	0.00
18.8.46	46.00	0.00	0.00	0.00	0.00	0.00
18.8.47	47.00	0.00	0.00	0.00	0.00	0.00
18.8.48	48.00	0.00	0.00	0.00	0.00	0.00
18.8.49	49.00	0.00	0.00	0.00	0.00	0.00
18.8.50	50.00	0.00	0.00	0.00	0.00	0.00
18.8.51	51.00	0.00	0.00	0.00	0.00	0.00
18.8.52	52.00	0.00	0.00	0.00	0.00	0.00
18.8.53	53.00	0.00	0.00	0.00	0.00	0.00
18.8.54	54.00	0.00	0.00	0.00	0.00	0.00
18.8.55	55.00	0.00	0.00	0.00	0.00	0.00
18.8.56	56.00	0.00	0.00	0.00	0.00	0.00
18.8.57	57.00	0.00	0.00	0.00	0.00	0.00
18.8.58	58.00	0.00	0.00	0.00	0.00	0.00
18.8.59	59.00	0.00	0.00	0.00	0.00	0.00
18.8.60	60.00	0.00	0.00	0.00	0.00	0.00
18.8.61	61.00	0.00	0.00	0.00	0.00	0.00
18.8.62	62.00	0.00	0.00	0.00	0.00	0.00
18.8.63	63.00	0.00	0.00	0.00	0.00	0.00
18.8.64	64.00	0.00	0.00	0.00	0.00	0.00
18.8.65	65.00	0.00	0.00	0.00	0.00	0.00
18.8.66	66.00	0.00	0.00	0.00	0.00	0.00
18.8.67	67.00	0.00	0.00	0.00	0.00	0.00

18.8.68	68.00	0.00	0.00	0.00	0.00	0.00
18.8.69	69.00	0.00	0.00	0.00	0.00	0.00
18.8.70	70.00	0.00	0.00	0.00	0.00	0.00
18.8.71	71.00	0.00	0.00	0.00	0.00	0.00
18.8.72	72.00	0.00	0.00	0.00	0.00	0.00
18.8.73	73.00	0.00	0.00	0.00	0.00	0.00
18.8.74	74.00	0.00	0.00	0.00	0.00	0.00
18.8.75	75.00	0.00	0.00	0.00	0.00	0.00
18.8.76	76.00	0.00	0.00	0.00	0.00	0.00
18.8.77	77.00	0.00	0.00	0.00	0.00	0.00
18.8.78	78.00	0.00	0.00	0.00	0.00	0.00
18.8.79	79.00	0.00	0.00	0.00	0.00	0.00
18.8.80	80.00	0.00	0.00	0.00	0.00	0.00
18.8.81	81.00	0.00	0.00	0.00	0.00	0.00
18.8.82	82.00	0.00	0.00	0.00	0.00	0.00
18.8.83	83.00	0.00	0.00	0.00	0.00	0.00
18.8.84	84.00	0.00	0.00	0.00	0.00	0.00
18.8.85	85.00	0.00	0.00	0.00	0.00	0.00
18.8.86	86.00	0.00	0.00	0.00	0.00	0.00
18.8.87	87.00	0.00	0.00	0.00	0.00	0.00
18.8.88	88.00	0.00	0.00	0.00	0.00	0.00
18.8.89	89.00	0.00	0.00	0.00	0.00	0.00
18.8.90	90.00	0.00	0.00	0.00	0.00	0.00
18.8.91	91.00	0.00	0.00	0.00	0.00	0.00

18.8.92	92.00	0.00	0.00	0.00	0.00	0.00
18.8.93	93.00	0.00	0.00	0.00	0.00	0.00
18.8.94	94.00	0.00	0.00	0.00	0.00	0.00
18.8.95	95.00	0.00	0.00	0.00	0.00	0.00
18.8.96	96.00	0.00	0.00	0.00	0.00	0.00
18.8.97	97.00	0.00	0.00	0.00	0.00	0.00
18.8.98	98.00	0.00	0.00	0.00	0.00	0.00
18.8.99	99.00	0.00	0.00	0.00	0.00	0.00
18.8.100	100.00	0.00	0.00	0.00	0.00	0.00
18.8.101	101.00	0.00	0.00	0.00	0.00	0.00
18.8.102	102.00	0.00	0.00	0.00	0.00	0.00
18.8.103	103.00	0.00	0.00	0.00	0.00	0.00
18.8.104	104.00	0.00	0.00	0.00	0.00	0.00
18.8.105	105.00	0.00	0.00	0.00	0.00	0.00
18.8.106	106.00	0.00	0.00	0.00	0.00	0.00
18.8.107	107.00	0.00	0.00	0.00	0.00	0.00
18.8.108	108.00	0.00	0.00	0.00	0.00	0.00
18.8.109	109.00	0.00	0.00	0.00	0.00	0.00
18.8.110	110.00	0.00	0.00	0.00	0.00	0.00
18.8.111	111.00	0.00	0.00	0.00	0.00	0.00
18.8.112	112.00	0.00	0.00	0.00	0.00	0.00
18.8.113	113.00	0.00	0.00	0.00	0.00	0.00
18.8.114	114.00	0.00	0.00	0.00	0.00	0.00
18.8.115	115.00	0.00	0.00	0.00	0.00	0.00

18.8.116	116.00	0.00	0.00	0.00	0.00	0.00
18.8.117	117.00	0.00	0.00	0.00	0.00	0.00
18.8.118	118.00	0.00	0.00	0.00	0.00	0.00
18.8.119	119.00	0.00	0.00	0.00	0.00	0.00
18.8.120	120.00	0.00	0.00	0.00	0.00	0.00
18.8.121	121.00	0.00	0.00	0.00	0.00	0.00
18.8.122	122.00	0.00	0.00	0.00	0.00	0.00
18.8.123	123.00	0.00	0.00	0.00	0.00	0.00
18.8.124	124.00	0.00	0.00	0.00	0.00	0.00
18.8.125	125.00	0.00	0.00	0.00	0.00	0.00
18.8.126	126.00	0.00	0.00	0.00	0.00	0.00
18.8.127	127.00	0.00	0.00	0.00	0.00	0.00

18.12 L2-Fabric (128B read requests per normUnit)

Metric_ID	Channel	Min	Q1	Median
Q3	Max			
18.12.0	0.00	34563072.00	34563177.25	35214257.00
37187905.25	38267843.00			
18.12.1	1.00	34563072.00	34563188.75	35214476.00
37186968.50	38268108.00			
18.12.2	2.00	34563072.00	34563175.50	35214430.50
37187754.75	38267612.00			
18.12.3	3.00	34563072.00	34563185.00	35214471.00
37187126.25	38267764.00			
18.12.4	4.00	34563072.00	34563183.50	35214258.00

37187921.25	38267905.00			
18.12.5	5.00	34563072.00	34563179.00	35214446.00
37187004.50	38268142.00			
18.12.6	6.00	34563072.00	34563187.75	35214440.00
37187746.25	38267616.00			
18.12.7	7.00	34563072.00	34563181.00	35214470.50
37187111.00	38267723.00			
18.12.8	8.00	34563072.00	34563182.00	35214160.00
37188127.50	38267727.00			
18.12.9	9.00	34563072.00	34563179.00	35214427.50
37187191.75	38268054.00			
18.12.10	10.00	34563072.00	34563185.75	35214427.50
37187952.00	38267580.00			
18.12.11	11.00	34563072.00	34563181.00	35214413.00
37187287.75	38267659.00			
18.12.12	12.00	34563072.00	34563177.25	35214180.00
37188121.25	38267723.00			
18.12.13	13.00	34563072.00	34563187.25	35214463.50
37187190.50	38268061.00			
18.12.14	14.00	34563072.00	34563175.50	35214429.00
37187947.25	38267593.00			
18.12.15	15.00	34563072.00	34563185.00	35214407.50
37187311.75	38267696.00			
18.12.16	16.00	34563181.00	34564027.25	35407754.00

37009594.00 39098741.00

18.12.17 17.00 34563185.00 34564036.75 35407872.00
37009266.25 39096395.00

18.12.18 18.00 34563176.00 34564028.25 35407821.50
37009498.00 39098523.00

18.12.19 19.00 34563180.00 34564025.00 35407965.50
37009327.00 39096852.00

18.12.20 20.00 34563183.00 34564042.50 35407770.50
37009577.25 39098749.00

18.12.21 21.00 34563192.00 34564038.00 35407895.00
37009271.25 39096387.00

18.12.22 22.00 34563182.00 34564048.50 35407814.00
37009538.50 39098510.00

18.12.23 23.00 34563189.00 34564045.00 35407985.00
37009326.50 39096822.00

18.12.24 24.00 34563183.00 34564045.50 35407796.00
37009548.25 39098472.00

18.12.25 25.00 34563192.00 34564038.50 35407834.50
37009279.50 39095954.00

18.12.26 26.00 34563182.00 34564048.00 35407795.50
37009465.75 39098159.00

18.12.27 27.00 34563188.00 34564045.50 35407924.00
37009361.00 39096481.00

18.12.28 28.00 34563181.00 34564029.00 35407790.50

37009566.75	39098430.00			
18.12.29	29.00	34563185.00	34564035.00	35407828.00
37009281.75	39095984.00			
18.12.30	30.00	34563176.00	34564029.25	35407798.50
37009442.25	39098126.00			
18.12.31	31.00	34563180.00	34564027.00	35407919.50
37009371.25	39096529.00			
18.12.32	32.00	34563072.00	34563142.75	35848745.50
37754531.25	39912993.00			
18.12.33	33.00	34563072.00	34563130.75	35849116.00
37754162.50	39913519.00			
18.12.34	34.00	34563072.00	34563145.75	35848882.00
37754351.25	39913560.00			
18.12.35	35.00	34563072.00	34563130.75	35849122.00
37754242.25	39913262.00			
18.12.36	36.00	34563072.00	34563132.25	35848729.00
37754544.75	39912934.00			
18.12.37	37.00	34563072.00	34563135.25	35849095.00
37754130.00	39913472.00			
18.12.38	38.00	34563072.00	34563132.50	35848867.00
37754336.75	39913518.00			
18.12.39	39.00	34563072.00	34563131.00	35849119.00
37754217.25	39913250.00			
18.12.40	40.00	34563072.00	34563133.00	35848816.00

37755831.25	39913410.00			
18.12.41	41.00	34563072.00	34563135.25	35849151.50
37755429.50	39913885.00			
18.12.42	42.00	34563072.00	34563132.50	35848987.00
37755607.00	39913890.00			
18.12.43	43.00	34563072.00	34563130.25	35849244.00
37755538.00	39913523.00			
18.12.44	44.00	34563072.00	34563142.00	35848801.00
37755823.50	39913428.00			
18.12.45	45.00	34563072.00	34563130.00	35849169.00
37755421.25	39913852.00			
18.12.46	46.00	34563072.00	34563145.00	35848994.00
37755678.00	39913869.00			
18.12.47	47.00	34563072.00	34563128.50	35849248.00
37755513.25	39913550.00			
18.12.48	48.00	34563186.00	34565575.25	34992911.50
36535371.00	38126064.00			
18.12.49	49.00	34563183.00	34565558.50	34992828.50
36535095.50	38126320.00			
18.12.50	50.00	34563190.00	34565553.00	34992941.50
36535484.25	38126320.00			
18.12.51	51.00	34563180.00	34565557.25	34992710.50
36535071.00	38126545.00			
18.12.52	52.00	34563199.00	34565576.50	34992972.00

36535370.25	38126071.00			
18.12.53	53.00	34563181.00	34565560.00	34992847.00
36535107.25	38126326.00			
18.12.54	54.00	34563195.00	34565571.25	34992961.00
36535473.25	38126339.00			
18.12.55	55.00	34563183.00	34565560.00	34992737.50
36535103.00	38126572.00			
18.12.56	56.00	34563199.00	34565574.75	34993006.50
36535465.75	38126517.00			
18.12.57	57.00	34563181.00	34565560.25	34992837.50
36535189.75	38126719.00			
18.12.58	58.00	34563195.00	34565571.25	34992975.00
36535546.50	38126836.00			
18.12.59	59.00	34563185.00	34565557.25	34992751.00
36535153.25	38126867.00			
18.12.60	60.00	34563185.00	34565575.50	34992970.50
36535465.00	38126509.00			
18.12.61	61.00	34563183.00	34565558.25	34992815.50
36535200.25	38126735.00			
18.12.62	62.00	34563190.00	34565551.75	34992956.50
36535565.25	38126846.00			
18.12.63	63.00	34563180.00	34565554.75	34992728.50
36535128.75	38126826.00			
18.12.64	64.00	34563072.00	34563113.25	35072859.00

37619723.50	38219533.00			
18.12.65	65.00	34563072.00	34563115.75	35072561.50
37619004.00	38219817.00			
18.12.66	66.00	34563072.00	34563114.50	35072897.00
37619946.50	38220088.00			
18.12.67	67.00	34563072.00	34563122.25	35072647.00
37618924.75	38219668.00			
18.12.68	68.00	34563072.00	34563123.00	35072860.00
37619723.75	38219521.00			
18.12.69	69.00	34563072.00	34563119.00	35072562.50
37619052.00	38219795.00			
18.12.70	70.00	34563072.00	34563122.50	35072922.50
37619904.25	38220060.00			
18.12.71	71.00	34563072.00	34563118.75	35072642.00
37618926.75	38219652.00			
18.12.72	72.00	34563072.00	34563123.00	35072779.50
37619405.00	38219295.00			
18.12.73	73.00	34563072.00	34563119.75	35072465.50
37618616.25	38219509.00			
18.12.74	74.00	34563072.00	34563122.50	35072803.00
37619613.75	38219825.00			
18.12.75	75.00	34563072.00	34563120.25	35072545.50
37618633.75	38219267.00			
18.12.76	76.00	34563072.00	34563114.00	35072779.00

37619414.25	38219279.00			
18.12.77	77.00	34563072.00	34563115.00	35072466.50
37618604.75	38219496.00			
18.12.78	78.00	34563072.00	34563114.50	35072802.50
37619612.00	38219839.00			
18.12.79	79.00	34563072.00	34563122.25	35072546.50
37618622.00	38219279.00			
18.12.80	80.00	34563127.00	34563972.00	35407336.50
37605160.50	38067820.00			
18.12.81	81.00	34563112.00	34563995.25	35407717.50
37605302.00	38066958.00			
18.12.82	82.00	34563128.00	34563968.25	35407433.50
37605017.25	38067562.00			
18.12.83	83.00	34563113.00	34563991.25	35407564.00
37605386.50	38066507.00			
18.12.84	84.00	34563123.00	34563970.25	35407333.50
37605136.00	38067779.00			
18.12.85	85.00	34563116.00	34563969.75	35407735.50
37605240.00	38066943.00			
18.12.86	86.00	34563126.00	34563973.50	35407442.50
37604981.25	38067536.00			
18.12.87	87.00	34563116.00	34563970.00	35407534.50
37605376.75	38066474.00			
18.12.88	88.00	34563120.00	34563971.75	35407320.00

37605561.75	38068234.00			
18.12.89	89.00	34563116.00	34563969.75	35407717.50
37605771.25	38067381.00			
18.12.90	90.00	34563128.00	34563972.25	35407400.00
37605473.75	38067890.00			
18.12.91	91.00	34563116.00	34563969.50	35407522.50
37605816.75	38066966.00			
18.12.92	92.00	34563130.00	34563973.75	35407298.00
37605580.50	38068245.00			
18.12.93	93.00	34563112.00	34563994.50	35407700.00
37605801.75	38067414.00			
18.12.94	94.00	34563128.00	34563970.00	35407407.50
37605420.75	38067871.00			
18.12.95	95.00	34563113.00	34563991.50	35407542.50
37605820.75	38066914.00			
18.12.96	96.00	34563072.00	34563137.00	35145902.50
36700427.50	39440245.00			
18.12.97	97.00	34563072.00	34563139.00	35145825.00
36700134.00	39439590.00			
18.12.98	98.00	34563072.00	34563125.25	35145779.00
36700290.50	39440513.00			
18.12.99	99.00	34563072.00	34563139.50	35145888.50
36699977.00	39439687.00			
18.12.100	100.00	34563072.00	34563128.75	35145901.50

36700439.75	39440270.00			
18.12.101	101.00	34563072.00	34563146.00	35145840.00
36700127.00	39439616.00			
18.12.102	102.00	34563072.00	34563126.50	35145802.50
36700283.75	39440543.00			
18.12.103	103.00	34563072.00	34563139.75	35145903.50
36700005.00	39439675.00			
18.12.104	104.00	34563072.00	34563128.75	35145820.50
36700582.00	39441508.00			
18.12.105	105.00	34563072.00	34563146.00	35145742.50
36700249.50	39440911.00			
18.12.106	106.00	34563072.00	34563126.50	35145750.00
36700411.50	39441679.00			
18.12.107	107.00	34563072.00	34563139.75	35145798.00
36700123.75	39440860.00			
18.12.108	108.00	34563072.00	34563137.25	35145828.50
36700569.00	39441468.00			
18.12.109	109.00	34563072.00	34563139.00	35145731.50
36700259.50	39440924.00			
18.12.110	110.00	34563072.00	34563125.25	35145732.50
36700395.50	39441648.00			
18.12.111	111.00	34563072.00	34563141.00	35145785.50
36700131.25	39440909.00			
18.12.112	112.00	34563235.00	34568709.00	35268662.00

36792496.25	38350592.00			
18.12.113	113.00	34563224.00	34568698.50	35268623.50
36792753.50	38350468.00			
18.12.114	114.00	34563231.00	34568703.00	35268712.50
36792424.50	38349913.00			
18.12.115	115.00	34563230.00	34568700.00	35268472.00
36792377.25	38350063.00			
18.12.116	116.00	34563222.00	34568686.50	35268633.00
36792497.50	38350591.00			
18.12.117	117.00	34563234.00	34568675.25	35268602.50
36792749.50	38350459.00			
18.12.118	118.00	34563221.00	34568678.25	35268696.00
36792402.75	38349931.00			
18.12.119	119.00	34563240.00	34568671.25	35268456.00
36792387.25	38350039.00			
18.12.120	120.00	34563222.00	34568687.25	35268170.00
36792357.00	38350473.00			
18.12.121	121.00	34563234.00	34568677.25	35268158.50
36792522.50	38350372.00			
18.12.122	122.00	34563221.00	34568679.75	35268199.50
36792240.75	38349656.00			
18.12.123	123.00	34563240.00	34568671.75	35268024.00
36792205.00	38350099.00			
18.12.124	124.00	34563235.00	34568709.25	35268214.00

36792354.75 38350486.00

18.12.125	125.00	34563224.00	34568698.25	35268180.50
36792532.25	38350334.00			

18.12.126	126.00	34563232.00	34568704.25	35268209.50
36792241.25	38349696.00			

18.12.127	127.00	34563229.00	34568700.75	35268037.50
36792207.00	38350148.00			

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